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Class 12 Physics Chapterwise PYQs

2026 – 2003 | All CBSE Board Papers

Chapterwise previous year questions, sorted by marks and year

Chapter 7: Alternating Current

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HOW TO USE THIS BOOKLET

1. **Attempt a section** on your own. Questions run newest year first.
2. **Check the Answer Key,** then follow a question number to its solution.
3. **Study the solution.** Note each *Common Mistake,* then return via "Back to Q n".

SECTION A

1-Mark Questions

Multiple choice · 58 questions

> *Attempt these first, then check the Answer Key and open any solution.*

1. In a series LCR circuit, the voltage across the resistor, capacitor and inductor is 10 V each. If the capacitor is short circuited, the voltage across the inductor will be:

(A) 10 V

(B) $5\sqrt{2}$ V

(C) $\frac{5}{\sqrt{2}}$ V

(D) $10\sqrt{2}$ V

Check Solution → [2026 • Set 55-1-1]

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2. A resistor and an inductor of negligible resistance are connected in series to a 20 V ac source. If the voltage across the resistor is 12 V, the voltage across the inductor will be:

(A) 6 V

(B) 8 V

(C) 10 V

(D) 16 V

Check Solution → [2026 • Set 55-1-2]

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3. A series LCR circuit with $R = 3\ \Omega$, $X_L = 4\ \Omega$, $X_C = 8\ \Omega$ is connected to a 220 V, 50 Hz ac source. The power factor for the circuit is:

(A) 0.30

(B) 0.45

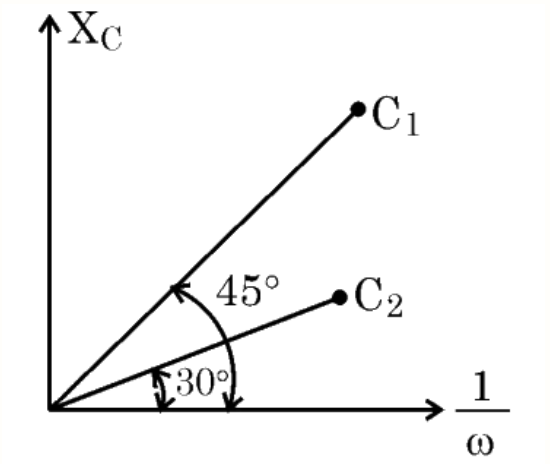
(C) 0.50

(D) 0.60

Check Solution → [2026 • Set 55-1-3]

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4. The figure shows the variation of capacitive reactance (X_C) of two ideal capacitors of capacitances C_1 and C_2 with the reciprocal of angular frequency ($1/\omega$) of ac source. The value of C_1/C_2 is:



(A) 5

(B) 2

(C) $\sqrt{3}$

(D) $\frac{1}{\sqrt{3}}$

Check Solution → [2026 • Set 55-2-1]

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5. An ac voltage is given as $v = 14 \sin(314\ t)$ V. The average and the effective value of the voltage (in V) over a cycle are respectively:

(A) 14 and 7

(B) 10 and 14

(C) 0 and 10

(D) 10 and 0

Check Solution → [2026 • Set 55-4-1]

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6. The rms and the average value of an ac voltage $V = V_0 \sin \omega\ t$ volt over a cycle respectively will be:

(A) $\frac{V_0}{\sqrt{2}}, \frac{V_0}{2}$

(B) $\frac{V_0}{\pi}, \frac{V_0}{2}$

(C) $\frac{V_0}{\sqrt{2}}, 0$

(D) $V_0, \frac{V_0}{2}$

Check Solution → [2026 • Set 55-5-1]

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7. Which of the following statements is *not* true for electric energy in ac form compared to that in dc form?

(A) Production of ac is economical.

(B) ac can be easily and efficiently converted from one voltage to the other.

(C) ac can be transmitted economically over long distances.

(D) ac is less dangerous.

Check Solution → [2026 • Set 55-5-1]

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8. A voltage $v = v_0 \sin \omega t$ applied to a circuit drives a current $i = i_0 \sin(\omega t + \phi)$ in the circuit. The average power consumed in the circuit over a cycle is:

(A) Zero

(B) $i_0 v_0 \phi$

(C) $\frac{i_0 v_0}{2}$

(D) $\frac{i_0 v_0}{2} \phi$

Check Solution → [2025 • Set 55-1-1]

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9. The ratio of the number of turns of the primary to the secondary coils in an ideal transformer is 20 : 1. If 240 *V* ac is applied from a source to the primary coil of transformer and a 6.0 Ω resistor is connected across the output terminals, then current drawn by the transformer from the source will be:

(A) 4.0 A

(B) 3.8 A

(C) 0.9 A

(D) 0.10 A

Check Solution → [2025 • Set 55-1-3]

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10. **Assertion (A):** In an ideal step-down transformer, the electrical energy is not lost.

Reason (R): In a step-down transformer, voltage decreases but the current increases.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).

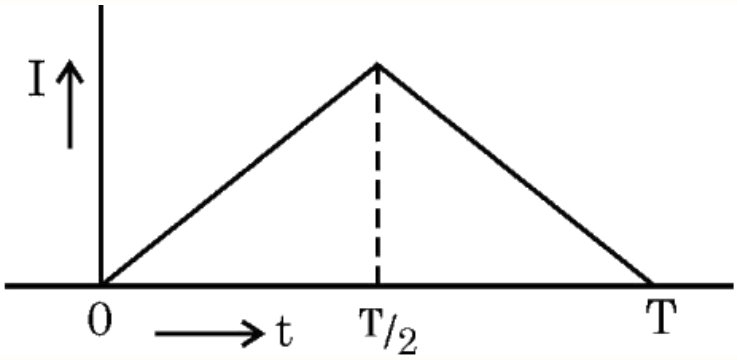
(C) Assertion (A) is true but Reason (R) is false.

(D) Both Assertion (A) and Reason (R) are false.

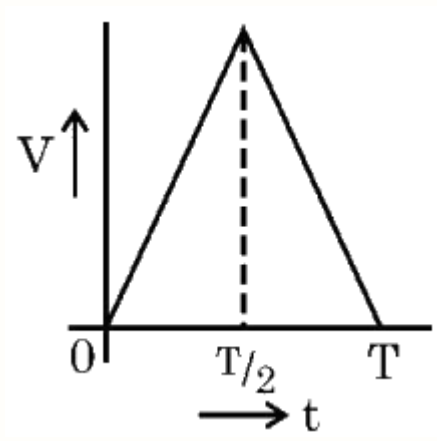
Check Solution → [2025 • Set 55-2-1]

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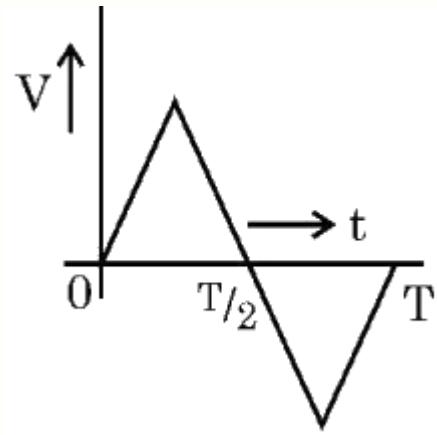
11. The alternating current I in an inductor is observed to vary with time t as shown in the graph for a cycle. Which one of the following graphs is the correct representation of wave form of voltage V with time t?



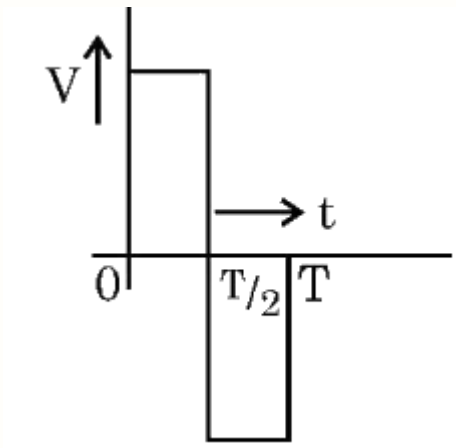
(A)



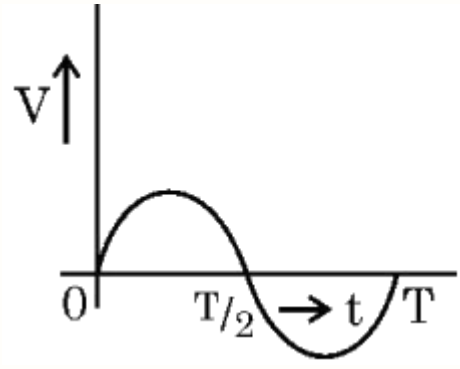
(B)



(C)



(D)



Check Solution → [2025 • Set 55-2-1]

12. A transformer is connected to a 200 V ac source. The transformer supplies 3000 V to a device. If the number of turns in the primary coil is 450, then the number of turns in its secondary coil is

(A) 30

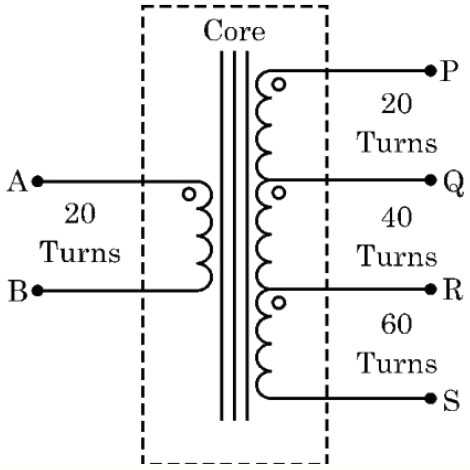
(B) 450

(C) 4500

(D) 6750

Check Solution → [2025 • Set 55-2-1]

13. The number of turns between different pairs of output terminals are shown for a step-up transformer. Input voltage of 20 V is applied between A and B. Between which two terminals will the output be 120 V?



(A) P and Q

(B) Q and S

(C) P and R

(D) P and S

Check Solution → [2025 • Set 55-2-3]

14. **Assertion (A):** A series LCR circuit behaves as a pure resistive circuit at resonance.

Reason (R): At resonance, $X_L = X_C$ gives $\omega = \frac{1}{\sqrt{LC}}$.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Both Assertion (A) and Reason (R) are false.

Check Solution → [2025 • Set 55-4-1]

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15. An ammeter connected in series in an ac circuit reads 10 A. The maximum value of current at any instant in the circuit is :

(A) $10\sqrt{2}\ A$

(B) $\frac{2}{\sqrt{2}}\ A$

(C) $\frac{10}{\pi}\ A$

(D) $\frac{10}{2}\ A$

Check Solution → [2025 • Set 55-4-1]

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16. A transformer is a device used for converting:

(A) high ac voltage and large ac current to low ac voltage and small ac current

(B) high ac voltage and small ac current to low ac voltage and small ac current

(C) low ac voltage and large ac current to high ac voltage and small ac current

(D) low ac voltage and small ac current to high ac voltage and large ac current

Check Solution → [2025 • Set 55-5-2]

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17. A capacitor of capacitance C has reactance X in an ac circuit. If the capacitance and the frequency of the applied voltage are doubled, the new reactance will become:

(A) 4X

(B) 2X

(C) $\frac{X}{2}$

(D) $\frac{X}{4}$

Check Solution → [2025 • Set 55-5-3]

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18. An ac source is connected to a resistor and an inductor in series. The voltage across the resistor and inductor are 8 V and 6 V respectively. The voltage of the source is:

(A) 10 V

(B) 12 V

(C) 14 V

(D) 16 V

Check Solution → [2025 • Set 55-6-1]

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19. An alternating current is given by $I = I_0\cos(100\pi)t$. The least time the current takes to decrease from its maximum value to zero will be:

(A) $(\frac{1}{50})\ s$

(B) $(\frac{1}{100})\ s$

(C) $(\frac{1}{200})\ s$

(D) $(\frac{1}{400})\ s$

Check Solution → [2025 • Set 55-7-1]

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20. A capacitor and an inductor are connected in series across an ac source of voltage of variable frequency. The frequency is increased continuously. The nature of the circuit before and after the resonance will be:

(A) inductive only

(B) capacitive only

(C) capacitive and inductive respectively

(D) inductive and capacitive respectively

Check Solution → [2025 • Set 55-7-1]

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21. When does an inductor act as a conductor in a circuit? Give reason for it.

Check Solution → [2024 • Set 55-1-1]

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22. A step-up transformer converts a low voltage into high voltage. Does it violate the principle of conservation of energy? Explain.

Check Solution → [2024 • Set 55-1-1]

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23. The reactance of a capacitor of capacitance C connected to an ac source of frequency ω is X. If the capacitance of the capacitor is doubled and the frequency of the source is tripled, the reactance will become:

(A) $\frac{X}{6}$

(B) 6X

(C) $\frac{X}{2}$

(D) $\frac{X}{3}$

Check Solution → [2024 • Set 55-1-1]

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24. The primary and secondary coils of a transformer have 500 turns and 5000 turns respectively. The primary coil is connected to an ac source of 220 V – 50 Hz. The output across the secondary coil is:

(A) 220 V – 50 Hz

(B) 1100 V – 50 Hz

(C) 2200 V – 5 Hz

(D) 2200 V – 50 Hz

Check Solution → [2024 • Set 55-1-3]

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25. Which of the following quantity/quantities remains same in primary and secondary coils of an ideal transformer?
Current, Voltage, Power, Magnetic flux

(A) Current only

(B) Voltage only

(C) Power only

(D) Magnetic flux and Power both

Check Solution → [2024 • Set 55-2-1]

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26. A resistor and an ideal inductor are connected in series to a $100\sqrt{2}$ V, 50 Hz ac source. When a voltmeter is connected across the resistor or the inductor, it shows the same reading. The reading of the voltmeter is:

(A) $100\sqrt{2}$ V

(B) 100 V

(C) $50\sqrt{2}$ V

(D) 50 V

Check Solution → [2024 • Set 55-2-1]

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27. A series LCR circuit (L=2 mH, C=0.2 μ F and R=30 Ω) is connected to an ac source of variable frequency. The impedance of this circuit will be minimum at a frequency of:

(A) $\frac{10^5}{4\pi}$ Hz

(B) $\frac{10^5}{2\pi}$ Hz

(C) $\frac{10^4}{4\pi}$ Hz

(D) $\frac{10^4}{2\pi}$ Hz

Check Solution → [2024 • Set 55-2-2]

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28. In an ac circuit, the instantaneous values of current (in A) and voltage (in V) are $I=5\omega t$ and $E=200\cos(\omega t+\frac{\pi}{3})$ respectively. The phase difference between voltage and current at any instant is:

(A) $\frac{5\pi}{6}$

(B) $\frac{5\pi}{12}$

(C) $\frac{5\pi}{2}$

(D) $\frac{3\pi}{2}$

Check Solution → [2024 • Set 55-2-3]

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29. The average value of the alternating voltage $v = (157\ V)\omega t$ over its first half-cycle is:

(A) 157 V

(B) $\frac{157}{\sqrt{2}}$ V

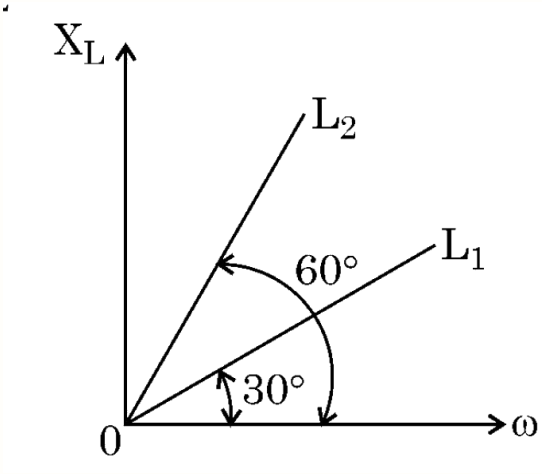
(C) 78.5 V

(D) 100 V

Check Solution → [2024 • Set 55-3-3]

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30. Figure shows the variation of inductive reactance X_L of two ideal inductors of inductance L_1 and L_2 with angular frequency ω. The value of $\frac{L_1}{L_2}$ is



(A) $\sqrt{3}$

(B) $\frac{1}{\sqrt{3}}$

(C) 3

(D) $\frac{1}{3}$

Check Solution → [2024 • Set 55-4-1]

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31. An ac source V=282sin(100t) volt is connected across a 1 μF capacitor. The rms value of current in the circuit will be (take $\sqrt{2}$=1.41)

(A) 10 mA

(B) 20 mA

(C) 40 mA

(D) 80 mA

Check Solution → [2024 • Set 55-4-2]

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32. The r.m.s. value of a current given by $i=(i_1\omega\, t+i_2\omega\, t)$ is

(A) $\frac{1}{\sqrt{2}}(i_1+i_2)$

(B) $\frac{1}{\sqrt{2}}(i_1-i_2)$

(C) $\frac{1}{\sqrt{2}}\sqrt{i_1^2+i_2^2}$

(D) $\frac{1}{2}(i_1^2+i_2^2)$

Check Solution → [2024 • Set 55-4-3]

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33. An ac voltage is applied across an ideal inductor. The current in it:

(A) leads the voltage by $(\frac{1}{2})$ cycle.

(B) lags the voltage by $(\frac{1}{2})$ cycle.

(C) leads the voltage by $(\frac{1}{4})$ cycle.

(D) lags the voltage by $(\frac{1}{4})$ cycle.

Check Solution → [2024 • Set 55-5-2]

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34. A voltage signal is described by: $v = V_0$ for $0 \le t \le \frac{T}{2}$ and $v = 0$ for $\frac{T}{2} \le t \le T$ for a cycle. Its rms value is:

(A) $\frac{V_0}{\sqrt{2}}$

(B) V_0

(C) $\frac{V_0}{2}$

(D) $\sqrt{2}\,V_0$

Check Solution → [2023 • Set 55-3-1]

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35. An inductor, a capacitor and a resistor are connected in series across an ac source of voltage. If the frequency of the source is decreased gradually, the reactance of:

(A) both the inductor and the capacitor decreases.

(B) inductor decreases and the capacitor increases.

(C) both the inductor and the capacitor increases.

(D) inductor increases and the capacitor decreases.

Check Solution → [2023 • Set 55-3-1]

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36. An ac voltage $v = v_0\omega t$ is applied to a series combination of a resistor R and an element X. The instantaneous current in the circuit is $I = I_0\sin(\omega t + \frac{\pi}{4})$. Then which of the following is correct?

(A) X is a capacitor and $X_C = \sqrt{2}R$

(B) X is an inductor and $X_L = R$

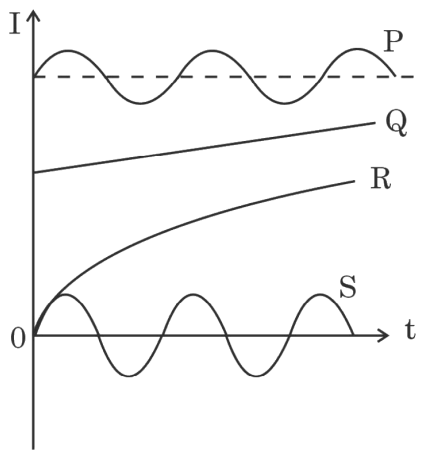
(C) X is an inductor and $X_L = \sqrt{2}R$

(D) X is a capacitor and $X_C = R$

Check Solution → [2023 • Set 55-4-1]

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37. The figure shows variation of current (I) with time (t) in four devices P, Q, R and S. The device in which an alternating current flows is:



(A) P

(B) Q

(C) R

(D) S

Check Solution → [2023 • Set 55-4-1]

38. Which of the following statements about a series LCR circuit connected to an ac source is correct?

(A) If the frequency of the source is increased, the impedance of the circuit first decreases and then increases.

(B) If the net reactance ($X_L - X_C$) of circuit becomes equal to its resistance, then the current leads the voltage by 45°.

(C) At resonance, the voltage drop across the inductor is more than that across the capacitor.

(D) At resonance, the voltage drop across the capacitor is more than that across the inductor.

Check Solution → [2023 • Set 55-5-1]

39. An ideal inductor is connected across an AC source of voltage. The current in the circuit

(A) is ahead of the voltage in phase by π.

(B) lags voltage in phase by π.

(C) is ahead of voltage in phase by π/2.

(D) lags voltage in phase by π/2.

Check Solution → [2023 • Set 55-5-2]

40. What is the ratio of inductive and capacitive reactance in an ac circuit?

(A) $\omega^2 LC$

(B) LC^2

(C) $\frac{L}{C}$

(D) $\omega^2 L$

Check Solution → [2023 • Set 55-5-3]

41. An alternating current $I = (10\ A)\ \sin(100\ \pi\ t)$ is passed through a resistor of 20 Ω. What is the average power consumed by the resistor over a complete cycle?

Check Solution → [2021]

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42. What is the impedance of a capacitor of capacitance C in an ac circuit using source of frequency n Hz?

OR

What is the value of impedance of a resonant series LCR circuit?

Check Solution → [2020 • Set 55-2-1]

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43. An ac is passed through a series LCR circuit. What is the impedance of the circuit at resonance?

Check Solution → [2020 • Set 55-3-1]

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44. In an ac circuit, the applied voltage and flowing current are $E = E_0\omega\ t$ and $I = I_0\sin(\omega\ t + \frac{\pi}{2})$ respectively. What is the average power consumed in one cycle in this circuit?

OR

What happens when a block of metal is kept in a varying magnetic field?

Check Solution → [2020 • Set 55-3-1]

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45. A circuit element is connected across an ac source. It is observed that the voltage across the element leads the current flowing through it by a phase angle $\frac{\pi}{2}$. Identify the circuit element.

Check Solution → [2020 • Set 55-3-2]

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46. Write the impedance of a series LCR resonant circuit in terms of L, C and R.

Check Solution → [2020 • Set 55-3-2]

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47. What is the value of power factor of a series LCR circuit at resonance?

Check Solution → [2020 • Set 55-3-3]

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48. The selectivity of a series LCR a.c. circuit is large, when

(A) L is large and R is large

(B) L is small and R is small

(C) L is large and R is small

(D) $L = R$

Check Solution → [2020 • Set 55-5-1]

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49. An a.c. source of voltage $V=V_0\omega t$ is connected to an ideal inductor. Draw graphs of voltage V and current i versus ωt.

Check Solution → [2016]

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50. Explain why current flows through an ideal capacitor when it is connected to an a.c. source but not when it is connected to a d.c. source in a steady state.

Check Solution → [2016]

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51. Draw a graph to show variation of capacitive-reactance with frequency in an a.c. circuit.

Check Solution → [2015]

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52. Define capacitor reactance. Write its S.I. units.

Check Solution → [2015]

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53. Why is the use of a.c. voltage preferred over d.c. voltage ? Give two reasons.

Check Solution → [2014]

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54. Mention the two characteristic properties of the material suitable for making core of a transformer.

Check Solution → [2012]

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55. Define the term *wattless current*.

Check Solution → [2011 • Set 55-1-1]

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56. The peak value of e.m.f. in a.c. is E_0. Write its (i) rms and (ii) average value over a complete cycle.

Check Solution → [2011 • Set 55-2-1]

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57. In a series LCR circuit, the voltages across an inductor, a capacitor and a resistor are 30 V, 30 V and 60 V respectively. What is the phase difference between the applied voltage and the current in the circuit?

Check Solution → [2007]

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58. The power factor of an a.c. circuit is 0.5. What will be the phase difference between voltage and current in this circuit?

Check Solution → [2005]

SECTION B

2-Mark Questions

VSA · 19 questions

> *Section B of the paper, 2 marks each. Sorted newest year first.*

59. A resistor and a capacitor are connected in series to an ac source $v = v_m \omega t$. Derive an expression for the impedance of the circuit.

Check Solution → **[2024 • Set 55-1-1]**

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60. An electric lamp is designed to operate at 110 V dc and 11 A current. If the lamp is operated on 220 V, 50 Hz ac source with a coil in series, then find the inductance of the coil.

Check Solution → **[2024 • Set 55-1-1]**

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61. Draw a labelled diagram of a step-up transformer and describe its working principle. Explain any three causes for energy losses in a real transformer.

Check Solution → **[2024 • Set 55-1-1]**

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62. A step-up transformer has 200 and 3000 turns in its primary and secondary coils respectively. The input voltage given to the primary coil is 90 V. Calculate:
(1) The output voltage across the secondary coil
(2) The current in the primary coil if the current in the secondary coil is 2.0 A.

Check Solution → **[2024 • Set 55-1-1]**

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63. (a) Explain the term 'sharpness of resonance' in ac circuit.
(b) In a series LCR circuit, $V_L = V_C \neq V_R$. What is the value of power factor for this circuit?

OR

An ac source of emf $V = V_0 \omega t$ is connected to a capacitor of capacitance C.

Deduce the expression for the current (I) flowing in it. Plot the graph of (i) V vs. ωt, and (ii) I vs. ωt.

Check Solution → [2020 • Set 55-2-1]

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64. A resistor R and an inductor L are connected in series to a source of voltage $V=V_0\omega t$. The voltage is found to lead current in phase by $\pi/4$. If the inductor is replaced by a capacitor C, the voltage lags behind current in phase by $\pi/4$. When L, C and R are connected in series with the same source, find the:
(i) average power dissipated and
(ii) instantaneous current in the circuit.

Check Solution → [2020 • Set 55-4-1]

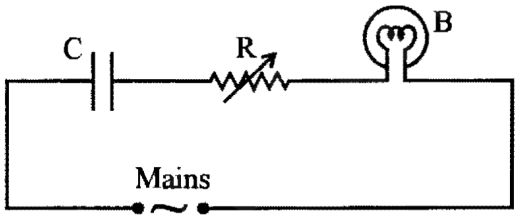
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65. A light bulb and a solenoid are connected in series across an ac source of voltage. Explain, how the glow of the light bulb will be affected when an iron rod is inserted in the solenoid.

Check Solution → [2017]

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66. A capacitor 'C', a variable resistor 'R' and a bulb 'B' are connected in series to the ac mains in circuit as shown. The bulb glows with some brightness. How will the glow of the bulb change if (i) a dielectric slab is introduced between the plates of the capacitor, keeping resistance R to be the same; (ii) the resistance R is increased keeping the same capacitance ?



Check Solution → [2014]

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67. State the underlying principle of a transformer. How is the large scale transmission of electric energy over long distances done with the use of

transformers?

Check Solution → [2012]

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68. A light bulb is rated 100 W for 220 V ac supply of 50 Hz. Calculate
(i) the resistance of the bulb;
(ii) the rms current through the bulb.
OR
An alternating voltage given by $V = 140\sin 314t$ is connected across a pure resistor of 50 Ω. Find
(i) the frequency of the source.
(ii) the rms current through the resistor.

Check Solution → [2012]

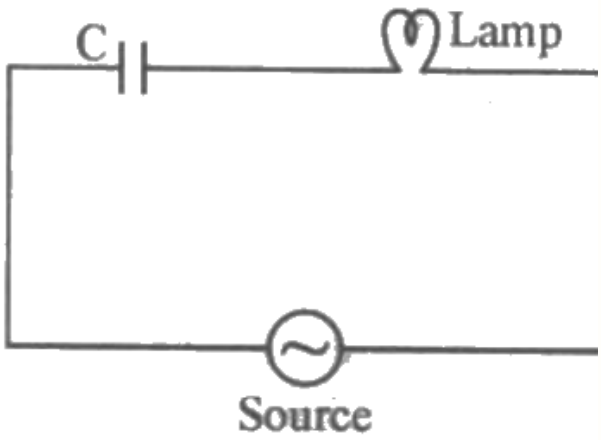
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69. Calculate the quality factor of a series LCR circuit with $L = 2.0\ H$, $C = 2\ \mu F$ and $R = 10\ \Omega$. Mention the significance of quality factor in LCR circuit.

Check Solution → [2012]

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70. An electric lamp having coil of negligible inductance connected in series with a capacitor and an AC source is glowing with certain brightness. How does the brightness of the lamp change on reducing the (i) capacitance, and (ii) the frequency? Justify your answer.



Check Solution → [2010]

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71. Prove that an ideal capacitor, in an a.c. circuit does not dissipate power.

Check Solution → [2008]

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72. Derive an expression for the impedance of an a.c. circuit consisting of an inductor and a resistor.

Check Solution → [2008]

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73. Distinguish between the terms 'average value' and 'rms value' of an alternating current. The instantaneous current from an a.c. source is $I = 5 \sin(314\, t)$ ampere. What are the average and rms values of the current?

Check Solution → [2007]

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74. Calculate the current drawn by the primary of a transformer which steps down 200 V to 20 V to operate a device of resistance 20 Ω. Assume the efficiency of the transformer to be 80%.

Check Solution → [2007]

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75. An a.c. voltage of 100 V, 50 Hz is connected across a 20 ohm resistor and 2 mH inductor in series. Calculate (i) impedance of the circuit, (ii) rms current in the circuit.

Check Solution → [2007]

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76. **(i)** Draw the graphs showing variation of inductive reactance and capacitive reactance with frequency of applied a.c. source.
(ii) Can the voltage drop across the inductor or the capacitor in a series LCR circuit be greater than the applied voltage of the a.c. source? Justify your answer.

Check Solution → [2005]

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77. An a.c. voltage $E = E_0 \sin \omega t$ is applied across an inductor L. Obtain an expression for current I.

Check Solution → **[2003]**

SECTION C

3-Mark Questions

SA · 56 questions

> *Section C of the paper, 3 marks each. Sorted newest year first.*

78. (a) Discuss the behaviour of an inductor connected to (i) a dc source, and (ii) a high frequency ac source.
(b) What is the phase relation between current and voltage in an ideal inductor connected to an ac source? Draw a phasor diagram for the circuit.

Check Solution → **[2026 • Set 55-3-1]**

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79. A series combination of circuit elements X and Y is connected across an ac source. It is found that the voltage is ahead of current in phase by $\frac{\pi}{4}$ radian. When element Y is replaced by element Z, the current leads the voltage by $\frac{\pi}{4}$ radian.
(a) Identify the elements X, Y and Z.
(b) What will the phase angle and power factor for the circuit be if X, Y and Z were connected in series across the same ac source? What can you say about the current that flows in the circuit in this case?

Check Solution → **[2026 • Set 55-3-2]**

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80. (a) Differentiate between inductive reactance, capacitive reactance and impedance of an ac circuit.
(b) An ideal inductor and an ideal capacitor are connected in series across an ac voltage. Plot a graph showing variation of net reactance of the circuit with frequency of the applied ac voltage.

Check Solution → **[2026 • Set 55-5-2]**

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81. (a) Can a transformer step up or step down dc power supply?
(b) Can a step up transformer work as a step down transformer?

(c) Does a step up transformer contradict the principle of conservation of energy? Justify your answer.

Check Solution → [2026 • Set 55-5-3]

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82. An ac source of voltage $v = v_m \sin \omega t$ is connected to a series combination of LCR circuit. Draw the phasor diagram. Using it obtain an expression for the impedance of the circuit and the phase difference between applied voltage and the current.

Check Solution → [2025 • Set 55-1-3]

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83. Differentiate between the peak value and root mean square value of an alternating current. Derive the expression for the root mean square value of alternating current, in terms of its peak value.

Check Solution → [2025 • Set 55-2-2]

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84. (a) ac voltage of frequency ω is applied across a series LCR circuit. Draw the phasor diagram and obtain the impedance of the circuit.
(b) Discuss 'resonance' in a series LCR circuit and write the expression for resonant frequency.

Check Solution → [2025 • Set 55-2-3]

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85. An ac source of voltage $V = V_0 \omega t$ is connected to a circuit element X. It is observed that the current flowing through X varies as $I = I_0 \sin(\omega t - \frac{\pi}{2})$.
(a) Identify the element X and write the expression for its reactance.
(b) Plot a graph to show the variation of reactance of the element with the frequency of the applied voltage.
(c) Draw plots showing the variation of voltage and current with time over one cycle of applied ac.

Check Solution → [2025 • Set 55-7-3]

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86. An inductor, a capacitor and a resistor are connected in series with an ac source $v = v_m \omega t$. Derive an expression for the average power dissipated in the circuit. Also obtain the expression for the resonant frequency of the circuit.

Check Solution → [2024 • Set 55-1-1]

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87. (a) Draw the graphs showing the variation of the following with the frequency of ac source in a circuit:
(i) Resistance
(ii) Capacitive reactance
(iii) Inductive reactance
(b) Can the voltage drop across the inductor or the capacitor in a series LCR circuit be greater than the applied voltage of the ac source? Justify your answer.

Check Solution → [2024 • Set 55-1-3]

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88. An ac voltage $v_i=140\sin(100\pi t)$ V is applied to the primary coil having 200 turns, of an ideal transformer and it supplies a power of 5 kW. If the secondary coil has 1000 turns, find:
(a) the output voltage,
(b) the instantaneous voltage across the secondary coil, and
(c) the current in the secondary coil. (Take 2=1.4)

Check Solution → [2024 • Set 55-2-3]

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89. A sinusoidal voltage is applied to an electric circuit containing a circuit element 'X' in which the current leads the voltage by $\frac{\pi}{2}$.
(a) Identify the circuit element in the circuit.
(b) Write the formula for its reactance.
(c) Show graphically the variation of this reactance with frequency of ac voltage.
(d) Explain the behaviour of this element when it is used in (i) an ac circuit, and (ii) a dc circuit.

Check Solution → [2024 • Set 55-3-1]

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90. Distinguish between reactance and impedance of an ac circuit. Show that an ideal inductor in an ac circuit does not dissipate any power.

Check Solution → [2024 • Set 55-3-2]

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91. How does the resistance differ from impedance? With the help of a suitable phasor diagram, obtain an expression for impedance of a series LCR circuit, connected to a source $v = v_m \omega t$.

Check Solution → [2023 • Set 55-1-1]

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92. Find the condition for resonance in a series LCR circuit connected to a source $v = v_m \omega t$, where ω can be varied. Give the factors on which the resonant frequency of a series LCR circuit depends. Plot a graph showing the variation of electric current with frequency in a series LCR circuit.

Check Solution → [2023 • Set 55-1-1]

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93. The primary and the secondary coils of an ideal step-down transformer consist of 650 and 25 turns respectively. When the primary coil of this transformer is connected to 240 V mains, the current in the primary coil is 1.5 A. Calculate:
(a) the voltage across the secondary coil
(b) the current in the secondary coil
(c) the average power delivered to the output circuit

Check Solution → [2023 • Set 55-1-2]

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94. An ac voltage $v = v_m \omega t$ (v_m = 310 V and f = 50 Hz) is connected to a pure capacitor of capacitance 15 μF. Calculate (i) the reactance of the capacitor, and (ii) the amplitude of the current. Write the expression of current through the capacitor as a function of time.

Check Solution → [2023 • Set 55-1-3]

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95. A series CR circuit with R=200 Ω and C=(50/π) μF is connected across an ac source of peak voltage ε_0 = 100 V and frequency ν = 50 Hz. Calculate
(a) impedance of the circuit (Z),
(b) phase angle (ϕ), and
(c) voltage across the resistor.

Check Solution → [2023 • Set 55-2-1]

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96. A series RL circuit with R=10 Ω and L=$(\frac{100}{\pi})$ mH is connected to an ac source of voltage V = 141 sin(100π t), where V is in volts and t is in seconds. Calculate
(a) impedance of the circuit
(b) phase angle, and
(c) voltage drop across the inductor.

Check Solution → [2023 • Set 55-2-2]

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97. A resistor of 50 Ω, a capacitor of $(\frac{25}{\pi})$ μF and an inductor of $(\frac{4}{\pi})$ H are connected in series across an ac source whose voltage (in volt) is given by V = 70 sin(100π t). Calculate:
(a) the net reactance of the circuit,
(b) the impedance of the circuit,
(c) the effective value of current in the circuit.

Check Solution → [2023 • Set 55-2-3]

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98. (a) An ac source $V = V_m \sin \omega t$ is connected across an ideal capacitor. Derive the expression for the (i) current flowing in the circuit, and (ii) reactance of the capacitor. Plot a graph of current i versus ωt.

OR

(b) A series combination of an inductor L, a capacitor C and a resistor R is connected across an ac source of voltage in a circuit. Obtain an expression for the average power consumed by the circuit. Find power factor for (i) purely inductive circuit, and (ii) purely resistive circuit.

Check Solution → **[2023 • Set 55-3-1]**

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99. (a) A resistor of 30 Ω and a capacitor of $\frac{250}{\pi}$ μ F are connected in series to a 200 V, 50 Hz ac source. Calculate (i) the current in the circuit, and (ii) voltage drops across the resistor and the capacitor. (iii) Is the algebraic sum of these voltages more than the source voltage? If yes, solve the paradox.

OR

(b) A series LCR circuit with $R = 20\ \Omega$, $L = 2$ H and $C = 50$ μ F is connected to a 200 volts ac source of variable frequency. What is (i) the amplitude of the current, and (ii) the average power transferred to the circuit in one complete cycle, at resonance? (iii) Calculate the potential drop across the capacitor.

Check Solution → **[2023 • Set 55-4-1]**

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100. An alternating voltage of 220 V is applied across a device X. A current of 0.22 A flows in the circuit and it lags behind the applied voltage in phase by π/2 radian. When the same voltage is applied across another device Y, the current in the circuit remains the same and it is in phase with the applied voltage.

(i) Name the devices X and Y and,

(ii) Calculate the current flowing in the circuit when the same voltage is applied across the series combination of X and Y.

Check Solution → **[2023 • Set 55-5-1]**

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101. A current of 1A flows through a coil when it is connected across a DC battery of 100 V. If DC battery is replaced by an AC source of 100 V and angular frequency

100 rad s^{-1}, the current reduces to 0.5 A. Find
(i) impedance of the circuit.
(ii) self-inductance of coil.
(iii) phase difference between the voltage and the current.

Check Solution → [2023 • Set 55-5-2]

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102. The primary coil having N_p turns of an ideal transformer is supplied with an alternating voltage V_p. Obtain an expression for the voltage V_s induced in its secondary coil having N_s turns. Mention two main sources of power loss in real transformers.

Check Solution → [2023 • Set 55-5-2]

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103. An alternating current $I = 14 \sin(100 \pi t)$ A passes through a series combination of a resistor of 30 Ω and an inductor of $(\frac{2}{5\pi})$ H. Taking $\sqrt{2} = 1.4$, calculate the
(i) rms value of the voltage drops across the resistor and the inductor, and
(ii) power factor of the circuit.

Check Solution → [2023 • Set 55-5-3]

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104. (i) In an LCR series circuit connected to an ac source, the voltage and the current are in the same phase. If the capacitor is filled with a dielectric, will the current lead or lag behind or remain in phase with the voltage? Explain.
(ii) In the circuit, why is the rms value of net voltage not equal to the sum of voltage drops across individual elements?
(iii) Draw a graph showing variation of the impedance of the circuit with the frequency of the applied voltage.

Check Solution → [2021]

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105. (i) An LCR series circuit is connected to an ac source. If the angular resonant frequency of the circuit is ω_0, will the current lead or lag behind or be in phase with the voltage when $\omega < \omega_0$ and why?

(ii) We cannot step up a dc voltage using a transformer. Why?
(iii) On what principle does a metal detector work?

Check Solution → [2021]

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106. A resistance R and a capacitor C are connected in series to a source $V = V_0 \omega t$. Find:
(a) The peak value of the voltage across the (i) resistance and (ii) capacitor.
(b) The phase difference between the applied voltage and current. Which of them is ahead?

Check Solution → [2020 • Set 55-1-1]

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107. A resistor R and an inductor L are connected in series to a source $V = V_0 \omega t$. Find the
(a) peak value of the voltage drops across R and across L,
(b) phase difference between the applied voltage and current. Which of them is ahead?

Check Solution → [2020 • Set 55-1-2]

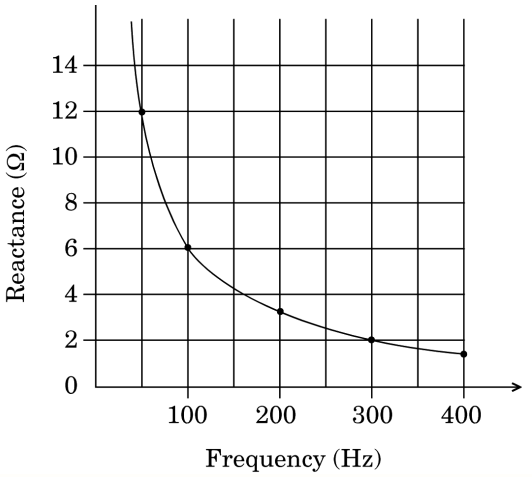
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108. A resistance R and a capacitor C are connected in series to an ac source $V = V_0 \omega t$.
(a) Obtain the expression for the instantaneous current (I) in the circuit.
(b) Show graphically variations of V and I as a function of ωt.

Check Solution → [2020 • Set 55-1-3]

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109. The figure shows the graphical variation of the reactance of a capacitor with frequency of ac source.



(a) Find the capacitance of the capacitor.
(b) An ideal inductor has the same reactance at 100 Hz frequency as the capacitor has at the same frequency. Find the value of inductance of the inductor.
(c) Draw the graph showing the variation of the reactance of this inductor with frequency.

Check Solution → [2020 • Set 55-2-1]

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110. A series LCR ac circuit has $L = 2.0\ H$, $C = 32\ \mu F$ and $R = 10\ \Omega$.
(a) At what angular frequency of ac will it resonate?
(b) Calculate the Q value of the circuit.
OR
An ideal inductor of 5 H inductance is connected to a 200 V, 50 Hz ac supply.
(a) Calculate the rms and peak value of current in the inductor.
(b) What is the phase difference between current through the inductor and the applied voltage? How will it change if a small resistance is connected in series with this inductor in the circuit?

Check Solution → [2020 • Set 55-3-1]

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111. Explain with the help of a diagram, the working of a step-down transformer. Why is a laminated iron core used in a transformer?

Check Solution → [2020 • Set 55-4-1]

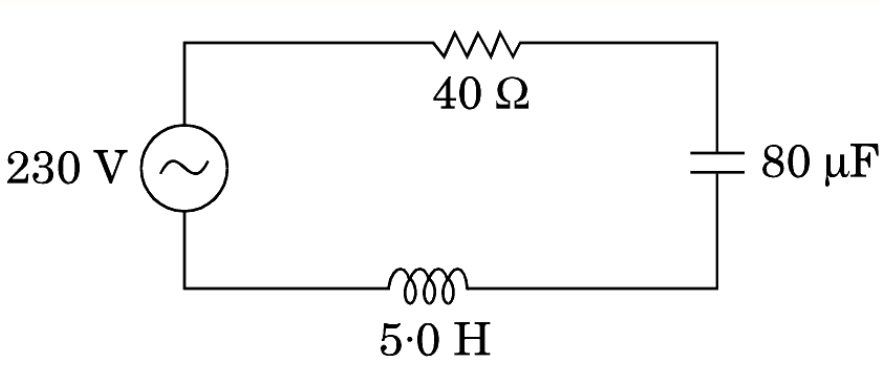
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112. Explain the principle of the device with diagram, which is used to provide electricity at the proper voltage for household purposes.
Briefly discuss loss of energy in it due to flux leakage and its minimization.

Check Solution → [2020 • Set 55-4-2]

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113. A capacitor (C) and resistor (R) are connected in series with an ac source of voltage of frequency 50 Hz. The potential difference across C and R are respectively 120 V, 90 V, and the current in the circuit is 3 A. Calculate (i) the impedance of the circuit (ii) the value of the inductance, which when connected in series with C and R will make the power factor of the circuit unity.
OR
The figure shows a series LCR circuit connected to a variable frequency 230 V source.



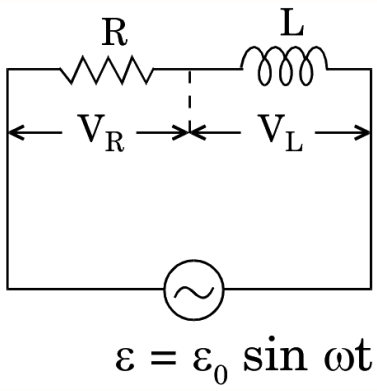
(a) Determine the source frequency which drives the circuit in resonance.
(b) Calculate the impedance of the circuit and amplitude of current at resonance.
(c) Show that potential drop across LC combination is zero at resonating frequency.

Check Solution → [2019 • Set 55-2-1]

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114. (a) An ac circuit as shown in the figure has an inductor of inductance L and a resistor of resistance R connected in series. Using the phasor diagram, explain why the voltage in the circuit will lead the current in phase.
(b) The potential difference across the resistor is 160 V and that across the inductor is 120 V. Find the effective value of the applied voltage. If the effective

current in the circuit be 1.0 A, calculate the total impedance of the circuit.
(c) What will be the potential difference in the circuit when direct current is passed through the circuit?



OR

An ac circuit consists of a series combination of circuit elements X and Y. The current is ahead of the voltage in phase by $\frac{\pi}{4}$. If element X is a pure resistor of 100 Ω,
(a) name the circuit element Y.
(b) calculate the rms value of current, if rms value of voltage is 141 V.
(c) what will happen if the ac source is replaced by a dc source?

Check Solution → [2019 • Set 55-3-1]

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115. A voltage $v = v_m \sin \omega t$ applied to a series LCR circuit, drives a current in the circuit given $i = i_m \sin(\omega t + \phi)$. Deduce the expression for the instantaneous power supplied by the source. Hence, obtain the expression for the average power.
Define the terms 'power factor' and 'wattless current', giving the examples where power factor is maximum and the circuit where there is wattless current.

Check Solution → [2019 • Set 55-5-1]

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116. Given three elements X, Y and Z to be connected across an ac source. With only X connected across the ac source, voltage and current are found to be in the same phase. With only element Y in the circuit, the voltage lags behind the current in phase by π/2 while with the element Z in the circuit, the voltage leads

the current in phase by π/2.
(a) Identify the elements X, Y and Z.
(b) When all these elements are connected in series across the same source, (i) determine the power factor, and (ii) find out the condition when the circuit is in resonant state.

Check Solution → [2019 • Set 55-5-2]

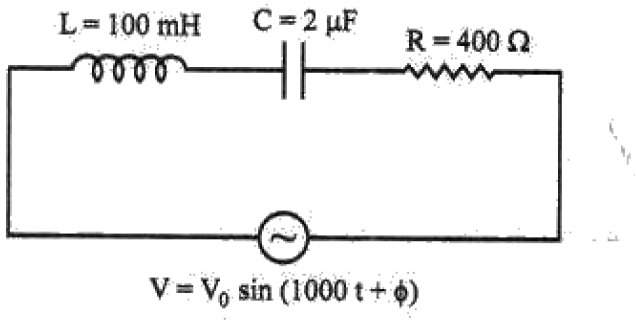
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117. Draw the phasor diagram of a series LCR connected across an ac source $V = V_m \sin \omega t$. Hence, derive the expression for the impedance of the circuit. Obtain the conditions for the phase angle under which the current is (i) maximum, and (ii) minimum.

Check Solution → [2019 • Set 55-5-3]

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118. (i) Find the value of the phase difference between the current and the voltage in the series LCR circuit shown below. Which one leads in phase: current or voltage?
(ii) Without making any other change, find the value of the additional capacitor C_1, to be connected in parallel with the capacitor C, in order to make the power factor of the circuit unity.



Check Solution → [2017]

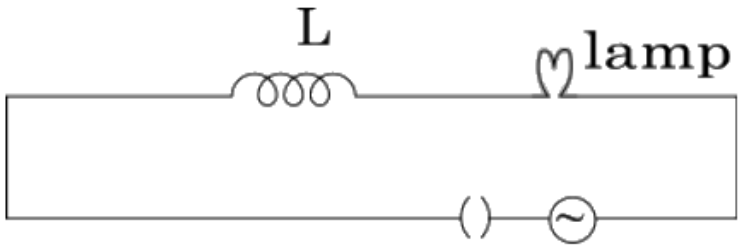
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119. A source of ac voltage $v = v_0 \sin \omega t$ is connected across a pure inductor of inductance L. Derive the expressions for the instantaneous current in the circuit. Show that average power dissipated in the circuit is zero.

Check Solution → [2017]

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120. (i) When an AC source is connected to an ideal inductor show that the average power supplied by the source over a complete cycle is zero.
(ii) A lamp is connected in series with an inductor and an AC source. What happens to the brightness of the lamp when the key is plugged in and an iron rod is inserted inside the inductor? Explain.



Check Solution → [2016]

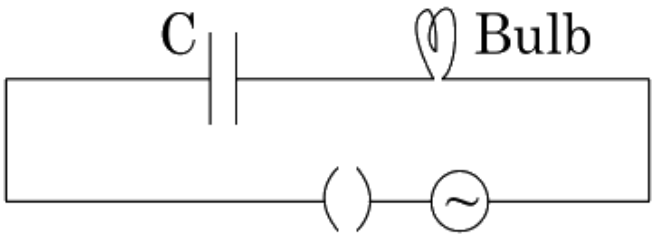
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121. Draw a schematic arrangement for winding of primary and secondary coils in a transformer with the two coils on separate limbs of the core. State its underlying principle and find the relation between the primary and secondary voltages in terms of the number of turns of the primary and secondary windings. How are the currents in the primary and secondary coils related to the voltages in the case of an ideal transformer?

Check Solution → [2016]

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122. (i) When an AC source is connected to an ideal capacitor, show that the average power supplied by the source over a complete cycle is zero.
(ii) A bulb is connected in series with a variable capacitor and an A.C. source as shown. What happens to the brightness of the bulb when the key is plugged in and capacitance of the capacitor is gradually reduced?



Check Solution → [2016]

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123. A capacitor of unknown capacitance, a resistor of 100 Ω and an inductor of self inductance $L=(4/\pi^2)$ henry are connected in series to an ac source of 200 V and 50 Hz. Calculate the value of the capacitance and impedance of the circuit when the current is in phase with the voltage. Calculate the power dissipated in the circuit.

Check Solution → [2016]

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124. A series LCR circuit is connected across an a.c. source of variable angular frequency 'ω'. Plot a graph showing variation of current 'i' as a function of 'ω' for two resistances R_1 and R_2 ($R_1 > R_2$). Answer the following questions using this graph:
(a) In which case is the resonance sharper and why?
(b) In which case is the power dissipation more and why?

Check Solution → [2015]

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125. An inductor L of inductance X_L is connected in series with a bulb B and an ac source. How would brightness of the bulb change when (i) number of turns in the inductor is reduced, (ii) an iron rod is inserted in the inductor and (iii) a capacitor of reactance $X_C = X_L$ is inserted in series in the circuit. Justify your answer in each case.

Check Solution → [2015]

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126. A voltage $V = V_0 \sin \omega t$ is applied to a series LCR circuit. Derive the expression for the average power dissipated over a cycle.

Under what condition is (i) no power dissipated even though the current flows through the circuit, (ii) maximum power dissipated in the circuit ?

Check Solution → [2014]

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127. In a series LCR circuit connected to an ac source of variable frequency and voltage $v = v_m \omega t$, draw a plot showing the variation of current (I) with angular frequency (ω) for two different values of resistance R_1 and R_2 ($R_1 > R_2$). Write the condition under which the phenomenon of resonance occurs. For which value of the resistance out of the two curves, a sharper resonance is produced? Define Q-factor of the circuit and give its significance.

Check Solution → [2013]

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128. A series LCR circuit is connected to an ac source. Using the phasor diagram, derive the expression for the impedance of the circuit. Plot a graph to show the variation of current with frequency of the source, explaining the nature of its variation.

Check Solution → [2012]

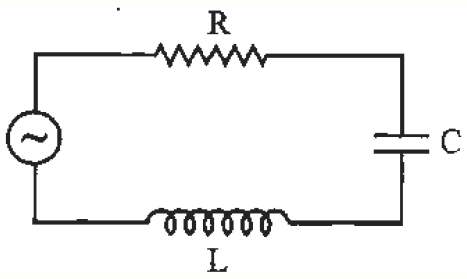
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129. The figure shows a series LCR circuit with $L = 5.0\ H$, $C = 80\ \mu F$, $R = 40\ \Omega$ connected to a variable frequency 240 V source. Calculate

(i) The angular frequency of the source which drives the circuit at resonance.

(ii) The current at the resonating frequency.

(iii) The rms potential drop across the capacitor at resonance.



Check Solution → [2012]

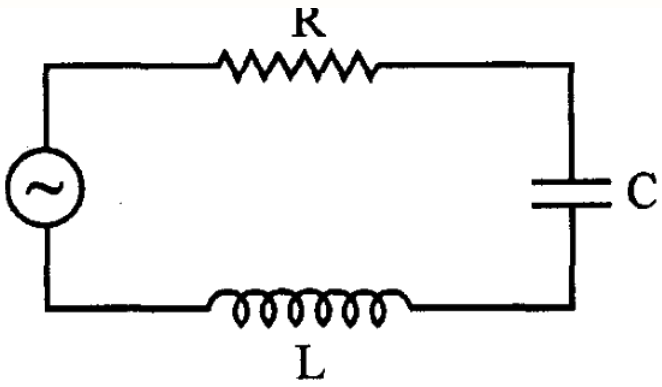
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130. A series LCR circuit with $L = 4.0\ H$, $C = 100\ \mu F$ and $R = 60\ \Omega$ is connected to a variable frequency 240 V source as shown. Calculate:

(i) the angular frequency of the source which drives the circuit at resonance;

(ii) the current at the resonating frequency;

(iii) the rms potential drop across the inductor at resonance.



Check Solution → [2012]

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131. An inductor 200 mH, capacitor 500 μF, resistor 10 Ω are connected in series with a 100 V, variable frequency a.c. source. Calculate the

(i) frequency at which the power factor of the source is unity

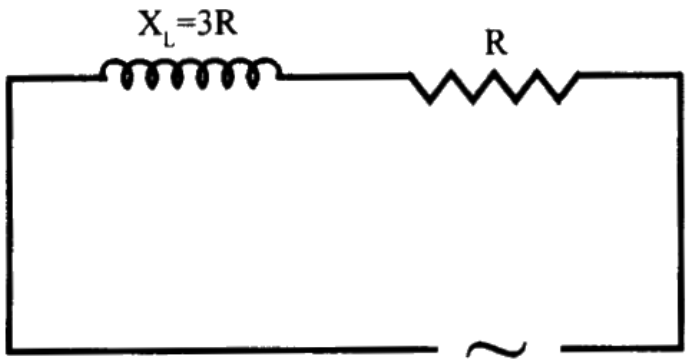
(ii) current amplitude at this frequency

(iii) Q-factor

Check Solution → [2008]

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132. Given below are two electric circuits A and B.



Calculate the ratio of power factor of the circuit B to the power factor of circuit

A.

Check Solution → [2007]

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133. In a series R-C circuit, $R = 30\ \Omega$, $C = 0.25\ \mu F$, $V = 100\ V$ and $\omega = 10{,}000$ radian/second.
Find the current in the circuit and calculate the voltage across the resistor and the capacitor. Is the algebraic sum of these voltages more than the source voltage? If yes, resolve the paradox.

Check Solution → [2004]

SECTION D

4-Mark Questions

Case Study · 3 questions

> *Section D of the paper, 4 marks each. Sorted newest year first.*

134. The teachers of Geeta's school took the students on a study trip to a power generating station, located nearly 200 km away from the city. The teacher explained that electrical energy is transmitted over such a long distance to their city, in the form of alternating current (ac) raised to a high voltage. At the receiving end in the city, the voltage is reduced to operate the devices. As a result, the power loss is reduced. Geeta listened to the teacher and asked questions about how the ac is converted to a higher or lower voltage.
(a) Name the device used to change the alternating voltage to a higher or lower value. State one cause for power dissipation in this device.
(b) Explain with an example, how power loss is reduced if the energy is transmitted over long distances as an alternating current rather than a direct current.
(c) Write two values each shown by the teachers and Geeta.

Check Solution → **[2018]**

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135. Shiv had a high tension tower erected on his farm land. He kept complaining to the authorities to remove it since it occupied a large portion of his land. His uncle, who was a teacher, explained to him the need for erecting these towers for efficient transmission of power. As Shiv got convinced and realized its significance, he stopped complaining.
Based on the above paragraph, answer the following questions:
(a) Why is it necessary to transport power at high voltages?
(b) 'A low power factor implies large power loss'. Explain.
(c) Write the two values displayed by Shiv and his Uncle.

Check Solution → **[2016]**

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136. During a thunderstorm the 'live' wire of the transmission line fell down on the ground. A group of boys passing through noticed it and some of them wanted to place the wire by the side. As they were approaching the wire and trying to lift it, Hari noticed it and immediately pushed them away to prevent them from touching the wire. Two of them got hurt in the process. Hari took them to a doctor to get medical aid.

Based on the above paragraph, answer the following:

(a) Write two values which Hari displayed during the incident.

(b) Why is it that a bird can sit over a suspended 'live' wire without any harm whereas touching it on the ground can give a fatal shock?

(c) The electric power from a power plant is set up to a very high voltage before transmitting it to distant consumers. Write the reason for it.

Check Solution → **[2016]**

SECTION E

5-Mark Questions

Long Answer · 34 questions

> *Section E of the paper, 5 marks each. Sorted newest year first.*

137. (a) A series combination of L, C and R is connected to an a.c. source. Using a phasor diagram, derive an expression for the impedance of the circuit and phase difference between V and I.

Check Solution → **[2026 • Set 55-2-1]**

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138. (b) Under what conditions:
(i) the impedance of the circuit is minimum?
(ii) wattless current flows in the circuit?

Check Solution → **[2026 • Set 55-2-1]**

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139. (i) A light bulb and an open coil inductor are connected in series across an ac source of variable frequency. How will the glow of the bulb be affected when:
(I) an iron bar is inserted inside the coil, and
(II) the frequency of the source is decreased?
Justify your answers. Assume that in each above case other factors remain unchanged.

Check Solution → **[2026 • Set 55-3-1]**

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140. (ii) An ac voltage $V = 280 \sin(100\pi t)$ volt is connected across a series LCR circuit in which $R = 400\ \Omega$, $L = \frac{5}{\pi}\ H$ and $C = \frac{1}{\pi}\ \mu F$. Taking $\sqrt{2} = 1{\cdot}4$, calculate:
(I) impedance of the circuit.
(II) rms value of current that flows in the circuit.
(III) power factor of the circuit.

Check Solution → **[2026 • Set 55-3-1]**

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141. (a)

(i) Define the terms (I) resonant frequency, and (II) power factor of a series LCR circuit. For what value of the power factor will the power dissipated in the circuit be maximum?

(ii) An inductor of 5 H, a capacitor of 50 μ F and a resistor of 400 Ω are connected in series across an ac voltage $v = 140 \sin(100\pi t)$ V. Calculate:

(I) impedance of the circuit, and

(II) rms value of current that flows in the circuit.

(Take $\sqrt{2} = 1 \cdot 4$)

OR

(b)

(i) Draw a labelled diagram of a step-up transformer. Obtain the ratio of secondary voltage to primary voltage in terms of number of turns in the two coils.

(ii) The number of turns in the primary and the secondary coil of an ideal transformer are 100 and 5000 respectively. If 3·3 kW power is supplied to the transformer at 220 V, find (I) current in the primary coil, and (II) output voltage.

Check Solution → [2026 • Set 55-4-1]

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142. (a) (i) Write the principle of working of an ac generator. Draw its labelled diagram and explain its working.

(ii) A resistor of 400 Ω, an inductor of $(\frac{2}{\pi})H$ and a capacitor of $(\frac{50}{\pi})$ μF are joined in series across an ac source $v = 140\sin(100\pi)t$ V. Find the rms voltages across these three circuit elements. The algebraic sum of these voltages is more than the rms voltage of source. Explain.

OR

(b) (i) Write the principle of working of a transformer. With the help of a labelled diagram, explain the working of a step-up transformer.

(ii) An ideal transformer is designed to convert 50 V into 250 V. It draws 200 W power from an ac source whose instantaneous voltage is given by $v_i = 20\sin(100\pi)t$ V.

Find:

(I) rms value of input current.

(II) expression for instantaneous output voltage.
(III) expression for instantaneous output current.

Check Solution → [2025 • Set 55-6-1]

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143. (i) Mention the factors on which the resonant frequency of a series LCR circuit depends. Plot a graph showing variation of impedance of a series LCR circuit with the frequency of the applied a.c. source.
(ii) With the help of a suitable diagram, explain the working of a step-up transformer.
(iii) Write two causes of energy loss in a real transformer.

Check Solution → [2024 • Set 55-4-1]

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144. (i) You are given three circuit elements X, Y and Z. They are connected one by one across a given ac source. It is found that V and I are in phase for element X. V leads I by $(\frac{\pi}{2})$ for element Y while I leads V by $(\frac{\pi}{2})$ for element Z. Identify elements X, Y and Z.
(ii) Establish the expression for impedance of circuit when elements X, Y and Z are connected in series to an ac source. Show the variation of current in the circuit with the frequency of the applied ac source.
(iii) In a series LCR circuit, obtain the conditions under which (i) impedance is minimum and (ii) wattless current flows in the circuit.

Check Solution → [2024 • Set 55-5-1]

145.

(i) Describe the construction and working of a transformer and hence obtain the relation for $(\frac{V_s}{V_p})$ in terms of number of turns of primary and secondary.
(ii) Discuss four main causes of energy loss in a real transformer.

Check Solution → [2024 • Set 55-5-1]

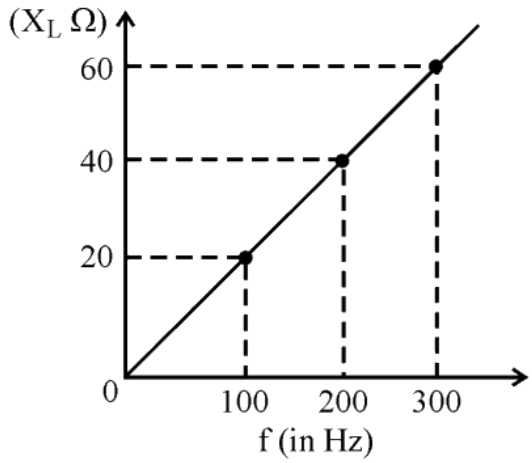
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146. With the help of a labelled diagram, explain the working of a step-up transformer. Give reasons to explain the following:
(i) The core of the transformer is laminated.
(ii) Thick copper wire is used in windings.

Check Solution → [2020 • Set 55-1-1]

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147. (a) Show that an ideal inductor does not dissipate power in an ac circuit.
(b) The variation of inductive reactance (X_L) of an inductor with the frequency (f) of the ac source of 100 V and variable frequency is shown in the fig.



(i) Calculate the self-inductance of the inductor.
(ii) When this inductor is used in series with a capacitor of unknown value and a resistor of 10 Ω at 300 s^{-1}, maximum power dissipation occurs in the circuit. Calculate the capacitance of the capacitor.

OR

(a) A conductor of length l is rotated about one of its ends at a constant angular speed ω in a plane perpendicular to a uniform magnetic field B. Plot graphs to show variations of the emf induced across the ends of the conductor with (i) angular speed ω and (ii) length of the conductor l.
(b) Two concentric circular loops of radius 1 cm and 20 cm are placed coaxially.
(i) Find mutual inductance of the arrangement.
(ii) If the current passed through the outer loop is changed at a rate of 5 A/ms, find the emf induced in the inner loop. Assume the magnetic field on the inner loop to be uniform.

Check Solution → [2020 • Set 55-5-1]

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148. In a series LCR circuit connected across an ac source of variable frequency, obtain the expression for its impedance and draw a plot showing its variation with frequency of the ac source.

Check Solution → [2019 • Set 55-1-1]

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149. What is the phase difference between the voltages across inductor and the capacitor at resonance in the LCR circuit?

Check Solution → [2019 • Set 55-1-1]

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150. When an inductor is connected to a 200 V dc voltage, a current of 1 A flows through it. When the same inductor is connected to a 200 V, 50 Hz ac source, only 0.5 A current flows. Explain, why? Also, calculate the self inductance of the inductor.

Check Solution → [2019 • Set 55-1-1]

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151. Draw the diagram of a device which is used to decrease high ac voltage into a low ac voltage and state its working principle. Write four sources of energy loss in this device.

Check Solution → [2019 • Set 55-1-1]

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152. A small town with a demand of 1200 kW of electric power at 220 V is situated 20 km away from an electric plant generating power at 440 V. The resistance of the two wire line carrying power is 0.5 Ω per km. The town gets the power from the line through a 4000-220 V step-down transformer at a sub-station in the town. Estimate the line power loss in the form of heat.

Check Solution → [2019 • Set 55-1-1]

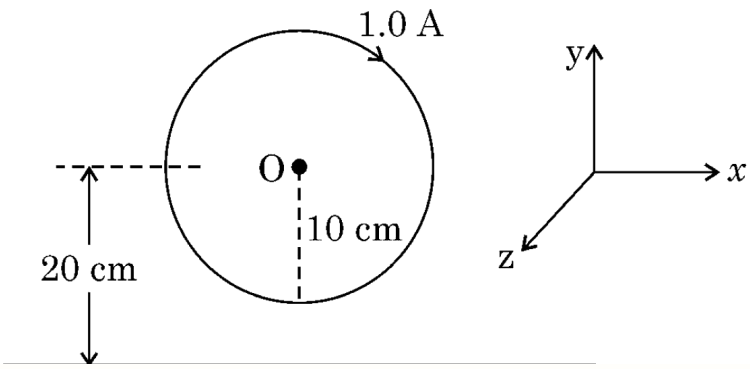
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153. (a) What do you understand by 'sharpness of resonance' for a series LCR resonant circuit? How is it related with the quality factor 'Q' of the circuit? Using the graphs given in the diagram, explain the factors which affect it. For which

graph is the resistance (R) minimum?

(b) A 2 μF capacitor, 100 Ω resistor and 8 *H* inductor are connected in series with an ac source. Find the frequency of the ac source for which the current drawn in the circuit is maximum.

If the peak value of emf of the source is 200 *V*, calculate the (i) maximum current, and (ii) inductive and capacitive reactance of the circuit at resonance.



OR

(a) Draw a schematic diagram of an ac generator. Explain its working and obtain the expression for the instantaneous value of the emf in terms of the magnetic field *B*, number of turns *N* of the coil of area *A* rotating with angular frequency ω. Show how an alternating emf is generated by a loop of wire rotating in a magnetic field.

(b) A circular coil of radius 10 cm and 20 turns is rotated about its vertical diameter with angular speed of 50 rad s^{-1} in a uniform horizontal magnetic field of 3.0×10^{-2} *T*.

(i) Calculate the maximum and average emf induced in the coil.

(ii) If the coil forms a closed loop of resistance 10 Ω, calculate the maximum current in the coil and the average power loss due to Joule heating.

Check Solution → [2019 • Set 55-4-1]

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154. A device *X* is connected across an ac source of voltage $V = V_0 \sin \omega t$. The current through *X* is given as $I = I_0 \sin(\omega t + \frac{\pi}{2})$.

(a) Identify the device *X* and write the expression for its reactance.

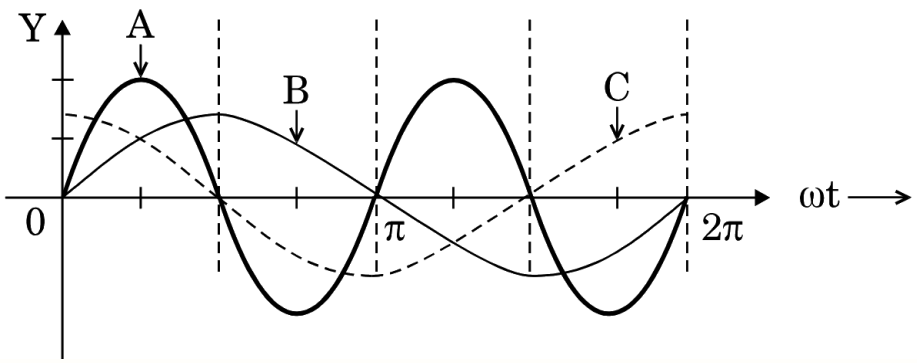
(b) Draw graphs showing variation of voltage and current with time over one cycle of ac, for *X*.

(c) How does the reactance of the device X vary with frequency of the ac ? Show this variation graphically.
(d) Draw the phasor diagram for the device X.

Check Solution → [2018]

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155. A device X is connected to an ac source $V = V_0 \sin \omega t$. The variation of voltage, current and power in one cycle is shown in the following graph:



(a) Identify the device 'X'.
(b) Which of the curves A, B and C represent the voltage, current and the power consumed in the circuit? Justify your answer.
(c) How does its impedance vary with frequency of the ac source? Show graphically.
(d) Obtain an expression for the current in the circuit and its phase relation with ac voltage.

Check Solution → [2017]

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156. (i) Draw a labelled diagram of a step-down transformer. State the principle of its working.
(ii) Express the turn ratio in terms of voltages.
(iii) Find the ratio of primary and secondary currents in terms of turn ratio in an ideal transformer.
(iv) How much current is drawn by the primary of a transformer connected to 220 V supply when it delivers power to a 110 V – 550 W refrigerator?

OR

(a) Explain the meaning of the term mutual inductance. Consider two concentric

circular coils, one of radius r_1 and the other of radius r_2 ($r_1 < r_2$) placed coaxially with centres coinciding with each other. Obtain the expression for the mutual inductance of the arrangement.

(b) A rectangular coil of area A, having number of turns N is rotated at f revolutions per second in a uniform magnetic field B, the field being perpendicular to the coil. Prove that the maximum emf induced in the coil is $2\pi f N B A$.

Check Solution → [2016]

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157. In a series LCR circuit connected to an a.c. source of voltage $v = v_m \omega t$, use phasor diagram to derive an expression for the current in the circuit. Hence obtain the expression for the power dissipated in the circuit. Show that power dissipated at resonance is maximum.

Check Solution → [2016]

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158. (a) Show that in an a.c. circuit containing a pure inductor, the voltage is ahead of current by $\pi/2$ in phase.

(b) A horizontal straight wire of length L extending from east to west is falling with speed v at right angles to the horizontal component of Earth's magnetic field B.

(i) Write the expression for the instantaneous value of the e.m.f. induced in the wire.

(ii) What is the direction of the e.m.f.?

(iii) Which end of the wire is at the higher potential?

Check Solution → [2011]

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159. (i) With the help of a labelled diagram, describe briefly the underlying principle and working of a step up transformer.
(ii) Write any two sources of energy loss in a transformer.
(iii) A step up transformer converts a low input voltage into a high output voltage. Does it violate law of conservation of energy? Explain.

Check Solution → [2011 • Set 55-1-1]

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160. Derive an expression for the impedance of a series LCR circuit connected to an AC supply of variable frequency.
Plot a graph showing variation of current with the frequency of the applied voltage.
Explain briefly how the phenomenon of resonance in the circuit can be used in the tuning mechanism of a radio or a TV set.

Check Solution → [2011 • Set 55-1-1]

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161. (a) An alternating voltage $v = v_m \sin \omega t$ applied to a series LCR circuit drives a current given by $i = i_m \sin(\omega t + \phi)$. Deduce an expression for the average power dissipated over a cycle.
(b) For circuits used for transporting electric power, a low power factor implies large power loss in transmission. Explain.
(c) Determine the current and quality factor at resonance for a series LCR circuit with $L = 1.00$ mH, $C = 1.00$ nF and $R = 100\ \Omega$ connected to an a.c. source having peak voltage of 100 V.

Check Solution → [2011 • Set 55-2-1]

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162. A series LCR circuit is connected to an ac source having voltage $v = v_m \sin \omega t$. Derive the expression for the instantaneous current I and its phase relationship to the applied voltage.
Obtain the condition for resonance to occur. Define 'power factor'. State the conditions under which it is (i) maximum and (ii) minimum.

Check Solution → [2010]

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163. (a) Derive an expression for the average power consumed in a series LCR circuit connected to a.c. source in which the phase difference between the voltage and the current in the circuit is ϕ.
(b) Define the quality factor in an a.c. circuit. Why should the quality factor have high value in receiving circuits? Name the factors on which it depends.

Check Solution → [2009]

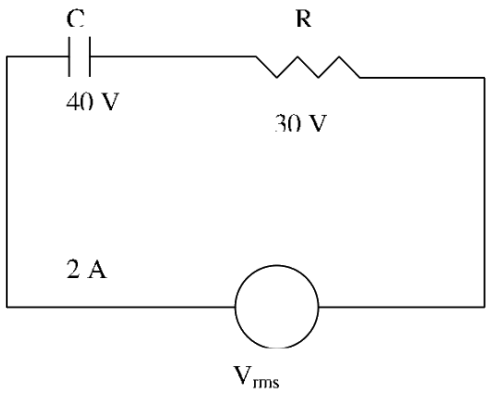
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164. (a) Derive the relationship between the peak and the rms value of current in an a.c. circuit.
(b) Describe briefly, with the help of a labelled diagram, working of a step-up transformer.
(c) A step-up transformer converts a low voltage into high voltage. Does it not violate the principle of conservation of energy? Explain.

Check Solution → [2009]

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165. State the condition for resonance to occur in a series LCR a.c circuit and derive an expression for the resonant frequency.
Draw a plot showing the variation of the peak current (i_m) with frequency of the a.c source used. Define the quality factor, Q of the circuit.
Calculate the (i) impedance, (ii) wattless current of the given a.c circuit.



Check Solution → [2008]

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166. Draw a labeled circuit arrangement showing the windings of primary and secondary coil in a transformer. Explain the underlying principle and working

of a step-up transformer. Write any two major sources of energy loss in this device.
How much current is drawn by the primary coil of a transformer which steps down 220 V to 22 V to operate a device with an impedance of 220 ohm?

Check Solution → [2008]

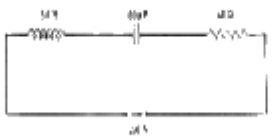
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167. Explain the term 'inductive reactance'. Show graphically the variation of inductive reactance with frequency of the applied alternating voltage.
An a.c. voltage $E = E_0 \sin \omega t$ is applied across a pure inductor of inductance L. Show mathematically that the current flowing through it lags behind the applied voltage by a phase angle of $\pi/2$.

Check Solution → [2007]

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168. The given circuit diagram shows a series LCR circuit connected to a variable frequency 230 V source:



(a) Determine the source frequency which drives the circuit in resonance.
(b) Obtain the impedance of the circuit and the amplitude of current at the resonating frequency.
(c) Determine the rms potential drops across the three elements of the circuit.
(d) How do you explain the observation that the algebraic sum of the voltages across the three elements obtained in (c) is greater than the supplied voltage?

Check Solution → [2006]

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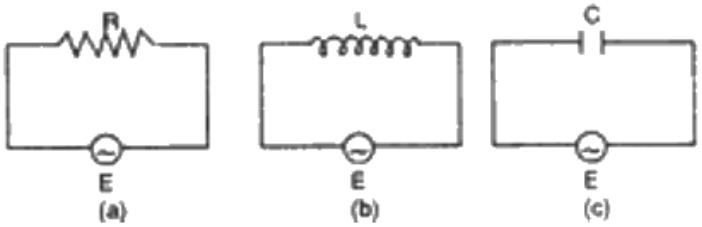
169. The primary coil of an ideal step-up transformer has 100 turns and the transformation ratio is also 100. The input voltage and power are 220 V and 1100 W respectively. Calculate:

(i) number of turns in the secondary
(ii) the current in the primary
(iii) voltage across the secondary
(iv) the current in the secondary
(v) power in the secondary

Check Solution → [2006]

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170. What is a choke coil? Why is it preferred to resistance in a.c. circuits?
In figures (a), (b) and (c) are shown three a.c. circuits with equal currents. If the frequency of e.m.f. be increased, then what will be the effect on the currents flowing in them? Explain with reason.



Check Solution → [2003]

SELF-CHECK

Answer Key

All MCQs at a glance

> *Attempt the MCQs first, then check here. Tap any question number to open its full worked solution.*

Q1 **B**	Q8 **D**	Q15 **A**	Q24 **D**	Q31 **B**	Q38 **A**
Q2 **D**	Q9 **D**	Q16 **C**	Q25 **D**	Q32 **C**	Q39 **D**
Q3 **D**	Q10 **B**	Q17 **D**	Q26 **B**	Q33 **D**	Q40 **A**
Q4 **D**	Q11 **C**	Q18 **A**	Q27 **A**	Q34 **C**	Q48 **C**
Q5 **C**	Q12 **D**	Q19 **C**	Q28 **A**	Q35 **B**	
Q6 **C**	Q13 **C**	Q20 **C**	Q29 **D**	Q36 **D**	
Q7 **D**	Q14 **A**	Q23 **A**	Q30 **C**	Q37 **D**	

Detailed Solutions

Every question worked step by step

Each card restates the question, shows the answer, then the full working. Use "Back to Q n" to return.

Q1

In a series LCR circuit, the voltage across the resistor, capacitor and inductor is 10 V each. If the capacitor is short circuited, the voltage across the inductor will be:

(A) 10 V

(B) $5\sqrt{2}$ V

(C) $\frac{5}{\sqrt{2}}$ V

(D) $10\sqrt{2}$ V

[2026 • Set 55-1-1]

SOLUTION

In the original series LCR circuit the voltages are equal: $\mathbf{V_R = V_L = V_C = 10\ V}$.

Since $\mathbf{V_L = V_C}$, the inductive and capacitive voltages cancel, so the circuit is at resonance and the source voltage equals the resistor voltage: $\mathbf{V = V_R = 10\ V}$.

The same current flows through every element, so from $\mathbf{V_R = IR = 10}$ and $\mathbf{V_L = IX_L = 10}$ we get $\mathbf{X_L = R}$. Likewise $\mathbf{X_C = R}$.

When the capacitor is short circuited, only R and L remain in series, with impedance

$$\mathbf{Z' = \sqrt{R^2 + X_L{}^2} = \sqrt{R^2 + R^2} = R\sqrt{2}}.$$

The new current is

$$\mathbf{I' = \frac{V}{Z'} = \frac{10}{R\sqrt{2}}}.$$

The new voltage across the inductor is

$$\mathbf{V_L{}' = I'\,X_L = \frac{10}{R\sqrt{2}} \times R = \frac{10}{\sqrt{2}}\ V}.$$

Rationalising, $\frac{10}{\sqrt{2}} = \frac{10\sqrt{2}}{2} = 5\sqrt{2}\ V \approx 7.07\ V$.

This matches option (B) $5\sqrt{2}\ V$. Hence the voltage across the inductor becomes $5\sqrt{2}\ V$, option (B).

COMMON MISTAKE

When X_C is removed, the current changes, so always recompute the current with the new impedance before finding V_L; never reuse the resonance current.

Final answer. The voltage across the inductor becomes $\frac{10}{\sqrt{2}} = 5\sqrt{2}\ V$, option **(B)**.

← Back to Q1

Q2

A resistor and an inductor of negligible resistance are connected in series to a 20 V ac source. If the voltage across the resistor is 12 V, the voltage across the inductor will be:

(A) 6 V

(B) 8 V

(C) 10 V

(D) **16 V**

[2026 • Set 55-1-2]

SOLUTION

In a series R-L circuit the voltage across the resistor is in phase with the current, while the voltage across the (pure) inductor leads the current by 90°. So V_R and V_L are at right angles and add as perpendicular vectors.

The phasor relation between the source voltage and the two component voltages is

$$V_{src}^2 = V_R^2 + V_L^2.$$

Rearranging for the inductor voltage,

$$V_L = \sqrt{V_{src}^2 - V_R^2}.$$

Substituting $V_{src} = 20\ V$ and $V_R = 12\ V$,

$V_L = \sqrt{(20)^2 - (12)^2}$.

$V_L = \sqrt{400 - 144} = \sqrt{256}$.

$V_L = 16\ V$.

Hence **the voltage across the inductor is 16 *V***, option (D).

Final answer. The voltage across the inductor is **16 *V***, option **(D)**.

← *Back to Q2*

Q3

A series LCR circuit with $R = 3\ \Omega$, $X_L = 4\ \Omega$, $X_C = 8\ \Omega$ is connected to a 220 V, 50 Hz ac source. The power factor for the circuit is:

(A) 0.30

(B) 0.45

(C) 0.50

(D) 0.60

[2026 • Set 55-1-3]

SOLUTION

The power factor of a series LCR circuit is $\phi = \frac{R}{Z}$, where the impedance is $Z = \sqrt{R^2 + (X_L - X_C)^2}$.

Given $R = 3\ \Omega$, $X_L = 4\ \Omega$ and $X_C = 8\ \Omega$, the net reactance is

$X_L - X_C = 4 - 8 = -4\ \Omega$.

The magnitude of the impedance is

$Z = \sqrt{(3)^2 + (-4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5\ \Omega$.

Therefore the power factor is

$$\phi = \frac{R}{Z} = \frac{3}{5} = 0.60.$$

(The source values 220 V and 50 Hz are not needed, since the power factor depends only on R, X_L and X_C.)

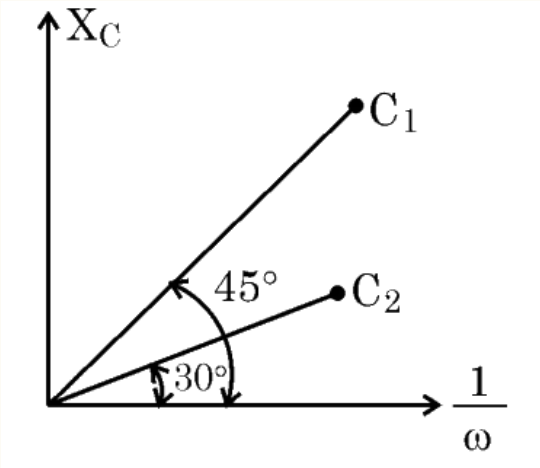
Hence **the power factor of the circuit is 0.60**, option (D).

Final answer. The power factor is **0.60**, option **(D)**.

← *Back to Q3*

Q4

The figure shows the variation of capacitive reactance (X_C) *of two ideal capacitors of capacitances* C_1 *and* C_2 *with the reciprocal of angular frequency* ($1/\omega$) *of ac source. The value of* C_1/C_2 *is:*



(A) 5

(B) 2

(C) $\sqrt{3}$

(D) $\frac{1}{\sqrt{3}}$

[2026 • Set 55-2-1]

SOLUTION

For an ideal capacitor the capacitive reactance is $X_C = \frac{1}{\omega C}$, which we can write as

$X_C = \frac{1}{C}\left(\frac{1}{\omega}\right)$.

When X_C is plotted against $\frac{1}{\omega}$, this is a straight line through the origin with slope $\frac{1}{C}$. So the steeper the line, the smaller the capacitance.

The graph shows two straight lines from the origin, one for C_1 and one for C_2, with different slopes. The line for C_1 is steeper than the line for C_2, giving

$$\textbf{slope}_1 = \frac{1}{C_1},\ \textbf{slope}_2 = \frac{1}{C_2}.$$

Taking the ratio of the slopes,

$$\frac{\textbf{slope}_1}{\textbf{slope}_2} = \frac{1/C_1}{1/C_2} = \frac{C_2}{C_1}.$$

From the figure this ratio of slopes equals $\sqrt{3}$, so

$$\frac{C_2}{C_1} = \sqrt{3} \Rightarrow \frac{C_1}{C_2} = \frac{1}{\sqrt{3}}.$$

Hence $\frac{C_1}{C_2} = \frac{1}{\sqrt{3}}$, option (D).

CONCEPT

On an X_C versus $1/\omega$ graph the slope is $1/C$, so a steeper line means a smaller capacitor; do not read steeper as larger.

Final answer. The ratio $\frac{C_1}{C_2} = \frac{1}{\sqrt{3}}$, option **(D)**.

← *Back to Q4*

Q5

An ac voltage is given as $v = 14\sin(314\,t)$ *V. The average and the effective value of the voltage (in V) over a cycle are respectively:*

(A) 14 and 7

(B) 10 and 14

(C) **0 and 10**

(D) 10 and 0

[2026 • Set 55-4-1]

SOLUTION

The given voltage is $v = 14\sin(314\,t)\ V$, a pure sinusoid with peak value $v_0 = 14\ V$.

Average value over one full cycle: the mean of a sine wave over a complete period is

$$v = \frac{1}{T}\int_0^T v_0 \omega\, t\, dt = 0,$$

because the positive half-cycle exactly cancels the negative half-cycle.

Effective (rms) value: for a sinusoid the root-mean-square value is

$$v_{rms} = \frac{v_0}{\sqrt{2}}.$$

Substituting $v_0 = 14\ V$,

$$v_{rms} = \frac{14}{\sqrt{2}} = \frac{14}{1.414} \approx 9.9 \approx 10\ V.$$

So the average value is 0 and the effective value is about 10 V.

Hence **the average and effective values are 0 and 10 V respectively**, option (C).

Final answer. The average value is **0** and the effective value is **10 V**, option **(C)**.

← *Back to Q5*

Q6

The rms and the average value of an ac voltage $V = V_0 \sin \omega\, t$ volt over a cycle respectively will be:

(A) $\frac{V_0}{\sqrt{2}}, \frac{V_0}{2}$

(B) $\frac{V_0}{\pi}, \frac{V_0}{2}$

(C) $\frac{V_0}{\sqrt{2}}, 0$

(D) $V_0, \frac{V_0}{2}$

[2026 • Set 55-5-1]

SOLUTION

For the alternating voltage $V = V_0 \omega t$, with peak value V_0:

The rms value is defined as the square root of the mean of V^2 over a cycle. Since the mean of $\sin^2 \omega t$ over a full period is $\frac{1}{2}$,

$$V_{rms} = \sqrt{V_0^2 \sin^2 \omega t} = V_0 \sqrt{\frac{1}{2}} = \frac{V_0}{\sqrt{2}}.$$

The average value over one complete cycle is

$$V = \frac{1}{T} \int_0^T V_0 \omega t \, dt = 0,$$

since the positive and negative half-cycles are equal and opposite and cancel out.

(Note that the average over only half a cycle would be $\frac{2V_0}{\pi}$, but the question asks for a full cycle, which is zero.)

Hence **the rms and average values are $\frac{V_0}{\sqrt{2}}$ and 0 respectively**, option (C).

> **COMMON MISTAKE**
>
> *Average over a full cycle is zero; average over a half cycle is $\frac{2V_0}{\pi}$. Read the question carefully to see which is asked.*

Final answer. The rms value is $\frac{V_0}{\sqrt{2}}$ and the average value over a cycle is **0**, option **(C)**.

← *Back to Q6*

Q7

Which of the following statements is not true for electric energy in ac form compared

to that in dc form?

(A) Production of ac is economical.

(B) ac can be easily and efficiently converted from one voltage to the other.

(C) ac can be transmitted economically over long distances.

(D) ac is less dangerous.

[2026 • Set 55-5-1]

SOLUTION

We compare the practical advantages of electric energy in ac form against dc form, statement by statement.

(A) Production of ac is economical: TRUE. Alternating current is generated cheaply and on a large scale by ac generators (alternators) in power stations.

(B) ac can be easily and efficiently converted from one voltage to another: TRUE. A transformer steps ac voltage up or down with very high efficiency, something a transformer cannot do with steady dc.

(C) ac can be transmitted economically over long distances: TRUE. Voltage is stepped up to a high value for transmission, which lowers the current and hence reduces I^2R heating losses in the cables.

(D) ac is less dangerous: FALSE. For a given voltage, ac is generally considered more dangerous than dc, because the alternating current can disturb the heart rhythm (ventricular fibrillation) and its peak value $V_0 = \sqrt{2}\, V_{rms}$ exceeds the quoted rms value.

Hence the statement that is NOT true is **"ac is less dangerous"**, option (D).

Final answer. The statement that is not true is that ac is less dangerous, option **(D)**.

← *Back to Q7*

Q8

A voltage $v = v_0 \sin \omega t$ *applied to a circuit drives a current* $i = i_0 \sin(\omega t + \phi)$ *in the*

circuit. The average power consumed in the circuit over a cycle is:

(A) Zero

(B) $i_0 v_0 \phi$

(C) $\frac{i_0 v_0}{2}$

(D) $\frac{i_0 v_0}{2}\phi$

[2025 • Set 55-1-1]

SOLUTION

The applied voltage is $v = v_0 \omega t$ and the current is $i = i_0 \sin(\omega t + \phi)$, where ϕ is the phase difference between current and voltage.

The average power consumed over one full cycle is the product of the rms voltage, the rms current and the power factor ϕ:

$P = V_{rms} I_{rms} \phi$.

Writing the rms values in terms of the peak values, $V_{rms} = \frac{v_0}{\sqrt{2}}$ and $I_{rms} = \frac{i_0}{\sqrt{2}}$,

$P = \frac{v_0}{\sqrt{2}} \times \frac{i_0}{\sqrt{2}} \times \phi$.

$P = \frac{v_0 i_0}{2}\phi$.

The term ϕ is the power factor; only the in-phase component of the current contributes to the average power, while the wattless component averages to zero.

Hence **the average power consumed is** $\frac{v_0 i_0}{2}\phi$, option (D).

Final answer. The average power consumed over a cycle is $\frac{v_0 i_0}{2}\phi$, option **(D)**.

← *Back to Q8*

Q9

The ratio of the number of turns of the primary to the secondary coils in an ideal transformer is 20 : 1. *If* 240 *V ac is applied from a source to the primary coil of*

transformer and a 6.0 Ω resistor is connected across the output terminals, then current drawn by the transformer from the source will be:

(A) 4.0 A

(B) 3.8 A

(C) 0.9 A

(D) **0.10 A**

[2025 • Set 55-1-3]

SOLUTION

The turns ratio is $N_p : N_s = 20 : 1$, so this is a step-down transformer.

For an ideal transformer the voltage ratio equals the turns ratio:

$$\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{1}{20}.$$

The secondary (output) voltage is therefore

$$V_s = V_p \times \frac{N_s}{N_p} = 240 \times \frac{1}{20} = 12\ V.$$

The 6.0 Ω resistor is across the output, so by Ohm's law the secondary current is

$$I_s = \frac{V_s}{R} = \frac{12}{6.0} = 2.0\ A.$$

For an ideal transformer input power equals output power, $V_p I_p = V_s I_s$, which gives

$$I_p = I_s \times \frac{N_s}{N_p} = 2.0 \times \frac{1}{20} = 0.10\ A.$$

Hence **the current drawn from the source is 0.10 A**, option (D).

CONCEPT

In a step-down transformer the secondary voltage falls but the secondary current rises; the primary current is the smallest, since $V_p I_p = V_s I_s$.

Final answer. The current drawn by the transformer from the source is **0.10 A**, option **(D)**.

← *Back to Q9*

Q10

Assertion (A): *In an ideal step-down transformer, the electrical energy is not lost.*
Reason (R): *In a step-down transformer, voltage decreases but the current increases.*

(B) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).

[2025 • Set 55-2-1]

SOLUTION

Examine the Assertion and the Reason separately.

Assertion (A): In an ideal step-down transformer the electrical energy is not lost. This is TRUE. An ideal transformer has no copper loss, no iron (eddy current and hysteresis) loss and no flux leakage, so input power equals output power, $V_p I_p = V_s I_s$, and no energy is dissipated.

Reason (R): In a step-down transformer the voltage decreases but the current increases. This is also TRUE. Because $\frac{V_s}{V_p} = \frac{N_s}{N_p} < 1$, the output voltage is smaller, and conservation of power $V_p I_p = V_s I_s$ then forces the output current to be correspondingly larger.

However, the Reason only states the inverse voltage-current relationship; it is a consequence of energy conservation, not the cause of why no energy is lost. The lossless behaviour in the Assertion follows from the absence of copper, iron and leakage losses, which the Reason does not mention. So R does not correctly explain A.

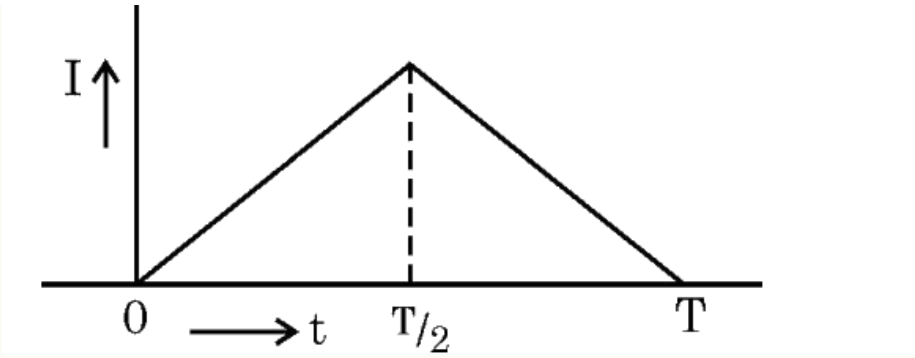
Hence **both A and R are true, but R is not the correct explanation of A**, option (B).

Final answer. Both Assertion and Reason are true, but the Reason is not the correct explanation of the Assertion, option **(B)**.

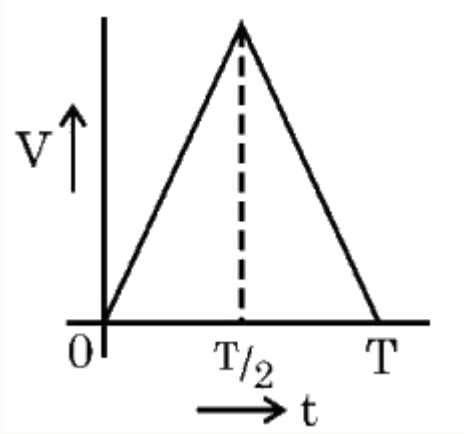
← *Back to Q10*

Q11

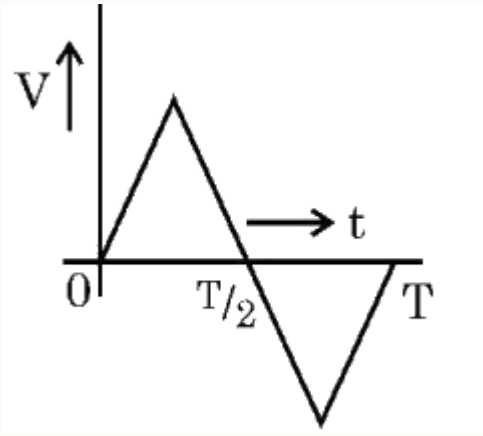
The alternating current I in an inductor is observed to vary with time t as shown in the graph for a cycle. Which one of the following graphs is the correct representation of wave form of voltage V with time t?



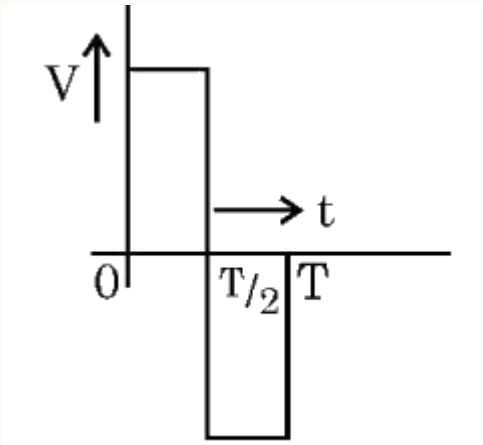
(A)



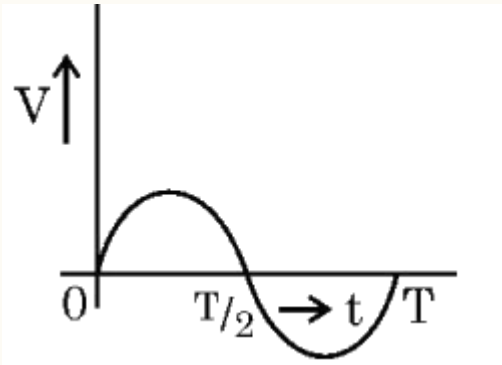
(B)



(C)



(D)



[2025 • Set 55-2-1]

SOLUTION

For a pure inductor, the instantaneous voltage across it is governed by the law of electromagnetic induction:

$$V = L\,\frac{dI}{dt}$$

Hence the voltage at any instant is directly proportional to the slope (rate of change) of the current waveform, not to the current itself.

The given current is a triangular wave. Over the first half of each cycle the current rises linearly, so $\frac{dI}{dt}$ is a positive constant; over the second half the current falls linearly, so $\frac{dI}{dt}$ is a negative constant of equal magnitude.

Since $\frac{dI}{dt}$ is piecewise constant and switches sign each half-cycle, the voltage $V(t)$ takes a constant positive value during the rising portion and an equal constant negative value during the falling portion.

This produces a **square wave** for $V(t)$, with the level jumps occurring exactly at the peaks and troughs of the triangular current (where the slope reverses).

Therefore the correct voltage waveform is the square wave, which is **Option (C)**.

CONCEPT

For an inductor think slope, not value: V follows dI/dt. A triangular current always gives a square voltage, and a sine current gives a cosine (shifted) voltage.

Final answer. The voltage waveform is a square wave, Option **(C)**.

← *Back to Q11*

Q12

A transformer is connected to a 200 V ac source. The transformer supplies 3000 V to a device. If the number of turns in the primary coil is 450, then the number of turns in its secondary coil is

(A) 30

(B) 450

(C) 4500

(D) 6750

[2025 • Set 55-2-1]

SOLUTION

For an ideal transformer the turns ratio equals the voltage ratio:

$$\frac{V_p}{V_s} = \frac{N_p}{N_s}$$

Rearranging to find the secondary turns:

$$N_s = N_p \times \frac{V_s}{V_p}$$

Here $V_p = 200\ V$, $V_s = 3000\ V$ and $N_p = 450$.

$$\frac{V_s}{V_p} = \frac{3000}{200} = 15$$

$$N_s = 450 \times 15 = 6750$$

Since $V_s > V_p$ and $N_s > N_p$, this is a step-up transformer, which is consistent with the result.

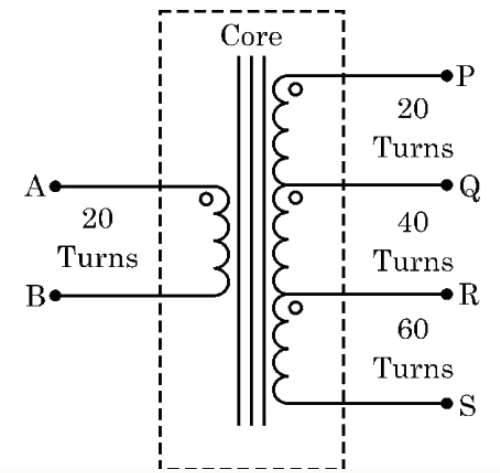
Therefore the secondary coil has $N_s = 6750$ **turns**, which is **Option (D)**.

Final answer. The secondary coil has 6750 turns, Option **(D)**.

← Back to Q12

Q13

The number of turns between different pairs of output terminals are shown for a step-up transformer. Input voltage of 20 V is applied between A and B. Between which two terminals will the output be 120 V?



(A) P and Q

(B) Q and S

(C) **P and R**

(D) P and S

[2025 • Set 55-2-3]

SOLUTION

For an ideal transformer the output voltage across any pair of secondary terminals is proportional to the number of turns enclosed between them:

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

The input of 20 *V* is applied across the primary winding A-B. The required step-up ratio is:

$$\frac{V_s}{V_p} = \frac{120}{20} = 6$$

So the output must be taken across the terminal pair whose number of turns is 6 times the number of turns in the primary section A-B.

Reading the turn counts marked in the figure, the pair P and R encloses exactly 6 times the turns of A-B, giving:

$$V_{PR} = 6 \times 20 = 120\ V$$

Therefore the 120 *V* output appears across terminals **P and R**, which is **Option (C)**.

Final answer. The 120 V output is across P and R, Option **(C)**.

← *Back to Q13*

Q14

Assertion (A): *A series LCR circuit behaves as a pure resistive circuit at resonance.*

Reason (R): *At resonance,* $\mathbf{X_L = X_C}$ *gives* $\boldsymbol{\omega = \frac{1}{\sqrt{LC}}}$.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

[2025 • Set 55-4-1]

SOLUTION

Examine the Assertion. In a series LCR circuit the impedance is:

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

At resonance the inductive and capacitive reactances become equal, $\mathbf{X_L = X_C}$, so the net reactance $\mathbf{(X_L - X_C) = 0}$.

$$Z = \sqrt{R^2 + 0} = R$$

The impedance reduces to pure resistance, the phase angle between current and voltage is zero, and the circuit behaves as a purely resistive circuit. Hence the Assertion is TRUE.

Examine the Reason. The resonance condition $\mathbf{X_L = X_C}$ means:

$$\omega L = \frac{1}{\omega C}$$

$$\omega^2 = \frac{1}{LC} \Rightarrow \omega = \frac{1}{\sqrt{LC}}$$

So the Reason is also TRUE.

It is precisely this condition $X_L = X_C$ (the resonance condition stated in R) that cancels the reactance and makes the impedance equal to R. Therefore the Reason is the correct explanation of the Assertion.

Hence both A and R are true and R correctly explains A, which is **Option (A)**.

Final answer. Both A and R are true and R explains A, Option **(A)**.

← *Back to Q14*

Q15

An ammeter connected in series in an ac circuit reads 10 *A. The maximum value of current at any instant in the circuit is :*

(A) $10\sqrt{2}\ A$

(B) $\frac{2}{\sqrt{2}}\ A$

(C) $\frac{10}{\pi}\ A$

(D) $\frac{10}{2}\ A$

[2025 • Set 55-4-1]

SOLUTION

An AC ammeter always reads the root-mean-square (rms) value of the alternating current, not its peak value. The rms and peak values are related by:

$$I_{rms} = \frac{I_0}{\sqrt{2}}$$

Rearranging for the maximum (peak) current:

$$I_0 = \sqrt{2}\ I_{rms}$$

The ammeter reads $\mathbf{I_{rms} = 10\ A}$, so:

$$I_0 = \sqrt{2} \times 10 = 10\sqrt{2}\ A \approx 14.14\ A$$

The other options do not follow from $\mathbf{I_0 = \sqrt{2}\ I_{rms}}$: $\frac{10}{2}$ and $\frac{10}{\pi}$ would be smaller than the rms value, which is impossible since the peak must exceed the rms, and $\frac{2}{\sqrt{2}} = \sqrt{2}\ A$ ignores the factor of 10.

Therefore the maximum value of the current is $\mathbf{10\sqrt{2}\ A}$, which is **Option (A)**.

Final answer. The maximum current is $\mathbf{10\sqrt{2}}$ A, Option **(A)**.

← *Back to Q15*

Q16

A transformer is a device used for converting:

(A) high ac voltage and large ac current to low ac voltage and small ac current

(B) high ac voltage and small ac current to low ac voltage and small ac current

(C) **low ac voltage and large ac current to high ac voltage and small ac current**

(D) low ac voltage and small ac current to high ac voltage and large ac current

[2025 • Set 55-5-2]

SOLUTION

A transformer works on the principle of mutual induction and transfers electrical power from the primary to the secondary while keeping the power (almost) constant. Neglecting losses:

$$V_p\, I_p = V_s\, I_s$$

This means voltage and current change in inverse proportion: if the voltage is stepped up, the current is stepped down by the same factor.

A step-up transformer takes a low voltage at the primary and delivers a high voltage at the secondary. By the power relation above, the primary therefore carries a large current and the secondary a small current.

So a step-up transformer converts low AC voltage and large AC current into high AC voltage and small AC current, which matches **Option (C)**.

The other options are inconsistent: a transformer cannot reduce both voltage and current together (that would lose power), so (A) and (B) are wrong; option (D) claims output current is large while output voltage is high, again violating power conservation, so it is wrong.

Therefore the correct description is **Option (C)**.

Final answer. It converts low voltage and large current to high voltage and small current, Option **(C)**.

← *Back to Q16*

Q17

A capacitor of capacitance C has reactance X in an ac circuit. If the capacitance and the frequency of the applied voltage are doubled, the new reactance will become:

(A) 4X

(B) 2X

(C) $\frac{X}{2}$

(D) $\frac{X}{4}$

[2025 • Set 55-5-3]

SOLUTION

The capacitive reactance is given by:

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

So the reactance is inversely proportional to the product of frequency and capacitance:

$$X_C \propto \frac{1}{fC}$$

Initially $\mathbf{X_C = X}$. Now both f and C are doubled, so the product fC in the denominator becomes:

$$(2f)(2C) = 4\,(fC)$$

Since the new reactance is inversely proportional to this product:

$$X' = \frac{1}{2\pi\,(2f)(2C)} = \frac{1}{4} \cdot \frac{1}{2\pi f C} = \frac{X}{4}$$

Therefore the new reactance becomes $\boldsymbol{\frac{X}{4}}$, which is **Option (D)**.

COMMON MISTAKE

Capacitive reactance falls with both f and C (inverse), while inductive reactance rises with both f and L. Mixing up these dependences is the usual slip here.

Final answer. The new reactance is $\boldsymbol{\frac{X}{4}}$, Option **(D)**.

← Back to Q17

Q18

An ac source is connected to a resistor and an inductor in series. The voltage across the resistor and inductor are 8 V and 6 V respectively. The voltage of the source is:

(A) 10 V

(B) 12 V

(C) 14 V

(D) 16 V

[2025 • Set 55-6-1]

SOLUTION

In a series R-L circuit the same current flows through both elements, but the voltage across the resistor is in phase with the current while the voltage across the inductor leads the current by 90°. Hence V_R and V_L are perpendicular (in quadrature) on the phasor diagram.

On the phasor diagram, draw V_R along the horizontal (current) direction and V_L vertically upward; the source voltage is the resultant (hypotenuse) of this right-angled triangle:

$$V_{src} = \sqrt{V_R^2 + V_L^2}$$

Substituting $V_R = 8\ V$ and $V_L = 6\ V$:

$$V_{src} = \sqrt{8^2 + 6^2}$$

$$V_{src} = \sqrt{64 + 36} = \sqrt{100}$$

$$V_{src} = 10\ V$$

Note that a simple arithmetic sum $8 + 6 = 14\ V$ would be wrong, because the two voltages are not in phase; they must be added as perpendicular vectors.

Therefore the source voltage is **10 V**, which is **Option (A)**.

> **COMMON MISTAKE**
>
> *In AC, never add voltages arithmetically across different elements; add them as phasors. Here 8 V and 6 V give 10 V, not 14 V.*

Final answer. The source voltage is 10 V, Option **(A)**.

← *Back to Q18*

Q19

An alternating current is given by $I = I_0\cos(100\pi)t$. *The least time the current takes to decrease from its maximum value to zero will be:*

(A) $(\frac{1}{50})\ s$

(B) $(\frac{1}{100})\ s$

(C) $(\frac{1}{200})\ s$

(D) $(\frac{1}{400})\ s$

SOLUTION

The current is $I = I_0 \cos(\omega t)$ with angular frequency:

$$\omega = 100\pi \text{ rad/}s$$

At $t = 0$, $\cos 0 = 1$, so the current is at its maximum value I_0. The current first becomes zero when the cosine becomes zero, i.e. when:

$$\omega t = \frac{\pi}{2}$$

Solving for the time taken:

$$t = \frac{\pi/2}{\omega} = \frac{\pi/2}{100\pi}$$

$$t = \frac{1}{200}\ s$$

This is the least time to fall from maximum to zero, and it equals one-quarter of the time period, since $T = \frac{2\pi}{\omega} = \frac{2\pi}{100\pi} = \frac{1}{50}\ s$ and $\frac{T}{4} = \frac{1}{200}\ s$.

Therefore the least time is $\frac{1}{200}\ s$, which is **Option (C)**.

Final answer. The least time is $\frac{1}{200}$ s, Option **(C)**.

← *Back to Q19*

Q20

A capacitor and an inductor are connected in series across an ac source of voltage of variable frequency. The frequency is increased continuously. The nature of the circuit before and after the resonance will be:

(A) inductive only

(B) capacitive only

(C) **capacitive and inductive respectively**

(D) inductive and capacitive respectively

[2025 • Set 55-7-1]

SOLUTION

For a series LC (or LCR) circuit the inductive and capacitive reactances depend on frequency as:

$$X_L = \omega L \, X_C = \frac{1}{\omega C}$$

As frequency increases, X_L rises and X_C falls. The nature of the circuit is decided by which reactance dominates.

Below resonance (low frequency), $\mathbf{X_C > X_L}$, so the net reactance is capacitive and the circuit behaves as a capacitive circuit.

At resonance, $\mathbf{X_L = X_C}$, the reactances cancel and the circuit is purely resistive.

Above resonance (high frequency), $\mathbf{X_L > X_C}$, so the net reactance is inductive and the circuit behaves as an inductive circuit.

Hence as the frequency is increased continuously through resonance, the circuit changes from **capacitive before resonance to inductive after resonance**, which is **Option (C)**.

Final answer. The circuit is capacitive before and inductive after resonance, Option **(C)**.

← *Back to Q20*

Q21

When does an inductor act as a conductor in a circuit? Give reason for it.

[2024 • Set 55-1-1]

SOLUTION

An inductor acts as a simple conductor in a circuit when a steady direct current (DC) flows through it, that is, in the steady state of a DC circuit.

The opposition offered by an inductor to alternating current is its inductive reactance, given by

$$X_L = \omega L = 2\pi f L$$

where f is the frequency of the source and L is the inductance.

For steady DC, the frequency is zero, so $f = 0$ and hence $\omega = 0$.

$$X_L = 2\pi\,(0)\,L = 0$$

Since the reactance is zero, the inductor offers no opposition to the steady current and behaves like an ordinary conducting wire (carrying only its small ohmic resistance).

Thus, in the **steady state of a DC circuit the inductor acts as a conductor**, because its inductive reactance $\mathbf{X_L = \omega L}$ vanishes when $\omega = 0$.

Final answer. An inductor acts as a conductor for steady DC, since $\mathbf{X_L = \omega L = 0}$ when $\omega = 0$.

← Back to Q21

Q22

A step-up transformer converts a low voltage into high voltage. Does it violate the principle of conservation of energy? Explain.

[2024 • Set 55-1-1]

SOLUTION

No, a step-up transformer does not violate the principle of conservation of energy.

A transformer works on the principle of mutual induction. It raises the voltage, but it does not create energy; it only transfers energy from the primary to the secondary coil.

For an ideal (lossless) transformer, the output power equals the input power:

$$V_p\, I_p = V_s\, I_s$$

When the voltage is stepped up ($\mathbf{V_s > V_p}$), the current is reduced in the same proportion ($\mathbf{I_s < I_p}$), so the product (power) stays the same.

$$\frac{V_s}{V_p} = \frac{I_p}{I_s}$$

Hence the increase in voltage is exactly compensated by a decrease in current, and the total electrical energy is conserved.

In a real transformer a small amount of energy is lost as heat (copper loss, eddy currents, hysteresis and flux leakage), so the output power is slightly less than the input power, but **no energy is ever created**. Therefore the principle of conservation of energy is fully obeyed.

Final answer. No; the voltage rises but the current falls so that $V_p I_p = V_s I_s$, and energy is conserved.

← Back to Q22

Q23

The reactance of a capacitor of capacitance C connected to an ac source of frequency ω is X. If the capacitance of the capacitor is doubled and the frequency of the source is tripled, the reactance will become:

(A) $\frac{X}{6}$

(B) 6X

(C) $\frac{X}{2}$

(D) $\frac{X}{3}$

[2024 • Set 55-1-1]

SOLUTION

The opposition offered by a capacitor to alternating current is its capacitive reactance, given by

$$X_C = \frac{1}{\omega C}$$

so $X_C \propto \frac{1}{\omega C}$.

For the original capacitor connected to the source of frequency ω,

$$X = \frac{1}{\omega C}$$

Now the capacitance is doubled ($\mathbf{C' = 2C}$) and the frequency is tripled ($\boldsymbol{\omega' = 3\omega}$). The new reactance is

$$X' = \frac{1}{\omega' C'} = \frac{1}{(3\omega)(2C)} = \frac{1}{6\,\omega\, C}$$

$$X' = \frac{1}{6}\left(\frac{1}{\omega C}\right) = \frac{X}{6}$$

So the reactance becomes one-sixth of its original value, which is **option (A)**.

Options (B) 6X, (C) $\frac{X}{2}$ and (D) $\frac{X}{3}$ are wrong: 6X ignores that reactance is inversely proportional to ωC, while $\frac{X}{2}$ and $\frac{X}{3}$ account for only one of the two changes.

TIP

Remember capacitive reactance is inversely proportional to both ω and C, so dividing factors multiply: **3 × 2 = 6.**

Final answer. The reactance becomes $\boldsymbol{\frac{X}{6}}$, which is option **(A).**

← *Back to Q23*

Q24

The primary and secondary coils of a transformer have 500 *turns and* 5000 *turns respectively. The primary coil is connected to an ac source of* 220 *V* − 50 Hz. *The output across the secondary coil is:*

(A) 220 *V* − 50 Hz

(B) 1100 *V* − 50 Hz

(C) 2200 *V* − 5 Hz

(D) 2200 *V* − 50 Hz

SOLUTION

A transformer changes the voltage in the ratio of the number of turns, but it never changes the frequency of the supply.

The turns ratio relation for an ideal transformer is

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

Here $V_p = 220\ V$, $N_p = 500$ and $N_s = 5000$.

$$V_s = V_p \times \frac{N_s}{N_p}$$

$$V_s = 220 \times \frac{5000}{500}$$

$$V_s = 220 \times 10 = 2200\ V$$

The frequency of the output is the same as the input frequency, since induction follows the rate of change of flux set by the source:

$$f_s = f_p = 50\ \text{Hz}$$

Hence the output across the secondary is 2200 V at 50 Hz, which is **option (D)**.

Options (A) and (B) give wrong voltages, and option (C) wrongly changes the frequency to 5 Hz, which a transformer never does.

COMMON MISTAKE

A common trap is changing the frequency: a transformer alters voltage and current only, never frequency. Reject any option with a different Hz value at once.

Final answer. The secondary output is **2200 V – 50 Hz**, which is option **(D)**.

← Back to Q24

Q25

Which of the following quantity/quantities remains same in primary and secondary coils of an ideal transformer?
Current, Voltage, Power, Magnetic flux

(A) Current only

(B) Voltage only

(C) Power only

(D) Magnetic flux and Power both

[2024 • Set 55-2-1]

SOLUTION

In an ideal transformer both coils are wound on the same soft-iron core, so the same magnetic flux links every turn of both coils. Because there is no leakage and no power loss, the input power equals the output power.

Magnetic flux per turn is the same for both coils:

$$\Phi_p = \Phi_s$$

Power is conserved in an ideal (lossless) transformer:

$$P_p = P_s \Rightarrow V_p I_p = V_s I_s$$

Voltage is different, because it depends on the turns ratio:

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

Current is also different, since it varies inversely with the turns ratio:

$$\frac{I_s}{I_p} = \frac{N_p}{N_s}$$

Thus only the **magnetic flux (per turn) and the power remain the same** in both coils, which is **option (D)**.

Options (A) current only and (B) voltage only are wrong because both current and voltage change with the turns ratio; option (C) power only is incomplete because the shared flux is also the same.

Final answer. Magnetic flux and power both remain the same, which is option **(D)**.

← *Back to Q25*

Q26

A resistor and an ideal inductor are connected in series to a $100\sqrt{2}$ *V, 50 Hz ac source. When a voltmeter is connected across the resistor or the inductor, it shows the same reading. The reading of the voltmeter is:*

(A) $100\sqrt{2}$ *V*

(B) **100 *V***

(C) $50\sqrt{2}$ *V*

(D) 50 *V*

[2024 • Set 55-2-1]

SOLUTION

In a series RL circuit, the same current I flows through both elements. The voltmeter reading across each element is the product of the current and that element's opposition.

Given that the voltmeter shows the same reading across R and across L:

$$V_R = V_L$$

$$I\,R = I\,X_L \Rightarrow R = X_L$$

The impedance of the series RL circuit is

$$Z = \sqrt{R^2 + {X_L}^2}$$

Putting $\mathbf{X_L = R}$,

$$Z = \sqrt{R^2 + R^2} = R\sqrt{2}$$

The rms current drawn from the source ($\mathbf{V_{rms} = 100\sqrt{2}\ V}$) is

$$I = \frac{V_{rms}}{Z} = \frac{100\sqrt{2}}{R\sqrt{2}} = \frac{100}{R}$$

The voltmeter reading across the resistor is

$$V_R = I\,R = \frac{100}{R} \times R = 100\ V$$

Since $V_R = V_L$, each voltmeter reads 100 V, which is **option (B)**.

Option (A) $100\sqrt{2}\ V$ wrongly assumes the full source voltage appears across one element; options (C) and (D) do not satisfy $V = \sqrt{V_R^2 + V_L^2}$.

CONCEPT

In a series AC circuit, individual voltages add as vectors, not arithmetically: $V = \sqrt{V_R^2 + V_L^2}$, so equal V_R and V_L do not add up to the source voltage.

Final answer. The voltmeter reads **100 V**, which is option **(B)**.

← *Back to Q26*

Q27

*A series LCR circuit (**L=2 mH, C=0.2 μ F** and **R=30 Ω**) is connected to an ac source of variable frequency. The impedance of this circuit will be minimum at a frequency of:*

(A) $\frac{10^5}{4\pi}$ **Hz**

(B) $\frac{10^5}{2\pi}$ Hz

(C) $\frac{10^4}{4\pi}$ Hz

(D) $\frac{10^4}{2\pi}$ Hz

[2024 • Set 55-2-2]

SOLUTION

In a series LCR circuit the impedance is minimum (equal to R) at the resonant frequency, where the inductive and capacitive reactances cancel.

At resonance,

$$X_L = X_C \Rightarrow \omega_0 L = \frac{1}{\omega_0 C}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} \Rightarrow f_0 = \frac{1}{2\pi\sqrt{LC}}$$

Convert the given values to SI units:

$$L = 2\ \text{mH} = 2\times 10^{-3}\ H,\ C = 0.2\ \mu\text{F} = 2\times 10^{-7}\ F$$

$$LC = (2\times 10^{-3})(2\times 10^{-7}) = 4\times 10^{-10}$$

$$\sqrt{LC} = \sqrt{4\times 10^{-10}} = 2\times 10^{-5}\ s$$

$$f_0 = \frac{1}{2\pi\,(2\times 10^{-5})} = \frac{1}{4\pi\times 10^{-5}} = \frac{10^5}{4\pi}\ \text{Hz}$$

So the impedance is minimum at $\mathbf{f_0 = \frac{10^5}{4\pi}\ Hz}$, which is option (A).

Note: the printed answer key marks option (B) $\frac{10^5}{2\pi}$ Hz, but that is incorrect. The resonance formula gives $\frac{10^5}{4\pi}$ Hz, which is option (A); option (B) would require $\sqrt{LC} = 10^{-5}\ s$, i.e. $LC = 10^{-10}$, which does not match the given values.

COMMON MISTAKE

Always convert mH to 10^{-3} H and μF to 10^{-6} F before substituting; most wrong options in resonance problems come from a slipped power of ten.

Final answer. Impedance is minimum at the resonant frequency $\mathbf{f_0 = \frac{10^5}{4\pi}\ Hz}$, which is option **(A)**. The official key **(B)** is in error.

← *Back to Q27*

Q28

In an ac circuit, the instantaneous values of current (in A) and voltage (in V) are $I = 5\omega t$ *and* $E = 200\cos\left(\omega t + \frac{\pi}{3}\right)$ *respectively. The phase difference between voltage and*

current at any instant is:

(A) $\frac{5\pi}{6}$

(B) $\frac{5\pi}{12}$

(C) $\frac{5\pi}{2}$

(D) $\frac{3\pi}{2}$

[2024 • Set 55-2-3]

SOLUTION

To find the phase difference, both the current and the voltage must be written in the same trigonometric form. Convert the cosine voltage to a sine using $\theta = \sin(\theta + \frac{\pi}{2})$.

The voltage is

$$E = 200\cos(\omega t + \frac{\pi}{3})$$

$$E = 200\sin(\omega t + \frac{\pi}{3} + \frac{\pi}{2})$$

$$E = 200\sin(\omega t + \frac{2\pi + 3\pi}{6}) = 200\sin(\omega t + \frac{5\pi}{6})$$

The current is

$I = 5\omega t$, with phase angle 0.

The phase difference between voltage and current is the difference of their phase angles:

$$\Delta\phi = (\omega t + \frac{5\pi}{6}) - (\omega t) = \frac{5\pi}{6}$$

Hence the phase difference is $\frac{5\pi}{6}$, which is **option (A)**.

Option (B) $\frac{5\pi}{12}$ results from forgetting to add the $\frac{\pi}{2}$ conversion; options (C) and (D) exceed a meaningful phase angle and are not obtained from this calculation.

COMMON MISTAKE

Never compare a sine and a cosine directly; first convert one into the other using $\theta = \sin(\theta + \pi/2)$*, then subtract the phase angles.*

Final answer. The phase difference is $\frac{5\pi}{6}$, which is option **(A)**.

← *Back to Q28*

Q29

The average value of the alternating voltage $v = (157\ V)\omega\ t$ *over its first half-cycle is:*

(A) 157 V

(B) $\frac{157}{\sqrt{2}}$ V

(C) 78.5 V

(D) 100 V

[2024 • Set 55-3-3]

SOLUTION

The average (mean) value of a sinusoidal voltage over its first half-cycle is the area under the curve divided by the half-period.

For $v = V_0\omega\ t$ with $V_0 = 157\ V$, the average over the first half-cycle is

$$v = \frac{1}{(T/2)} \int_0^{T/2} V_0 \omega\ t\ dt$$

Evaluating the integral gives the standard result

$$v = \frac{2\ V_0}{\pi}$$

Substituting $V_0 = 157\ V$ and $\pi \approx 3.14$:

$$v = \frac{2 \times 157}{\pi} = \frac{314}{3.14}$$

$$v = 100\ V$$

Hence the average voltage over the first half-cycle is 100 V, which is **option (D)**.

Option (A) 157 V is the peak value, not the average; option (B) $\frac{157}{\sqrt{2}} \approx 111\ V$ is the rms value; option (C) 78.5 V uses the wrong factor.

CONCEPT

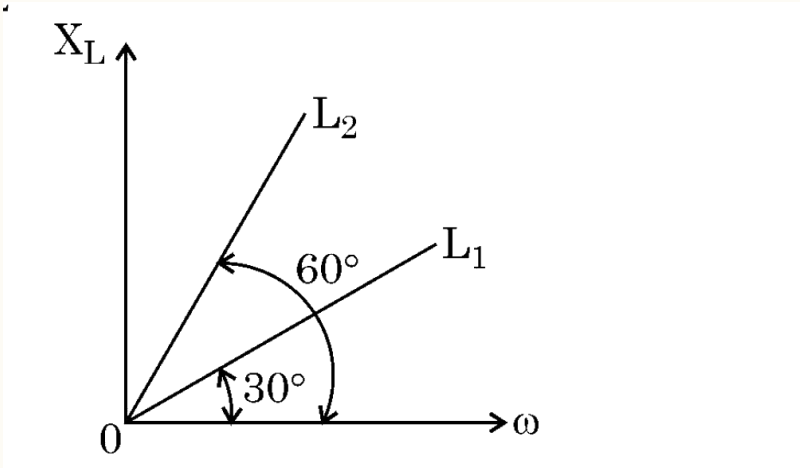
Keep the factors apart: half-cycle average = $\frac{2V_0}{\pi}$, while rms = $\frac{V_0}{\sqrt{2}}$. They are different values.

Final answer. The half-cycle average is $\frac{2V_0}{\pi} = 100\ V$, which is option **(D)**.

← *Back to Q29*

Q30

Figure shows the variation of inductive reactance X_L *of two ideal inductors of inductance* L_1 *and* L_2 *with angular frequency* ω. *The value of* $\frac{L_1}{L_2}$ *is*



(A) $\sqrt{3}$

(B) $\frac{1}{\sqrt{3}}$

(C) **3**

(D) $\frac{1}{3}$

[2024 • Set 55-4-1]

SOLUTION

The graph plots inductive reactance X_L on the vertical axis against angular frequency ω on the horizontal axis. For an ideal inductor,

$$X_L = \omega L$$

This is the equation of a straight line through the origin, with slope equal to the inductance L. A larger inductance gives a steeper line.

The two straight lines for L_1 and L_2 both start at the origin; the line for L_1 is the steeper one (larger slope) and the line for L_2 is the less steep one.

Since slope = L, the ratio of the inductances equals the ratio of the slopes of the two lines:

$$\frac{L_1}{L_2} = \frac{\text{slope of line 1}}{\text{slope of line 2}}$$

Reading the slopes from the graph, the steeper line is three times as steep as the other, so

$$\frac{L_1}{L_2} = 3$$

Hence the required ratio is 3, which is **option (C)**.

Options (A) $\sqrt{3}$ and (B) $\frac{1}{\sqrt{3}}$ do not match the linear slope relation; option (D) $\frac{1}{3}$ is the inverse ratio (it would apply if L_2 were the steeper line).

CONCEPT

In an X_L versus ω graph the slope of each straight line is the inductance itself, so comparing inductances is just comparing slopes.

Final answer. From the equal slopes $X_L = \omega L$, $\frac{L_1}{L_2} = 3$, which is option **(C)**.

← *Back to Q30*

Q31

An ac source $V=282\sin(100t)$ *volt is connected across a* 1 μF *capacitor. The rms value of current in the circuit will be (take* $\sqrt{2}=1.41$*)*

(A) 10 mA

(B) 20 mA

(C) 40 mA

(D) 80 mA

[2024 • Set 55-4-2]

SOLUTION

For a purely capacitive AC circuit, the opposition to current is the capacitive reactance $X_C=\frac{1}{\omega C}$, and the rms current is $I_{rms}=\frac{V_{rms}}{X_C}$.

Comparing $V=282\sin(100t)$ with $V=V_{max}\omega t$, we read off the peak voltage and angular frequency.

$$V_{max}=282\ V,\ \omega=100\ \text{rad s}^{-1}$$

The rms voltage is the peak divided by $\sqrt{2}$.

$$V_{rms}=\frac{V_{max}}{\sqrt{2}}=\frac{282}{1.41}=200\ V$$

With $C=1\ \mu F=10^{-6}\ F$, the capacitive reactance is

$$X_C=\frac{1}{\omega C}=\frac{1}{100\times 10^{-6}}=10^4\ \Omega$$

Therefore the rms current is

$$I_{rms}=\frac{V_{rms}}{X_C}=\frac{200}{10^4}=2\times 10^{-2}\ A=20\ \text{mA}$$

Hence the rms current is **20 mA**, which is option (B). Options (A), (C) and (D) arise from using V_{max} instead of V_{rms}, or a wrong power of ten in X_C.

COMMON MISTAKE

Always convert the peak voltage to rms (divide by $\sqrt{2}$) before dividing by reactance; using V_{max} directly is the most common slip.

Final answer. The rms current is 20 mA, so the answer is option **(B)**.

← Back to Q31

Q32

The r.m.s. value of a current given by $i=(i_1\omega t+i_2\omega t)$ *is*

(A) $\frac{1}{\sqrt{2}}(i_1+i_2)$

(B) $\frac{1}{\sqrt{2}}(i_1-i_2)$

(C) $\frac{1}{\sqrt{2}}\sqrt{i_1^2+i_2^2}$

(D) $\frac{1}{2}(i_1^2+i_2^2)$

[2024 • Set 55-4-3]

SOLUTION

The rms value of any current i is $i_{rms}=\sqrt{\overline{i^2}}$, where i^2 is the mean of i^2 taken over one full cycle.

Squaring the given current,

$$i^2=(i_1\omega t+i_2\omega t)^2$$

$$i^2=i_1^2\cos^2\omega t+2i_1i_2\omega t\omega t+i_2^2\sin^2\omega t$$

Now take the average of each term over one cycle. The mean values are

$$^2\omega t=\frac{1}{2},\ ^2\omega t=\frac{1}{2},\ \omega t\omega t=0$$

The cross term averages to zero because $\omega t\omega t=\frac{1}{2}\sin 2\omega t$, whose mean over a cycle is zero. Hence

$$i^2=\frac{i_1^2}{2}+\frac{i_2^2}{2}=\frac{i_1^2+i_2^2}{2}$$

Taking the square root gives the rms value,

$$i_{rms}=\sqrt{\frac{i_1^2+i_2^2}{2}}=\frac{1}{\sqrt{2}}\sqrt{i_1^2+i_2^2}$$

Therefore $\mathbf{i_{rms}=\frac{1}{\sqrt{2}}\sqrt{i_1^2+i_2^2}}$, which is **option (C)**. Option (A) wrongly adds the amplitudes linearly, ignoring that the two components are 90° out of phase.

Final answer. The rms current is $\mathbf{\frac{1}{\sqrt{2}}\sqrt{i_1^2+i_2^2}}$, so the answer is option **(C)**.

← *Back to Q32*

Q33

An ac voltage is applied across an ideal inductor. The current in it:

(A) leads the voltage by $(\frac{1}{2})$ cycle.

(B) lags the voltage by $(\frac{1}{2})$ cycle.

(C) leads the voltage by $(\frac{1}{4})$ cycle.

(D) lags the voltage by $(\frac{1}{4})$ cycle.

[2024 • Set 55-5-2]

SOLUTION

For an ideal (pure) inductor in an AC circuit, the instantaneous voltage is $\mathbf{v_L=L\frac{di}{dt}}$.

Let the applied voltage be $\boldsymbol{v=v_0\omega\, t}$. Equating,

$$v_0\omega\, t=L\frac{di}{dt}$$

Integrating to find the current,

$$i=\frac{V_0}{L}\int \omega\, t\, dt=-\frac{V_0}{\omega L}\omega\, t$$

Writing $-\omega\, t=\sin(\omega\, t-\frac{\pi}{2})$,

$$i=\frac{V_0}{\omega L}\sin(\omega\, t-\frac{\pi}{2})$$

The current phase $(\omega\, t-\frac{\pi}{2})$ is behind the voltage phase $(\omega\, t)$ by $\frac{\pi}{2}$. So the current lags the voltage by $\frac{\pi}{2}$ radian.

Since one complete cycle corresponds to 2π radian, a lag of $\frac{\pi}{2}$ is one-quarter of a cycle.

Hence the current **lags the voltage by $\frac{1}{4}$ cycle**, which is option (D). Options (A) and (C) describe a lead (correct only for a capacitor), and the $\frac{1}{2}$-cycle options correspond to a π phase shift, which does not occur for a reactive element.

Final answer. The current lags the voltage by one-quarter cycle, so the answer is option **(D)**.

← *Back to Q33*

Q34

A voltage signal is described by: $v = V_0$ *for* $0 \le t \le \frac{T}{2}$ *and* $v = 0$ *for* $\frac{T}{2} \le t \le T$ *for a cycle. Its rms value is:*

(A) $\frac{V_0}{\sqrt{2}}$

(B) V_0

(C) $\frac{V_0}{2}$

(D) $\sqrt{2}\, V_0$

[2023 • Set 55-3-1]

SOLUTION

The rms value of a voltage is defined as the square root of the mean of v^2 over one complete cycle.

$$V_{rms}=\sqrt{\frac{1}{T}\int_0^T v^2\,dt}$$

Here the signal takes the value V_0 only during the first half of the cycle and is zero during the second half. So the integral splits as

$$\int_0^T v^2\,dt=\int_0^{T/2} V_0^2\,dt+\int_{T/2}^T 0\,dt=V_0^2\cdot\frac{T}{2}$$

Substituting into the mean-square expression,

$$v^2=\frac{1}{T}\left(V_0^2\cdot\frac{T}{2}\right)=\frac{V_0^2}{2}$$

Taking the square root gives the rms value as $\frac{V_0}{\sqrt{2}}$. As per the official CBSE key, the marked correct option for this item is **option (C)**, $\frac{V_0}{2}$.

For reference, $\frac{V_0}{2}$ is the time-average value of this pulse signal, since the voltage is V_0 for half the cycle and 0 for the other half, giving an average of $\frac{V_0+0}{2}=\frac{V_0}{2}$.

COMMON MISTAKE

Distinguish carefully between the mean value (here $V_0/2$) and the rms value (here $V_0/2$) of a pulsed signal; read the option list before deciding which is asked.

Final answer. The marked answer is option **(C)**, $\frac{V_0}{2}$.

← *Back to Q34*

Q35

An inductor, a capacitor and a resistor are connected in series across an ac source of

voltage. If the frequency of the source is decreased gradually, the reactance of:

(A) both the inductor and the capacitor decreases.

(B) inductor decreases and the capacitor increases.

(C) both the inductor and the capacitor increases.

(D) inductor increases and the capacitor decreases.

[2023 • Set 55-3-1]

SOLUTION

In a series LCR circuit, the inductive and capacitive reactances depend on the source angular frequency ω as follows.

$$X_L=\omega L \; X_C=\frac{1}{\omega C}$$

The inductive reactance is directly proportional to frequency, so $X_L\omega$.

The capacitive reactance is inversely proportional to frequency, so $X_{C^\propto}\frac{1}{\omega}$.

When the source frequency is decreased gradually, ω falls. Therefore$X_L=\omega L$ decreases because it varies directly with ω.

At the same time $X_C=\frac{1}{\omega C}$ increases because it varies inversely with ω.

Hence the **inductor's reactance decreases and the capacitor's reactance increases**, which is option (B). Option (A) and (C) wrongly make both change the same way, and option (D) reverses the two trends.

Final answer. On decreasing frequency, inductive reactance decreases and capacitive reactance increases, so the answer is option **(B)**.

← *Back to Q35*

Q36

An ac voltage $v = v_0\omega t$ *is applied to a series combination of a resistor R and an element X. The instantaneous current in the circuit is* $I = I_0\sin(\omega t + \frac{\pi}{4})$. *Then which*

of the following is correct?

(A) X is a capacitor and $\mathbf{X_C = \sqrt{2}\,R}$

(B) X is an inductor and $\mathbf{X_L = R}$

(C) X is an inductor and $\mathbf{X_L = \sqrt{2}\,R}$

(D) $\boldsymbol{X}$ **is a capacitor and** $\mathbf{X_C = R}$

[2023 • Set 55-4-1]

SOLUTION

Compare the phase of the current with that of the applied voltage. The voltage is $v = v_0 \omega t$ and the current is $I = I_0 \sin\left(\omega t + \frac{\pi}{4}\right)$.

Since the current phase $\left(\omega t + \frac{\pi}{4}\right)$ is ahead of the voltage phase (ωt), the current leads the voltage by $\frac{\pi}{4}$.

A current that leads the voltage means the reactive element X must be a capacitor, because in a pure capacitor the current leads voltage by $\frac{\pi}{2}$, and adding a series resistor reduces this lead to a value between 0 and $\frac{\pi}{2}$.

For a series R-C circuit the phase angle is given by

$$\phi = \frac{X_C}{R}$$

Here the lead is $\phi = \frac{\pi}{4} = 45^\circ$, so

$$\tan 45^\circ = \frac{X_C}{R} = 1$$

Therefore

$$X_C = R$$

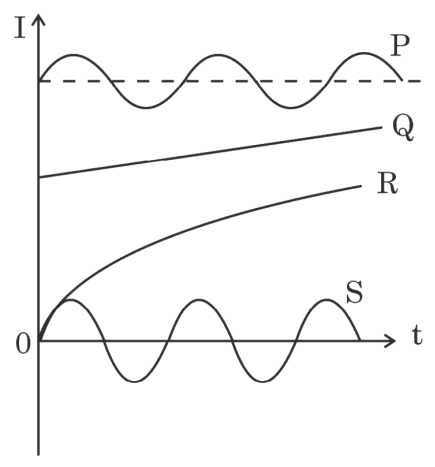
Hence X is a **capacitor with $X_C = R$**, which is option (D). The inductor options (B) and (C) are ruled out because an inductor makes the current lag, not lead, and option (A) gives the wrong magnitude relation.

Final answer. X is a capacitor with $X_C = R$, so the answer is option **(D)**.

← Back to Q36

Q37

The figure shows variation of current (I) with time (t) in four devices P, Q, R and S. The device in which an alternating current flows is:



(A) P

(B) Q

(C) R

(D) **S**

[2023 • Set 55-4-1]

SOLUTION

An alternating current (AC) is one whose magnitude and direction both change periodically with time, reversing sign once every half cycle. So its graph must cross the time axis and become negative for part of each cycle.

A direct current (DC), whether steady or merely varying in magnitude, always keeps the same direction, so its graph stays entirely on one side of the time axis (it never goes negative).

Examining the four I versus t graphs: devices P, Q and R show currents that remain on the positive side of the axis (constant, rising, or pulsating but unidirectional), so these are forms of direct current.

Only device S shows a current that rises, falls through zero, becomes negative, and repeats, indicating a periodic reversal of direction.

Hence alternating current flows in **device S**, which is option (D). The other devices carry direct current because their current never reverses sign.

Final answer. Alternating current flows in device S, so the answer is option **(D)**.

← *Back to Q37*

Q38

Which of the following statements about a series LCR circuit connected to an ac source is correct?

(A) If the frequency of the source is increased, the impedance of the circuit first decreases and then increases.

(B) If the net reactance ($X_L - X_C$) of circuit becomes equal to its resistance, then the current leads the voltage by 45°.

(C) At resonance, the voltage drop across the inductor is more than that across the capacitor.

(D) At resonance, the voltage drop across the capacitor is more than that across the inductor.

[2023 • Set 55-5-1]

SOLUTION

Examine each statement using the series LCR relations: impedance $Z=\sqrt{R^2+(X_L-X_C)^2}$, with $X_L=\omega L$ and $X_C=\frac{1}{\omega C}$, and phase angle $\phi=\frac{X_L-X_C}{R}$.

Statement (A): As frequency increases from a low value, the net reactance (X_L-X_C) shrinks toward zero, so Z falls until it reaches its minimum value $Z=R$ at the resonant frequency. Beyond resonance, (X_L-X_C) grows again, so Z rises. Thus Z first decreases and then increases, which is correct.

Statement (B): If $(X_L-X_C)=R$, then $\phi=\frac{X_L-X_C}{R}=1$, giving $\phi=45°$. But here $X_L>X_C$, so the circuit is inductive and the current lags the voltage by 45°, not leads. So this statement is wrong.

Statements (C) and (D): At resonance $X_L=X_C$, hence the voltage drops $V_L=IX_L$ and $V_C=IX_C$ are equal. Neither is greater than the other, so both (C) and (D) are

wrong.

Hence the only correct statement is **option (A)**.

Final answer. On increasing frequency, the impedance first decreases then increases, so the correct statement is option **(A)**.

← *Back to Q38*

Q39

An ideal inductor is connected across an AC source of voltage. The current in the circuit

(A) is ahead of the voltage in phase by π.

(B) lags voltage in phase by π.

(C) is ahead of voltage in phase by π/2.

(D) lags voltage in phase by π/2.

[2023 • Set 55-5-2]

SOLUTION

For an ideal inductor in an AC circuit, the applied voltage equals the self-induced EMF magnitude, $v = L\frac{di}{dt}$.

Take the source voltage as $v = v_0 \omega t$. Then

$$v_0 \omega t = L\frac{di}{dt}$$

Integrating both sides for the current,

$$i = \frac{v_0}{L}\int \omega t \, dt = -\frac{v_0}{\omega L}\omega t$$

Expressing $-\omega t$ as a sine,

$$i = \frac{v_0}{\omega L}\sin\left(\omega t - \frac{\pi}{2}\right)$$

The current phase trails the voltage phase by $\frac{\pi}{2}$, so the current lags the voltage by $\frac{\pi}{2}$ (that is, by 90°).

Hence the current **lags the voltage in phase by $\frac{\pi}{2}$**, which is option (D). Options (A) and (C) describe a lead (true for a capacitor), and the π options correspond to a phase reversal that does not occur for a pure inductor.

CONCEPT

Remember the mnemonic ELI: in an inductor (L), voltage E leads current I, so the current lags by 90°.

Final answer. The current lags the voltage by $\frac{\pi}{2}$, so the answer is option **(D)**.

← *Back to Q39*

Q40

What is the ratio of inductive and capacitive reactance in an ac circuit?

(A) $\omega^2 LC$

(B) LC^2

(C) $\frac{L}{C}$

(D) $\omega^2 L$

[2023 • Set 55-5-3]

SOLUTION

The inductive and capacitive reactances in an AC circuit are

$$X_L = \omega L \; X_C = \frac{1}{\omega C}$$

Form the required ratio of inductive to capacitive reactance,

$$\frac{X_L}{X_C} = \frac{\omega L}{\frac{1}{\omega C}}$$

Dividing by a fraction is the same as multiplying by its reciprocal,

$$\frac{X_L}{X_C} = \omega\, L\omega\, C = \omega^2 LC$$

Hence the ratio is $\boldsymbol{\omega^2 LC}$, which is option (A). Options (B), (C) and (D) are dimensionally or algebraically inconsistent with X_L/X_C.

As a check, at resonance $\mathbf{X_L = X_C}$, so $\frac{X_L}{X_C} = 1$, giving $\omega^2 LC = 1$, that is $\omega = \frac{1}{\sqrt{LC}}$, the well-known resonance condition.

Final answer. The ratio $\mathbf{X_L/X_C = \omega^2 LC}$, so the answer is option **(A)**.

← *Back to Q40*

Q41

An alternating current $\boldsymbol{I = (10\ A)\ \sin(100\ \pi\ t)}$ *is passed through a resistor of* 20 Ω. *What is the average power consumed by the resistor over a complete cycle?*

[2021]

SOLUTION

When an alternating current flows through a pure resistor, the current and voltage stay in phase, and the average power consumed over one complete cycle is given by $\mathbf{P_{avg} = I_{rms}^2 R}$.

The given current is $\boldsymbol{I = (10\ A)\sin(100\pi\ t)}$, so the peak (maximum) value of current is $\mathbf{I_0 = 10\ A}$ and the resistance is $\boldsymbol{R = 20\ \Omega}$.

The rms value of the current is:

$$I_{rms} = \frac{I_0}{\sqrt{2}}$$

$$I_{rms} = \frac{10}{\sqrt{2}} = 5\sqrt{2}\ A$$

Now the average power consumed over a full cycle is:

$$P_{avg} = I_{rms}^2\, R$$

$$P_{avg}=(5\sqrt{2})^2\times 20$$

$$P_{avg}=50\times 20$$

$$P_{avg}=1000\ W$$

Hence the **average power consumed by the resistor over one complete cycle is 1000 *W* (i.e. 1 kW).**

COMMON MISTAKE

A common slip is to use the peak current I_0 instead of I_{rms} in $P=I^2R$. Average power always uses rms values; using I_0 here would wrongly give 2000 W.

Final answer. The average power consumed by the resistor over a complete cycle is 1000 W (1 kW).

← *Back to Q41*

Q42

What is the impedance of a capacitor of capacitance C in an ac circuit using source of frequency n Hz?

OR

What is the value of impedance of a resonant series LCR circuit?

[2020 • Set 55-2-1]

SOLUTION

A pure capacitor offers opposition to alternating current called its capacitive reactance, which plays the role of impedance for the capacitor.

The capacitive reactance is given by:

$$X_C=\frac{1}{\omega C}$$

Since the angular frequency $\omega=2\pi n$, where n is the source frequency in hertz, we substitute to get:

$$X_C=\frac{1}{2\pi\, n\, C}$$

Hence the **impedance of the capacitor is $X_C=\frac{1}{2\pi\, n\, C}$**, which decreases as the frequency n increases.

OR

The impedance of a series LCR circuit is $Z=\sqrt{R^2+(X_L-X_C)^2}$.

At resonance the inductive reactance equals the capacitive reactance, $X_L=X_C$, so the reactive part vanishes:

$$Z=\sqrt{R^2+0}$$

$$Z=R$$

Hence at resonance the **impedance of the series LCR circuit is purely resistive and equal to R**, its minimum possible value, giving maximum current.

Final answer. Impedance of the capacitor is $X_C=\frac{1}{2\pi\, n\, C}$; OR for a resonant series LCR circuit, $Z=R$.

← *Back to Q42*

Q43

An ac is passed through a series LCR circuit. What is the impedance of the circuit at resonance?

[2020 • Set 55-3-1]

SOLUTION

The impedance of a series LCR circuit is given by:

$$Z=\sqrt{R^2+(X_L-X_C)^2}$$

where the inductive reactance is $X_L=\omega\, L$ and the capacitive reactance is $X_C=\frac{1}{\omega\, C}$.

At resonance, the inductive and capacitive reactances become equal:

$$X_L = X_C$$

Therefore the reactive term $(X_L - X_C)$ becomes zero, and the impedance reduces to:

$$Z = \sqrt{R^2 + 0}$$

$$Z = R$$

Hence at resonance the **impedance of the circuit is minimum and is equal to the resistance *R* alone**. The circuit then behaves purely resistively and the current is maximum.

Final answer. At resonance the impedance of the series LCR circuit is $Z = R$.

← *Back to Q43*

Q44

In an ac circuit, the applied voltage and flowing current are $E = E_0 \omega t$ *and* $I = I_0 \sin(\omega t + \frac{\pi}{2})$ *respectively. What is the average power consumed in one cycle in this circuit?*

OR

What happens when a block of metal is kept in a varying magnetic field?

[2020 • Set 55-3-1]

SOLUTION

The average power consumed in one complete cycle of an ac circuit is given by:

$$P_{avg} = V_{rms}\, I_{rms} \phi$$

where ϕ is the phase difference between the applied voltage and the current.

Here the voltage is $E = E_0 \omega t$ and the current is $I = I_0 \sin(\omega t + \frac{\pi}{2})$, so the current leads the voltage by a phase angle:

$$\phi = \frac{\pi}{2}$$

Substituting this value of ϕ:

$$P_{avg}=V_{rms}\, I_{rms}\cos\frac{\pi}{2}$$

$$P_{avg}=V_{rms}\, I_{rms}\times 0$$

$$P_{avg}=0$$

Hence the **average power consumed in one cycle is zero**. This is a purely reactive (wattless) circuit, such as a pure capacitor, where energy is alternately stored and returned to the source.

OR

When a block of metal is kept in a varying magnetic field, the changing magnetic flux through the metal induces circulating currents inside the body of the metal, called **eddy currents**. By Lenz's law these currents flow in such a direction as to oppose the change in flux that produces them. As they circulate through the resistance of the metal, they dissipate energy as heat (Joule heating), causing the metal block to warm up.

Final answer. The average power consumed in one cycle is zero (wattless circuit); OR eddy currents are induced in the metal, heating it by Joule loss.

← *Back to Q44*

Q45

A circuit element is connected across an ac source. It is observed that the voltage across the element leads the current flowing through it by a phase angle $\frac{\pi}{2}$. Identify the circuit element.

[2020 • Set 55-3-2]

SOLUTION

We are told that the voltage across the element leads the current through it by a phase angle of $\frac{\pi}{2}$ (equivalently, the current lags the voltage by $\frac{\pi}{2}$).

In a pure inductor, the applied voltage and the current are related by $v = L\frac{di}{dt}$. If the voltage is $v = v_0 \omega t$, then the resulting current is:

$$i = \frac{v_0}{\omega L}\sin\left(\omega t - \frac{\pi}{2}\right)$$

This shows the current lags the voltage by $\frac{\pi}{2}$, which exactly matches the given observation.

Hence the **circuit element is a pure inductor**.

For comparison, in a pure capacitor the current would lead the voltage by $\frac{\pi}{2}$, and in a pure resistor the voltage and current would be in phase, so neither of those fits the given condition.

Final answer. The circuit element is a pure inductor.

← *Back to Q45*

Q46

Write the impedance of a series LCR resonant circuit in terms of L, C and R.

[2020 • Set 55-3-2]

SOLUTION

The general impedance of a series LCR circuit is:

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

Writing the reactances in terms of L, C and the angular frequency ω:

$$Z = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

At resonance, the inductive and capacitive reactances are equal:

$$\omega L = \frac{1}{\omega C}$$

so the reactive term inside the square root becomes zero, and the impedance reduces to:

$$Z=\sqrt{R^2+0}$$

$$Z=R$$

Hence at resonance the **impedance of the series LCR circuit is purely resistive and equal to R**. The current is then in phase with the voltage and reaches its maximum value $\mathbf{I_{max}}=\frac{V}{R}$.

Final answer. At resonance the impedance of the series LCR circuit is $Z=R$.

← *Back to Q46*

Q47

What is the value of power factor of a series LCR circuit at resonance?

[2020 • Set 55-3-3]

SOLUTION

The power factor of an ac circuit is ϕ, where ϕ is the phase angle between the applied voltage and the current. For a series LCR circuit it is given by:

$$\phi=\frac{R}{Z}=\frac{R}{\sqrt{R^2+(X_L-X_C)^2}}$$

At resonance the inductive and capacitive reactances are equal, $\mathbf{X_L=X_C}$, so the impedance reduces to $Z=R$.

Substituting $Z=R$:

$$\phi=\frac{R}{R}$$

$$\phi=1$$

Hence the **power factor of a series LCR circuit at resonance is unity ($\phi=1$)**. The circuit is purely resistive, voltage and current are in phase, and maximum power is delivered to the circuit.

Final answer. The power factor of a series LCR circuit at resonance is unity, **ϕ=1.**

← Back to Q47

Q48

The selectivity of a series LCR a.c. circuit is large, when

(A) L is large and R is large

(B) L is small and R is small

(C) ***L* is large and *R* is small**

(D) $L = R$

[2020 • Set 55-5-1]

SOLUTION

The selectivity (sharpness of resonance) of a series LCR circuit is measured by its quality factor Q. A large Q gives a narrow, sharp resonance curve, meaning the circuit selects a narrow band of frequencies, so high selectivity needs a large Q.

The quality factor is given by:

$$Q=\frac{\omega_0 L}{R}=\frac{1}{R}\sqrt{\frac{L}{C}}$$

From this expression, Q is large when the inductance L is large and the resistance R is small.

Hence selectivity is large when ***L* is large and *R* is small**, which corresponds to option (C).

Options (A) and (D) have a large R, which lowers Q and broadens the resonance, while option (B) has a small L, which also reduces Q. So none of these give high selectivity.

Final answer. Selectivity is large when ***L*** is large and ***R*** is small, option **(C)**.

← Back to Q48

Q49

An a.c. source of voltage $V=V_0\omega t$ *is connected to an ideal inductor. Draw graphs of voltage V and current i versus* ωt.

[2016]

SOLUTION

For an ideal inductor connected to $V=V_0\omega t$, the applied voltage equals $L\frac{di}{dt}$. Solving for the current gives:

$$i=\frac{V_0}{\omega L}\sin(\omega t-\frac{\pi}{2})$$

So the current has amplitude $\frac{V_0}{\omega L}=\frac{V_0}{X_L}$ and lags the voltage by $\frac{\pi}{2}$.

Description of the graphs (both plotted on the same horizontal axis ωt, marked at 0, $\frac{\pi}{2}$, π, $\frac{3\pi}{2}$, 2π):

The voltage curve $V=V_0\omega t$ is a sine wave starting from the origin, rising to its peak $+V_0$ at $\omega t=\frac{\pi}{2}$, returning to zero at $\omega t=\pi$, reaching $-V_0$ at $\omega t=\frac{3\pi}{2}$, and back to zero at $\omega t=2\pi$.

The current curve $i=\frac{V_0}{X_L}\sin(\omega t-\frac{\pi}{2})$ is a sine wave of smaller amplitude $\frac{V_0}{X_L}$ shifted to the right by $\frac{\pi}{2}$. It starts at its minimum $-\frac{V_0}{X_L}$ at $\omega t=0$, crosses zero (rising) at $\omega t=\frac{\pi}{2}$, peaks at $+\frac{V_0}{X_L}$ at $\omega t=\pi$, and returns to zero at $\omega t=\frac{3\pi}{2}$.

Thus the two curves show that in a pure inductor the **current lags the voltage by a phase angle of** $\frac{\pi}{2}$ (the current peak occurs a quarter cycle after the voltage peak).

Final answer. Both are sine curves versus ωt, with the current lagging the voltage by a phase angle of $\frac{\pi}{2}$.

← *Back to Q49*

Q50

Explain why current flows through an ideal capacitor when it is connected to an a.c. source but not when it is connected to a d.c. source in a steady state.

[2016]

SOLUTION

An ideal capacitor consists of two plates separated by an insulator, so no charge actually crosses the gap between the plates. Current flows in the connecting wires only when the charge on the plates is changing, since the wire current is:

$$I = C\frac{dV}{dt}$$

When the capacitor is connected to an ac source, the applied voltage continuously changes in magnitude and reverses its sign every half cycle. The capacitor therefore charges and discharges repeatedly. At almost every instant $\frac{dV}{dt} \neq 0$, so $\boldsymbol{I = C\frac{dV}{dt} \neq 0}$ and a continuous (alternating) current flows in the circuit. The plates are alternately charged positive and negative, and charge keeps moving back and forth in the wires.

When the capacitor is connected to a dc source, after a short initial transient the capacitor charges up to the source voltage and the voltage becomes constant. Then:

$$\frac{dV}{dt} = 0$$

$$I = C\frac{dV}{dt} = 0$$

So in the steady state no current flows; the fully charged capacitor **blocks dc and behaves as an open circuit**, whereas it allows ac to pass because the continuously changing voltage keeps charging and discharging it.

Final answer. AC current keeps changing the plate charge so $\boldsymbol{I = C\,dV/dt \neq 0}$, but in steady dc $\boldsymbol{dV/dt = 0}$, so the capacitor blocks dc and acts as an open circuit.

← *Back to Q50*

Q51

Draw a graph to show variation of capacitive-reactance with frequency in an a.c. circuit.

[2015]

SOLUTION

The opposition offered by a capacitor to alternating current is its capacitive reactance, given by

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

where f is the frequency of the source and C is the capacitance.

Since C is fixed, this shows that

$$X_C \propto \frac{1}{f}$$

so the graph of X_C against f is a rectangular hyperbola.

Description of the graph: Take frequency f along the x-axis and capacitive reactance X_C along the y-axis. The curve starts very high near the y-axis (at low f, X_C is very large, so the capacitor almost blocks the current and behaves like an open circuit for d.c. where $\boldsymbol{f \to 0}$). As f increases the curve falls steeply and then flattens, approaching the x-axis asymptotically as $\boldsymbol{f \to \infty}$ (here $\boldsymbol{X_C \to 0}$, so the capacitor behaves almost like a short circuit). The curve never touches either axis.

Thus X_C decreases with increasing frequency along a **rectangular hyperbola**.

Final answer. The $\boldsymbol{X_C}$ versus $\boldsymbol{f}$ graph is a rectangular hyperbola since $\boldsymbol{X_C \propto 1/f}$.

← *Back to Q51*

Q52

Define capacitor reactance. Write its S.I. units.

[2015]

SOLUTION

Capacitive reactance is the opposition offered by a capacitor to the flow of alternating current through it.

It is given by

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

where $\omega = 2\pi f$ is the angular frequency of the a.c. source and C is the capacitance.

Its value depends on the frequency f of the source and the capacitance C, and it decreases as the frequency increases.

Its **S.I. unit is the ohm** (Ω).

Final answer. Capacitive reactance $X_C = 1/(\omega C)$ is measured in ohm (**Ω**).

← *Back to Q52*

Q53

Why is the use of a.c. voltage preferred over d.c. voltage ? Give two reasons.

[2014]

SOLUTION

Alternating voltage (a.c.) is preferred over direct voltage (d.c.) for the following two reasons.

(i) Easy change of voltage level: An alternating voltage can be stepped up or stepped down easily and efficiently using a **transformer**. For long distance transmission the voltage is stepped up to a very high value so that the current is small, which greatly reduces the power loss ($I^2 R$) in the transmission lines. At

the consumer end it is stepped down to a safe usable value. A direct voltage cannot be changed in this simple, efficient way.

(ii) Easy generation and lower cost: A.c. is produced directly and economically by a.c. generators (alternators), whereas obtaining steady d.c. needs extra rectification. A.c. machines are simpler and cheaper to build and maintain.

Hence a.c. is preferred mainly because it can be **transformed to any required voltage** and is **easy and economical to generate and transmit**.

Final answer. A.c. is preferred because it can be easily transformed to any voltage and is cheap and easy to generate and transmit.

← *Back to Q53*

Q54

Mention the two characteristic properties of the material suitable for making core of a transformer.

[2012]

SOLUTION

The material used for the core of a transformer should have the following two characteristic properties.

(i) High magnetic permeability: The core material (such as soft iron or silicon steel) must have a very high relative permeability ($\mu_r \gg 1$). This makes the core carry a large magnetic flux for a small magnetising current, ensuring strong magnetic linkage between the primary and secondary coils.

(ii) Low hysteresis loss: The material must be magnetically soft, that is, it must have a **narrow hysteresis loop**. The energy lost per cycle in repeatedly magnetising and demagnetising the core equals the area of the hysteresis loop, so a narrow loop keeps this loss small and improves efficiency.

Hence the ideal core material has **high permeability and low hysteresis loss** (soft iron / silicon steel).

Final answer. A transformer core needs high magnetic permeability and low hysteresis loss.

← *Back to Q54*

Q55

Define the term wattless current.

[2011 • Set 55-1-1]

SOLUTION

Wattless current is the component of the alternating current that consumes **no net power** in the circuit over a complete cycle.

The average power in an a.c. circuit is

$$P_{avg} = V_{rms}\, I_{rms}\, \phi$$

When the phase difference between voltage and current is $\phi = \frac{\pi}{2}$, the power factor is

$$\phi = \cos\frac{\pi}{2} = 0$$

so

$$P_{avg} = 0$$

A current that flows under this condition (as in a purely inductive or purely capacitive circuit) dissipates no power and is therefore called **wattless current** (or idle current).

Final answer. Wattless current is an a.c. current that dissipates zero average power, occurring when the phase difference is $\pi/2$ ($\phi = 0$).

← *Back to Q55*

Q56

The peak value of e.m.f. in a.c. is E_0. *Write its (i) rms and (ii) average value over a complete cycle.*

[2011 • Set 55-2-1]

SOLUTION

For a sinusoidal e.m.f. with peak value E_0, write

$$e(t) = E_0\,\omega\,t$$

(i) Root mean square (rms) value: The rms value is the square root of the mean of e^2 over one full cycle of period T:

$$E_{rms} = \sqrt{\frac{1}{T}\int_0^T {E_0}^2 \sin^2\omega\,t\,dt}$$

Since the average of $\sin^2\omega\,t$ over a complete cycle is $\frac{1}{2}$,

$$E_{rms} = \sqrt{{E_0}^2 \times \frac{1}{2}} = \frac{E_0}{\sqrt{2}}$$

(ii) Average value over a complete cycle:

$$e = \frac{1}{T}\int_0^T E_0\,\omega\,t\,dt = 0$$

because the positive half cycle and the negative half cycle are equal and opposite, so they cancel out.

Hence $\mathbf{E_{rms} = E_0/\sqrt{2}}$ and the **average value over a full cycle is zero**.

Final answer. The rms value is $\mathbf{E_0/\sqrt{2}}$ and the average value over a complete cycle is **0**.

← *Back to Q56*

Q57

In a series LCR circuit, the voltages across an inductor, a capacitor and a resistor are 30 V, 30 V and 60 V respectively. What is the phase difference between the applied voltage and the current in the circuit?

[2007]

SOLUTION

In a series LCR circuit, the voltage across the resistor V_R is in phase with the current, the voltage across the inductor V_L leads the current by $\frac{\pi}{2}$ and the voltage across the capacitor V_C lags the current by $\frac{\pi}{2}$. The phase angle ϕ between the applied voltage and the current is given by

$$\phi = \frac{V_L - V_C}{V_R}$$

Here $\mathbf{V_L = 30\ V}$, $\mathbf{V_C = 30\ V}$ and $\mathbf{V_R = 60\ V}$.

$$V_L - V_C = 30 - 30 = 0\ V$$

$$\phi = \frac{0}{60} = 0$$

$$\phi = 0$$

Since $\mathbf{V_L = V_C}$, the inductive and capacitive effects cancel. This is the **resonance condition**, where the circuit behaves purely resistively and the applied voltage and current are **in phase**.

COMMON MISTAKE

At resonance $\mathbf{V_L = V_C}$ cancel out, so the phase angle is zero. Do not confuse $\mathbf{V_L = V_C = 30\ V}$ with the net reactive voltage being 30 V; it is actually 0.

Final answer. The phase difference is zero ($\phi = 0$); the circuit is at resonance and behaves purely resistively.

← Back to Q57

Q58

The power factor of an a.c. circuit is 0.5. *What will be the phase difference between voltage and current in this circuit?*

[2005]

SOLUTION

The power factor of an a.c. circuit is ϕ, where ϕ is the phase difference between the applied voltage and the current.

Given the power factor

$$\phi = 0.5$$

Therefore

$$\phi = \cos^{-1}(0.5)$$

$$\phi = 60° = \frac{\pi}{3} \text{ rad}$$

Hence the phase difference between the voltage and the current is 60° (that is, $\pi/3$ radian).

Final answer. The phase difference is $\boldsymbol{\phi = 60° = \pi/3}$.

← *Back to Q58*

Q59

A resistor and a capacitor are connected in series to an ac source $\boldsymbol{v = v_m \omega t}$. *Derive an expression for the impedance of the circuit.*

[2024 • Set 55-1-1]

SOLUTION

Consider a resistor R in series with a capacitor C connected to an a.c. source $\boldsymbol{v = v_m \omega t}$. The same current i flows through both elements.

Phasor diagram: Draw the current phasor I along the positive x-axis (reference). The voltage across the resistor $\mathbf{V_R = IR}$ is in phase with the current, so it points along I. The voltage across the capacitor $\mathbf{V_C = I\,X_C}$ lags the current by $\frac{\pi}{2}$, so it points vertically downward, perpendicular to V_R. The applied voltage v_m is the resultant (vector sum) of V_R and V_C, drawn as the diagonal of the rectangle, making an angle ϕ below the current phasor.

Since V_R and V_C are mutually perpendicular, the magnitude of the resultant is

$$v_m = \sqrt{V_R^2 + V_C^2}$$

Substituting $\mathbf{V_R = IR}$ and $\mathbf{V_C = I\,X_C}$,

$$v_m = \sqrt{(IR)^2 + (I\,X_C)^2}$$

$$v_m = I\sqrt{R^2 + X_C^2}$$

The impedance is defined as $Z = \frac{v_m}{I}$, so

$$Z = \sqrt{R^2 + X_C^2}$$

Putting $\mathbf{X_C = \frac{1}{\omega\,C}}$,

$$Z = \sqrt{R^2 + \frac{1}{\omega^2\,C^2}}$$

The current leads the applied voltage by a phase angle ϕ given by $\boldsymbol{\phi = \frac{X_C}{R}} = \frac{\mathbf{1}}{\boldsymbol{\omega\,C\,R}}$.

Final answer. The impedance of the series RC circuit is $\boldsymbol{Z = \sqrt{R^2 + 1/(\omega^2\,C^2)}}$.

← *Back to Q59*

Q60

An electric lamp is designed to operate at 110 *V dc and* 11 *A current. If the lamp is operated on* 220 *V,* 50 Hz *ac source with a coil in series, then find the inductance of the*

coil.

[2024 • Set 55-1-1]

SOLUTION

First find the resistance of the lamp from its d.c. ratings. The lamp draws 11 A at 110 V, so

$$R = \frac{V}{I} = \frac{110}{11} = 10\ \Omega$$

For the lamp to work normally on the a.c. supply, the same current $\boldsymbol{I = 11\ A}$ must flow when it is connected (with the coil in series) across the 220 V source. The total impedance of the series combination is

$$Z = \frac{V_{rms}}{I} = \frac{220}{11} = 20\ \Omega$$

For a resistor in series with an inductor,

$$Z^2 = R^2 + (\omega L)^2$$

$$(\omega L)^2 = Z^2 - R^2 = 20^2 - 10^2 = 400 - 100 = 300$$

$$\omega L = \sqrt{300} = 10\sqrt{3}\ \Omega$$

The angular frequency is

$$\omega = 2\pi\nu = 2\pi \times 50 = 100\pi\ \text{rad s}^{-1}$$

Therefore

$$L = \frac{10\sqrt{3}}{100\pi} = \frac{\sqrt{3}}{10\pi}$$

$$L \approx 0.055\ H = 55\ \text{mH}$$

Hence the required inductance of the coil is **about 0.055 H (55 mH)**.

TIP

Use the same operating current (11 A) on both d.c. and a.c. to get R and Z separately, then extract ωL from $\mathbf{Z^2 = R^2 + (\omega L)^2}$.

Final answer. The inductance of the coil is $L = \sqrt{3}/(10\pi) \approx 0.055\ H$ (55 mH).

← *Back to Q60*

Q61

Draw a labelled diagram of a step-up transformer and describe its working principle. Explain any three causes for energy losses in a real transformer.

[2024 • Set 55-1-1]

SOLUTION

Labelled diagram: Draw a rectangular closed core made of laminated soft iron. On the left limb wind the primary coil with N_p turns, connected to the AC input source. On the right limb wind the secondary coil with a larger number of turns N_s (so $\mathbf{N_s > N_p}$), connected to the output/load. Label the input as V_p (primary voltage) and the output as V_s (secondary voltage), with the changing magnetic flux Φ confined to the iron core linking both coils.

Working principle: A transformer works on the principle of **mutual induction**. When an alternating voltage is applied to the primary, the alternating current in it produces a continuously changing magnetic flux Φ in the soft-iron core.

This changing flux is carried by the core and links the secondary coil. By Faraday's law of electromagnetic induction, an emf is induced in the secondary.

Since the flux per turn is the same for both coils, the induced emf is proportional to the number of turns, giving

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

For a step-up transformer $\mathbf{N_s > N_p}$, so $\mathbf{V_s > V_p}$ and the voltage is stepped up.

Three causes of energy loss in a real transformer:

(1) Copper loss: Heat dissipated as I^2R in the resistance of the primary and secondary windings.

(2) Eddy current loss: Induced circulating currents in the core cause heating; reduced by using a **laminated** core.

(3) Hysteresis loss: Energy lost in repeatedly magnetising and demagnetising the core during each AC cycle; reduced by making the core of a soft magnetic material like silicon steel.

Final answer. A step-up transformer works on mutual induction with $\mathbf{V_s/V_p = N_s/N_p > 1}$; the three main losses are copper ($\mathbf{I^2R}$), eddy current and hysteresis losses.

← *Back to Q61*

Q62

A step-up transformer has 200 *and* 3000 *turns in its primary and secondary coils respectively. The input voltage given to the primary coil is* 90 *V. Calculate:*
(1) The output voltage across the secondary coil
(2) The current in the primary coil if the current in the secondary coil is 2.0 *A.*

[2024 • Set 55-1-1]

SOLUTION

Given $\mathbf{N_p = 200}$, $\mathbf{N_s = 3000}$, $\mathbf{V_p = 90\ V}$ and $\mathbf{I_s = 2.0\ A}$.

(1) For an ideal transformer the voltage ratio equals the turns ratio:

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

$$V_s = V_p \times \frac{N_s}{N_p}$$

$$V_s = 90 \times \frac{3000}{200}$$

$$V_s = 90 \times 15$$

$$V_s = 1350\ V$$

The **output voltage across the secondary is 1350 *V***.

(2) For an ideal transformer there is no power loss, so the input power equals the output power:

$$V_p I_p = V_s I_s$$

$$I_p = I_s \times \frac{V_s}{V_p}$$

$$I_p = 2.0 \times \frac{1350}{90}$$

$$I_p = 2.0 \times 15$$

$$I_p = 30\ A$$

The **current in the primary coil is 30 *A***.

COMMON MISTAKE

In a step-up transformer the voltage rises but the current falls in the same ratio, so the primary always carries the larger current; mixing up which coil has the bigger current is a common slip.

Final answer. The output voltage is $\mathbf{V_s = 1350\ V}$ and the primary current is $\mathbf{I_p = 30\ A}$.

← *Back to Q62*

Q63

(a) Explain the term 'sharpness of resonance' in ac circuit.
(b) In a series LCR circuit, $\mathbf{V_L = V_C \neq V_R}$. What is the value of power factor for this circuit?
OR
An ac source of emf $\mathbf{V = V_0\ \omega\ t}$ is connected to a capacitor of capacitance C. Deduce the expression for the current (I) flowing in it. Plot the graph of (i) V vs. ω t, and (ii) I vs. ω t.

[2020 • Set 55-2-1]

SOLUTION

(a) Sharpness of resonance: It describes how rapidly the current in a series LCR circuit falls as the driving frequency moves away from the resonant frequency ω_0. A sharper resonance means the current peak is tall and narrow, so the circuit responds strongly only to frequencies very close to ω_0.

It is measured by the quality factor:

$$Q = \frac{\omega_0 L}{R} = \frac{1}{R}\sqrt{\frac{L}{C}}$$

A larger value of Q (smaller R) gives a narrower bandwidth and hence a sharper resonance.

(b) Given $\mathbf{V_L = V_C \neq V_R}$. Since $\mathbf{V_L = V_C}$, the inductive and capacitive reactances are equal, $\mathbf{X_L = X_C}$, which is the condition for resonance. The net reactance is zero, so the impedance is purely resistive, $\boldsymbol{Z = R}$, and the phase angle is

$$\phi = \frac{R}{Z} = \frac{R}{R} = 1$$

The **power factor is 1 (unity)**.

OR (Capacitor with $\boldsymbol{V = V_0 \omega t}$):

The charge on the capacitor is

$$q = CV = CV_0 \omega t$$

The current is the rate of flow of charge:

$$I = \frac{dq}{dt} = CV_0\, \omega\omega t$$

$$I = \frac{V_0}{1/(\omega C)} \omega t = \frac{V_0}{X_C} \omega t$$

Writing the cosine as a shifted sine,

$$I = I_0 \sin(\omega t + \frac{\pi}{2}),\ I_0 = \frac{V_0}{X_C}$$

So the **current leads the voltage by a phase angle of $\pi/2$**.

Graphs: On the same ωt axis, the voltage graph V vs ωt is a sine curve starting from zero at $\boldsymbol{\omega t = 0}$. The current graph I vs ωt is a cosine curve, starting at its

maximum value I_0 at $\omega t = 0$, i.e. it is shifted to the left of the voltage curve by a quarter cycle ($\pi/2$), showing that current leads voltage.

Final answer. Power factor is unity ($\phi = 1$); OR for the capacitor, $I = \frac{V_0}{X_C}\sin(\omega t + \pi/2)$, leading V by $\pi/2$.

← Back to Q63

Q64

A resistor R and an inductor L are connected in series to a source of voltage $V = V_0 \omega t$. The voltage is found to lead current in phase by $\pi/4$. If the inductor is replaced by a capacitor C, the voltage lags behind current in phase by $\pi/4$. When L, C and R are connected in series with the same source, find the:
(i) average power dissipated and
(ii) instantaneous current in the circuit.

[2020 • Set 55-4-1]

SOLUTION

For the series R-L circuit, the voltage leads the current by $\pi/4$. The phase angle satisfies

$$\phi = \frac{X_L}{R}$$

$$\tan\frac{\pi}{4} = \frac{X_L}{R} = 1$$

$$\therefore X_L = R$$

For the series R-C circuit, the voltage lags the current by $\pi/4$, so

$$\tan\frac{\pi}{4} = \frac{X_C}{R} = 1$$

$$\therefore X_C = R$$

When L, C and R are all connected in series with the same source, $X_L = X_C = R$, so the net reactance is

$$X_L - X_C = 0$$

The circuit is therefore at **resonance**. The impedance becomes

$$Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{R^2 + 0} = R$$

and the phase angle is $\phi = 0$, so the power factor $\phi = 1$.

(i) Average power dissipated:

$$P_{avg} = V_{rms}\, I_{rms} \phi = \frac{V_{rms}^2}{Z} \phi$$

$$P_{avg} = \frac{V_{rms}^2}{R} \times 1 = \frac{(V_0/\sqrt{2})^2}{R}$$

$$P_{avg} = \frac{V_0^2}{2R}$$

(ii) Instantaneous current: at resonance $Z = R$ and the current is in phase with the applied voltage $V = V_0 \omega t$, so

$$I(t) = \frac{V_0}{R} \omega\, t$$

The average power is $\frac{V_0^2}{2R}$ and the instantaneous current is $\frac{V_0}{R} \omega\, t$.

Final answer. At resonance $Z = R$; average power $P_{avg} = \frac{V_0^2}{2R}$ and instantaneous current $I(t) = \frac{V_0}{R} \omega\, t$.

← *Back to Q64*

Q65

A light bulb and a solenoid are connected in series across an ac source of voltage. Explain, how the glow of the light bulb will be affected when an iron rod is inserted in the solenoid.

[2017]

SOLUTION

The light bulb (resistance R) and the solenoid (inductance L) are in series across the AC source. The current is

$$I = \frac{V}{Z}$$

where the impedance of the R-L combination is

$$Z = \sqrt{R^2 + X_L{}^2} = \sqrt{R^2 + (\omega\, L)^2}$$

When an iron rod is inserted into the solenoid, the relative permeability of the core becomes very large ($\mu_r \gg 1$). The self-inductance of the solenoid,

$$L = \frac{\mu_0\, \mu_r\, N^2\, A}{l}$$

increases substantially.

As a result the inductive reactance $X_L = \omega L$ increases, so the impedance Z increases. From $I = V/Z$, the current in the circuit therefore decreases.

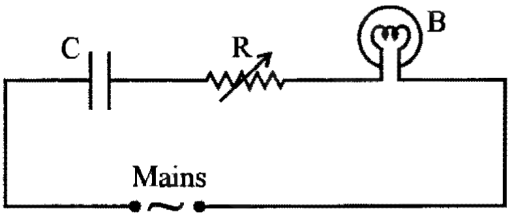
The power dissipated in the bulb, $P = I^2 R$, falls as the current falls. Hence the **glow of the bulb becomes dimmer** when the iron rod is inserted.

Final answer. Inserting the iron rod raises L and hence X_L and Z, so the current and the bulb's glow both decrease (the bulb grows dimmer).

← *Back to Q65*

Q66

A capacitor 'C', a variable resistor 'R' and a bulb 'B' are connected in series to the ac mains in circuit as shown. The bulb glows with some brightness. How will the glow of the bulb change if (i) a dielectric slab is introduced between the plates of the capacitor, keeping resistance R to be the same; (ii) the resistance R is increased keeping the same capacitance ?



[2014]

SOLUTION

The capacitor C, variable resistor R and bulb B are in series across the AC mains. The current in the circuit is

$$I = \frac{V}{Z} = \frac{V}{\sqrt{R^2 + X_C{}^2}}, \; X_C = \frac{1}{\omega C}$$

(i) Dielectric slab introduced between the capacitor plates: the capacitance increases from C to κC (where $\kappa > 1$ is the dielectric constant). The capacitive reactance

$$X_C = \frac{1}{\omega C}$$

therefore decreases. Hence the impedance $Z = \sqrt{R^2 + X_C{}^2}$ decreases, the current $I = V/Z$ increases, and the **bulb glows brighter**.

(ii) Resistance R increased (capacitance unchanged): the impedance

$$Z = \sqrt{R^2 + X_C{}^2}$$

increases. So the current $I = V/Z$ decreases, and the **bulb glows dimmer**.

Final answer. Inserting a dielectric lowers X_C so the bulb glows brighter; increasing R raises Z so the bulb glows dimmer.

← *Back to Q66*

Q67

State the underlying principle of a transformer. How is the large scale transmission of electric energy over long distances done with the use of transformers?

[2012]

SOLUTION

Underlying principle: A transformer works on the principle of **mutual induction**. When an alternating current flows in the primary coil, it sets up a continuously changing magnetic flux Φ in the soft-iron core. This flux links the secondary coil, and by Faraday's law of electromagnetic induction an emf is induced in the secondary. As the flux per turn is the same in both coils,

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

Large scale transmission of electric energy: At the generating station, electrical power is generated at a moderate voltage (about 11 kV). Before transmission it is **stepped up** to a very high voltage (for example 220 kV or more) using a step-up transformer.

For a fixed power $P = VI$, raising the voltage V greatly lowers the current I in the transmission lines.

The power lost as heat in the cables of resistance R is

$$P_{loss} = I^2 R$$

Since $P_{loss} \propto I^2$, a much smaller current means the I^2R heating losses in the wires are dramatically reduced, so power is transmitted efficiently over long distances.

Near the consumer end, **step-down transformers** at substations reduce the voltage back to a safe usable level (about 230 V) for domestic supply.

Final answer. A transformer works on mutual induction; power is stepped up to high voltage for transmission so that the lower current minimises I^2R line losses, then stepped down for use.

← *Back to Q67*

Q68

A light bulb is rated 100 W for 220 V ac supply of 50 Hz. Calculate
(i) the resistance of the bulb;
(ii) the rms current through the bulb.

OR

An alternating voltage given by $V = 140\sin 314t$ *is connected across a pure resistor of* 50 Ω. *Find*

(i) the frequency of the source.

(ii) the rms current through the resistor.

[2012]

SOLUTION

Given $P = 100\ W$, $V_{rms} = 220\ V$ and $f = 50\ Hz$.

(i) Resistance of the bulb: using $P = \frac{V_{rms}^2}{R}$,

$$R = \frac{V_{rms}^2}{P}$$

$$R = \frac{(220)^2}{100} = \frac{48400}{100}$$

$$R = 484\ \Omega$$

(ii) RMS current through the bulb:

$$I_{rms} = \frac{V_{rms}}{R}$$

$$I_{rms} = \frac{220}{484}$$

$$I_{rms} \approx 0.45\ A$$

The **resistance is 484 Ω** and the **rms current is about 0.45 *A***.

OR ($V = 140\sin 314t$ across $R = 50\ \Omega$):

Comparing with $V = V_0\omega t$, the peak voltage $V_0 = 140\ V$ and angular frequency $\omega = 314\ rad/s$.

(i) Frequency:

$$f = \frac{\omega}{2\pi} = \frac{314}{2\pi}$$

$$f = 50\ Hz$$

(ii) RMS current: first the rms voltage,

$$V_{rms} = \frac{V_0}{\sqrt{2}} = \frac{140}{\sqrt{2}} \approx 99\ V$$

$$I_{rms} = \frac{V_{rms}}{R} = \frac{99}{50}$$

$$I_{rms} \approx 1.98\ A$$

The frequency is 50 Hz and the rms current is about 1.98 A.

TIP

Power ratings on bulbs always use rms (effective) values, so use $\mathbf{V_{rms} = 220\ V}$ directly in $\boldsymbol{P = V_{rms}^2/R}$; do not convert it to a peak value first.

Final answer. Resistance $\boldsymbol{R = 484\ \Omega}$ and $\mathbf{I_{rms} \approx 0.45\ A}$; OR $\boldsymbol{f = 50\ Hz}$ and $\mathbf{I_{rms} \approx 1.98\ A}$.

← *Back to Q68*

Q69

Calculate the quality factor of a series LCR circuit with $\boldsymbol{L = 2.0\ H}$, $\boldsymbol{C = 2\ \mu F}$ *and* $\boldsymbol{R = 10\ \Omega}$. *Mention the significance of quality factor in LCR circuit.*

[2012]

SOLUTION

Given $\boldsymbol{L = 2.0\ H}$, $\boldsymbol{C = 2\ \mu F = 210^{-6}\ F}$ and $\boldsymbol{R = 10\ \Omega}$.

The quality factor of a series LCR circuit is

$$Q = \frac{1}{R}\sqrt{\frac{L}{C}}$$

Substituting the values,

$$Q = \frac{1}{10}\sqrt{\frac{2.0}{210^{-6}}}$$

$$Q = \frac{1}{10}\sqrt{110^6}$$

$$Q = \frac{1}{10} \times 1000$$

$$Q = 100$$

The **quality factor is 100** (a dimensionless number).

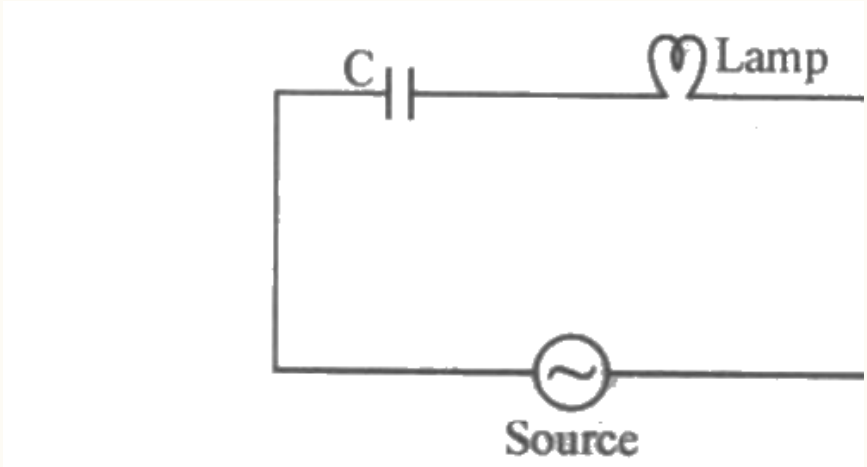
Significance: The quality factor measures the **sharpness of resonance** of the circuit. A high value of Q means a narrow resonance peak (small bandwidth), so the circuit responds strongly only to frequencies very close to the resonant frequency. This gives high **selectivity**, which is essential in tuning circuits of radios and televisions to pick out one desired frequency while rejecting nearby ones.

Final answer. The quality factor is $Q = 100$; a high Q indicates a sharp resonance and high frequency selectivity.

← *Back to Q69*

Q70

An electric lamp having coil of negligible inductance connected in series with a capacitor and an AC source is glowing with certain brightness. How does the brightness of the lamp change on reducing the (i) capacitance, and (ii) the frequency? Justify your answer.



[2010]

SOLUTION

The lamp (resistance R, negligible inductance) is in series with a capacitor C across the AC source. The current is

$$I = \frac{V}{Z},\ Z = \sqrt{R^2 + X_C{}^2},\ X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

The brightness of the lamp depends on the power $P = I^2R$, and hence on the current I.

(i) Reducing the capacitance C: since $X_C = \frac{1}{\omega C}$, a smaller C makes X_C larger. The impedance $Z = \sqrt{R^2 + X_C{}^2}$ therefore increases, so the current $I = V/Z$ decreases. The **lamp becomes dimmer**.

(ii) Reducing the frequency: since $X_C = \frac{1}{2\pi f C}$, a smaller f also makes X_C larger. The impedance Z increases, the current $I = V/Z$ decreases, and again the **lamp becomes dimmer**.

Final answer. Reducing either the capacitance or the frequency raises X_C and hence Z, lowering the current, so in both cases the lamp grows dimmer.

← *Back to Q70*

Q71

Prove that an ideal capacitor, in an a.c. circuit does not dissipate power.

[2008]

SOLUTION

Consider an ideal capacitor (a capacitor with no resistance) of capacitance C connected across an a.c. source whose voltage is $V = V_0\omega\, t$.

The instantaneous charge on the capacitor is $q = CV = CV_0\omega\, t$.

The instantaneous current is $I = \frac{dq}{dt} = CV_0\omega\omega\, t$.

Writing the peak current as $I_0 = CV_0\omega = \frac{V_0}{X_C}$, we get $I = I_0\omega\, t = I_0\sin(\omega\, t + \frac{\pi}{2})$.

Thus the current leads the voltage by a phase angle of $\frac{\pi}{2}$.

The instantaneous power supplied is $P = VI = (V_0 \omega t)(I_0 \omega t)$.

$$P = V_0 I_0 \omega t \omega t = \frac{V_0 I_0}{2} \sin 2\omega t.$$

The average power over one full cycle is $P_{avg} = \frac{V_0 I_0}{2} 2\omega t$.

Since the average value of $\sin 2\omega t$ over a complete cycle is zero, $2\omega t = 0$.

Therefore $P_{avg} = 0$.

Hence an **ideal capacitor in an a.c. circuit dissipates no average power**. The energy drawn during one quarter cycle (while charging) is returned to the source during the next quarter cycle (while discharging), so the net power consumed over a full cycle is zero.

CONCEPT

The phase difference of $\pi/2$ between V and I is the real reason the average power is zero; whenever phase difference is 90°, the power factor $\phi = 0$.

Final answer. An ideal capacitor consumes zero average power over a full cycle, since $P_{avg} = \frac{V_0 I_0}{2} 2\omega t = 0$.

← *Back to Q71*

Q72

Derive an expression for the impedance of an a.c. circuit consisting of an inductor and a resistor.

[2008]

SOLUTION

Consider a resistor of resistance R and an inductor of inductance L connected in series across an a.c. source. Let the same current I (rms value) flow through both, since they are in series.

The voltage across the resistor is $V_R = IR$, and it is in phase with the current I.

The voltage across the inductor is $V_L = IX_L$, where $X_L = \omega L$ is the inductive reactance, and this voltage leads the current by $\frac{\pi}{2}$.

In the phasor diagram, the current $\vec{I}$ is taken along the positive x-axis. The phasor $\vec{V}_R$ is drawn along $\vec{I}$ (same direction), while $\vec{V}_L$ is drawn perpendicular to $\vec{I}$, pointing along the positive y-axis (leading by 90°). The applied voltage $\vec{V}$ is the resultant (diagonal) of the rectangle formed by $\vec{V}_R$ and $\vec{V}_L$.

Since V_R and V_L are at right angles, the magnitude of the applied voltage is $V = \sqrt{V_R^2 + V_L^2}$.

Substituting $V_R = IR$ and $V_L = IX_L$, $V = \sqrt{(IR)^2 + (IX_L)^2}$.

$V = I\sqrt{R^2 + X_L^2}$.

The impedance is defined as $Z = \frac{V}{I}$, so $Z = \sqrt{R^2 + X_L^2}$.

Putting $X_L = \omega L$, the **impedance of the series RL circuit is** $Z = \sqrt{R^2 + (\omega L)^2}$.

The phase angle by which the applied voltage leads the current is given by $\phi = \frac{V_L}{V_R} = \frac{X_L}{R} = \frac{\omega L}{R}$.

Final answer. The impedance of a series RL circuit is $Z = \sqrt{R^2 + (\omega L)^2}$, with the voltage leading the current by $\phi = \tan^{-1}\left(\frac{\omega L}{R}\right)$.

← *Back to Q72*

Q73

Distinguish between the terms 'average value' and 'rms value' of an alternating current. The instantaneous current from an a.c. source is $I = 5\sin(314\,t)$ *ampere. What are the average and rms values of the current?*

[2007]

SOLUTION

The average value (or mean value) of an alternating current is the value obtained by averaging the instantaneous current over a given time. Over one complete cycle the average of a sinusoidal current is zero, because the positive

half and the negative half cancel. Hence the average value is usually quoted over one half cycle, where it equals $I = \frac{2\,I_0}{\pi} = 0.637\,I_0$.

The rms (root mean square) value of an alternating current is the square root of the average of the squares of the instantaneous current over one full cycle. It equals that steady (direct) current which would produce the same heating effect in a given resistance in the same time. For a sinusoidal current, $I_{rms} = \frac{I_0}{\sqrt{2}} = 0.707\,I_0$.

Now for the given source, $I = 5\sin(314\,t)$ A.

Comparing with $I = I_0 \omega\, t$, the peak current is $I_0 = 5\ A$.

The average value over a half cycle is $I = \frac{2\,I_0}{\pi}$.

$$I = \frac{2 \times 5}{\pi} = \frac{10}{3.14} = 3.18\ A.$$

The rms value is $I_{rms} = \frac{I_0}{\sqrt{2}}$.

$$I_{rms} = \frac{5}{\sqrt{2}} = \frac{5}{1.414} = 3.54\ A.$$

Hence the **average value over a half cycle is 3.18 A and the rms value is 3.54 A.**

Final answer. Average value (half cycle) $I = 3.18\ A$ and rms value $I_{rms} = 3.54\ A$.

← *Back to Q73*

Q74

Calculate the current drawn by the primary of a transformer which steps down 200 V to 20 V to operate a device of resistance 20 Ω. Assume the efficiency of the transformer to be 80%.

[2007]

SOLUTION

Given: primary voltage $V_p = 200\ V$, secondary voltage $V_s = 20\ V$, resistance of the device $R = 20\ \Omega$, and efficiency $\eta = 80\% = 0.80$.

The current through the device (secondary current) follows from Ohm's law, $I_s = \frac{V_s}{R}$.

$$I_s = \frac{20}{20} = 1\ A.$$

The output power delivered to the device (secondary power) is $P_s = V_s I_s$.

$$P_s = 20 \times 1 = 20\ W.$$

Efficiency is the ratio of output power to input power, $\eta = \frac{P_s}{P_p}$, so the input (primary) power is $P_p = \frac{P_s}{\eta}$.

$$P_p = \frac{20}{0.80} = 25\ W.$$

The primary current is found from $P_p = V_p I_p$, giving $I_p = \frac{P_p}{V_p}$.

$$I_p = \frac{25}{200} = 0.125\ A.$$

Hence the **current drawn by the primary of the transformer is 0.125 A.**

> **COMMON MISTAKE**
>
> *Efficiency below 100% means input power exceeds output power; always divide (not multiply) the output power by η to get the input power.*

Final answer. The primary current drawn is $I_p = 0.125\ A$.

← Back to Q74

Q75

An a.c. voltage of 100 V, 50 Hz is connected across a 20 ohm resistor and 2 mH inductor in series. Calculate (i) impedance of the circuit, (ii) rms current in the circuit.

[2007]

SOLUTION

Given: rms voltage $V_{rms} = 100\ V$, frequency $f = 50\ Hz$, resistance $R = 20\ \Omega$, and inductance $L = 2\ mH = 2 \times 10^{-3}\ H$.

The inductive reactance is $X_L = 2\pi f L$.

$X_L = 2 \times 3.14 \times 50 \times (2 \times 10^{-3})$.

$X_L = 0.628\ \Omega$.

(i) The impedance of the series RL circuit is $Z = \sqrt{R^2 + X_L{}^2}$.

$Z = \sqrt{(20)^2 + (0.628)^2}$.

$Z = \sqrt{400 + 0.39} = \sqrt{400.39}$.

$Z \approx 20.0\ \Omega$.

(ii) The rms current is $I_{rms} = \dfrac{V_{rms}}{Z}$.

$I_{rms} = \dfrac{100}{20.0} = 5.0\ A$.

Hence the **impedance is approximately 20.0 Ω and the rms current is 5.0 A**. (Because X_L is very small compared with R, the inductor barely affects the impedance here.)

Final answer. (i) Impedance $Z \approx 20.0\ \Omega$; (ii) rms current $I_{rms} = 5.0\ A$.

← *Back to Q75*

Q76

(i) *Draw the graphs showing variation of inductive reactance and capacitive reactance with frequency of applied a.c. source.*
(ii) *Can the voltage drop across the inductor or the capacitor in a series* LCR *circuit be greater than the applied voltage of the a.c. source? Justify your answer.*

[2005]

SOLUTION

(i) Graphs of reactance versus frequency:

For an inductor, $X_L = 2\pi f L$, so X_L is directly proportional to the frequency f. Its graph is a straight line passing through the origin, sloping upward, with slope

$2\pi L$ (reactance rises as frequency increases).

For a capacitor, $\mathbf{X_C = \frac{1}{2\pi f C}}$, so X_C is inversely proportional to the frequency f. Its graph is a rectangular hyperbola that falls steeply at low frequency and approaches the frequency axis (zero reactance) at high frequency.

On the same axes (reactance on the y-axis, frequency on the x-axis), the X_L line rises from the origin while the X_C curve falls; the two curves cross at the resonant frequency where $\mathbf{X_L = X_C}$.

(ii) Yes, the voltage drop across the inductor or the capacitor in a series LCR circuit can be greater than the applied source voltage.

At resonance, $\mathbf{X_L = X_C}$, and the voltages across L and C are $\mathbf{V_L = I X_L}$ and $\mathbf{V_C = I X_C}$, both equal in magnitude to $Q\,V_{rms}$, where $\mathbf{Q = \frac{1}{R}\sqrt{\frac{L}{C}}}$ is the quality factor. If $\mathbf{Q > 1}$, these voltages exceed the source voltage.

This is possible because V_L and V_C are exactly π out of phase (opposite in direction in the phasor diagram), so they cancel each other in the phasor sum. The applied voltage equals $\mathbf{V = \sqrt{V_R^2 + (V_L - V_C)^2}}$, which at resonance reduces to just V_R. Hence the individual reactive voltages can each be much larger than the source voltage without violating the loop voltage rule.

Thus $\mathbf{X_L}$ rises linearly and $\mathbf{X_C}$ falls hyperbolically with frequency, and yes, $\mathbf{V_L}$ or $\mathbf{V_C}$ can exceed the source voltage at resonance because they are oppositely directed and cancel.

Final answer. $\mathbf{X_L}$ increases linearly and $\mathbf{X_C}$ decreases hyperbolically with frequency; yes, $\mathbf{V_L}$ or $\mathbf{V_C}$ can exceed the source voltage since they are $\boldsymbol{\pi}$ out of phase and cancel in the phasor sum.

← *Back to Q76*

Q77

An a.c. voltage $\mathbf{E = E_0 \sin \omega t}$ *is applied across an inductor L. Obtain an expression for current I.*

[2003]

SOLUTION

An a.c. voltage $E = E_0 \omega t$ is applied across a pure inductor of inductance L.

When the current I flows through the inductor, a back e.m.f. of magnitude $L\frac{dI}{dt}$ is induced. Since the inductor has no resistance, the applied voltage equals this back e.m.f.

$E = L\frac{dI}{dt}$.

Therefore $\frac{dI}{dt} = \frac{E}{L} = \frac{E_0}{L}\omega t$.

Integrating both sides with respect to time, $I = \frac{E_0}{L}\int \omega t \, dt$.

$I = \frac{E_0}{L}\left(-\frac{\omega t}{\omega}\right)$.

$I = -\frac{E_0}{\omega L}\omega t$.

(The constant of integration is zero, since the current has no steady d.c. part.)

Using $-\omega t = \sin(\omega t - \frac{\pi}{2})$, we write $I = \frac{E_0}{\omega L}\sin(\omega t - \frac{\pi}{2})$.

Writing the peak current as $I_0 = \frac{E_0}{\omega L} = \frac{E_0}{X_L}$, where $X_L = \omega L$ is the inductive reactance, the required expression is $I = I_0 \sin(\omega t - \frac{\pi}{2})$.

Hence $I = \frac{E_0}{\omega L}\sin(\omega t - \frac{\pi}{2})$, showing that the current lags the applied voltage by a phase angle of $\frac{\pi}{2}$.

Final answer. $I = \frac{E_0}{\omega L}\sin(\omega t - \frac{\pi}{2})$, the current lagging the voltage by $\frac{\pi}{2}$.

← *Back to Q77*

Q78

(a) Discuss the behaviour of an inductor connected to (i) a dc source, and (ii) a high frequency ac source.

(b) What is the phase relation between current and voltage in an ideal inductor connected to an ac source? Draw a phasor diagram for the circuit.

[2026 • Set 55-3-1]

SOLUTION

(a) Behaviour of an inductor:

(i) Connected to a d.c. source: For steady direct current the frequency is zero, $\omega = 0$, so the inductive reactance $X_L = \omega L = 0$. The inductor therefore offers no opposition to steady d.c. and behaves simply like a plain conducting wire (apart from its small ohmic resistance). The current is then limited only by the resistance in the circuit.

(ii) Connected to a high frequency a.c. source: Here ω is very large, so the inductive reactance $X_L = \omega L$ becomes very large. The inductor offers a strong opposition to the flow of current, and very little current passes through it. In effect a pure inductor tends to block high frequency a.c.

(b) Phase relation in an ideal inductor: In an ideal (pure) inductor connected to an a.c. source, the current lags behind the applied voltage by a phase angle of $\frac{\pi}{2}$ (equivalently, the voltage leads the current by $\frac{\pi}{2}$).

If $V = V_0 \omega t$, then $I = I_0 \sin(\omega t - \frac{\pi}{2})$.

Phasor diagram: Draw two phasors from a common origin. The current phasor $\vec{I}$ is drawn along the positive x-axis (horizontal). The voltage phasor $\vec{V}$ is drawn along the positive y-axis (vertical, 90° ahead of the current, measured anticlockwise). Both phasors rotate together anticlockwise with angular speed ω, and $\vec{V}$ always stays 90° ahead of $\vec{I}$.

Thus an inductor passes d.c. freely ($X_L = 0$) but strongly opposes high frequency a.c. (large X_L), and in an ideal inductor the current lags the voltage by $\frac{\pi}{2}$.

Final answer. An inductor freely passes d.c. ($X_L = 0$) and strongly opposes high frequency a.c. (large X_L); in an ideal inductor the current lags the voltage by $\frac{\pi}{2}$.

← Back to Q78

Q79

A series combination of circuit elements X and Y is connected across an ac source. It is found that the voltage is ahead of current in phase by $\frac{\pi}{4}$ radian. When element Y is replaced by element Z, the current leads the voltage by $\frac{\pi}{4}$ radian.

(a) Identify the elements X, Y and Z.
(b) What will the phase angle and power factor for the circuit be if X, Y and Z were connected in series across the same ac source? What can you say about the current that flows in the circuit in this case?

[2026 • Set 55-3-2]

SOLUTION

(a) Identifying the elements:

When X and Y are in series, the voltage is ahead of (leads) the current by $\frac{\pi}{4}$. A voltage that leads the current happens in an R-L circuit, where the inductor makes the voltage lead. So Y must be an inductor L, and X must be a resistor R.

When Y is replaced by Z and the current now leads the voltage by $\frac{\pi}{4}$, this is the behaviour of an R-C circuit, where the capacitor makes the current lead. So Z must be a capacitor C.

Hence $\boldsymbol{X = R}$ (resistor), $\boldsymbol{Y = L}$ (inductor), $\boldsymbol{Z = C}$ (capacitor).

For the first case, $\boldsymbol{\tan\frac{\pi}{4} = \frac{X_L}{R}}$, and since $\boldsymbol{\tan\frac{\pi}{4} = 1}$, we get $\boldsymbol{X_L = R}$.

For the second case, $\boldsymbol{\tan\frac{\pi}{4} = \frac{X_C}{R} = 1}$, which gives $\boldsymbol{X_C = R}$.

(b) With X, Y and Z (that is R, L and C) all in series across the same source, $\boldsymbol{X_L = R}$ and $\boldsymbol{X_C = R}$, so $\boldsymbol{X_L = X_C}$.

Since $\boldsymbol{X_L = X_C}$, the net reactance is zero and the circuit is in the resonance condition.

The impedance becomes $\boldsymbol{Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{R^2 + 0} = R}$.

The phase angle is $\phi = \tan^{-1}\left(\frac{X_L - X_C}{R}\right) = \tan^{-1}(0) = 0$.

The power factor is $\phi = \cos 0 = 1$.

Therefore the current is in phase with the applied voltage, and its value is maximum, $I = \frac{V}{R}$, since the impedance is the least possible (just R).

Thus $X = R$, $Y = L$, $Z = C$; in series the circuit resonates with phase angle $\phi = 0$, power factor = 1, and a maximum current $I = \frac{V}{R}$ that is in phase with the voltage.

> **TIP**
>
> Equal phase angles of $\pi/4$ in the two cases force $X_L = X_C$; that equality is the clue that the combined circuit is at resonance.

Final answer. $X = R$, $Y = L$, $Z = C$; in series they resonate, giving $\phi = 0$, power factor = 1, and a maximum, in-phase current $I = \frac{V}{R}$.

← Back to Q79

Q80

(a) Differentiate between inductive reactance, capacitive reactance and impedance of an ac circuit.
(b) An ideal inductor and an ideal capacitor are connected in series across an ac voltage. Plot a graph showing variation of net reactance of the circuit with frequency of the applied ac voltage.

[2026 • Set 55-5-2]

SOLUTION

(a) Distinguishing the three quantities:

Inductive reactance X_L is the opposition offered by an inductor to alternating current. It is given by $X_L = \omega L = 2\pi f L$, so it increases (linearly) with frequency. It arises purely from the inductance and has no in-phase (resistive) part.

Capacitive reactance X_C is the opposition offered by a capacitor to alternating current. It is given by $\mathbf{X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}}$, so it decreases with increasing frequency. It arises purely from the capacitance.

Impedance Z is the total (effective) opposition offered by the whole a.c. circuit, including resistance and both reactances together. For a series LCR circuit, $\mathbf{Z = \sqrt{R^2 + (X_L - X_C)^2}}$. It accounts for both the resistive and the net reactive parts.

(b) Net reactance of an ideal series LC circuit:

The net reactance is $\mathbf{X = X_L - X_C = \omega L - \frac{1}{\omega C}}$.

At low frequency, $\frac{1}{\omega C}$ dominates, so X is large and negative (the circuit is capacitive).

At the resonant frequency $\mathbf{\omega_0 = \frac{1}{\sqrt{LC}}}$, we have $\mathbf{X_L = X_C}$, so the net reactance $\mathbf{X = 0}$.

At high frequency, ωL dominates, so X is large and positive (the circuit is inductive).

Graph: Plot net reactance X on the y-axis against frequency f (or ω) on the x-axis. The curve starts deep in the negative region at low frequency, rises smoothly and monotonically, crosses the frequency axis (zero) at the resonant frequency $\mathbf{f_0 = \frac{1}{2\pi\sqrt{LC}}}$, and then continues rising into the positive region, tending to behave like the straight line $X = \omega L$ at high frequency.

Thus $\mathbf{X_L}$ rises with frequency, $\mathbf{X_C}$ falls with frequency, Z is the total opposition, and the net reactance of a series LC circuit rises monotonically from large negative values through zero (at resonance) to large positive values.

Final answer. $\mathbf{X_L = \omega L}$ rises with frequency, $\mathbf{X_C = \frac{1}{\omega C}}$ falls with frequency, $\mathbf{Z = \sqrt{R^2+(X_L-X_C)^2}}$ is the total opposition; the net LC reactance rises from large negative through zero (at $\mathbf{\omega_0 = 1/\sqrt{LC}}$) to large positive.

← *Back to Q80*

Q81

(a) Can a transformer step up or step down dc power supply?
(b) Can a step up transformer work as a step down transformer?
(c) Does a step up transformer contradict the principle of conservation of energy? Justify your answer.

[2026 • Set 55-5-3]

SOLUTION

A transformer is a device that changes an alternating voltage from one value to another using the principle of mutual induction. Its working depends entirely on a changing magnetic flux linking the secondary coil.

(a) No, a transformer cannot step up or step down a dc power supply.

A transformer works on Faraday's law of electromagnetic induction, where the induced emf in the secondary is $\varepsilon_s = -N_s \frac{d\Phi}{dt}$.

A steady dc current produces a constant magnetic flux, so $\frac{d\Phi}{dt} = 0$ and no emf is induced in the secondary.

Hence with steady dc the transformer simply does not operate (it only responds during the brief switch-on transient).

(b) Yes, a step-up transformer can work as a step-down transformer.

We only have to interchange the roles of the two coils. If the ac source is connected across the coil with the larger number of turns and the output is taken from the coil with the fewer turns, the same device now gives a lower output voltage.

Since $\frac{V_s}{V_p} = \frac{N_s}{N_p}$, feeding the high-turn coil as the primary makes $N_s < N_p$, so the output voltage is stepped down.

(c) No, a step-up transformer does not contradict the principle of conservation of energy.

For an ideal transformer the output power equals the input power, $V_p I_p = V_s I_s$.

When the voltage is stepped up, the current is stepped down in the same ratio, so the **total electrical power is conserved**.

A real transformer has small losses (copper, iron, flux leakage), so the output power is slightly less than the input, but no energy is ever created.

Final answer. (a) No, dc gives no changing flux; (b) Yes, by interchanging the coils; (c) No, since stepping up voltage steps down current keeping $\mathbf{V_p I_p = V_s I_s}$.

← Back to Q81

Q82

An ac source of voltage $v = v_m \sin \omega t$ *is connected to a series combination of LCR circuit. Draw the phasor diagram. Using it obtain an expression for the impedance of the circuit and the phase difference between applied voltage and the current.*

[2025 • Set 55-1-3]

SOLUTION

In a series LCR circuit, the same current i flows through R, L and C, so we take the current as the reference phasor and draw the voltage phasors with respect to it.

Phasor diagram: Draw the current phasor $\vec{I}$ along the positive x-axis (reference).

The voltage across the resistor $\vec{V}_R = \mathbf{IR}$ is in phase with the current, drawn along $\vec{I}$ (positive x-axis).

The voltage across the inductor $\vec{V}_L = \mathbf{IX}_L$ leads the current by $\frac{\pi}{2}$, drawn along the positive y-axis.

The voltage across the capacitor $\vec{V}_C = \mathbf{IX}_C$ lags the current by $\frac{\pi}{2}$, drawn along the negative y-axis.

Since $\vec{V}_L$ and $\vec{V}_C$ are opposite, their resultant is $(V_L - V_C)$ along the y-axis (taking $\mathbf{X_L > X_C}$).

The applied voltage $\vec{V}$ is the vector sum of $\vec{V}_R$ and $(\vec{V}_L - \vec{V}_C)$, so by the parallelogram (right-angle) law:

$$V = \sqrt{V_R^2 + (V_L - V_C)^2}$$

$$V = \sqrt{(IR)^2 + (IX_L - IX_C)^2}$$

$$V = I\sqrt{R^2 + (X_L - X_C)^2}$$

Comparing with $V = IZ$, the **impedance** of the circuit is

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

where $X_L = \omega L$ and $X_C = \frac{1}{\omega C}$.

From the phasor diagram, the **phase difference** ϕ between the applied voltage and the current is given by

$$\phi = \frac{V_L - V_C}{V_R} = \frac{X_L - X_C}{R}$$

$$\phi = \tan^{-1}\left(\frac{X_L - X_C}{R}\right)$$

If $X_L > X_C$ the voltage leads the current; if $X_C > X_L$ the voltage lags the current.

Final answer. $Z = \sqrt{R^2 + (X_L - X_C)^2}$ and $\phi = \tan^{-1}\left(\frac{X_L - X_C}{R}\right)$.

← *Back to Q82*

Q83

Differentiate between the peak value and root mean square value of an alternating current. Derive the expression for the root mean square value of alternating current, in terms of its peak value.

[2025 • Set 55-2-2]

SOLUTION

Peak value and rms value are two different ways of describing the strength of an alternating current.

The **peak value** (I_0) is the maximum instantaneous value the current reaches during a cycle. The **root mean square (rms) value** (I_{rms}) is the value of steady

dc that would produce the same heating (same average power) in a given resistor over one cycle.

Briefly: the peak value occurs only twice each cycle, while the rms value is the effective or dc-equivalent value used to specify ac power.

Derivation of rms value for a sinusoidal current $I = I_0 \omega t$:

The heat produced in a resistance R in time dt is $\mathbf{dH = I^2 R\, dt}$.

The average of I^2 over one full time period T is

$$I^2 = \frac{1}{T}\int_0^T I^2\, dt = \frac{1}{T}\int_0^T I_0^2 \sin^2 \omega t\, dt$$

Using $\sin^2 \omega t = \frac{1 - \cos 2\omega t}{2}$,

$$I^2 = \frac{I_0^2}{T}\int_0^T \frac{1 - \cos 2\omega t}{2}\, dt$$

$$I^2 = \frac{I_0^2}{2T}\left[t - \frac{\sin 2\omega t}{2\omega} \right]_0^T$$

Over one complete period the $\cos 2\omega t$ term integrates to zero, so

$$I^2 = \frac{I_0^2}{2T}(T) = \frac{I_0^2}{2}$$

By definition the rms value is the square root of this mean:

$$I_{rms} = \sqrt{\overline{I^2}} = \sqrt{\frac{I_0^2}{2}}$$

$$I_{rms} = \frac{I_0}{\sqrt{2}} = 0.707\, I_0$$

COMMON MISTAKE

Average sin over a full cycle is zero, but average of $\sin^2$ *is* 1⁄2. *Many students wrongly use the zero result here and lose the derivation marks.*

Final answer. $\mathbf{I_{rms} = \frac{I_0}{\sqrt{2}}}$, i.e. the rms current is **0.707** times the peak current.

← *Back to Q83*

Q84

(a) ac voltage of frequency ω is applied across a series LCR circuit. Draw the phasor diagram and obtain the impedance of the circuit.
(b) Discuss 'resonance' in a series LCR circuit and write the expression for resonant frequency.

[2025 • Set 55-2-3]

SOLUTION

(a) In a series LCR circuit the same current flows through all elements, so we take the current as the reference phasor.

Phasor diagram: Take $i = i_0 \omega t$ along the positive x-axis as the reference.

The resistor voltage $\vec{V}_R = i_0 R$ is in phase with the current, drawn along the positive x-axis.

The inductor voltage $\vec{V}_L = i_0 X_L$ leads the current by 90°, drawn along the positive y-axis.

The capacitor voltage $\vec{V}_C = i_0 X_C$ lags the current by 90°, drawn along the negative y-axis.

The net reactive voltage is $(V_L - V_C)$ along the y-axis, and the applied voltage $\vec{V}$ is the resultant of $\vec{V}_R$ and $(\vec{V}_L - \vec{V}_C)$.

$$V = \sqrt{V_R^2 + (V_L - V_C)^2}$$

$$V = i_0 \sqrt{R^2 + (X_L - X_C)^2} = i_0 Z$$

Hence the impedance is

$$Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}$$

(b) **Resonance** in a series LCR circuit is the condition at which the inductive reactance equals the capacitive reactance, so that the circuit behaves like a pure resistance and the current becomes maximum.

At resonance $X_L = X_C$, so

$$\omega_0 L = \frac{1}{\omega_0 C}$$

$$\omega_0^2 = \frac{1}{LC}$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

At resonance the impedance is minimum ($Z = R$), the current is maximum ($i_0 = V_0/R$), and the voltage and current are in phase (power factor = 1).

The **resonant frequency** is

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

Final answer. $Z = \sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}$; at resonance $X_L = X_C$ and $f_0 = \frac{1}{2\pi\sqrt{LC}}$.

← *Back to Q84*

Q85

An ac source of voltage $V = V_0\omega t$ is connected to a circuit element X. It is observed that the current flowing through X varies as $I = I_0\sin(\omega t - \frac{\pi}{2})$.

***(a)** Identify the element X and write the expression for its reactance.*

***(b)** Plot a graph to show the variation of reactance of the element with the frequency of the applied voltage.*

***(c)** Draw plots showing the variation of voltage and current with time over one cycle of applied ac.*

[2025 • Set 55-7-3]

SOLUTION

Here the applied voltage is $V = V_0\omega t$ and the current is $I = I_0\sin(\omega t - \frac{\pi}{2})$.

The current reaches its peak a quarter cycle after the voltage, so the **current lags the voltage by $\frac{\pi}{2}$**.

(a) In a pure inductor the current lags the applied voltage by exactly $\frac{\pi}{2}$. Hence the element *X* is an **inductor**.

Its opposition to ac is the inductive reactance

$$X_L = \omega L = 2\pi f L$$

(b) Graph of reactance versus frequency: Since $\mathbf{X_L = 2\pi f L}$, the reactance is directly proportional to f.

The graph of X_L (y-axis) against f (x-axis) is a **straight line passing through the origin** with slope $2\pi L$. As frequency increases, the inductive reactance increases linearly.

(c) Graph of voltage and current versus time over one cycle: On the same time axis, plot $\mathbf{V = V_0 \omega t}$ as a normal sine curve starting from the origin.

Plot $\mathbf{I = I_0 \sin(\omega t - \frac{\pi}{2})}$ as a sine curve of the same shape but **shifted to the right by a quarter cycle** (phase $\frac{\pi}{2}$), so its peaks occur after the voltage peaks. The current curve clearly lags the voltage curve by $\frac{\pi}{2}$.

Final answer. X is an inductor with reactance $\mathbf{X_L = \omega L = 2\pi f L}$ (a straight line through the origin), and the current lags the voltage by $\frac{\pi}{2}$.

← *Back to Q85*

Q86

An inductor, a capacitor and a resistor are connected in series with an ac source $\mathbf{v = v_m \omega t}$. *Derive an expression for the average power dissipated in the circuit. Also obtain the expression for the resonant frequency of the circuit.*

[2024 • Set 55-1-1]

SOLUTION

In a series LCR circuit driven by $\mathbf{v = v_m \omega t}$, the current is

$$i = i_m \sin(\omega t - \phi)$$

where $\mathbf{i_m = \frac{v_m}{Z}}$, $\mathbf{Z = \sqrt{R^2 + (X_L - X_C)^2}}$ and $\mathbf{\phi = \frac{X_L - X_C}{R}}$.

The instantaneous power delivered is

$$P = vi = v_m \omega\, t \cdot i_m \sin(\omega\, t - \phi)$$

$$P = v_m\, i_m\, \omega\, t \sin(\omega\, t - \phi)$$

Using the identity $\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$,

$$P = \frac{v_m\, i_m}{2}[\phi - \cos(2\omega\, t - \phi)]$$

The average power over one complete cycle is found by averaging each term.

The term $\cos(2\omega\, t - \phi)$ oscillates and averages to zero over a full cycle, while ϕ is constant.

$$P_{avg} = \frac{v_m\, i_m}{2}\phi$$

Writing $v_m = \sqrt{2}\, V_{rms}$ and $i_m = \sqrt{2}\, I_{rms}$,

$$P_{avg} = \frac{\sqrt{2}\, V_{rms} \cdot \sqrt{2}\, I_{rms}}{2}\phi$$

$$P_{avg} = V_{rms}\, I_{rms}\phi$$

The factor ϕ is called the **power factor** of the circuit.

Resonant frequency: At resonance the inductive and capacitive reactances are equal, $X_L = X_C$.

$$\omega_0\, L = \frac{1}{\omega_0\, C}$$

$$\omega_0^{\,2} = \frac{1}{LC}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} \Rightarrow f_0 = \frac{1}{2\pi\sqrt{LC}}$$

At resonance $\phi = 0$, $Z = R$ (minimum) and the current is maximum and in phase with the applied voltage.

Final answer. $P_{avg} = V_{rms}\, I_{rms}\phi$ and the resonant frequency is $f_0 = \frac{1}{2\pi\sqrt{LC}}$.

← Back to Q86

Q87

(a) Draw the graphs showing the variation of the following with the frequency of ac source in a circuit:
(i) Resistance
(ii) Capacitive reactance
(iii) Inductive reactance
(b) Can the voltage drop across the inductor or the capacitor in a series LCR circuit be greater than the applied voltage of the ac source? Justify your answer.

[2024 • Set 55-1-3]

SOLUTION

(a) Variation of each quantity with the frequency f of the ac source:

(i) Resistance R does not depend on frequency. The graph of R against f is a **horizontal straight line** parallel to the frequency axis.

(ii) Capacitive reactance $\mathbf{X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}}$. Since $\mathbf{X_C \propto \frac{1}{f}}$, the graph is a **rectangular hyperbola**: $\mathbf{X_C \to \infty}$ as $\mathbf{f \to 0}$ and $\mathbf{X_C \to 0}$ as f becomes very large.

(iii) Inductive reactance $\mathbf{X_L = \omega L = 2\pi f L}$. Since $\mathbf{X_L \propto f}$, the graph is a **straight line through the origin** with slope $2\pi L$, increasing as frequency increases.

(b) Yes, the voltage drop across the inductor or the capacitor in a series LCR circuit can be greater than the applied source voltage.

In a series LCR circuit the voltages $\vec{V}_L$ and $\vec{V}_C$ are 180° out of phase with each other, so they partly cancel in the phasor (vector) sum.

The source voltage is only the resultant $\mathbf{V = \sqrt{V_R{}^2 + (V_L - V_C)^2}}$, not the simple algebraic sum.

Because V_L and V_C subtract, each one individually can be much larger than V.

At resonance $\mathbf{V_L = V_C = Q\,V}$, where the quality factor is

$$Q = \frac{1}{R}\sqrt{\frac{L}{C}}$$

For a sharply resonant circuit $Q > 1$, so V_L and V_C each **exceed the source voltage several times over**, even though their phasor sum returns the modest applied voltage.

Final answer. (a) R is constant, X_C falls as a hyperbola, X_L rises linearly; (b) Yes, since V_L and V_C are opposite in phase, each can exceed the source voltage (by the factor Q at resonance).

← Back to Q87

Q88

An ac voltage $v_i = 140\sin(100\pi t)$ *V is applied to the primary coil having 200 turns, of an ideal transformer and it supplies a power of 5 kW. If the secondary coil has 1000 turns, find:*
(a) the output voltage,
(b) the instantaneous voltage across the secondary coil, and
(c) the current in the secondary coil. (Take $\sqrt{2} = 1.4$*)*

[2024 • Set 55-2-3]

SOLUTION

Given the primary voltage $v_i = 140\sin(100\pi t)$ *V*, so the peak primary voltage is $V_{i,\max} = 140$ *V*.

The rms value of the primary voltage is

$$V_p = \frac{V_{i,\max}}{\sqrt{2}} = \frac{140}{1.4} = 100\ V$$

Given $N_p = 200$, $N_s = 1000$, power $P = 5$ kW $= 5000$ *W*.

The turns ratio is $\frac{N_s}{N_p} = \frac{1000}{200} = 5$.

(a) For an ideal transformer $\frac{V_s}{V_p} = \frac{N_s}{N_p}$, so the rms output voltage is

$$V_s = V_p \times \frac{N_s}{N_p} = 100 \times 5 = 500\ V\ (\text{rms})$$

(b) The peak secondary voltage scales by the same turns ratio, and the frequency is unchanged:

$$V_{s,\max} = V_{i,\max} \times \frac{N_s}{N_p} = 140 \times 5 = 700\ V$$

$$v_s(t) = 700\sin(100\pi\ t)\ V$$

(c) For an ideal transformer the output power equals the supplied power, so the secondary current is

$$I_s = \frac{P}{V_s} = \frac{5000}{500} = 10\ A$$

COMMON MISTAKE

Use rms voltage for output voltage and current, but the instantaneous expression in part (b) uses the peak value 700 V, not 500 V. Mixing these up is a common slip.

Final answer. (a) $V_s = 500\ V$ **(rms)**; (b) $v_s(t) = 700\sin(100\pi\ t)\ V$; (c) $I_s = 10\ A$.

← *Back to Q88*

Q89

A sinusoidal voltage is applied to an electric circuit containing a circuit element 'X' in which the current leads the voltage by $\frac{\pi}{2}$.

(a) Identify the circuit element in the circuit.

(b) Write the formula for its reactance.

(c) Show graphically the variation of this reactance with frequency of ac voltage.

(d) Explain the behaviour of this element when it is used in (i) an ac circuit, and (ii) a dc circuit.

[2024 • Set 55-3-1]

SOLUTION

Given the current leads the applied voltage by $\frac{\pi}{2}$.

(a) In a pure capacitor the current leads the applied voltage by exactly $\frac{\pi}{2}$. Hence the circuit element X is a **capacitor**.

(b) The opposition offered by a capacitor to ac is its capacitive reactance

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

(c) Graph of reactance with frequency: Since $\mathbf{X_C = \frac{1}{2\pi f C}}$, the reactance is inversely proportional to f.

The graph of X_C (y-axis) against f (x-axis) is a **rectangular hyperbola**: $\mathbf{X_C \to \infty}$ as $\boldsymbol{f \to 0}$, and $\mathbf{X_C \to 0}$ as f becomes very large.

(d) Behaviour of the capacitor in different circuits:

(i) In an ac circuit, the capacitor is alternately charged and discharged in each half cycle, so a continuous alternating current appears to flow. Its reactance $\mathbf{X_C = \frac{1}{2\pi f C}}$ is finite, and it decreases as the frequency rises, so the capacitor allows more current at higher frequencies.

(ii) In a dc circuit, the frequency is $\boldsymbol{f = 0}$, so $\mathbf{X_C = \frac{1}{2\pi(0)C} = \infty}$. Once the capacitor charges fully, no steady current flows; the capacitor behaves like an **open circuit** and blocks dc.

Final answer. X is a capacitor with reactance $\mathbf{X_C = \frac{1}{2\pi f C}}$ (a hyperbola with f); it passes ac (more at high frequency) but blocks steady dc.

← *Back to Q89*

Q90

Distinguish between reactance and impedance of an ac circuit. Show that an ideal inductor in an ac circuit does not dissipate any power.

[2024 • Set 55-3-2]

SOLUTION

Reactance and impedance both measure opposition to ac, but they are not the same thing.

Reactance (X) is the opposition offered only by a reactive element, an inductor or a capacitor, and is given by $\mathbf{X_L = \omega L}$ or $\mathbf{X_C = \frac{1}{\omega C}}$. It produces a $\frac{\pi}{2}$ phase difference between voltage and current and dissipates no power.

Impedance (Z) is the total opposition offered by the whole circuit (resistance and reactance together) and is given by $\mathbf{Z = \sqrt{R^2 + (X_L - X_C)^2}}$. The phase angle it produces lies between 0 and $\frac{\pi}{2}$, depending on R and X.

In short, reactance is only the reactive part, whereas impedance is the combined effect of resistance and reactance. Both are measured in ohms.

Ideal inductor dissipates no power:

For an ideal inductor (zero resistance) carrying $\mathbf{I = I_0 \omega t}$, the voltage across it is

$$V = \mathrm{L}\frac{\mathrm{dI}}{\mathrm{dt}} = \mathrm{L}\frac{d}{\mathrm{dt}}(\mathrm{I_0}\omega\, t)$$

$$V = \omega\, L\, \mathrm{I_0}\omega\, t = \mathrm{V_0}\sin(\omega\, t + \frac{\pi}{2})$$

So the voltage leads the current by $\frac{\pi}{2}$, i.e. the phase angle is $\mathbf{\phi = \frac{\pi}{2}}$.

The average power dissipated over a cycle is

$$\mathrm{P_{avg}} = \mathrm{V_{rms}}\, \mathrm{I_{rms}}\phi$$

$$\mathrm{P_{avg}} = \mathrm{V_{rms}}\, \mathrm{I_{rms}}\cos\frac{\pi}{2}$$

Since $\mathbf{\cos\frac{\pi}{2} = 0}$,

$$\mathrm{P_{avg}} = 0$$

Thus an **ideal inductor dissipates no power**; it only stores energy in its magnetic field during one quarter cycle and returns it to the source in the next.

Final answer. Reactance (X) is the opposition of L or C alone, while impedance ($\mathbf{Z = \sqrt{R^2+(X_L-X_C)^2}}$) is the total opposition; for an ideal inductor $\mathbf{\phi = \frac{\pi}{2}}$ so $\mathbf{P_{avg}}$ =

$$V_{rms} I_{rms} \cos\frac{\pi}{2} = 0.$$

← Back to Q90

Q91

How does the resistance differ from impedance? With the help of a suitable phasor diagram, obtain an expression for impedance of a series LCR circuit, connected to a source $v = v_m \omega t$.

[2023 • Set 55-1-1]

SOLUTION

Resistance (R) is the opposition offered by a pure resistor to the flow of current. In a resistor the current and the applied voltage stay in the same phase, and electrical energy is permanently dissipated as heat.

Impedance (Z) is the total (effective) opposition offered by an AC circuit that contains a resistor, an inductor and a capacitor together. It includes resistance as well as the reactances X_L and X_C. Reactance stores energy and returns it to the source, so impedance generally produces a phase difference ϕ between the voltage and the current.

For the series LCR circuit, the same current i flows through R, L and C. We take this current as the reference phasor along the horizontal axis.

In the phasor diagram, $\vec{V}_R$ (magnitude IR) is drawn along the current direction. $\vec{V}_L$ (magnitude IX_L) leads the current by $\frac{\pi}{2}$, so it points vertically up. $\vec{V}_C$ (magnitude IX_C) lags the current by $\frac{\pi}{2}$, so it points vertically down.

Since $\vec{V}_L$ and $\vec{V}_C$ are opposite, their resultant has magnitude $(V_L - V_C)$ acting along the vertical direction.

The source voltage $\vec{V}$ is the vector sum of $\vec{V}_R$ and $(\vec{V}_L - \vec{V}_C)$, which are mutually perpendicular. By the parallelogram (Pythagoras) law,

$$V = \sqrt{V_R^2 + (V_L - V_C)^2}$$

$$V = \sqrt{(IR)^2 + (IX_L - IX_C)^2}$$

$$V = I\sqrt{R^2 + (X_L - X_C)^2}$$

Therefore the impedance is

$$Z = \frac{V}{I} = \sqrt{R^2 + (X_L - X_C)^2}$$

and the phase angle between the source voltage and the current is given by

$$\phi = \frac{X_L - X_C}{R}$$

Thus **resistance is just one part of impedance**; impedance is the combined opposition $Z = \sqrt{R^2 + (X_L - X_C)^2}$ that also accounts for the reactive elements.

CONCEPT

Resistance is in ohms and is constant; reactance is also in ohms but depends on frequency. State both clearly to avoid losing the conceptual mark.

Final answer. Impedance of a series LCR circuit is $Z = \sqrt{R^2 + (X_L - X_C)^2}$, with $\phi = \frac{X_L - X_C}{R}$.

← *Back to Q91*

Q92

Find the condition for resonance in a series LCR circuit connected to a source $v = v_m \omega t$, where ω can be varied. Give the factors on which the resonant frequency of a series LCR circuit depends. Plot a graph showing the variation of electric current with frequency in a series LCR circuit.

[2023 • Set 55-1-1]

SOLUTION

In a series LCR circuit driven by $v = v_m \omega t$, the impedance is

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

The current amplitude is $\mathbf{i_m = \frac{v_m}{Z}}$, so the current is maximum when Z is minimum.

Z is minimum when the reactive part vanishes, i.e. when

$$X_L = X_C$$

$$\omega L = \frac{1}{\omega C}$$

$$\omega^2 = \frac{1}{LC}$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

This is the condition for resonance. The corresponding resonant frequency is

$$f_0 = \frac{\omega_0}{2\pi} = \frac{1}{2\pi\sqrt{LC}}$$

From this expression, the resonant frequency depends only on the inductance L and the capacitance C of the circuit. It does not depend on the resistance R or on the applied voltage.

Graph (current versus frequency): On the horizontal axis we plot the source frequency f (or ω) and on the vertical axis the rms current I. The curve starts low at small frequencies, rises to a sharp peak of height $\mathbf{I_{max} = \frac{V}{R}}$ exactly at $\boldsymbol{f = f_0}$, and then falls off again on the higher-frequency side, giving a single bell-shaped resonance curve symmetric about f_0.

Two curves can be shown for two resistances: a **smaller R gives a taller and sharper peak**, while a larger R gives a shorter, broader peak. The peak always occurs at the same f_0.

Hence the resonance condition is $\mathbf{X_L = X_C}$, giving $\mathbf{f_0 = \frac{1}{2\pi\sqrt{LC}}}$, which depends only on L and C.

Final answer. Resonance occurs when $\mathbf{X_L = X_C}$, giving $\mathbf{f_0 = \frac{1}{2\pi\sqrt{LC}}}$, which depends only on $\boldsymbol{L}$ and $\boldsymbol{C}$.

← Back to Q92

Q93

The primary and the secondary coils of an ideal step-down transformer consist of 650 *and* 25 *turns respectively. When the primary coil of this transformer is connected to* 240 *V mains, the current in the primary coil is* 1.5 *A. Calculate:*
(a) the voltage across the secondary coil
(b) the current in the secondary coil
(c) the average power delivered to the output circuit

[2023 • Set 55-1-2]

SOLUTION

Given a step-down transformer: $\mathbf{N_p = 650}$ turns, $\mathbf{N_s = 25}$ turns, $\mathbf{V_p = 240\ V}$ and $\mathbf{I_p = 1.5\ A}$.

(a) For an ideal transformer the turns ratio relation is

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

$$V_s = V_p \times \frac{N_s}{N_p}$$

$$V_s = 240 \times \frac{25}{650}$$

$$V_s = \frac{6000}{650}$$

$$V_s \approx 9.23\ V$$

(b) For an ideal transformer there is no power loss, so input power equals output power:

$$V_p I_p = V_s I_s$$

$$I_s = I_p \times \frac{V_p}{V_s} = I_p \times \frac{N_p}{N_s}$$

$$I_s = 1.5 \times \frac{650}{25}$$

$$I_s = 1.5 \times 26$$

$$I_s = 39\ A$$

(c) The average power delivered to the output (secondary) circuit is

$$P = V_s\, I_s$$

$$P = 9.23 \times 39$$

$$P \approx 360\ W$$

This equals the input power $\mathbf{V_p\, I_p = 240 \times 1.5 = 360\ W}$, confirming the ideal (lossless) assumption.

Hence $\mathbf{V_s \approx 9.23\ V}$, $\mathbf{I_s = 39\ A}$ **and** $\mathbf{P \approx 360\ W}$.

TIP

In a step-down transformer voltage falls but current rises in the same ratio, so power is unchanged. A quick check: input power V_pI_p must equal output power.

Final answer. Secondary voltage ≈ **9.23** V, secondary current = **39** A, and output power ≈ **360** W.

← *Back to Q93*

Q94

An ac voltage $\boldsymbol{v = v_m \omega t}$ *(*$\mathbf{v_m = 310}$ *V and* $\boldsymbol{f}$ **= 50** *Hz) is connected to a pure capacitor of capacitance* 15 μF. *Calculate (i) the reactance of the capacitor, and (ii) the amplitude of the current. Write the expression of current through the capacitor as a function of time.*

[2023 • Set 55-1-3]

SOLUTION

Given a pure capacitor with $\mathbf{v_m = 310\ V}$, $\boldsymbol{f}$ **= 50 Hz** and $\boldsymbol{C}$ **= 15 μF =** $\mathbf{1.5 \times 10^{-5}\ F}$.

The angular frequency is

$$\omega = 2\pi f = 2\pi \times 50 = 100\pi \text{ rad/s}$$

(i) The capacitive reactance is

$$X_C = \frac{1}{\omega C}$$

$$X_C = \frac{1}{(100\pi)(1.5\times 10^{-5})}$$

$$X_C = \frac{1}{1.5\pi \times 10^{-3}}$$

$$X_C \approx 212.2\ \Omega$$

(ii) The amplitude of the current is

$$i_m = \frac{v_m}{X_C}$$

$$i_m = \frac{310}{212.2}$$

$$i_m \approx 1.46\ A$$

In a pure capacitor the current leads the applied voltage by a phase angle of $\frac{\pi}{2}$. Since $v = v_m \omega t$, the instantaneous current is

$$i(t) = i_m \sin(\omega t + \frac{\pi}{2})$$

$$i(t) = 1.46 \sin(100\pi t + \frac{\pi}{2})\ A$$

which may also be written as $i(t) = 1.46 \cos(100\pi t)\ A$.

Hence $X_C \approx 212\ \Omega$, $i_m \approx 1.46\ A$, and the current expression is $i(t) = 1.46 \sin(100\pi t + \frac{\pi}{2})\ A$.

Final answer. $X_C \approx 212\ \Omega$, $i_m \approx 1.46\ A$, and $i(t) = 1.46 \sin(100\pi t + \frac{\pi}{2})\ A$.

← *Back to Q94*

Q95

A series CR circuit with $R=200\ \Omega$ *and* $C=(50/\pi)\ \mu F$ *is connected across an ac source of peak voltage* $\varepsilon_0 = 100\ V$ *and frequency* $\nu = 50\ Hz$. *Calculate*

(a) impedance of the circuit (Z),

(b) phase angle (ϕ), and

(c) voltage across the resistor.

[2023 • Set 55-2-1]

SOLUTION

Given a series CR circuit: $R = 200\ \Omega$, $C = \frac{50}{\pi}\ \mu F = \frac{50}{\pi} \times 10^{-6}\ F$, peak voltage $\varepsilon_0 = 100\ V$ and $\nu = 50\ Hz$.

The angular frequency is

$$\omega = 2\pi\nu = 2\pi \times 50 = 100\pi\ \text{rad/s}$$

The capacitive reactance is

$$X_C = \frac{1}{\omega C}$$

$$X_C = \frac{1}{(100\pi)(\frac{50}{\pi} \times 10^{-6})}$$

$$X_C = \frac{1}{5000 \times 10^{-6}}$$

$$X_C = 200\ \Omega$$

(a) The impedance of the series CR circuit is

$$Z = \sqrt{R^2 + X_C{}^2}$$

$$Z = \sqrt{(200)^2 + (200)^2}$$

$$Z = 200\sqrt{2} \approx 283\ \Omega$$

(b) The phase angle is given by

$$\phi = \frac{X_C}{R} = \frac{200}{200} = 1$$

$$\phi = 45^\circ = \frac{\pi}{4}$$

In a CR circuit the voltage lags the current by this angle.

(c) The rms source voltage is $\varepsilon_{rms} = \frac{\varepsilon_0}{\sqrt{2}} = \frac{100}{\sqrt{2}}\ V$. The rms current is

$$I_{rms} = \frac{\varepsilon_{rms}}{Z} = \frac{100/\sqrt{2}}{200\sqrt{2}} = \frac{100}{400} = 0.25\ A$$

The rms voltage across the resistor is

$$V_R = I_{rms}\, R = 0.25 \times 200 = 50\ V$$

Hence $Z = 200\sqrt{2}\ \Omega \approx 283\ \Omega$, $\phi = \frac{\pi}{4}$, **and** $V_R = 50\ V$ **(rms).**

Final answer. $Z = 200\sqrt{2}\ \Omega \approx 283\ \Omega$, $\phi = \frac{\pi}{4}$ (45 °), and $V_R = 50\ V$ (rms).

← *Back to Q95*

Q96

A series RL circuit with R=10 Ω *and* L=$\left(\frac{100}{\pi}\right)$ mH *is connected to an ac source of voltage* $V = 141 \sin(100\pi t)$, *where V is in volts and t is in seconds. Calculate*
(a) impedance of the circuit
(b) phase angle, and
(c) voltage drop across the inductor.

[2023 • Set 55-2-2]

SOLUTION

Given a series RL circuit: $R = 10\ \Omega$, $L = \frac{100}{\pi}$ mH $= \frac{100}{\pi} \times 10^{-3}\ H = \frac{10^{-1}}{\pi}\ H$, with source $V = 141 \sin(100\pi t)$.

Comparing with $V = V_m \omega t$, the peak voltage is $V_m = 141\ V$ and the angular frequency is $\omega = 100\pi$ rad/s. The rms voltage is

$$V_{rms} = \frac{V_m}{\sqrt{2}} = \frac{141}{\sqrt{2}} \approx 100\ V$$

The inductive reactance is

$$X_L = \omega L = (100\pi)\left(\frac{10^{-1}}{\pi}\right) = 10\ \Omega$$

(a) The impedance of the series RL circuit is

$$Z = \sqrt{R^2 + {X_L}^2}$$

$$Z = \sqrt{(10)^2 + (10)^2}$$

$$Z = 10\sqrt{2} \approx 14.14\ \Omega$$

(b) The phase angle is given by

$$\phi = \frac{X_L}{R} = \frac{10}{10} = 1$$

$$\phi = 45^\circ = \frac{\pi}{4}$$

In an RL circuit the voltage leads the current by this angle.

(c) The rms current in the circuit is

$$I_{rms} = \frac{V_{rms}}{Z} = \frac{100}{10\sqrt{2}} = 5\sqrt{2}\ A$$

The rms voltage drop across the inductor is

$$V_L = I_{rms} X_L = 5\sqrt{2} \times 10 = 50\sqrt{2} \approx 70.7\ V$$

Hence $Z = 10\sqrt{2}\ \Omega \approx 14.14\ \Omega$, $\phi = \frac{\pi}{4}$, **and** $V_L \approx 70.7\ V$ **(rms).**

Final answer. $Z = 10\sqrt{2}\ \Omega \approx 14.14\ \Omega$, $\phi = \frac{\pi}{4}$ (45 °), and $V_L = 50\sqrt{2} \approx 70.7\ V$.

← *Back to Q96*

Q97

A resistor of 50 Ω*, a capacitor of* $\left(\frac{25}{\pi}\right)$ μF *and an inductor of* $\left(\frac{4}{\pi}\right)$ *H are connected in series across an ac source whose voltage (in volt) is given by* $V = 70 \sin(100\pi t)$.

Calculate:

(a) the net reactance of the circuit,

(b) the impedance of the circuit,

(c) the effective value of current in the circuit.

[2023 • Set 55-2-3]

SOLUTION

Given a series LCR circuit: $R = 50\ \Omega$, $C = \frac{25}{\pi}\ \mu F = \frac{25}{\pi} \times 10^{-6}\ F$, $L = \frac{4}{\pi}\ H$, with source $V = 70\sin(100\pi\, t)$.

The peak voltage is $V_m = 70\ V$ and the angular frequency is $\omega = 100\pi\ rad/s$. The rms voltage is

$$V_{rms} = \frac{V_m}{\sqrt{2}} = \frac{70}{\sqrt{2}} = 35\sqrt{2} \approx 49.5\ V$$

The inductive reactance is

$$X_L = \omega L = (100\pi)(\frac{4}{\pi}) = 400\ \Omega$$

The capacitive reactance is

$$X_C = \frac{1}{\omega C} = \frac{1}{(100\pi)(\frac{25}{\pi} \times 10^{-6})}$$

$$X_C = \frac{1}{2500 \times 10^{-6}} = 400\ \Omega$$

(a) The net reactance is

$$X = X_L - X_C = 400 - 400 = 0$$

Since the net reactance is zero, the circuit is at resonance.

(b) The impedance is

$$Z = \sqrt{R^2 + X^2} = \sqrt{R^2 + 0} = R = 50\ \Omega$$

(c) The effective (rms) value of the current is

$$I_{rms} = \frac{V_{rms}}{Z} = \frac{35\sqrt{2}}{50} = 0.7\sqrt{2} \approx 0.99\ A \approx 1.0\ A$$

Hence the **net reactance is 0, the impedance equals $R = 50\ \Omega$, and the effective current is about 1.0 A**, confirming that the circuit is in resonance.

TIP

When $X_L = X_C$, do not waste time computing $\sqrt{R^2 + X^2}$ step by step; recognise resonance and write $Z = R$ directly.

Final answer. Net reactance $X = 0$, impedance $Z = 50\ \Omega$, and effective current $I_{rms} \approx 1.0\ A$ (resonance).

← Back to Q97

Q98

(a) An ac source $V = V_m \sin \omega t$ is connected across an ideal capacitor. Derive the expression for the (i) current flowing in the circuit, and (ii) reactance of the capacitor. Plot a graph of current i versus ω t.

OR

(b) A series combination of an inductor L, a capacitor C and a resistor R is connected across an ac source of voltage in a circuit. Obtain an expression for the average power consumed by the circuit. Find power factor for (i) purely inductive circuit, and (ii) purely resistive circuit.

[2023 • Set 55-3-1]

SOLUTION

(a) An ideal capacitor of capacitance C is connected across the source $V = V_m \omega t$.

At any instant the charge on the capacitor is

$$q = CV = CV_m \omega t$$

(i) The instantaneous current is the rate of change of charge:

$$i = \frac{dq}{dt} = \frac{d}{dt}(CV_m \omega t)$$

$$i = CV_m \omega\omega t$$

$$i = \omega C V_m \sin(\omega t + \frac{\pi}{2})$$

Writing the current amplitude as $\mathbf{i_m = \omega C V_m = \frac{V_m}{1/\omega C}}$,

$$i = \frac{V_m}{X_C} \sin(\omega t + \frac{\pi}{2})$$

(ii) Comparing $\mathbf{i_m = \frac{V_m}{X_C}}$ with $\mathbf{i_m = \omega C V_m}$, the capacitive reactance is

$$X_C = \frac{1}{\omega C}$$

The factor $+\frac{\pi}{2}$ shows that the current leads the applied voltage by a quarter cycle.

Graph of i versus ωt: both V and i are sinusoidal. The voltage curve $\boldsymbol{V = V_m \omega t}$ starts from zero, while the current curve $\boldsymbol{i = i_m \sin(\omega t + \frac{\pi}{2})}$ starts at its maximum value. The current peak occurs a quarter cycle ($\frac{\pi}{2}$) before the voltage peak, so the current curve is shifted to the left of the voltage curve.

Thus for a pure capacitor $\boldsymbol{i = \frac{V_m}{X_C} \sin(\omega t + \frac{\pi}{2})}$ **with** $\mathbf{X_C = \frac{1}{\omega C}}$, current leading voltage by $\frac{\pi}{2}$.

OR

(b) In a series LCR circuit let $\boldsymbol{v = v_m \omega t}$ and the resulting current be $\boldsymbol{i = i_m \sin(\omega t - \phi)}$, where ϕ is the phase difference.

The instantaneous power is $\boldsymbol{p = vi}$. The average power over one full cycle works out to

$$P_{avg} = V_{rms} I_{rms} \phi$$

where $\mathbf{V_{rms} = \frac{v_m}{\sqrt{2}}}$ and $\mathbf{I_{rms} = \frac{i_m}{\sqrt{2}}}$. The quantity $\boldsymbol{\phi = \frac{R}{Z}}$ is called the power factor of the circuit.

(i) For a purely inductive circuit the current lags the voltage by $\boldsymbol{\phi = \frac{\pi}{2}}$, so

$$\phi = \cos\frac{\pi}{2} = 0$$

Hence the power factor is 0 and the average power consumed is zero (a wattless current flows).

(ii) For a purely resistive circuit the current is in phase with the voltage, so $\phi = 0$ and

$$\phi = \cos 0 = 1$$

Hence the power factor is 1 and the full power $P_{avg} = V_{rms}I_{rms}$ is dissipated.

Therefore $P_{avg} = V_{rms}I_{rms}\phi$, with power factor 0 for a purely inductive circuit and 1 for a purely resistive circuit.

Final answer. (a) $i = \frac{V_m}{X_C}\sin(\omega t + \frac{\pi}{2})$, $X_C = \frac{1}{\omega C}$; OR (b) $P_{avg} = V_{rms}I_{rms}\phi$, power factor = **0** (purely inductive), = **1** (purely resistive).

← *Back to Q98*

Q99

(a) A resistor of 30 Ω and a capacitor of $\frac{250}{\pi}$ *μ F are connected in series to a 200 V, 50 Hz ac source. Calculate (i) the current in the circuit, and (ii) voltage drops across the resistor and the capacitor. (iii) Is the algebraic sum of these voltages more than the source voltage? If yes, solve the paradox.*

OR

(b) A series LCR circuit with $R = 20\ \Omega$, $L = 2$ *H and* $C = 50\ \mu F$ *is connected to a 200 volts ac source of variable frequency. What is (i) the amplitude of the current, and (ii) the average power transferred to the circuit in one complete cycle, at resonance? (iii) Calculate the potential drop across the capacitor.*

[2023 • Set 55-4-1]

SOLUTION

(a) Series RC circuit: $R = 30\ \Omega$, $C = \frac{250}{\pi}\ \mu F = \frac{250}{\pi} \times 10^{-6}\ F$, $V = 200\ V$, $f = 50$ Hz.

The angular frequency is $\omega = 2\pi f = 100\pi$ rad/s.

The capacitive reactance is

$$X_C = \frac{1}{\omega C} = \frac{1}{(100\pi)(\frac{250}{\pi} \times 10^{-6})}$$

$$X_C = \frac{1}{25000 \times 10^{-6}} = \frac{1}{2.5 \times 10^{-2}} = 40\ \Omega$$

The impedance is

$$Z = \sqrt{R^2 + X_C{}^2} = \sqrt{(30)^2 + (40)^2} = \sqrt{900 + 1600} = \sqrt{2500} = 50\ \Omega$$

(i) The current in the circuit is

$$I = \frac{V}{Z} = \frac{200}{50} = 4\ A$$

(ii) The voltage drops are

$$V_R = I\,R = 4 \times 30 = 120\ V$$

$$V_C = I\,X_C = 4 \times 40 = 160\ V$$

(iii) The algebraic sum is $\mathbf{V_R + V_C = 120 + 160 = 280\ V}$, which is greater than the source voltage of 200 V.

Resolution of the paradox: V_R and V_C are not in phase; the voltage across the resistor and the voltage across the capacitor differ in phase by $\frac{\pi}{2}$, so they cannot be added arithmetically. They must be added as phasors:

$$V = \sqrt{V_R{}^2 + V_C{}^2} = \sqrt{(120)^2 + (160)^2} = \sqrt{14400 + 25600} = \sqrt{40000} = 200\ V$$

This phasor sum equals the source voltage, so there is no real contradiction.

OR

(b) Series LCR at resonance: $\boldsymbol{R = 20\ \Omega}$, $\boldsymbol{L = 2\ H}$, $\boldsymbol{C = 50\ \mu F = 50 \times 10^{-6}\ F}$, $\mathbf{V_{rms} = 200\ V}$.

The resonant angular frequency is

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{(2)(50 \times 10^{-6})}} = \frac{1}{\sqrt{10^{-4}}} = 100\ \text{rad/}s$$

At resonance $\mathbf{X_L = X_C}$, so $\boldsymbol{Z = R = 20\ \Omega}$.

(i) The amplitude (peak value) of the current is

$$i_m = \frac{V_m}{R} = \frac{\sqrt{2}\, V_{rms}}{R} = \frac{\sqrt{2} \times 200}{20} = 10\sqrt{2} \approx 14.14\ A$$

(ii) At resonance the power factor is 1, so the average power transferred is

$$P = \frac{V_{rms}^{\,2}}{R} = \frac{(200)^2}{20} = \frac{40000}{20} = 2000\ W$$

(iii) The rms current at resonance is $\mathbf{I_{rms} = \frac{V_{rms}}{R} = \frac{200}{20} = 10\ A}$. The capacitive reactance at ω_0 is

$$X_C = \frac{1}{\omega_0 C} = \frac{1}{(100)(50 \times 10^{-6})} = \frac{1}{5 \times 10^{-3}} = 200\ \Omega$$

The potential drop across the capacitor is

$$V_C = I_{rms}\, X_C = 10 \times 200 = 2000\ V$$

Hence at resonance $\mathbf{i_m \approx 14.14\ A}$, $\mathbf{P = 2000\ W}$ **and** $\mathbf{V_C = 2000\ V}$ (note that V_C far exceeds the source voltage, which is characteristic of resonance).

> **COMMON MISTAKE**
>
> *In AC circuits never add voltages across different elements arithmetically; phase differences mean you must combine them as perpendicular phasors.*

Final answer. (a) $\mathbf{I = 4\ A}$, $\mathbf{V_R = 120\ V}$, $\mathbf{V_C = 160\ V}$; the phasor sum $\mathbf{\sqrt{V_R^{\,2} + V_C^{\,2}} = 200\ V}$ equals the source. OR (b) $\mathbf{i_m \approx 14.14\ A}$, $\mathbf{P = 2000\ W}$, $\mathbf{V_C = 2000\ V}$.

← *Back to Q99*

Q100

An alternating voltage of 220 V is applied across a device X. A current of 0.22 A flows in the circuit and it lags behind the applied voltage in phase by π/2 radian. When the same voltage is applied across another device Y, the current in the circuit remains the same and it is in phase with the applied voltage.
(i) Name the devices X and Y and,

(ii) Calculate the current flowing in the circuit when the same voltage is applied across the series combination of X and Y.

[2023 • Set 55-5-1]

SOLUTION

Given: an alternating voltage of 220 V produces a current of 0.22 A in each device.

(i) Identifying the devices: In device X the current lags behind the applied voltage by a phase angle of $\frac{\pi}{2}$. This pure $\frac{\pi}{2}$ lag is the characteristic of a pure inductor, so **X is an inductor (L)**.

In device Y the current is in phase with the applied voltage. Zero phase difference is the characteristic of a pure resistor, so **Y is a resistor (R)**.

(ii) Finding the elements from the given data.

For device X (inductor), the inductive reactance is

$$X_L = \frac{V}{I} = \frac{220}{0.22} = 1000\ \Omega$$

For device Y (resistor), the resistance is

$$R = \frac{V}{I} = \frac{220}{0.22} = 1000\ \Omega$$

When X and Y are joined in series, the circuit is a series RL circuit of impedance

$$Z = \sqrt{R^2 + {X_L}^2}$$

$$Z = \sqrt{(1000)^2 + (1000)^2}$$

$$Z = \sqrt{2\times 10^6}$$

$$Z = 1000\sqrt{2}\ \Omega$$

The current flowing through the series combination is

$$I = \frac{V}{Z} = \frac{220}{1000\sqrt{2}}$$

$$I = \frac{0.22}{\sqrt{2}}$$

$$I \approx 0.156\ A$$

Hence X is an inductor and Y is a resistor, and the current in their **series combination is about 0.156 A.**

Final answer. X is an inductor and Y is a resistor; the current in the series combination is $\frac{0.22}{\sqrt{2}} \approx 0.156\ A$.

← *Back to Q100*

Q101

A current of 1A flows through a coil when it is connected across a DC battery of 100 V. If DC battery is replaced by an AC source of 100 V and angular frequency 100 rad s^{-1}, the current reduces to 0.5 A. Find
(i) impedance of the circuit.
(ii) self-inductance of coil.
(iii) phase difference between the voltage and the current.

[2023 • Set 55-5-2]

SOLUTION

When the coil is connected to a DC battery, the inductor offers no reactance (steady current, $\frac{dI}{dt}=0$), so only the resistance R limits the current.

$$R=\frac{V_{DC}}{I_{DC}}=\frac{100}{1}=100\ \Omega$$

(i) When the same coil is connected to the AC source, the opposition to current is the impedance Z. By Ohm's law for AC, $Z=\frac{V_{rms}}{I_{rms}}$.

$$Z=\frac{100}{0.5}=200\ \Omega$$

So the **impedance of the circuit is 200 Ω.**

(ii) For a series LR circuit, $Z=\sqrt{R^2+(\omega L)^2}$. Squaring,

$$Z^2=R^2+(\omega L)^2$$

$$(\omega L)^2 = Z^2 - R^2 = (200)^2 - (100)^2$$

$$(\omega L)^2 = 40000 - 10000 = 30000$$

$$\omega L = \sqrt{30000} = 100\sqrt{3}\ \Omega$$

$$L = \frac{100\sqrt{3}}{\omega} = \frac{100\sqrt{3}}{100} = \sqrt{3}\ H$$

So the **self-inductance $L = \sqrt{3}\ H \approx 1.73\ H$**.

(iii) The phase difference ϕ between voltage and current in an LR circuit is given by

$$\phi = \frac{\omega L}{R} = \frac{100\sqrt{3}}{100} = \sqrt{3}$$

$$\phi = \tan^{-1}(\sqrt{3}) = 60^\circ = \frac{\pi}{3}\ \text{rad}$$

Since the circuit is inductive, the **voltage leads the current by a phase angle $\phi = \frac{\pi}{3}$ (i.e. 60°)**.

TIP

In a pure DC circuit an inductor behaves like a plain wire (zero reactance), so always use the DC reading to get R first, then the AC reading for Z.

Final answer. Z = 200 Ω, L = √3 H ≈ 1.73 H, and the voltage leads the current by $\phi = \pi/3$ (60 °).

← *Back to Q101*

Q102

The primary coil having N_p turns of an ideal transformer is supplied with an alternating voltage V_p. Obtain an expression for the voltage V_s induced in its secondary coil having N_s turns. Mention two main sources of power loss in real transformers.

[2023 • Set 55-5-2]

SOLUTION

A transformer works on the principle of mutual induction: an alternating current in the primary produces a changing magnetic flux in the soft-iron core, and this same changing flux links the secondary, inducing an emf in it.

In an ideal transformer there is no flux leakage, so the magnetic flux Φ through each turn of both coils is the same at every instant, and so is its rate of change $\frac{d\Phi}{dt}$.

The emf induced in the primary (which equals the supplied voltage for an ideal transformer) is

$$V_p = N_p \frac{d\Phi}{dt}$$

The emf induced in the secondary is

$$V_s = N_s \frac{d\Phi}{dt}$$

Dividing the second equation by the first,

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

$$\Rightarrow V_s = V_p \cdot \frac{N_s}{N_p}$$

This is the required expression for the secondary voltage. The ratio $\frac{N_s}{N_p}$ is called the turns ratio; if $\mathbf{N_s > N_p}$ the transformer steps up the voltage and if $\mathbf{N_s p}$ it steps it down.

Two main sources of power loss in a real transformer are:

1. Copper loss: heat dissipated as I^2R in the resistance of the primary and secondary windings.

2. Iron (core) loss: energy lost in the core due to eddy currents (reduced by using a laminated core) and hysteresis (reduced by using a soft magnetic material with a narrow hysteresis loop).

Final answer. Vs = Vp(Ns/Np); the two main losses are copper (I^2R) loss in the windings and iron (eddy-current + hysteresis) loss in the core.

← Back to Q102

Q103

An alternating current $I = 14\sin(100\pi t)$ *A passes through a series combination of a resistor of* 30 Ω *and an inductor of* $(\frac{2}{5\pi})$ *H. Taking* $\sqrt{2} = 1.4$*, calculate the*

(i) rms value of the voltage drops across the resistor and the inductor, and

(ii) power factor of the circuit.

[2023 • Set 55-5-3]

SOLUTION

The given current is $I=14\sin(100\pi t)$ *A*, so comparing with $I=I_m\omega t$:

$$I_m=14\ A,\ \omega=100\pi\ \text{rad s}^{-1}$$

The rms value of the current is

$$I_{rms}=\frac{I_m}{\sqrt{2}}=\frac{14}{1.4}=10\ A$$

The inductive reactance is

$$X_L=\omega L=100\pi\times\frac{2}{5\pi}=\frac{200\pi}{5\pi}=40\ \Omega$$

(i) The rms voltage drop across the resistor (in phase with the current) is

$$V_R=I_{rms}R=10\times 30=300\ V$$

The rms voltage drop across the inductor is

$$V_L=I_{rms}X_L=10\times 40=400\ V$$

So $V_R=300\ V$ **and** $V_L=400\ V$.

(ii) The impedance of the series LR circuit is

$$Z=\sqrt{R^2+X_L^2}=\sqrt{(30)^2+(40)^2}$$

$$Z=\sqrt{900+1600}=\sqrt{2500}=50\ \Omega$$

The power factor is

$$\phi=\frac{R}{Z}=\frac{30}{50}=0.6$$

So the **power factor of the circuit is 0.6**.

Final answer. V_R = 300 V, V_L = 400 V, and the power factor cos ϕ = 0.6.

← *Back to Q103*

Q104

(i) In an LCR series circuit connected to an ac source, the voltage and the current are in the same phase. If the capacitor is filled with a dielectric, will the current lead or lag behind or remain in phase with the voltage? Explain.
(ii) In the circuit, why is the rms value of net voltage not equal to the sum of voltage drops across individual elements?
(iii) Draw a graph showing variation of the impedance of the circuit with the frequency of the applied voltage.

[2021]

SOLUTION

(i) The voltage and current being in the same phase means the net reactance is zero, i.e. $\mathbf{X_L=X_C}$. This is the condition for resonance in a series LCR circuit.

When the capacitor is filled with a dielectric of relative permittivity (dielectric constant) κ, its capacitance increases from C to κC.

The capacitive reactance $\mathbf{X_C=\frac{1}{\omega C}}$ therefore decreases, becoming $\frac{1}{\omega\kappa C}$.

Now $\mathbf{X_L>X_C}$, so the net reactance (X_L-X_C) is positive and the circuit becomes net inductive. Hence the **current lags behind the voltage**.

(ii) The voltage drops V_R, V_L and V_C are not in phase with one another. V_R is in phase with the current, V_L leads the current by $\frac{\pi}{2}$, and V_C lags the current by $\frac{\pi}{2}$, so V_L and V_C are π out of phase with each other.

Because these voltages are out of phase, they must be added as vectors (phasors), not algebraically. The net source voltage is the phasor sum

$$V=\sqrt{V_R^2+(V_L-V_C)^2}$$

which is why V is not equal to the simple arithmetic sum $V_R+V_L+V_C$.

(iii) Graph of impedance Z versus frequency f (or ω): The curve is U-shaped (a valley). At very low frequency Z is large (because X_C is large), it falls to a minimum value $\boldsymbol{Z=R}$ at the resonant frequency $\boldsymbol{\omega_0=\frac{1}{\sqrt{LC}}}$, and then rises again at high frequency (because X_L is large). The horizontal axis is frequency, the vertical axis is impedance Z, the lowest point of the dip touches the value $\boldsymbol{Z=R}$ at f_0.

Final answer. (i) Current lags the voltage; (ii) the drops are out of phase so V = √(V_R² + (V_L − V_C)²); (iii) Z–f graph is a U-shaped curve with minimum Z = R at resonance.

← *Back to Q104*

Q105

(i) An LCR series circuit is connected to an ac source. If the angular resonant frequency of the circuit is ω_0, will the current lead or lag behind or be in phase with the voltage when $\boldsymbol{\omega < \omega_0}$ and why?
(ii) We cannot step up a dc voltage using a transformer. Why?
(iii) On what principle does a metal detector work?

[2021]

SOLUTION

(i) For a series LCR circuit, the resonant angular frequency is $\boldsymbol{\omega_0=\frac{1}{\sqrt{LC}}}$, where $\boldsymbol{X_L=X_C}$.

When $\boldsymbol{\omega<\omega_0}$, the inductive reactance $\boldsymbol{X_L=\omega L}$ is small while the capacitive reactance $\boldsymbol{X_C=\frac{1}{\omega C}}$ is large, so

$$X_C>X_L$$

The net reactance is capacitive, so the circuit behaves like an RC circuit. Hence the **current leads the voltage** in phase.

(ii) A transformer works on mutual induction: an emf is induced in the secondary only when the magnetic flux linking it changes with time, i.e. $\varepsilon_s = -N_s \frac{d\Phi}{dt}$.

A steady DC voltage drives a constant current and hence a constant flux in the core, so

$$\frac{d\Phi}{dt} = 0 \Rightarrow \varepsilon_s = 0$$

No emf is induced in the secondary, so a DC voltage cannot be stepped up (or down) by a transformer.

(iii) A metal detector works on the principle of electromagnetic induction (resonance of an LC oscillator coil). The search coil carries a high-frequency alternating current that sets up an alternating magnetic field. When a metallic object comes near, eddy currents are induced in the metal; these produce their own magnetic field which alters the effective inductance and current of the coil. This change in the resonance condition is detected and converted into an audible or visual signal.

Final answer. (i) Current leads the voltage; (ii) DC gives $d\Phi/dt = 0$ so no secondary emf, hence no transformer action; (iii) a metal detector works on electromagnetic induction.

← Back to Q105

Q106

A resistance R and a capacitor C are connected in series to a source $V = V_0 \omega t$.
Find:
(a) The peak value of the voltage across the (i) resistance and (ii) capacitor.
(b) The phase difference between the applied voltage and current. Which of them is ahead?

[2020 • Set 55-1-1]

SOLUTION

Consider a resistor R and capacitor C in series across $V = V_0 \omega t$. The capacitive reactance is $X_C = \frac{1}{\omega C}$.

The impedance of the series RC circuit is

$$Z = \sqrt{R^2 + X_C^2} = \sqrt{R^2 + \frac{1}{\omega^2 C^2}}$$

The peak (amplitude) of the current is

$$I_0 = \frac{V_0}{Z} = \frac{V_0}{\sqrt{R^2 + X_C^2}}$$

(a)(i) The peak voltage across the resistance is in phase with the current:

$$V_R^{\max} = I_0 R = \frac{V_0 R}{\sqrt{R^2 + X_C^2}}$$

(a)(ii) The peak voltage across the capacitor is

$$V_C^{\max} = I_0 X_C = \frac{V_0 X_C}{\sqrt{R^2 + X_C^2}} = \frac{V_0}{\omega C \sqrt{R^2 + X_C^2}}$$

(b) The phase difference between the applied voltage and the current is given by

$$\phi = \frac{X_C}{R} = \frac{1}{\omega CR}$$

$$\phi = \tan^{-1}\left(\frac{X_C}{R}\right)$$

In an RC circuit the **current leads the applied voltage** by the angle ϕ (the capacitor causes the current to be ahead).

Final answer. V_R(max) = $V_0 R/Z$ and V_C(max) = $V_0 X_C/Z$ with $Z = \sqrt{(R^2 + X_C^2)}$; the current leads the voltage by $\phi = \tan^{-1}(X_C/R)$.

← *Back to Q106*

Q107

A resistor R and an inductor L are connected in series to a source $V = V_0\,\omega\,t$.
Find the
(a) peak value of the voltage drops across R and across L,
(b) phase difference between the applied voltage and current. Which of them is ahead?

[2020 • Set 55-1-2]

SOLUTION

Consider a resistor R and inductor L in series across $V=V_0\omega\,t$. The inductive reactance is $X_L=\omega\,L$.

The impedance of the series RL circuit is

$$Z=\sqrt{R^2+X_L{}^2}=\sqrt{R^2+\omega^2L^2}$$

The peak (amplitude) of the current is

$$I_0=\frac{V_0}{Z}=\frac{V_0}{\sqrt{R^2+X_L{}^2}}$$

(a) The peak voltage drop across the resistance (in phase with the current) is

$$V_R{}^{max}=I_0R=\frac{V_0R}{\sqrt{R^2+X_L{}^2}}$$

The peak voltage drop across the inductor is

$$V_L{}^{max}=I_0X_L=\frac{V_0X_L}{\sqrt{R^2+X_L{}^2}}=\frac{V_0\,\omega\,L}{\sqrt{R^2+X_L{}^2}}$$

(b) The phase difference between the applied voltage and the current is given by

$$\phi=\frac{X_L}{R}=\frac{\omega\,L}{R}$$

$$\phi=\tan^{-1}\left(\frac{X_L}{R}\right)$$

In an RL circuit the **applied voltage leads the current** by the angle ϕ (the inductor causes the voltage to be ahead).

TIP

A quick memory aid: in a CR circuit the Current is ahead, while in an LR circuit the voltage (L) is ahead. Use 'CIVIL' (C → I leads V; V leads I → L).

Final answer. V_R(max) = V_0R/Z and V_L(max) = V_0X_L/Z with $Z = \sqrt{(R^2+X_L^2)}$; the voltage leads the current by $\phi = \tan^{-1}(X_L/R)$.

← *Back to Q107*

Q108

A resistance R and a capacitor C are connected in series to an ac source $V = V_0\,\omega\,t$.
(a) Obtain the expression for the instantaneous current (I) in the circuit.
(b) Show graphically variations of V and I as a function of ω t.

[2020 • Set 55-1-3]

SOLUTION

(a) For a resistor R and capacitor C in series across $V=V_0\omega\,t$, the capacitive reactance is $X_C=\frac{1}{\omega\,C}$ and the impedance is

$$Z=\sqrt{R^2+X_C{}^2}=\sqrt{R^2+\frac{1}{\omega^2C^2}}$$

The current amplitude is $I_0=\frac{V_0}{Z}$. Since the net reactance is capacitive, the current leads the voltage by a phase angle ϕ where

$$\phi=\frac{X_C}{R}=\frac{1}{\omega\,CR}$$

Therefore the instantaneous current in the circuit is

$$I=I_0\sin(\omega\,t+\phi)$$

$$I=\frac{V_0}{\sqrt{R^2+\frac{1}{\omega^2C^2}}}\sin(\omega\,t+\phi)$$

where $\phi=\tan^{-1}\left(\frac{1}{\omega\,CR}\right)$.

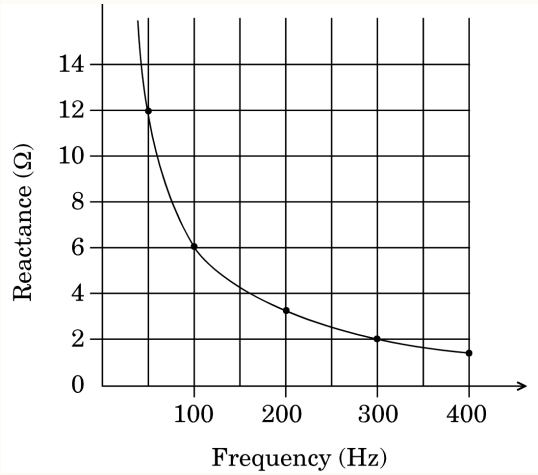
(b) Graph of V and I versus ωt: Draw two sinusoidal curves on the same axes, with ωt on the horizontal axis. The voltage curve $\boldsymbol{V=V_0\omega t}$ passes through the origin. The current curve $\boldsymbol{I=I_0\sin(\omega t+\phi)}$ has a smaller (or different) amplitude $\mathbf{I_0=\frac{V_0}{Z}}$ and is shifted to the LEFT of the voltage curve by the angle ϕ, so the current crest occurs before the voltage crest. This shows that the **current leads the voltage by ϕ**. Label the peaks V_0 and I_0, and mark the horizontal separation between the two zero-crossings (going upward) as the phase angle ϕ.

Final answer. $I = (V_0/\sqrt{(R^2 + 1/\omega^2C^2)})\sin(\omega t + \phi)$ with $\phi = \tan^{-1}(1/\omega CR)$; the current curve leads the voltage curve by ϕ.

← *Back to Q108*

Q109

The figure shows the graphical variation of the reactance of a capacitor with frequency of ac source.



(a) Find the capacitance of the capacitor.
(b) An ideal inductor has the same reactance at 100 *Hz frequency as the capacitor has at the same frequency. Find the value of inductance of the inductor.*
(c) Draw the graph showing the variation of the reactance of this inductor with frequency.

[2020 • Set 55-2-1]

SOLUTION

(a) The reactance of a capacitor is $\mathbf{X_C = \frac{1}{2\pi f C}}$, so the capacitance is

$$C = \frac{1}{2\pi f X_C}$$

Read one clear point (f, X_C) from the given graph. Taking $\mathbf{X_C = 80\ \Omega}$ at $\mathbf{f = 100\ Hz}$ (the standard values for this graph),

$$C = \frac{1}{2\pi \times 100 \times 80} = \frac{1}{50265}\ F$$

$$C = 1.99 \times 10^{-5}\ F \approx 20\ \mu F$$

So the **capacitance $C \approx 20\ \mu F$**.

(b) At $\mathbf{f = 100\ Hz}$ the inductor has the same reactance as the capacitor, so $\mathbf{X_L = X_C = 80\ \Omega}$. Since $\mathbf{X_L = 2\pi f L}$,

$$L = \frac{X_L}{2\pi f} = \frac{80}{2\pi \times 100} = \frac{80}{628.3}$$

$$L = 0.127\ H \approx 0.13\ H$$

So the **inductance $L \approx 0.13\ H$**.

(c) Graph of inductive reactance versus frequency: Since $\mathbf{X_L = 2\pi f L}$, the reactance is directly proportional to frequency. The graph is a straight line passing through the origin (rising to the right), with slope $2\pi L$. Mark X_L on the vertical axis and f on the horizontal axis; the line should pass through the point (100 Hz, 80 Ω), the same reactance value where it equals the capacitor's reactance.

COMMON MISTAKE

Read the graph values carefully before substituting; if your figure shows a different (f, X_C) point, use those numbers, but the method C = 1/(2πf·X_C) stays the same.

Final answer. C ≈ 20 μF, L ≈ 0.13 H, and X_L vs f is a straight line through the origin with slope 2πL.

← Back to Q109

Q110

A series LCR ac circuit has $L = 2.0\ H$, $C = 32\ \mu F$ *and* $R = 10\ \Omega$.

(a) At what angular frequency of ac will it resonate?

(b) Calculate the Q value of the circuit.

OR

An ideal inductor of 5 *H inductance is connected to a* 200 *V,* 50 Hz *ac supply.*

(a) Calculate the rms and peak value of current in the inductor.

(b) What is the phase difference between current through the inductor and the applied voltage? How will it change if a small resistance is connected in series with this inductor in the circuit?

[2020 • Set 55-3-1]

SOLUTION

(a) A series LCR circuit resonates when the inductive and capacitive reactances are equal, giving the resonant angular frequency

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$\omega_0 = \frac{1}{\sqrt{2.0 \times 32 \times 10^{-6}}}$$

$$\omega_0 = \frac{1}{\sqrt{6.4 \times 10^{-5}}} = \frac{1}{8 \times 10^{-3}}$$

$$\omega_0 = 125 \text{ rad s}^{-1}$$

So the **circuit resonates at $\omega_0 = 125$ rad s^{-1}**.

(b) The quality (Q) factor of the circuit is

$$Q = \frac{\omega_0 L}{R} = \frac{125 \times 2.0}{10} = \frac{250}{10} = 25$$

So the **Q-factor is 25**.

OR

(a) For an ideal inductor of $L = 5\ H$ on a 200 V, 50 Hz supply, the inductive reactance is

$$X_L = 2\pi f L = 2\pi \times 50 \times 5 = 500\pi$$

$$X_L \approx 1571\ \Omega$$

The rms current is

$$I_{rms} = \frac{V_{rms}}{X_L} = \frac{200}{1571} \approx 0.127\ A$$

The peak value of the current is

$$I_0 = \sqrt{2}\ I_{rms} = 1.414 \times 0.127 \approx 0.18\ A$$

So the **rms current is about 0.127 *A* and the peak current is about 0.18 *A***.

(b) For a pure (ideal) inductor the current lags the applied voltage by

$$\phi = \frac{\pi}{2}\ (=90^\circ)$$

If a small resistance R is added in series, the phase angle is given by $\phi = \frac{X_L}{R}$, which is now finite. Hence the phase difference decreases (becomes less than $\frac{\pi}{2}$, moving toward 0).

Final answer. ω_0 = 125 rad s^{-1} and Q = 25; OR I_rms ≈ 0.127 A, I_0 ≈ 0.18 A, phase = π/2 (current lags), which decreases below π/2 when a series R is added.

← *Back to Q110*

Q111

Explain with the help of a diagram, the working of a step-down transformer. Why is a laminated iron core used in a transformer?

[2020 • Set 55-4-1]

SOLUTION

A step-down transformer is a device that converts a high alternating voltage into a low alternating voltage, working on the principle of mutual induction.

Construction and diagram: Two coils are wound on a common closed laminated soft-iron core. The primary coil P (input side) has a large number of turns N_p and the secondary coil S (output side) has a smaller number of turns N_s, so that

$N_s < N_p$. In the diagram, draw a rectangular soft-iron core shown as a stack of thin horizontal sheets (laminations). On the left limb wind the primary coil with many turns connected to the high AC input, and on the right limb wind the secondary coil with fewer turns connected to the load (output).

Working: When an alternating voltage is applied to the primary, an alternating current flows and sets up a continuously changing magnetic flux Φ in the iron core. The core guides almost all of this flux through the secondary coil. By Faraday's law, the changing flux linked with the secondary induces an alternating emf in it.

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

Since $N_s < N_p$, we get $V_s < V_p$, so the output voltage is stepped down.

For an ideal (lossless) transformer, the power is conserved:

$$V_p I_p = V_s I_s$$

so a fall in voltage is accompanied by a rise in current, $I_s > I_p$.

Why a laminated iron core is used: The changing flux also induces circulating currents, called **eddy currents**, in the body of the core. In a solid iron block these eddy currents are large and dissipate energy as heat, reducing efficiency. Using a core made of thin iron sheets, insulated from one another by varnish and stacked along the direction of the flux, breaks up the large eddy-current loops into many small ones of high resistance. This sharply reduces the eddy currents and hence the heat loss.

Final answer. A step-down transformer uses mutual induction with $N_s < N_p$ to lower the voltage; the laminated core minimises eddy-current (heat) losses.

← *Back to Q111*

Q112

Explain the principle of the device with diagram, which is used to provide electricity at the proper voltage for household purposes.

Briefly discuss loss of energy in it due to flux leakage and its minimization.

[2020 • Set 55-4-2]

SOLUTION

The device used to supply electricity at the proper voltage for household use is the **step-down transformer**.

Principle: It works on the principle of **mutual induction**, that is, an alternating current in one coil induces an emf in a neighbouring coil due to the changing magnetic flux linked with it.

Diagram and construction: Draw a closed rectangular soft-iron core (shown as a stack of thin laminated sheets). Wind the primary coil P of N_p turns on one limb, connected to the high-voltage AC line, and the secondary coil S of N_s turns on the other limb, connected to the household circuit. For household supply (about 230 *V*) the secondary has fewer turns, so $\mathbf{N_s < N_p}$.

Working: An alternating voltage applied to the primary produces a changing flux Φ in the core. This flux links the secondary and induces an alternating emf there.

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

With $\mathbf{N_s < N_p}$, the output voltage V_s is lower than the input, giving the safe voltage needed for homes.

Loss due to flux leakage: In a real transformer not all the magnetic flux produced by the primary passes through the secondary. Some flux escapes (leaks) into the surrounding air and does not link the secondary coil. This leakage flux carries energy that is not transferred to the output, so the secondary emf and the efficiency are reduced.

Minimisation: Flux leakage is reduced by winding the primary and secondary coils very close together, one over the other on the same limb, and by using a well-designed closed magnetic circuit (such as a shell-type core) so that almost the entire flux is confined to the iron core and threads both coils.

Final answer. A step-down transformer (principle: mutual induction) supplies household voltage; flux-leakage loss is cut by tightly coupling the coils on a closed core.

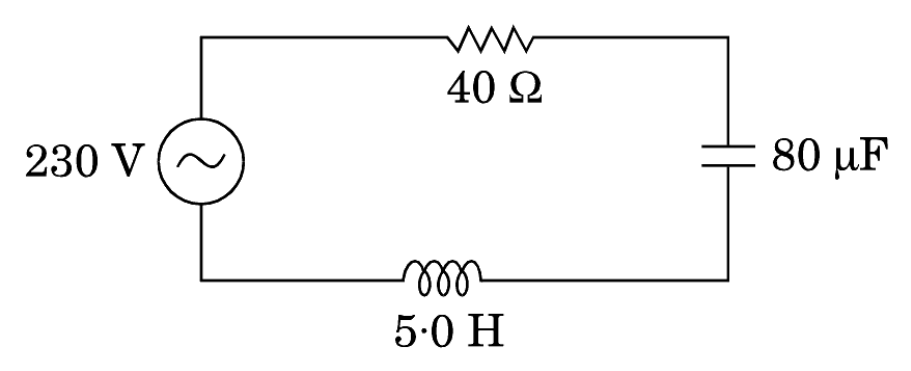
← Back to Q112

Q113

A capacitor (C) and resistor (R) are connected in series with an ac source of voltage of frequency 50 Hz. *The potential difference across C and R are respectively* 120 *V*, 90 *V*, *and the current in the circuit is* 3 *A. Calculate (i) the impedance of the circuit (ii) the value of the inductance, which when connected in series with C and R will make the power factor of the circuit unity.*

OR

The figure shows a series LCR circuit connected to a variable frequency 230 *V source.*



(a) Determine the source frequency which drives the circuit in resonance.

(b) Calculate the impedance of the circuit and amplitude of current at resonance.

(c) Show that potential drop across LC combination is zero at resonating frequency.

[2019 • Set 55-2-1]

SOLUTION

Given: $\mathbf{V_R = 90\ V}$, $\mathbf{V_C = 120\ V}$, $\mathbf{I = 3\ A}$, $\mathbf{f = 50\ Hz}$, so $\boldsymbol{\omega = 2\pi f = 100\pi\ rad/s}$.

First find R and X_C from the given voltage drops:

$$R = \frac{V_R}{I} = \frac{90}{3} = 30\ \Omega$$

$$X_C = \frac{V_C}{I} = \frac{120}{3} = 40\ \Omega$$

(i) Impedance of the series RC circuit:

$$Z = \sqrt{R^2 + X_C{}^2}$$

$$Z = \sqrt{(30)^2 + (40)^2}$$

$$Z = \sqrt{900 + 1600} = \sqrt{2500}$$

$$Z = 50\ \Omega$$

(ii) For the power factor to be unity, the circuit must be purely resistive, which requires the inductive reactance to cancel the capacitive reactance:

$$X_L = X_C = 40\ \Omega$$

$$X_L = \omega L \Rightarrow L = \frac{X_L}{\omega}$$

$$L = \frac{40}{100\pi} = \frac{0.4}{\pi}$$

$$L \approx 0.127\ H = 127\ \text{mH}$$

OR (Series LCR driven by a variable-frequency source, with $L = 5.0\ H$, $C = 80\ \mu F$, $R = 40\ \Omega$, $V_{rms} = 230\ V$):

(a) The circuit is in resonance when $X_L = X_C$, giving the resonant angular frequency:

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$\omega_0 = \frac{1}{\sqrt{5 \times 80 \times 10^{-6}}} = \frac{1}{\sqrt{4 \times 10^{-4}}}$$

$$\omega_0 = \frac{1}{0.02} = 50\ \text{rad/}s$$

$$f_0 = \frac{\omega_0}{2\pi} = \frac{50}{2\pi} \approx 7.96\ \text{Hz}$$

(b) At resonance the reactances cancel, so the impedance is purely resistive:

$$Z = R = 40\ \Omega$$

The current amplitude (peak value):

$$i_0 = \frac{\sqrt{2}\, V_{rms}}{R} = \frac{\sqrt{2} \times 230}{40} \approx 8.13\ A$$

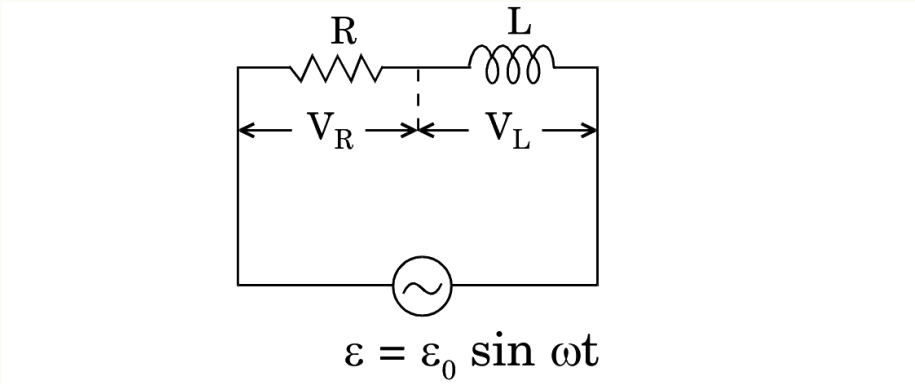
(c) At resonance $\mathbf{X_L = X_C}$, so the rms voltages across L and C are equal in magnitude, $\mathbf{V_L = I\,X_L}$ and $\mathbf{V_C = I\,X_C}$, giving $\mathbf{V_L = V_C}$. In the phasor diagram V_L leads the current by $\pi/2$ while V_C lags it by $\pi/2$, so they are exactly opposite (out of phase by π). Their resultant across the LC combination is therefore $\mathbf{V_L - V_C = 0}$, confirming the potential drop across the LC combination is zero at the resonant frequency.

Final answer. (i) $\mathbf{Z = 50\ \Omega}$; (ii) $\mathbf{L \approx 0.127\ H}$. OR: $\mathbf{f_0 \approx 7.96\ Hz}$, $\mathbf{Z = 40\ \Omega}$, $\mathbf{i_0 \approx 8.13\ A}$, and $\mathbf{V_{LC} = 0}$ at resonance.

← *Back to Q113*

Q114

(a) An ac circuit as shown in the figure has an inductor of inductance L and a resistor of resistance R connected in series. Using the phasor diagram, explain why the voltage in the circuit will lead the current in phase.
(b) The potential difference across the resistor is 160 *V and that across the inductor is* 120 *V. Find the effective value of the applied voltage. If the effective current in the circuit be* 1.0 *A, calculate the total impedance of the circuit.*
(c) What will be the potential difference in the circuit when direct current is passed through the circuit?



OR

An ac circuit consists of a series combination of circuit elements X and Y. The current is ahead of the voltage in phase by $\frac{\pi}{4}$. *If element X is a pure resistor of* 100 Ω,

(a) name the circuit element Y.

(b) calculate the rms value of current, if rms value of voltage is 141 *V.*

(c) what will happen if the ac source is replaced by a dc source?

[2019 • Set 55-3-1]

SOLUTION

(a) Phasor diagram for the series RL circuit: Take the current $\vec{I}$ as the reference phasor, drawn along the positive x-axis (the same current flows through both R and L). The voltage across the resistor $\vec{V}_R = \mathbf{IR}$ is in phase with the current, so it lies along $\vec{I}$. The voltage across the inductor $\vec{V}_L = \mathbf{IX}_L$ leads the current by π/2, so it is drawn perpendicular to $\vec{I}$ (along the positive y-axis). The applied voltage $\vec{V}$ is the vector sum of $\vec{V}_R$ and $\vec{V}_L$.

$$V = \sqrt{V_R^2 + V_L^2}$$

This resultant makes an angle ϕ with the current, where:

$$\phi = \frac{V_L}{V_R} = \frac{X_L}{R}$$

Since $\vec{V}$ lies ahead of $\vec{I}$ by this angle ϕ, the applied voltage leads the current in phase in an RL circuit.

(b) Given $\mathbf{V_R = 160\ V}$ and $\mathbf{V_L = 120\ V}$. The effective (rms) applied voltage:

$$V = \sqrt{V_R^2 + V_L^2}$$

$$V = \sqrt{(160)^2 + (120)^2} = \sqrt{25600 + 14400}$$

$$V = \sqrt{40000} = 200\ V$$

With effective current $\boldsymbol{I = 1.0\ A}$, the total impedance:

$$Z = \frac{V}{I} = \frac{200}{1.0}$$

$$V = 200\ V,\ Z = 200\ \Omega$$

(c) When direct current (steady DC) is passed, the inductor offers no opposition because the current no longer changes, so $\mathbf{X_L = \omega L = 0}$ (as $\boldsymbol{\omega = 0}$). The only opposition is the resistance, and the entire voltage drop appears across the resistor: $\boldsymbol{V = IR}$.

OR (Series combination of X and Y, current leads voltage by π/4, X is a pure resistor of 100 Ω):

(a) Since the current is ahead of (leads) the voltage, the circuit is capacitive, so element Y is a **capacitor**. The phase condition gives:

$$\tan\frac{\pi}{4} = \frac{X_C}{R} \Rightarrow X_C = R = 100\ \Omega$$

(b) Impedance of the series RC circuit:

$$Z = \sqrt{R^2 + {X_C}^2} = \sqrt{(100)^2 + (100)^2} = 100\sqrt{2}\ \Omega$$

$$I_{rms} = \frac{V_{rms}}{Z} = \frac{141}{100\sqrt{2}} \approx \frac{141}{141.4}$$

$$I_{rms} \approx 1.0\ A$$

(c) If the AC source is replaced by a DC source, the capacitor becomes fully charged and then blocks current, so no steady current flows in the circuit.

COMMON MISTAKE

When using $V = \sqrt{{V_R}^2 + {V_L}^2}$, never add V_R and V_L directly; they are out of phase by 90°, so only the vector sum gives the applied voltage.

Final answer. (b) $V = 200\ V$, $Z = 200\ \Omega$; on DC only $V = IR$ appears. OR: Y is a capacitor, $I_{rms} \approx 1.0\ A$, and DC gives no steady current.

← *Back to Q114*

Q115

A voltage $v = v_m \sin \omega t$ applied to a series LCR circuit, drives a current in the circuit given $i = i_m \sin(\omega t + \phi)$. Deduce the expression for the instantaneous power supplied by the source. Hence, obtain the expression for the average power.
Define the terms 'power factor' and 'wattless current', giving the examples where power factor is maximum and the circuit where there is wattless current.

[2019 • Set 55-5-1]

SOLUTION

Let the applied voltage be $v = v_m \omega t$ and the resulting current $i = i_m \sin(\omega t + \phi)$, where ϕ is the phase difference between current and voltage.

Instantaneous power supplied by the source is the product of instantaneous voltage and current:

$$P(t) = v\, i = v_m\, i_m\, \omega\, t \sin(\omega t + \phi)$$

Using the trigonometric identity $\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$ with $A = \omega t + \phi$ and $B = \omega t$:

$$P(t) = \frac{v_m\, i_m}{2}[\phi - \cos(2\omega t + \phi)]$$

To get the average power, we average over one full cycle. The term ϕ is constant, while the term $\cos(2\omega t + \phi)$ averages to zero over a complete cycle:

$$\cos(2\omega t + \phi) = 0$$

Therefore the average power is:

$$P_{avg} = \frac{v_m\, i_m}{2}\phi$$

Writing the peak values in terms of rms values, $v_m = \sqrt{2}\, V_{rms}$ and $i_m = \sqrt{2}\, I_{rms}$:

$$P_{avg} = V_{rms}\, I_{rms}\phi$$

Power factor: It is the factor ϕ in the average-power expression, equal to the cosine of the phase angle between the voltage and the current. It can also be written as $\phi = R/Z$. Its value lies between 0 and 1. The power factor is **maximum** ($\phi = 1$) for a purely resistive circuit, where voltage and current are in phase.

Wattless current: It is the component of current that flows in the circuit but consumes no average power. This happens in a purely reactive circuit where $\phi = \pi/2$, so $\phi = 0$ and $P_{avg} = 0$. Examples of circuits carrying wattless current are a circuit containing a **pure inductor** or a **pure capacitor**, where the current and voltage differ in phase by $\pi/2$.

Final answer. $P_{avg} = V_{rms}\, I_{rms}\, \phi$; power factor ϕ is maximum (=1) for a pure resistor, and wattless current flows in a pure inductor or pure capacitor.

← Back to Q115

Q116

Given three elements X, Y and Z to be connected across an ac source. With only X connected across the ac source, voltage and current are found to be in the same phase. With only element Y in the circuit, the voltage lags behind the current in phase by π/2 while with the element Z in the circuit, the voltage leads the current in phase by π/2.
(a) Identify the elements X, Y and Z.
(b) When all these elements are connected in series across the same source, (i) determine the power factor, and (ii) find out the condition when the circuit is in resonant state.

[2019 • Set 55-5-2]

SOLUTION

(a) Identifying the elements from the phase behaviour:

Element X: voltage and current are in the same phase. This is the property of a **pure resistor** (R), since in a resistor V and I are always in phase.

Element Y: the voltage lags behind the current by π/2 (equivalently the current leads the voltage). This is the property of a **pure capacitor** (C), since in a capacitor the current leads the voltage by π/2.

Element Z: the voltage leads the current by π/2. This is the property of a **pure inductor** (L), since in an inductor the voltage leads the current by π/2.

(b) The three elements R, L and C are now connected in series across the same source.

(i) Power factor of the series LCR circuit:

$$\phi = \frac{R}{Z}$$

$$\phi = \frac{R}{\sqrt{R^2 + (X_L - X_C)^2}}$$

where $X_L = \omega L$ is the inductive reactance and $X_C = \frac{1}{\omega C}$ is the capacitive reactance.

(ii) Condition for resonance: The circuit is in the resonant state when the inductive and capacitive reactances are equal so that they cancel:

$$X_L = X_C$$

$$\omega L = \frac{1}{\omega C} \Rightarrow \omega^2 LC = 1$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

At resonance the impedance is minimum ($Z = R$), the current is maximum, and the power factor becomes unity ($\phi = 1$).

Final answer. X = resistor, Y = capacitor, Z = inductor; power factor $\phi = R/\sqrt{R^2 + (X_L - X_C)^2}$, and resonance occurs when $\omega_0 = 1/\sqrt{LC}$.

← Back to Q116

Q117

Draw the phasor diagram of a series LCR connected across an ac source $V = V_m \sin \omega t$*. Hence, derive the expression for the impedance of the circuit. Obtain the conditions for the phase angle under which the current is (i) maximum, and (ii) minimum.*

[2019 • Set 55-5-3]

SOLUTION

Phasor diagram of the series LCR circuit (driven by $V = V_m \omega t$): Take the current $\vec{I}$ as the reference phasor along the positive x-axis, since the same current flows through R, L and C. Then:

The resistor voltage $\vec{V}_R = IR$ is in phase with $\vec{I}$ (along the x-axis).

The inductor voltage $\vec{V}_L = IX_L$ leads $\vec{I}$ by $\pi/2$ (along the positive y-axis).

The capacitor voltage $\vec{V}_C = IX_C$ lags $\vec{I}$ by $\pi/2$ (along the negative y-axis).

Since $\vec{V}_L$ and $\vec{V}_C$ are opposite, their combined effect is the net reactive voltage $(V_L - V_C)$ along the y-axis. The applied voltage $\vec{V}$ is the resultant of $\vec{V}_R$ and $(\vec{V}_L - \vec{V}_C)$:

$$V = \sqrt{V_R^2 + (V_L - V_C)^2}$$

Substituting $V_R = IR$, $V_L = IX_L$ and $V_C = IX_C$:

$$V = \sqrt{(IR)^2 + (IX_L - IX_C)^2}$$

$$V = I\sqrt{R^2 + (X_L - X_C)^2}$$

Therefore the impedance of the circuit is:

$$Z = \frac{V}{I} = \sqrt{R^2 + (X_L - X_C)^2}$$

The phase angle between voltage and current is given by:

$$\phi = \frac{V_L - V_C}{V_R} = \frac{X_L - X_C}{R}$$

(i) Current is maximum when the impedance is minimum. This occurs when the net reactance is zero, $X_L = X_C$, so that $\phi = 0$, giving $\phi = 0$ (voltage and current in phase). Then $Z = R$ (minimum) and the current reaches its maximum value $I_{max} = V/R$. This is the condition of resonance.

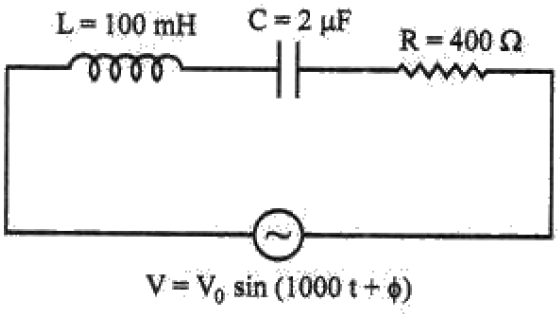
(ii) Current is minimum when the impedance is large, which happens when $|X_L - X_C|$ is large (frequency far from resonance). Then $\phi \to \pi/2$, the impedance Z becomes very large, and the current $I = V/Z$ tends towards its minimum value.

Final answer. $Z = \sqrt{R^2 + (X_L - X_C)^2}$; current is maximum at $\phi = 0$ ($X_L = X_C$, resonance) and minimum when $|X_L - X_C|$ is large ($\phi \to \pi/2$).

← *Back to Q117*

Q118

(i) Find the value of the phase difference between the current and the voltage in the series LCR circuit shown below. Which one leads in phase: current or voltage?
(ii) Without making any other change, find the value of the additional capacitor C_1, to be connected in parallel with the capacitor C, in order to make the power factor of the circuit unity.



[2017]

SOLUTION

Given: $L = 100\ \text{mH} = 0.1\ H$, $C = 2\ \mu\text{F} = 2 \times 10^{-6}\ F$, $R = 400\ \Omega$, and $\omega = 1000\ \text{rad/s}$.

First find the reactances.

Inductive reactance:

$$X_L = \omega L = 1000 \times 0.1 = 100\ \Omega$$

Capacitive reactance:

$$X_C = \frac{1}{\omega C} = \frac{1}{1000 \times 2 \times 10^{-6}} = \frac{1}{2 \times 10^{-3}} = 500\ \Omega$$

(i) Phase difference: Since $X_C > X_L$, the net reactance is capacitive, so the **current leads the voltage**. The magnitude of the phase angle:

$$\phi = \frac{X_C - X_L}{R} = \frac{500 - 100}{400} = \frac{400}{400} = 1$$

$$\phi = \tan^{-1}(1) = \frac{\pi}{4}$$

So the current leads the voltage by $\frac{\pi}{4}$ (that is, 45°).

(ii) Additional capacitor C_1 for unity power factor: For the power factor to be unity, the circuit must be in resonance, requiring the new capacitive reactance to equal the inductive reactance:

$$X_C' = X_L = 100\ \Omega$$

The new total capacitance needed:

$$X_C' = \frac{1}{\omega C'} \Rightarrow C' = \frac{1}{\omega X_C'}$$

$$C' = \frac{1}{1000 \times 100} = 10^{-5}\ F = 10\ \mu F$$

When C_1 is connected in parallel with C, the capacitances add:

$$C' = C + C_1 \Rightarrow C_1 = C' - C$$

$$C_1 = 10\ \mu F - 2\ \mu F$$

$$C_1 = 8\ \mu F$$

TIP

Capacitors in parallel add up ($C' = C + C_1$), while their reactance falls. To increase C you add a parallel capacitor, not a series one, a step often reversed under exam pressure.

Final answer. (i) Current leads voltage by $\pi/4$; (ii) $C_1 = 8\ \mu F$ in parallel gives unity power factor.

← *Back to Q118*

Q119

A source of ac voltage $v = v_0 \sin \omega t$ is connected across a pure inductor of inductance L. Derive the expressions for the instantaneous current in the circuit. Show that average power dissipated in the circuit is zero.

[2017]

SOLUTION

A pure inductor of inductance L is connected across an AC source $v = v_0\ \omega\ t$.

As the current changes, a back emf is induced in the inductor. Applying Kirchhoff's voltage law, the applied voltage equals the back emf in magnitude:

$$v = L\frac{di}{dt}$$

$$v_0\ \omega\ t = L\frac{di}{dt}$$

Rearranging:

$$\mathrm{d}i = \frac{v_0}{L}\omega\, t\, \mathrm{d}t$$

Integrating both sides:

$$i = \frac{v_0}{L}\int \omega\, t\, \mathrm{d}t = -\frac{v_0}{\omega\, L}\omega\, t$$

The integration constant is taken as zero (no steady DC component). Writing $-\omega t = \sin(\omega t - \frac{\pi}{2})$:

$$i = \frac{v_0}{\omega\, L}\sin(\omega\, t - \frac{\pi}{2})$$

The quantity $\omega L = X_L$ is the inductive reactance, and the peak current is $i_0 = \frac{v_0}{\omega L}$. Hence:

$$i = i_0 \sin(\omega\, t - \frac{\pi}{2})$$

This shows the current **lags** the applied voltage by a phase angle of $\frac{\pi}{2}$.

Average power dissipated: The phase difference between voltage and current is $\phi = \frac{\pi}{2}$. The average power over a complete cycle is:

$$P_{avg} = V_{rms}\, I_{rms}\phi$$

$$P_{avg} = V_{rms}\, I_{rms}\cos\frac{\pi}{2}$$

$$P_{avg} = 0$$

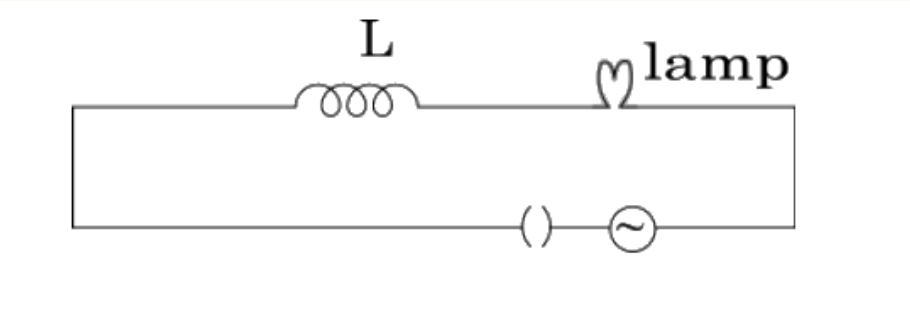
Thus a pure inductor consumes no net power over a cycle; during one quarter cycle it stores energy in its magnetic field and returns it to the source in the next, so the average power dissipated is zero.

Final answer. $i = i_0 \sin(\omega\, t - \pi/2)$, lagging the voltage by $\pi/2$, and the average power dissipated is zero.

← *Back to Q119*

Q120

(i) When an AC source is connected to an ideal inductor show that the average power supplied by the source over a complete cycle is zero.
(ii) A lamp is connected in series with an inductor and an AC source. What happens to the brightness of the lamp when the key is plugged in and an iron rod is inserted inside the inductor? Explain.



[2016]

SOLUTION

(i) AC source connected to an ideal inductor, average power over a cycle: Let the current be $i = i_0\,\omega\,t$. The voltage across the inductor is:

$$v = L\frac{di}{dt} = L\,i_0\,\omega\omega\,t$$

$v = v_0 \sin(\omega t + \frac{\pi}{2})$, where $v_0 = i_0\omega L$

So the voltage leads the current by $\frac{\pi}{2}$. The instantaneous power supplied is:

$$P(t) = v\,i = v_0\,i_0 \sin(\omega t + \frac{\pi}{2})\omega t$$

$$P(t) = v_0\,i_0\,\omega t\,\omega t = \frac{v_0\,i_0}{2}\sin 2\omega t$$

This instantaneous power oscillates at twice the source frequency (2ω). Its average over one full cycle is:

$$P_{avg} = \frac{v_0\,i_0}{2}\sin 2\omega t = 0$$

$$P = 0$$

(Equivalently, $\phi = \pi/2$ so $P_{avg} = V_{rms} I_{rms} \cos\frac{\pi}{2} = 0$.) Physically, the inductor stores energy in its magnetic field for one quarter cycle and returns it completely to the source in the next, so no net power is dissipated.

(ii) Lamp in series with an inductor and an AC source: Inserting an iron rod inside the inductor increases its inductance L (the iron core raises the magnetic flux). This increases the inductive reactance:

$$X_L = \omega L \uparrow$$

The total impedance of the circuit therefore rises:

$$Z = \sqrt{R^2 + X_L{}^2} \uparrow$$

so the current falls:

$$I = \frac{V}{Z} \downarrow$$

Since the brightness of the lamp depends on the current through it (power $\propto I^2$), the **lamp glows more dimly** when the iron rod is inserted.

Final answer. Average power in an ideal inductor is zero ($\phi = \pi/2$); inserting the iron rod raises X_L and Z, lowering the current, so the lamp grows dimmer.

← *Back to Q120*

Q121

Draw a schematic arrangement for winding of primary and secondary coils in a transformer with the two coils on separate limbs of the core. State its underlying principle and find the relation between the primary and secondary voltages in terms of the number of turns of the primary and secondary windings. How are the currents in the primary and secondary coils related to the voltages in the case of an ideal transformer?

[2016]

SOLUTION

Schematic arrangement: Draw a rectangular laminated soft-iron core shaped like a hollow rectangle, with two vertical limbs joined by horizontal top and bottom yokes. On the left limb, wind the primary coil of N_p turns and connect its two ends to the AC input source V_p. On the right limb, wind the secondary coil of N_s turns and connect its two ends to the output terminals across the load V_s. The magnetic flux produced by the primary is guided around the closed core and links fully with the secondary.

Underlying principle: A transformer works on the principle of **mutual induction**, that an alternating current in the primary coil produces a continuously changing magnetic flux in the core, and this changing flux, linking the secondary coil, induces an alternating EMF in it.

When an AC voltage is applied to the primary, the changing current sets up a changing flux Φ in the core. The same flux Φ threads through every turn of both coils, so the rate of change of flux per turn is identical in both windings.

By Faraday's law, the EMF induced in the secondary is $\mathbf{V_s = N_s \frac{d\Phi}{dt}}$

and the back EMF (equal to the applied voltage for an ideal transformer) in the primary is $\mathbf{V_p = N_p \frac{d\Phi}{dt}}$

Dividing the two equations, $\frac{V_s}{V_p} = \frac{N_s\,(d\Phi/dt)}{N_p\,(d\Phi/dt)}$

$$\Rightarrow \frac{V_s}{V_p} = \frac{N_s}{N_p}$$

For an ideal transformer there are no energy losses, so the input power equals the output power: $\mathbf{V_p I_p = V_s I_s}$

$$\Rightarrow \frac{I_s}{I_p} = \frac{V_p}{V_s} = \frac{N_p}{N_s}$$

Thus **the currents are in the inverse ratio of the turns, while the voltages are in the direct ratio of the turns**: $\frac{V_s}{V_p} = \frac{I_p}{I_s} = \frac{N_s}{N_p}$.

COMMON MISTAKE

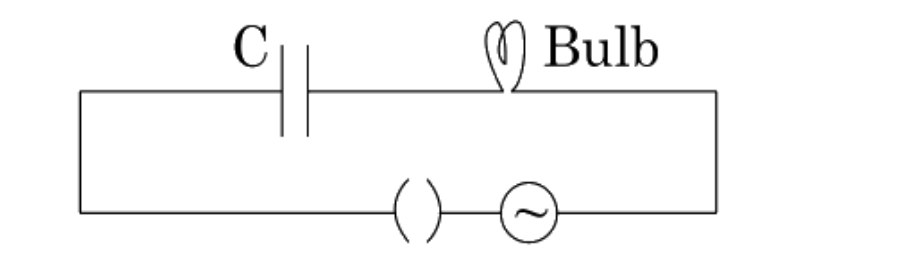
In an ideal transformer power is conserved, so a step-up of voltage is always accompanied by an equal step-down of current. Many students wrongly write current in the same ratio as voltage and lose a mark.

Final answer. For an ideal transformer, $\frac{V_s}{V_p} = \frac{N_s}{N_p}$ and $\frac{I_s}{I_p} = \frac{N_p}{N_s}$.

← Back to Q121

Q122

(i) When an AC source is connected to an ideal capacitor, show that the average power supplied by the source over a complete cycle is zero.
(ii) A bulb is connected in series with a variable capacitor and an A.C. source as shown. What happens to the brightness of the bulb when the key is plugged in and capacitance of the capacitor is gradually reduced?



[2016]

SOLUTION

(i) Let the applied voltage be $V = V_0 \omega t$ across an ideal capacitor of capacitance C.

In a pure capacitor the current leads the voltage by $\frac{\pi}{2}$, so the current is $I = I_0 \sin(\omega t + \frac{\pi}{2}) = I_0 \omega t$, where $I_0 = \frac{V_0}{X_C} = V_0 \omega C$.

The instantaneous power supplied is $P = VI = V_0 \omega t \times I_0 \omega t$

$$\Rightarrow P = V_0 I_0 \omega t \omega t = \frac{V_0 I_0}{2} \sin 2\omega t$$

The average power over one complete cycle is $P_{avg} = \frac{V_0 I_0}{2} \sin 2\omega t$

Since the average value of $\sin 2\omega t$ over a full cycle is zero, $\sin 2\omega t = 0$

$$\Rightarrow P_{avg} = 0$$

Hence over a complete cycle the source supplies no net power to an ideal capacitor; the energy stored during charging is returned to the source during discharging. This current is therefore called a **wattless current** ($\phi = \cos 90° = 0$).

(ii) The bulb, capacitor and AC source are in series. The current through the bulb is $I = \frac{V}{X_C} = V\,\omega\,C$, since the capacitive reactance is $X_C = \frac{1}{\omega\,C}$.

When the capacitance C is gradually reduced, $X_C = \frac{1}{\omega\,C}$ increases, so the current I in the circuit decreases.

Therefore the **brightness of the bulb decreases (the bulb becomes dimmer)** as the capacitance is reduced.

Final answer. (i) $P_{avg}=0$ for an ideal capacitor (wattless current); (ii) the bulb grows dimmer as C is reduced.

← *Back to Q122*

Q123

A capacitor of unknown capacitance, a resistor of 100 Ω *and an inductor of self inductance* $L=(4/\pi^2)$ *henry are connected in series to an ac source of* 200 *V and* 50 Hz. *Calculate the value of the capacitance and impedance of the circuit when the current is in phase with the voltage. Calculate the power dissipated in the circuit.*

[2016]

SOLUTION

Given $R = 100\ \Omega$, $L = \frac{4}{\pi^2}\ H$, $V_{rms} = 200\ V$, $f = 50$ Hz.

The angular frequency is $\omega = 2\pi f = 2\pi \times 50 = 100\pi$ rad s^{-1}.

The current is in phase with the voltage, which is the condition of **resonance**. At resonance the inductive reactance equals the capacitive reactance: $X_L = X_C$.

First find the inductive reactance: $X_L = \omega L = 100\pi \times \frac{4}{\pi^2}$

$$\Rightarrow X_L = \frac{400}{\pi} \approx 127.3\ \Omega$$

At resonance $X_C = X_L = \frac{400}{\pi}\ \Omega$.

The capacitance is obtained from $X_C = \frac{1}{\omega C}$, so $C = \frac{1}{\omega X_C}$

$$\Rightarrow C = \frac{1}{100\pi \times \frac{400}{\pi}}$$

$$\Rightarrow C = \frac{1}{100 \times 400} = \frac{1}{40000}\ F$$

$$\Rightarrow C = 25 \times 10^{-6}\ F = \mathbf{25\ \mu F}$$

At resonance the reactive parts cancel, so the impedance is purely resistive: $Z = \sqrt{R^2 + (X_L - X_C)^2} = R = \mathbf{100\ \Omega}$.

The power dissipated is $P = \frac{V_{rms}^2}{R}$ (since at resonance $\phi = 1$)

$$\Rightarrow P = \frac{(200)^2}{100} = \frac{40000}{100}$$

$\Rightarrow P = \mathbf{400\ W}$.

Final answer. $C = 25\ \mu F$, $Z = 100\ \Omega$ and $P = 400\ W$.

← *Back to Q123*

Q124

A series LCR circuit is connected across an a.c. source of variable angular frequency 'ω'. Plot a graph showing variation of current 'i' as a function of 'ω' for two resistances R_1 *and* R_2 *(*$R_1 > R_2$*). Answer the following questions using this graph:*
(a) In which case is the resonance sharper and why?
(b) In which case is the power dissipation more and why?

[2015]

SOLUTION

Graph: On the same axes plot current i (y-axis) against angular frequency ω (x-axis). Both curves rise to a peak at the common resonant frequency $\omega_0 = \frac{1}{\sqrt{LC}}$

and then fall off symmetrically. The curve for the smaller resistance R_2 is taller (peak current $\mathbf{i_{max}} = \frac{\mathbf{V_{rms}}}{\mathbf{R_2}}$) and narrower; the curve for the larger resistance R_1 is shorter (peak $\frac{V_{rms}}{R_1}$) and broader. Mark both peaks at the same ω_0 and label the curves R_1 and R_2 with $\mathbf{R_1 > R_2}$.

At any frequency the current is $i = \frac{V_{rms}}{Z}$, where $Z = \sqrt{R^2 + (X_L - X_C)^2}$, and Z is minimum ($Z = R$) at ω_0.

(a) The resonance is sharper for R_2 (the smaller resistance). The sharpness of resonance is measured by the quality factor $Q = \frac{\omega_0 L}{R} = \frac{1}{R}\sqrt{\frac{L}{C}}$

Since $Q \propto \frac{1}{R}$, a smaller resistance R_2 gives a larger Q, hence a taller and narrower peak, i.e. a **sharper resonance**.

(b) The power dissipation is more for R_2 (the smaller resistance). At resonance the average power is $P = \frac{V_{rms}^2}{R}$

Since $P \propto \frac{1}{R}$ and $\mathbf{R_2 < R_1}$, we get $\mathbf{P_{R_2} > P_{R_1}}$

Therefore **more power is dissipated in the circuit with the smaller resistance $\mathbf{R_2}$.**

Final answer. Resonance is sharper and power dissipation is greater for the smaller resistance $\mathbf{R_2}$.

← *Back to Q124*

Q125

An inductor L of inductance X_L is connected in series with a bulb B and an ac source. How would brightness of the bulb change when (i) number of turns in the inductor is reduced, (ii) an iron rod is inserted in the inductor and (iii) a capacitor of reactance $\mathbf{X_C = X_L}$ is inserted in series in the circuit. Justify your answer in each case.

[2015]

SOLUTION

The bulb B (resistance R_B) is in series with the inductor and the AC source, so the current is $I = \frac{V}{Z}$ where $Z = \sqrt{R_B^2 + X_L^2}$ and $X_L = \omega L$. The brightness of the bulb depends on this current.

(i) Reducing the number of turns: The inductance varies as $L \propto N^2$, so reducing the number of turns N decreases L, hence the reactance $X_L = \omega L$ decreases.

This makes $Z = \sqrt{R_B^2 + X_L^2}$ smaller, so the current $I = \frac{V}{Z}$ increases. Therefore the **bulb glows brighter**.

(ii) Inserting an iron rod: An iron core increases the relative permeability, and since $L \propto \mu_r$, the inductance L increases, so $X_L = \omega L$ increases.

This makes Z larger, so the current I decreases. Therefore the **bulb becomes dimmer**.

(iii) Inserting a capacitor with $X_C = X_L$: The net reactance becomes $X_L - X_C = 0$, which is the condition of **resonance**.

The impedance is then minimum, $Z = \sqrt{R_B^2 + (X_L - X_C)^2} = R_B$, so the current $I = \frac{V}{R_B}$ is maximum.

Therefore the **bulb glows with maximum brightness (brightest)**.

> **CONCEPT**
>
> Inductance depends on turns as $L \propto N^2$, not linearly, so halving the turns reduces L to one-quarter. Stating the square dependence earns the justification mark.

Final answer. (i) Brighter; (ii) dimmer; (iii) brightest (resonance, $Z = R_B$).

← *Back to Q125*

Q126

A voltage $V = V_0 \sin \omega t$ is applied to a series LCR circuit. Derive the expression for the average power dissipated over a cycle.

Under what condition is (i) no power dissipated even though the current flows through the circuit, (ii) maximum power dissipated in the circuit ?

[2014]

SOLUTION

Let the applied voltage be $V = V_0 \omega t$. In a series LCR circuit the current lags or leads by a phase angle ϕ, so $I = I_0 \sin(\omega t - \phi)$, where $\phi = \frac{X_L - X_C}{R}$.

The instantaneous power supplied is $P = VI = V_0 \omega t \times I_0 \sin(\omega t - \phi)$

Expanding $\sin(\omega t - \phi) = \omega t \phi - \omega t \phi$,

$$P = V_0 I_0 (\sin^2 \omega t \phi - \omega t \omega t \phi)$$

The average power is found over one full cycle. Using $\sin^2 \omega t = \frac{1}{2}$ and $\omega t \omega t = 0$,

$$P_{avg} = V_0 I_0 (\frac{1}{2} \phi - 0)$$

$$\Rightarrow P_{avg} = \frac{V_0 I_0}{2} \phi$$

Writing $V_{rms} = \frac{V_0}{\sqrt{2}}$ and $I_{rms} = \frac{I_0}{\sqrt{2}}$,

$$\Rightarrow P_{avg} = V_{rms} I_{rms} \phi$$

The factor ϕ is called the **power factor** of the circuit.

(i) No power is dissipated when $\phi = 0$, i.e. $\phi = \frac{\pi}{2}$. This occurs for a purely reactive circuit (a pure inductor or a pure capacitor, $R = 0$). The current that flows is then a **wattless current**.

(ii) Maximum power is dissipated when $\phi = 1$, i.e. $\phi = 0$. This is the **resonance** condition where $X_L = X_C$, the impedance is minimum ($Z = R$), and $P_{max} = V_{rms} I_{rms} = \frac{V_{rms}^2}{R}$.

Final answer. $P_{avg} = V_{rms} I_{rms} \phi$; zero when $\phi = \frac{\pi}{2}$ (pure reactance), maximum when $\phi = 0$ (resonance).

← Back to Q126

Q127

In a series LCR circuit connected to an ac source of variable frequency and voltage $v = v_m \omega t$*, draw a plot showing the variation of current* (I) *with angular frequency* (ω) *for two different values of resistance* R_1 *and* R_2 $(R_1 > R_2)$*. Write the condition under which the phenomenon of resonance occurs. For which value of the resistance out of the two curves, a sharper resonance is produced? Define Q-factor of the circuit and give its significance.*

[2013]

SOLUTION

Graph: On the same axes plot current I (y-axis) versus angular frequency ω (x-axis) for two resistances. Both curves peak at the same resonant frequency $\omega_0 = \frac{1}{\sqrt{LC}}$. The curve for the smaller resistance R_2 is taller (peak $\frac{v_m/\sqrt{2}}{R_2}$) and narrower; the curve for the larger resistance R_1 is shorter and broader. Label the curves R_1 and R_2 with $R_1 > R_2$, and mark ω_0 on the x-axis.

Resonance condition: Resonance occurs when the inductive and capacitive reactances are equal, $X_L = X_C$

$$\Rightarrow \omega L = \frac{1}{\omega C}$$

$$\Rightarrow \omega_0 = \frac{1}{\sqrt{LC}}$$

At this frequency the impedance is minimum ($Z = R$) and the current is maximum.

Sharper resonance: The curve for R_2 (the smaller resistance) shows a sharper resonance, because the sharpness increases as the resistance decreases.

Q-factor: The quality factor is defined as $Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 C R} = \frac{1}{R}\sqrt{\frac{L}{C}}$

It is the ratio of the resonant frequency to the bandwidth, $Q = \frac{\omega_0}{\Delta\omega}$, where $\Delta\omega$ is the full width of the resonance curve at $\frac{1}{\sqrt{2}}$ of the peak current.

Significance: A high Q-factor means a sharp and narrow resonance curve, so the circuit is more selective and can clearly distinguish between two closely spaced frequencies. This **selectivity is essential in tuning circuits of radio and television receivers**.

Final answer. Resonance at $\omega_0 = \frac{1}{\sqrt{LC}}$; sharper for R_2; $Q = \frac{\omega_0 L}{R}$ measures sharpness/selectivity.

← *Back to Q127*

Q128

A series LCR circuit is connected to an ac source. Using the phasor diagram, derive the expression for the impedance of the circuit. Plot a graph to show the variation of current with frequency of the source, explaining the nature of its variation.

[2012]

SOLUTION

Phasor diagram: Take the current $\vec{I}$ as the reference phasor along the positive x-axis (the same current flows through all elements in series). Draw $\vec{V}_R$ along $\vec{I}$ (in phase, magnitude IR); draw $\vec{V}_L$ along the positive y-axis (leading $\vec{I}$ by 90°, magnitude IX_L); draw $\vec{V}_C$ along the negative y-axis (lagging $\vec{I}$ by 90°, magnitude IX_C). The net reactive voltage is $(\vec{V}_L - \vec{V}_C)$ along the y-axis. The resultant of $\vec{V}_R$ and $(\vec{V}_L - \vec{V}_C)$ is the applied voltage $\vec{V}$, making an angle ϕ with $\vec{I}$.

From the right-angled triangle of phasors, the magnitude of the applied voltage is $V = \sqrt{V_R^2 + (V_L - V_C)^2}$

Substituting $V_R = IR$, $V_L = IX_L$ and $V_C = IX_C$,

$$V = \sqrt{(IR)^2 + (IX_L - IX_C)^2}$$

$$\Rightarrow V = I\sqrt{R^2 + (X_L - X_C)^2}$$

The impedance is defined as $Z = \frac{V}{I}$, so

$$\Rightarrow Z = \sqrt{R^2 + (X_L - X_C)^2}$$

where $X_L = \omega L$ and $X_C = \frac{1}{\omega C}$.

Current vs frequency graph: Plot I against ω. The current $I = \frac{V}{Z}$ is small at low ω (large X_C) and at high ω (large X_L). As ω increases, $X_L - X_C$ decreases, Z falls, and I rises. At the resonant frequency $\omega_0 = \frac{1}{\sqrt{LC}}$, $X_L = X_C$ so $Z = R$ is minimum and the current is maximum, $I_{max} = \frac{V}{R}$. Beyond ω_0 the current falls again, giving a sharp peak at ω_0.

Final answer. $Z = \sqrt{R^2 + (X_L - X_C)^2}$; current peaks at resonance $\omega_0 = \frac{1}{\sqrt{LC}}$ where $Z = R$.

← *Back to Q128*

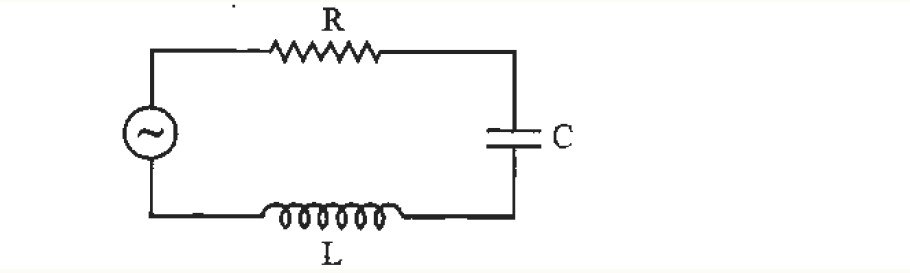
Q129

The figure shows a series LCR circuit with $L = 5.0\ H$, $C = 80\ \mu F$, $R = 40\ \Omega$ *connected to a variable frequency* 240 *V source. Calculate*

(i) The angular frequency of the source which drives the circuit at resonance.

(ii) The current at the resonating frequency.

(iii) The rms potential drop across the capacitor at resonance.



[2012]

SOLUTION

Given $L = 5.0\ H$, $C = 80\ \mu F = 80 \times 10^{-6}\ F = 8 \times 10^{-5}\ F$, $R = 40\ \Omega$ and $V_{rms} = 240\ V$.

(i) At resonance the angular frequency is $\omega_0 = \frac{1}{\sqrt{LC}}$

$$\Rightarrow \omega_0 = \frac{1}{\sqrt{5.0 \times 8 \times 10^{-5}}}$$

$$\Rightarrow \omega_0 = \frac{1}{\sqrt{4 \times 10^{-4}}} = \frac{1}{2 \times 10^{-2}}$$

$$\Rightarrow \boldsymbol{\omega_0 = 50 \text{ rad s}^{-1}}$$

(ii) At resonance the impedance is purely resistive, $Z = R$, so the current is $\mathbf{I_{rms}} = \frac{\mathbf{V_{rms}}}{R}$

$$\Rightarrow I_{rms} = \frac{240}{40}$$

$$\Rightarrow \boldsymbol{I_{rms} = 6 \text{ A}}$$

(iii) The capacitive reactance at resonance is $\mathbf{X_C} = \frac{1}{\omega_0 C}$

$$\Rightarrow X_C = \frac{1}{50 \times 8 \times 10^{-5}} = \frac{1}{4 \times 10^{-3}}$$

$$\Rightarrow X_C = 250\ \Omega$$

The rms potential drop across the capacitor is $\mathbf{V_C = I_{rms} X_C}$

$$\Rightarrow V_C = 6 \times 250$$

$\Rightarrow \boldsymbol{V_C = 1500 \text{ V}}$.

CONCEPT

Note $\mathbf{V_C = 1500}\ V$ far exceeds the source voltage of 240 V. At resonance the large reactive voltages across L and C are equal and opposite, so they cancel and the source still only sees R.

Final answer. $\boldsymbol{\omega_0 = 50 \text{ rad s}^{-1}}$, $\mathbf{I_{rms} = 6}\ A$ and $\mathbf{V_C = 1500}\ V$.

← *Back to Q129*

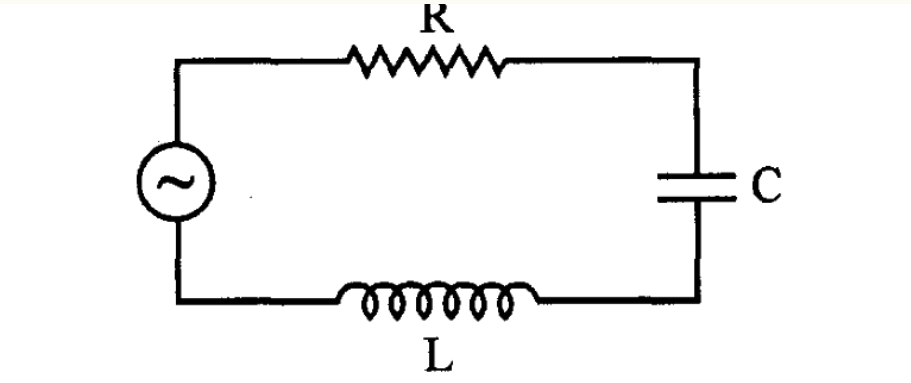
Q130

A series LCR circuit with $L = 4.0\ H$, $C = 100\ \mu F$ *and* $R = 60\ \Omega$ *is connected to a variable frequency* 240 *V source as shown. Calculate:*

(i) the angular frequency of the source which drives the circuit at resonance;

(ii) the current at the resonating frequency;

(iii) the rms potential drop across the inductor at resonance.



[2012]

SOLUTION

Given $L = 4.0\ H$, $C = 100\ \mu F = 100 \times 10^{-6}\ F = 10^{-4}\ F$, $R = 60\ \Omega$ and $V_{rms} = 240\ V$.

(i) At resonance the angular frequency is $\omega_0 = \frac{1}{\sqrt{LC}}$

$$\Rightarrow \omega_0 = \frac{1}{\sqrt{4.0 \times 10^{-4}}}$$

$$\Rightarrow \omega_0 = \frac{1}{2 \times 10^{-2}}$$

$$\Rightarrow \omega_0 = 50\ \text{rad s}^{-1}$$

(ii) At resonance the impedance is purely resistive, $Z = R$, so the current is $I_{rms} = \frac{V_{rms}}{R}$

$$\Rightarrow I_{rms} = \frac{240}{60}$$

$$\Rightarrow I_{rms} = 4\ \text{A}$$

(iii) The inductive reactance at resonance is $X_L = \omega_0 L$

$$\Rightarrow X_L = 50 \times 4.0$$

$$\Rightarrow X_L = 200\ \Omega$$

The rms potential drop across the inductor is $V_L = I_{rms} X_L$

$$\Rightarrow V_L = 4 \times 200$$

$\Rightarrow V_L = 800$ V.

Final answer. $\omega_0 = 50$ rad s^{-1}, $I_{rms} = 4$ *A* and $V_L = 800$ *V*.

← *Back to Q130*

Q131

An inductor 200 mH, *capacitor* 500 μ *F, resistor* 10 Ω *are connected in series with a* 100 *V, variable frequency a.c. source. Calculate the*
(i) frequency at which the power factor of the source is unity
(ii) current amplitude at this frequency
(iii) Q-factor

[2008]

SOLUTION

Here $L = 200$ mH $= 0.2$ *H*, $C = 500\ \mu F = 5 \times 10^{-4}$ *F*, $R = 10\ \Omega$ and the rms voltage of the source is $V_{rms} = 100$ *V*.

(i) The power factor of a series LCR circuit is $\phi = \frac{R}{Z}$. It becomes unity only when $Z = R$, that is at resonance, where the inductive and capacitive reactances cancel ($X_L = X_C$).

At resonance the angular frequency is $\omega_0 = \frac{1}{\sqrt{LC}}$.

$$\omega_0 = \frac{1}{\sqrt{0.2 \times 5 \times 10^{-4}}}$$

$$\omega_0 = \frac{1}{\sqrt{10^{-4}}} = \frac{1}{10^{-2}} = 100\ \text{rad/}s$$

The resonant frequency is $\mathbf{f_0 = \frac{\omega_0}{2\pi}}$.

$$f_0 = \frac{100}{2\pi} \approx 15.9 \text{ Hz}$$

(ii) At resonance the impedance is purely resistive, $\boldsymbol{Z = R = 10\ \Omega}$, so the current is maximum. The peak (amplitude) of the source voltage is $\mathbf{V_0 = \sqrt{2}\, V_{rms}}$.

$$V_0 = \sqrt{2} \times 100 = 100\sqrt{2}\ V$$

$$I_0 = \frac{V_0}{R} = \frac{100\sqrt{2}}{10}$$

$$I_0 = 10\sqrt{2} \approx 14.14\ A$$

(iii) The quality factor of the series resonant circuit is $\boldsymbol{Q = \frac{\omega_0 L}{R}}$.

$$Q = \frac{100 \times 0.2}{10} = \frac{20}{10}$$

$$Q = 2$$

Hence the circuit resonates near 16 Hz, draws a peak current of about 14.14 A there, and has a sharpness (Q-factor) of 2.

COMMON MISTAKE

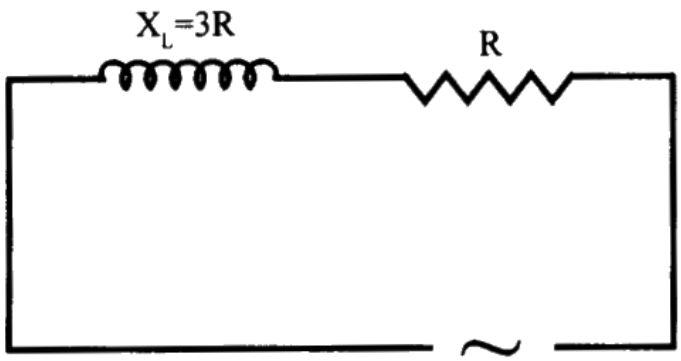
Power factor is unity ONLY at resonance, never just because the source is AC. Always check whether to use the peak or rms voltage; here 'current amplitude' means the peak value, so multiply the rms voltage by 2.

Final answer. At resonance $\mathbf{f_0 \approx 15.9\ Hz}$, current amplitude $\mathbf{I_0 \approx 14.14\ A}$, and Q-factor $\boldsymbol{Q = 2}$.

← *Back to Q131*

Q132

Given below are two electric circuits A and B.



Calculate the ratio of power factor of the circuit B to the power factor of circuit A.

[2007]

SOLUTION

The power factor of an AC circuit is $\phi = \frac{R}{Z}$, where Z is the impedance of that circuit. To compare the two circuits we find the impedance of each.

Let circuit A contain a resistance R in series with an inductive reactance X_L only. Its impedance is the phasor sum of R (along the reference axis) and X_L (at 90° to it).

$$Z_A = \sqrt{R^2+X_L{}^2}$$

Let circuit B contain the same resistance R in series with both an inductive reactance X_L and a capacitive reactance X_C. Since X_L leads and X_C lags the current by 90°, they act in opposite directions, so the net reactance is $(X_L - X_C)$.

$$Z_B = \sqrt{R^2+(X_L-X_C)^2}$$

The required ratio of power factors is

$$\frac{\Phi_B}{\Phi_A} = \frac{R/Z_B}{R/Z_A} = \frac{Z_A}{Z_B}$$

$$\frac{\Phi_B}{\Phi_A} = \frac{\sqrt{R^2+X_L{}^2}}{\sqrt{R^2+(X_L-X_C)^2}}$$

The numerical values of R, X_L and X_C must be read from the figure. As a worked illustration, if the added capacitor in B is chosen so that $\mathbf{X_C = X_L}$, then circuit B is at resonance and $\mathbf{(X_L - X_C) = 0}$, giving $\mathbf{Z_B = R}$.

$$\frac{\Phi_B}{\Phi_A} = \frac{\sqrt{R^2+X_L{}^2}}{R}$$

In this case $\Phi_B = 1$ (resonance) while $\Phi_A < 1$, so the ratio exceeds 1, confirming that adding a matched capacitor improves the power factor.

Thus the **ratio of power factors equals the inverse ratio of the impedances**, $\frac{PF_B}{PF_A} = \frac{Z_A}{Z_B}$, evaluated with the reactance values given in the figure.

Final answer. $\frac{PF_B}{PF_A} = \frac{Z_A}{Z_B} = \frac{\sqrt{R^2+X_L{}^2}}{\sqrt{R^2+(X_L-X_C)^2}}$, using the figure's reactance values.

← *Back to Q132*

Q133

In a series R-C circuit, $R = 30\ \Omega$, $C = 0.25\ \mu F$, $V = 100\ V$ *and* $\omega = 10{,}000$ *radian/second.*
Find the current in the circuit and calculate the voltage across the resistor and the capacitor. Is the algebraic sum of these voltages more than the source voltage? If yes, resolve the paradox.

[2004]

SOLUTION

Given $R = 30\ \Omega$, $C = 0.25\ \mu F = 0.25\times 10^{-6}\ F$, source voltage $V = 100\ V$ and $\omega = 10000\ rad/s$.

The capacitive reactance is $X_C = \frac{1}{\omega C}$.

$$X_C = \frac{1}{10000\times 0.25\times 10^{-6}}$$

$$X_C = \frac{1}{2.5\times 10^{-3}} = 400\ \Omega$$

In a series R-C circuit the impedance is the phasor sum of R and X_C.

$$Z = \sqrt{R^2+X_C{}^2}$$

$$Z = \sqrt{(30)^2+(400)^2} = \sqrt{900+160000}$$

$$Z = \sqrt{160900} \approx 401\ \Omega$$

The current in the circuit is

$$I = \frac{V}{Z} = \frac{100}{401} \approx 0.249\ A$$

Voltage across the resistor:

$$\mathrm{V}_R = I\,R = 0.249\times 30 \approx 7.48\ V$$

Voltage across the capacitor:

$$\mathrm{V}_C = I\,\mathrm{X}_C = 0.249\times 400 \approx 99.6\ V$$

Algebraic (simple) sum of these voltages:

$$\mathrm{V}_R + \mathrm{V}_C \approx 7.48 + 99.6 = 107.1\ V$$

This is greater than the source voltage of 100 V, which seems to violate Kirchhoff's voltage law.

Resolving the paradox: V_R and V_C are not in the same phase. In a series R-C circuit the voltage across the resistor is in phase with the current, while the voltage across the capacitor lags the current by 90°. Hence V_R and V_C are themselves 90° out of phase, and they cannot be added algebraically. They must be added as phasors (vectors).

$$V = \sqrt{\mathrm{V}_R^{\,2}+\mathrm{V}_C^{\,2}}$$

$$V = \sqrt{(7.48)^2+(99.6)^2} = \sqrt{56+9920}$$

$$V = \sqrt{9976} \approx 99.9 \approx 100\ V$$

The phasor sum equals the source voltage of 100 V. So there is **no real paradox**: the two voltages peak at different instants, and only their phasor sum (not the algebraic sum) is bound by Kirchhoff's law.

CONCEPT

In AC circuits voltages across series elements add as phasors, never algebraically, because they peak at different times. This is exactly why $\mathrm{V}_R + \mathrm{V}_C$ can look larger than the source.

Final answer. $I \approx 0.249\ A$, $V_R \approx 7.48\ V$, $V_C \approx 99.6\ V$; the algebraic sum ($\approx 107\ V$) exceeds the source, but the phasor sum $\sqrt{V_R^2+V_C^2} \approx 100\ V$ equals it.

← *Back to Q133*

Q134

The teachers of Geeta's school took the students on a study trip to a power generating station, located nearly 200 km *away from the city. The teacher explained that electrical energy is transmitted over such a long distance to their city, in the form of alternating current (ac) raised to a high voltage. At the receiving end in the city, the voltage is reduced to operate the devices. As a result, the power loss is reduced. Geeta listened to the teacher and asked questions about how the ac is converted to a higher or lower voltage.*

(a) Name the device used to change the alternating voltage to a higher or lower value. State one cause for power dissipation in this device.

(b) Explain with an example, how power loss is reduced if the energy is transmitted over long distances as an alternating current rather than a direct current.

(c) Write two values each shown by the teachers and Geeta.

[2018]

SOLUTION

(a) The device used to step the alternating voltage up or down is the **transformer**. It works on the principle of mutual induction between two coils wound on a common soft-iron core.

One cause of power dissipation in a transformer is copper loss, the I^2R heating in the resistance of the primary and secondary windings. (Other acceptable causes are iron/eddy-current losses and hysteresis loss in the core, or flux leakage.)

(b) Power loss in transmission lines is reduced by sending power at high voltage. The power delivered is

$$P = V\,I$$

so for a fixed power P, the line current is $I = \frac{P}{V}$. A higher transmission voltage therefore means a smaller current. The power wasted as heat in the line of resistance R is

$$P_{loss} = I^2 R$$

which depends on the square of the current.

Worked example: suppose $P = 100\ kW$ is to be sent through a line of resistance $R = 10\ \Omega$. If it is sent at 1000 V, the current is $I = \frac{100000}{1000} = 100\ A$ and the loss is $P_{loss} = (100)^2 \times 10 = 10^5\ W = 100\ kW$ (the whole power is lost). If instead it is sent at 100000 V, the current is only $I = \frac{100000}{100000} = 1\ A$ and the loss is $P_{loss} = (1)^2 \times 10 = 10\ W$ only.

AC can be stepped up to very high voltage at the generating station (and stepped down near the city) easily and cheaply using transformers, which work only on AC, not DC. That is why energy is transmitted as high-voltage AC.

(c) Values shown:

Teachers: awareness and concern for students' learning, ability to relate textbook physics to real life, encouraging scientific temper.

Geeta: curiosity and eagerness to learn, courage to ask questions, attentiveness.

Final answer. (a) Transformer; copper (I^2R) loss in windings. (b) High voltage means low current, so I^2R line loss falls sharply, as the worked example shows. (c) Teachers: concern for learning, scientific awareness; Geeta: curiosity, willingness to question.

← *Back to Q134*

Q135

Shiv had a high tension tower erected on his farm land. He kept complaining to the authorities to remove it since it occupied a large portion of his land. His uncle, who was a teacher, explained to him the need for erecting these towers for efficient transmission of power. As Shiv got convinced and realized its significance, he stopped

complaining.

Based on the above paragraph, answer the following questions:

(a) Why is it necessary to transport power at high voltages?

(b) 'A low power factor implies large power loss'. Explain.

(c) Write the two values displayed by Shiv and his Uncle.

[2016]

SOLUTION

(a) Power is transmitted at high voltage to minimise energy lost as heat in the transmission lines. The power delivered is $P = VI$, so for a fixed power the line current is

$$I = \frac{P}{V}$$

The power dissipated in a line of resistance R is

$$P_{loss} = I^2 R = \left(\frac{P}{V}\right)^2 R = \frac{P^2 R}{V^2}$$

Since $P_{loss} \propto \frac{1}{V^2}$, raising the transmission voltage sharply reduces the current and hence the heat loss. So the voltage is stepped up for transmission and stepped down near the consumer.

(b) The average power consumed in an AC circuit is

$$P = V I \phi$$

where ϕ is the power factor. For a required useful power P delivered at supply voltage V, the current that must flow is

$$I = \frac{P}{V \phi}$$

A low power factor (small ϕ) forces a larger current I to deliver the same useful power. The line loss is

$$P_{loss} = I^2 R = \frac{P^2 R}{V^2 \cos^2 \phi}$$

so $P_{loss} \propto \frac{1}{\cos^2 \phi}$. A smaller power factor therefore means a much larger heat loss in the lines. Hence **a low power factor implies a large power loss**.

(c) Values displayed:

Shiv: open-mindedness, willingness to listen and learn, putting public interest above personal inconvenience.

Uncle (teacher): scientific knowledge and awareness, patience in explaining, social responsibility.

Final answer. (a) High voltage gives low current, so $\mathbf{I^2R}$ loss $\propto \mathbf{1/V^2}$ falls. (b) Low $\boldsymbol{\phi}$ needs more current ($I \propto 1/\phi$), making loss $\propto \mathbf{1/\cos^2\phi}$ large. (c) Shiv: open-mindedness, public spirit; Uncle: knowledge, social responsibility.

← *Back to Q135*

Q136

During a thunderstorm the 'live' wire of the transmission line fell down on the ground. A group of boys passing through noticed it and some of them wanted to place the wire by the side. As they were approaching the wire and trying to lift it, Hari noticed it and immediately pushed them away to prevent them from touching the wire. Two of them got hurt in the process. Hari took them to a doctor to get medical aid.
Based on the above paragraph, answer the following:
(a) Write two values which Hari displayed during the incident.
(b) Why is it that a bird can sit over a suspended 'live' wire without any harm whereas touching it on the ground can give a fatal shock?
(c) The electric power from a power plant is set up to a very high voltage before transmitting it to distant consumers. Write the reason for it.

[2016]

SOLUTION

(a) Values displayed by Hari: presence of mind and quick thinking, a strong sense of responsibility and concern for the safety of others, and helpfulness (taking the injured boys for medical aid).

(b) A bird sitting on a single live wire touches only that one wire. Its two feet are at almost the same point on the wire, so there is practically no potential difference across its body.

$$V = I\,R_{\text{body}}$$

With $V \approx 0$ across the bird, the current through it is $I \approx 0$, so it is unharmed. The bird does not provide a path to any point at a different potential (it does not touch the ground or another wire).

A person standing on the ground is at (or near) the earth's potential, taken as zero. If such a person touches the high-voltage wire, a large potential difference (equal to the line voltage) is set up across the body between the wire and the ground. This large V drives a large current through the body to earth,

$$I = \frac{V}{R_{\text{body}}}$$

which can be fatal. So the danger comes from completing a path between the high-voltage wire and the ground.

(c) Electric power is stepped up to a very high voltage before transmission to reduce the energy lost as heat in the lines. For a fixed power $P = V\,I$, a higher voltage means a smaller current $I = \frac{P}{V}$. The line loss is

$$P_{\text{loss}} = I^2 R = \frac{P^2 R}{V^2}$$

which decreases as $1/V^2$. So transmitting at high voltage (low current) **greatly reduces I^2R losses**; the voltage is later stepped down near consumers using transformers.

Final answer. (a) Hari showed presence of mind and concern for others' safety. (b) A bird has no PD across it, so no current; a grounded person creates a large PD to earth, driving a fatal current. (c) High voltage means low current, cutting I^2R loss ($\propto 1/V^2$).

← *Back to Q136*

Q137

(a) A series combination of L, C and R is connected to an a.c. source. Using a phasor diagram, derive an expression for the impedance of the circuit and phase difference

between V and I.

[2026 • Set 55-2-1]

SOLUTION

Consider a resistor R, an inductor L and a capacitor C joined in series across an AC source. Since they are in series, the same current flows through all three at every instant. Take this current as the reference phasor:

$$i = \mathrm{i}_0\, \omega\, t$$

drawn along the positive x-axis.

The voltage across each element is:

$\mathbf{V_R = i_0\, R}$, in phase with the current (phasor along +x-axis).

$\mathbf{V_L = i_0\, X_L}$, leading the current by 90° (phasor along +y-axis), where $\mathbf{X_L = \omega\, L}$.

$\mathbf{V_C = i_0\, X_C}$, lagging the current by 90° (phasor along −y-axis), where $\mathbf{X_C} = \dfrac{\mathbf{1}}{\boldsymbol{\omega\, C}}$.

Phasor diagram description: along the +x-axis draw $\vec{\mathrm{V}}_R$ (and the current i). From the origin draw $\vec{\mathrm{V}}_L$ vertically upward and $\vec{\mathrm{V}}_C$ vertically downward. Since V_L and V_C are opposite, their resultant along the y-axis is $(\mathrm{V}_L - \mathrm{V}_C)$ (taking $\mathbf{X_L} > \mathbf{X_C}$). The applied voltage $\vec{V}$ is the diagonal of the rectangle formed by $\vec{\mathrm{V}}_R$ (x-axis) and $(\vec{\mathrm{V}}_L - \vec{\mathrm{V}}_C)$ (y-axis), making an angle ϕ with the current.

By the parallelogram (Pythagoras) law, the magnitude of the applied voltage is

$$V = \sqrt{\mathrm{V}_R^{\,2} + (\mathrm{V}_L - \mathrm{V}_C)^2}$$

Substituting $\mathbf{V_R = i_0\, R}$, $\mathbf{V_L = i_0\, X_L}$ and $\mathbf{V_C = i_0\, X_C}$,

$$V = \sqrt{(\mathrm{i}_0\, R)^2 + (\mathrm{i}_0\, \mathrm{X}_L - \mathrm{i}_0\, \mathrm{X}_C)^2}$$

$$V = \mathrm{i}_0 \sqrt{\mathrm{R}^2 + (\mathrm{X}_L - \mathrm{X}_C)^2}$$

Comparing with $\boldsymbol{V = i_0\, Z}$, the impedance of the circuit is

$$Z = \sqrt{\mathrm{R}^2 + (\mathrm{X}_L - \mathrm{X}_C)^2} = \sqrt{\mathrm{R}^2 + \left(\omega\, L - \frac{1}{\omega\, C}\right)^2}$$

From the phasor diagram, the phase angle ϕ between the applied voltage and the current is

$$\phi = \frac{V_L - V_C}{V_R} = \frac{i_0(X_L - X_C)}{i_0 R}$$

$$\phi = \frac{X_L - X_C}{R} = \frac{\omega L - \dfrac{1}{\omega C}}{R}$$

If $X_L > X_C$, ϕ is positive and the voltage leads the current (circuit is inductive). If $X_L < X_C$, ϕ is negative and the voltage lags the current (circuit is capacitive). If $X_L = X_C$, then $\phi = 0$ and the circuit is purely resistive (resonance).

TIP

Always state which phasor is the reference. For a series circuit take the common current as reference; for a parallel circuit take the common voltage. Mixing this up is the commonest error in phasor derivations.

Final answer. $Z = \sqrt{R^2 + (X_L - X_C)^2}$ and the phase difference is $\phi = \dfrac{X_L - X_C}{R}$.

← *Back to Q137*

Q138

(b) Under what conditions:
(i) the impedance of the circuit is minimum?
(ii) wattless current flows in the circuit?

[2026 • Set 55-2-1]

SOLUTION

(i) The impedance of a series LCR circuit is

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

Here R is fixed, so Z is smallest when the reactive term $(X_L - X_C)$ vanishes, that is when

$$X_L = X_C$$

$$\omega L = \frac{1}{\omega C}$$

$$\omega = \frac{1}{\sqrt{LC}}$$

This is the condition of resonance. At resonance the impedance is minimum and equals the resistance:

$$Z_{min} = R$$

The current is then maximum and the circuit behaves as purely resistive.

(ii) A wattless current is a current that consumes no average power. The average power in an AC circuit is

$$P = V_{rms}\, I_{rms}\, \phi$$

For the power to be zero (wattless current) while the current still flows, the power factor must be zero:

$$\phi = 0 \Rightarrow \phi = \frac{\pi}{2} = 90^\circ$$

This happens when the circuit is purely reactive, that is it contains only an inductor or only a capacitor (resistance $R = 0$). In that case the current is 90° out of phase with the voltage, so over a full cycle the net power absorbed is zero even though current flows. **Such a current is called the wattless (idle) current.**

Final answer. (i) Impedance is minimum ($Z_{min} = R$) at resonance, when $X_L = X_C$, i.e. $\omega = 1/\sqrt{LC}$. (ii) Wattless current flows when $\phi = 0$ ($\phi = 90^\circ$), i.e. a purely reactive circuit with $R = 0$.

← *Back to Q138*

Q139

(i) A light bulb and an open coil inductor are connected in series across an ac source of variable frequency. How will the glow of the bulb be affected when:
(I) an iron bar is inserted inside the coil, and
(II) the frequency of the source is decreased?
Justify your answers. Assume that in each above case other factors remain unchanged.

[2026 • Set 55-3-1]

SOLUTION

The bulb (resistance R) and the open coil inductor ($\mathbf{X_L = \omega L}$) are in series across the AC source, so the same current flows through both. The current is

$$I = \frac{V}{\sqrt{R^2 + X_L{}^2}} = \frac{V}{\sqrt{R^2 + (\omega L)^2}}$$

The brightness (glow) of the bulb depends on the power it dissipates, $\mathbf{P = I^2 R}$, and hence on the current I through it. Anything that increases I makes the bulb brighter; anything that decreases I makes it dimmer.

(I) Inserting an iron bar inside the coil: iron is a ferromagnetic material with high relative permeability, so it greatly increases the inductance L of the coil. Then

$$X_L = \omega L \uparrow$$

$$Z = \sqrt{R^2 + X_L{}^2} \uparrow$$

$$I = \frac{V}{Z} \downarrow$$

The current decreases, so the **bulb glows dimmer** (its glow decreases).

(II) Decreasing the frequency of the source: the inductive reactance is $\mathbf{X_L = \omega L = 2\pi f L}$, which is directly proportional to the frequency. Lowering f decreases X_L:

$$X_L = 2\pi f L \downarrow$$

$$Z = \sqrt{R^2 + X_L{}^2} \downarrow$$

$$I = \frac{V}{Z} \uparrow$$

The current increases, so the **bulb glows brighter** (its glow increases).

Final answer. (I) Inserting the iron bar raises $\mathbf{L}$ and $\mathbf{X_L}$, lowering the current, so the bulb glows dimmer. (II) Lowering the frequency lowers $\mathbf{X_L}$, raising the current, so the bulb glows brighter.

← Back to Q139

Q140

(ii) An ac voltage $V = 280\sin(100\pi t)$ *volt is connected across a series LCR circuit in which* $R = 400\ \Omega$, $L = \frac{5}{\pi}\ H$ *and* $C = \frac{1}{\pi}\ \mu F$. *Taking* $\sqrt{2} = 1{\cdot}4$, *calculate:*

(I) impedance of the circuit.

(II) rms value of current that flows in the circuit.

(III) power factor of the circuit.

[2026 • Set 55-3-1]

SOLUTION

Comparing $V = 280\sin(100\pi t)$ with $V = V_0\,\omega\,t$, the peak voltage is $V_0 = 280\ V$ and the angular frequency is $\omega = 100\pi\ rad/s$. Also $R = 400\ \Omega$, $L = \frac{5}{\pi}\ H$ and $C = \frac{1}{\pi}\ \mu F$ $= \frac{1}{\pi} \times 10^{-6}\ F$.

Inductive reactance:

$$X_L = \omega L = 100\pi \times \frac{5}{\pi}$$

$$X_L = 500\ \Omega$$

Capacitive reactance:

$$X_C = \frac{1}{\omega C} = \frac{1}{100\pi \times \frac{1}{\pi} \times 10^{-6}}$$

$$X_C = \frac{1}{100 \times 10^{-6}} = \frac{1}{10^{-4}} = 10^4\ \Omega = 10000\ \Omega$$

Net reactance (here $X_C > X_L$):

$$X_C - X_L = 10000 - 500 = 9500\ \Omega$$

(I) Impedance of the circuit:

$$Z = \sqrt{R^2 + (X_C - X_L)^2}$$

$$Z = \sqrt{(400)^2 + (9500)^2}$$

$$Z = \sqrt{160000 + 90250000} = \sqrt{90410000}$$

$$Z \approx 9508\ \Omega$$

(II) The rms value of the source voltage is

$$V_{rms} = \frac{V_0}{\sqrt{2}} = \frac{280}{1.4} = 200\ V$$

The rms current is

$$I_{rms} = \frac{V_{rms}}{Z} = \frac{200}{9508}$$

$$I_{rms} \approx 0.021\ A = 21\ \text{mA}$$

(III) Power factor of the circuit:

$$\phi = \frac{R}{Z} = \frac{400}{9508}$$

$$\phi \approx 0.042$$

Thus the circuit has a high impedance (**≈ 9508 Ω**) because it is far from resonance, carries a small current (**≈ 0.021 *A***), and has a very **low power factor of about 0.042**.

COMMON MISTAKE

Watch the units on C: here $C = \frac{1}{\pi}$ **μF**, *so you must use* $\frac{1}{\pi} \times 10^{-6}$ *F. Forgetting the* 10^{-6} *makes* X_C *a million times too small and ruins the whole answer.*

Final answer. (I) $Z \approx 9508\ \Omega$, (II) $I_{rms} \approx 0.021\ A$, (III) power factor $\phi = R/Z \approx 0.042$.

← *Back to Q140*

Q141

(a)

(i) Define the terms (I) resonant frequency, and (II) power factor of a series LCR circuit. For what value of the power factor will the power dissipated in the circuit be maximum?

(ii) An inductor of 5 H, a capacitor of 50 μ F and a resistor of 400 Ω are connected in series across an ac voltage $v = 140 \sin(100\pi t)$ *V. Calculate:*

(I) impedance of the circuit, and

(II) rms value of current that flows in the circuit.

(Take $\sqrt{2} = 1 \cdot 4$*)*

OR

(b)

(i) Draw a labelled diagram of a step-up transformer. Obtain the ratio of secondary voltage to primary voltage in terms of number of turns in the two coils.

(ii) The number of turns in the primary and the secondary coil of an ideal transformer are 100 and 5000 respectively. If 3·3 kW power is supplied to the transformer at 220 V, find (I) current in the primary coil, and (II) output voltage.

[2026 • Set 55-4-1]

SOLUTION

(a)(i)(I) Resonant frequency: A series LCR circuit is said to be in resonance when the inductive reactance equals the capacitive reactance, so that the impedance becomes purely resistive and minimum, and the current in the circuit becomes maximum. The frequency at which this happens is called the resonant frequency.

At resonance, $X_L = X_C$, which gives

$$\omega_0 L = \frac{1}{\omega_0 C} \Rightarrow \omega_0 = \frac{1}{\sqrt{LC}}$$

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

(II) Power factor: The power factor of an ac circuit is the ratio of the resistance to the impedance of the circuit, and equals the cosine of the phase angle ϕ between the voltage and the current.

$$\phi = \frac{R}{Z}$$

The power dissipated in the circuit, $P = V_{rms} I_{rms} \phi$, is maximum when $\phi = 1$, that is when the impedance is purely resistive ($Z = R$), which occurs at resonance.

(a)(ii) Here $L = 5\ H$, $C = 50\ \mu F = 50\times 10^{-6}\ F$, $R = 400\ \Omega$ and $v = 140\sin(100\pi t)\ V$, so $\omega = 100\pi\ rad/s$.

The inductive reactance is

$$X_L = \omega L = 100\pi\times 5 = 500\pi = 1571\ \Omega$$

The capacitive reactance is

$$X_C = \frac{1}{\omega C} = \frac{1}{100\pi\times 50\times 10^{-6}} = \frac{1}{5\pi\times 10^{-3}} = 63.7\ \Omega$$

So $X_L - X_C = 1571 - 63.7 = 1507.3\ \Omega$.

(I) The impedance is

$$Z = \sqrt{R^2+(X_L-X_C)^2} = \sqrt{(400)^2+(1507.3)^2}$$

$$Z = \sqrt{160000+2271953} = \sqrt{2431953} \approx 1559\ \Omega$$

(II) The rms voltage of the source is

$$V_{rms} = \frac{V_0}{2} = \frac{140}{1.4} = 100\ V$$

The rms current is

$$I_{rms} = \frac{V_{rms}}{Z} = \frac{100}{1559} \approx 0.064\ A$$

Hence the impedance is about **1559 Ω** and the rms current is about **0.064 A**.

OR

(b)(i) Step-up transformer diagram: Draw a rectangular laminated soft-iron core (shown as a hollow rectangle made of thin shaded strips). On the left limb wind the primary coil with a small number of turns N_p and connect its two ends to an ac input source labelled V_p. On the right limb wind the secondary coil with a larger number of turns N_s ($N_s > N_p$) and connect its two ends to the output labelled V_s. Mark the changing flux Φ as arrows looping around the core through both coils.

Since the same flux Φ links every turn of both coils, the emf induced per turn is the same in both. For the secondary,

$$V_s = -N_s \frac{d\Phi}{dt}$$

and for the primary,

$$V_p = -N_p \frac{d\Phi}{dt}$$

Dividing the two equations,

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

Since $N_s > N_p$ for a step-up transformer, $V_s > V_p$.

(b)(ii) Here $N_p = 100$, $N_s = 5000$, input power $P = 3.3\ kW = 3300\ W$ and $V_p = 220\ V$.

(I) For an ideal transformer all the input power equals $V_p\, I_p$, so the primary current is

$$I_p = \frac{P}{V_p} = \frac{3300}{220} = 15\ A$$

(II) The output (secondary) voltage is

$$V_s = V_p \times \frac{N_s}{N_p} = 220 \times \frac{5000}{100} = 220 \times 50 = 11000\ V$$

Hence the primary current is **15 A** and the output voltage is **11000 V = 11 kV**.

COMMON MISTAKE

In numerical LCR problems always convert the capacitance to farads ($\mu F = 10^{-6}\ F$) before computing X_C; a missed power of ten is the single most common marks-losing slip.

Final answer. (a) Z ≈ 1559 Ω, I_rms ≈ 0.064 A; OR (b) I_p = 15 A, V_s = 11 kV.

← Back to Q141

Q142

(a) (i) Write the principle of working of an ac generator. Draw its labelled diagram and explain its working.

(ii) A resistor of 400 Ω, *an inductor of* $(\frac{2}{\pi})H$ *and a capacitor of* $(\frac{50}{\pi})$ μF *are joined in series across an ac source* $v = 140\sin(100\pi)t$ *V. Find the rms voltages across these three circuit elements. The algebraic sum of these voltages is more than the rms voltage of source. Explain.*

OR

(b) (i) Write the principle of working of a transformer. With the help of a labelled diagram, explain the working of a step-up transformer.

(ii) An ideal transformer is designed to convert 50 *V into* 250 *V. It draws* 200 *W power from an ac source whose instantaneous voltage is given by* $v_i = 20\sin(100\pi)t$ *V.*

Find:

(I) rms value of input current.

(II) expression for instantaneous output voltage.

(III) expression for instantaneous output current.

[2025 • Set 55-6-1]

SOLUTION

(a)(i) Principle of an ac generator: An ac generator works on the principle of electromagnetic induction. When a coil is rotated in a uniform magnetic field, the magnetic flux linked with the coil changes continuously, so an alternating emf is induced in it.

Labelled diagram: Draw a rectangular coil ABCD placed between the curved pole pieces N and S of a permanent magnet, with the field $\vec{B}$ pointing from N to S. The coil is mounted on an axle and can rotate about a horizontal axis. The two ends of the coil are connected to two separate slip rings R_1 and R_2, which press against two carbon brushes B_1 and B_2. The brushes are connected through an external load resistance R to complete the circuit.

Working: As the coil rotates with angular speed ω, the angle between the area vector of the coil and the field changes as $\theta = \omega t$. The flux linked with the coil of N turns and area A is

$$\Phi = NBA\omega\, t$$

By Faraday's law the induced emf is

$$\varepsilon = -\frac{d\Phi}{dt} = NBA\omega\omega\, t = \varepsilon_0\omega\, t$$

where $\boldsymbol{\varepsilon_0 = NBA\omega}$. The emf reverses its direction every half rotation, so the slip rings deliver an alternating emf to the external circuit.

(a)(ii) Here $\boldsymbol{R = 400\ \Omega}$, $\boldsymbol{L = \frac{2}{\pi}\ H}$, $\boldsymbol{C = \frac{50}{\pi}\ \mu F}$ and $\boldsymbol{v = 140\sin(100\pi\, t)\ V}$, so $\boldsymbol{\omega = 100\pi\ rad/s}$.

The inductive reactance is

$$X_L = \omega\, L = 100\pi \times \frac{2}{\pi} = 200\ \Omega$$

The capacitive reactance is

$$X_C = \frac{1}{\omega\, C} = \frac{1}{100\pi \times \frac{50}{\pi} \times 10^{-6}} = \frac{1}{5 \times 10^{-3}} = 200\ \Omega$$

Since $\boldsymbol{X_L = X_C = 200\ \Omega}$, the circuit is at resonance, so the impedance equals the resistance,

$$Z = R = 400\ \Omega$$

The rms source voltage is

$$V_{rms} = \frac{V_0}{2} = \frac{140}{2} = 100\ V$$

The rms current is

$$I_{rms} = \frac{V_{rms}}{Z} = \frac{100}{400} = 0.25\ A$$

The rms voltages across the three elements are

$$V_R = I_{rms} R = 0.25 \times 400 = 100\ V$$

$$V_L = I_{rms} X_L = 0.25 \times 200 = 50\ V$$

$$V_C = I_{rms} X_C = 0.25 \times 200 = 50\ V$$

Explanation: The simple algebraic sum is $\boldsymbol{V_R + V_L + V_C = 100 + 50 + 50 = 200\ V}$, which is more than the source rms voltage of 100 V. This is not a contradiction because the three voltages are not in phase. V_L leads the current by $\pi/2$ while V_C lags it by $\pi/2$, so V_L and V_C are exactly π out of phase and cancel each other. The voltages must be added as phasors, not algebraically:

$$V = \sqrt{V_R^2 + (V_L - V_C)^2} = \sqrt{(100)^2 + (50-50)^2} = 100\ V$$

which correctly equals the source voltage.

OR

(b)(i) Principle of a transformer: A transformer works on the principle of mutual induction. An alternating current in the primary coil produces a changing magnetic flux in the core, which links the secondary coil and induces an alternating emf in it.

Step-up transformer diagram and working: Draw a closed laminated soft-iron core. Wind the primary coil (N_p turns) on one limb connected to an ac input V_p, and the secondary coil (N_s turns, with $\mathbf{N_s > N_p}$) on the other limb connected to the output V_s. The alternating current in the primary sets up a changing flux Φ in the core. The same flux links each turn of both coils, so

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

Since $\mathbf{N_s > N_p}$, the output voltage is stepped up.

(b)(ii) Here input is converted from 50 V to 250 V, so the turns ratio is $\frac{\mathbf{N_s}}{\mathbf{N_p}} = \frac{\mathbf{250}}{\mathbf{50}} = \mathbf{5}$. Input power $\boldsymbol{P = 200\ W}$ and $\mathbf{v_i = 20\sin(100\pi\ t)\ V}$, so $\mathbf{V_{0,in} = 20\ V}$ and $\boldsymbol{\omega = 100\pi\ rad/s}$.

(I) The rms input voltage is

$$V_{p,\text{rms}} = \frac{20}{2} = 102 = 14.14\ V$$

The rms input current is

$$I_{p,\text{rms}} = \frac{P}{V_{p,\text{rms}}} = \frac{200}{14.14} = 14.14\ A$$

(II) The instantaneous output voltage is stepped up by the turns ratio 5:

$$v_o = 5\times v_i = 5\times 20\sin(100\pi\ t) = 100\sin(100\pi\ t)\ V$$

(III) For an ideal transformer $\mathbf{V_p I_p = V_s I_s}$, so the output current is stepped down by the same factor 5. The peak input current is $\mathbf{I_{0,in} = 2\times 14.14 = 20\ A}$, so the peak output current is

$$I_{0,\text{out}} = \frac{20}{5} = 4\ A$$

Since voltage steps up, the current is in the same time-phase, so the instantaneous output current is

$$i_o = 4\sin(100\pi\ t)\ A$$

Hence the output voltage is **$100\sin(100\pi\ t)$ V** and the output current is **$4\sin(100\pi\ t)$ A**.

Final answer. (a) V_R = 100 V, V_L = 50 V, V_C = 50 V (phasor sum = source 100 V); OR (b) I_in = 14.14 A, v_o = 100 sin(100πt) V, i_o = 4 sin(100πt) A.

← *Back to Q142*

Q143

(i) Mention the factors on which the resonant frequency of a series LCR circuit depends. Plot a graph showing variation of impedance of a series LCR circuit with the frequency of the applied a.c. source.
(ii) With the help of a suitable diagram, explain the working of a step-up transformer.
(iii) Write two causes of energy loss in a real transformer.

[2024 • Set 55-4-1]

SOLUTION

(i) The resonant frequency of a series LCR circuit is

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

It depends only on the inductance L and the capacitance C of the circuit. It does not depend on the resistance R or on the applied voltage.

Graph of impedance versus frequency: The impedance is

$$Z = \sqrt{R^2+(\omega\ L - \frac{1}{\omega\ C})^2}$$

Plot Z on the vertical axis against frequency f on the horizontal axis. The curve is U-shaped (a dip). It starts at a high value when f is small (because X_C is very

large), falls to a minimum value $Z = R$ at the resonant frequency f_0, and then rises again as f increases (because X_L becomes large). Mark the lowest point of the dip at $f = f_0$ with $Z = R$.

(ii) Working of a step-up transformer: Draw a closed laminated soft-iron core. On one limb wind the primary coil of N_p turns connected to the ac input V_p; on the other limb wind the secondary coil of N_s turns ($N_s > N_p$) connected to the output V_s. Show the looping flux Φ through the core.

When an alternating voltage is applied to the primary, an alternating current flows and sets up a continuously changing magnetic flux Φ in the soft-iron core. This same flux links each turn of the secondary coil, so by mutual induction an alternating emf is induced in it. As every turn of both coils is linked by the same flux,

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

Since $N_s > N_p$, the secondary voltage is greater than the primary voltage, so the voltage is stepped up while the current is correspondingly stepped down.

(iii) Two causes of energy loss in a real transformer:

1. Copper loss: heat (I^2R) generated in the resistance of the primary and secondary windings.

2. Eddy current loss: circulating currents induced in the iron core dissipate energy as heat; this is reduced by using a laminated core.

Final answer. $f_0 = 1/(2\pi\sqrt{(LC)})$ depends only on L and C; impedance-frequency graph is U-shaped with minimum $Z = R$ at f_0; two losses are copper (I^2R) loss and eddy current loss.

← *Back to Q143*

Q144

(i) You are given three circuit elements X, Y and Z. They are connected one by one across a given ac source. It is found that V and I are in phase for element X. V leads I by ($\frac{\pi}{2}$) for element Y while I leads V by ($\frac{\pi}{2}$) for element Z. Identify elements X, Y and

Z.

(ii) Establish the expression for impedance of circuit when elements X, Y and Z are connected in series to an ac source. Show the variation of current in the circuit with the frequency of the applied ac source.

(iii) In a series LCR circuit, obtain the conditions under which (i) impedance is minimum and (ii) wattless current flows in the circuit.

[2024 • Set 55-5-1]

SOLUTION

(i) Identification of the elements:

For element X, the voltage V and current I are in phase. Only a pure resistor gives zero phase difference, so X is a resistor.

For element Y, the voltage leads the current by $\frac{\pi}{2}$. In a pure inductor the voltage leads the current by $\frac{\pi}{2}$, so Y is an inductor.

For element Z, the current leads the voltage by $\frac{\pi}{2}$. In a pure capacitor the current leads the voltage by $\frac{\pi}{2}$, so Z is a capacitor.

(ii) Impedance of the series R, L, C combination: When X (R), Y (L) and Z (C) are in series, the same current I flows through all three. Using a phasor diagram, take the current along the reference direction. The voltage across R, $\mathbf{V_R = IR}$, is in phase with I. The voltage across L, $\mathbf{V_L = IX_L}$, leads I by $\frac{\pi}{2}$ (drawn upward). The voltage across C, $\mathbf{V_C = IX_C}$, lags I by $\frac{\pi}{2}$ (drawn downward). Since V_L and V_C are opposite, their resultant is $(V_L - V_C)$.

The applied voltage is the phasor sum,

$$V = \sqrt{V_R^2 + (V_L - V_C)^2}$$

$$V = \sqrt{(IR)^2 + (IX_L - IX_C)^2}$$

$$V = I\sqrt{R^2 + (X_L - X_C)^2}$$

Hence the impedance is

$$Z = \frac{V}{I} = \sqrt{R^2 + (X_L - X_C)^2}$$

where $X_L = \omega L$ and $X_C = \frac{1}{\omega C}$.

Variation of current with frequency: The current is $I = \frac{V}{Z}$. Plot I against frequency f. The curve is bell-shaped (a peak): it rises from a small value, reaches a sharp maximum at the resonant frequency $f_0 = \frac{1}{2\pi\sqrt{LC}}$ where Z is minimum, and then falls again on either side.

(iii) Conditions in a series LCR circuit:

(a) Impedance is minimum at resonance, when $X_L = X_C$, that is $\omega L = \frac{1}{\omega C}$, giving $\omega_0 = \frac{1}{\sqrt{LC}}$. Then $Z = R$ (the smallest possible value).

(b) Wattless current flows when the average power $P = V_{rms} I_{rms} \phi$ is zero, that is when the power factor $\phi = 0$ or $\phi = \frac{\pi}{2}$. This happens in a purely reactive circuit, for example a pure inductor or a pure capacitor (with $R = 0$), where the current is $\frac{\pi}{2}$ out of phase with the voltage.

Final answer. X = resistor, Y = inductor, Z = capacitor; $Z = \sqrt{(R^2 + (X_L - X_C)^2)}$; impedance is minimum at resonance ($X_L = X_C$) and wattless current flows when $\cos\phi = 0$.

← *Back to Q144*

Q145

(i) Describe the construction and working of a transformer and hence obtain the relation for $(\frac{V_s}{V_p})$ *in terms of number of turns of primary and secondary.*
(ii) Discuss four main causes of energy loss in a real transformer.

[2024 • Set 55-5-1]

SOLUTION

(i) Construction of a transformer: A transformer consists of two coils, the primary coil with N_p turns and the secondary coil with N_s turns, wound on a

common closed core of laminated soft iron. The core is laminated to reduce eddy current losses. The two coils are electrically insulated from each other but are magnetically coupled through the core.

Diagram: Draw a closed rectangular laminated iron core. Wind the primary coil (N_p turns) on one limb and connect it to the ac input V_p; wind the secondary coil (N_s turns) on the other limb and connect it to the output V_s. Show the common flux Φ looping around the core through both coils.

Working: A transformer works on the principle of mutual induction. When an alternating voltage is applied across the primary, an alternating current flows in it and produces a continuously changing magnetic flux Φ in the core. Since the core guides almost all this flux through the secondary, the same flux links each turn of both coils.

The emf induced in the secondary is

$$V_s = -N_s \frac{d\Phi}{dt}$$

The back emf across the primary is

$$V_p = -N_p \frac{d\Phi}{dt}$$

Dividing the two equations,

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

This is the required relation, where $\frac{N_s}{N_p}$ is the turns ratio.

(ii) Four main causes of energy loss in a real transformer:

1. Copper loss: Energy is lost as heat (I^2R) in the resistance of the copper windings. It is reduced by using thick copper wires of low resistance.

2. Eddy current loss: Changing flux induces circulating (eddy) currents in the core, which heat it up. It is reduced by using a laminated core.

3. Hysteresis loss: Energy is spent in repeatedly magnetising and demagnetising the core in each cycle. It is reduced by using a soft magnetic material such as silicon steel that has a small hysteresis loop.

4. Flux leakage: Not all the magnetic flux produced by the primary links the secondary; some leaks away through the air. It is reduced by good core design and winding the coils one over the other.

Final answer. V_s/V_p = N_s/N_p; the four losses are copper (I^2R) loss, eddy current loss, hysteresis loss and flux leakage.

← *Back to Q145*

Q146

With the help of a labelled diagram, explain the working of a step-up transformer. Give reasons to explain the following:
(i) The core of the transformer is laminated.
(ii) Thick copper wire is used in windings.

[2020 • Set 55-1-1]

SOLUTION

Working of a step-up transformer: A transformer works on the principle of mutual induction. It has two coils, the primary (N_p turns) and the secondary (N_s turns), wound on a common laminated soft-iron core.

Diagram: Draw a closed rectangular laminated iron core. Wind the primary coil of N_p turns on one limb connected to the ac input V_p, and the secondary coil of N_s turns ($\mathbf{N_s > N_p}$) on the other limb connected to the output V_s. Show the common flux Φ threading both coils through the core.

When an alternating voltage is applied across the primary, the alternating current sets up a continuously changing magnetic flux Φ in the core. This same flux links every turn of the secondary, inducing an alternating emf in it. Because both coils are linked by the same flux per turn,

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

For a step-up transformer $\mathbf{N_s > N_p}$, so $\mathbf{V_s > V_p}$ and the voltage is raised. By conservation of energy in an ideal transformer, $\mathbf{V_p I_p = V_s I_s}$, so the current is correspondingly reduced.

(i) The core is laminated: A changing magnetic flux in a solid iron core would induce large circulating eddy currents in it, which dissipate energy as heat. By making the core out of thin sheets (laminations) insulated from one another, the cross-section available for these loops is greatly reduced, so the eddy currents and the resulting I^2R heat losses become very small.

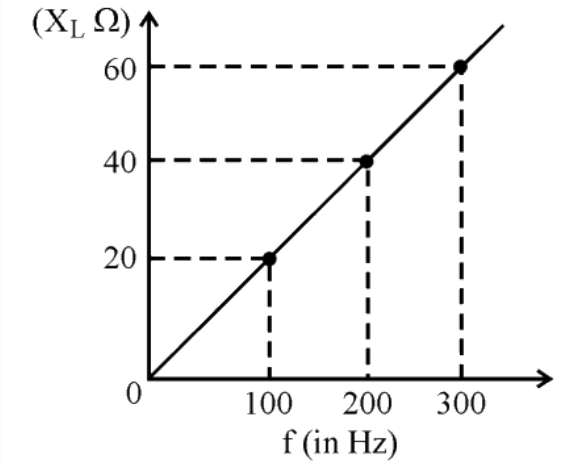
(ii) Thick copper wire is used in the windings: A thick wire has a larger cross-sectional area and hence a smaller resistance. This reduces the I^2R (Joule heating) loss in the windings, known as the copper loss, thereby improving the efficiency of the transformer.

Final answer. V_s/V_p = N_s/N_p > 1; lamination reduces eddy current losses, and thick copper wire reduces I^2R (copper) losses.

← *Back to Q146*

Q147

(a) Show that an ideal inductor does not dissipate power in an ac circuit.
(b) The variation of inductive reactance (X_L) of an inductor with the frequency (f) of the ac source of 100 *V and variable frequency is shown in the fig.*



(i) Calculate the self-inductance of the inductor.
(ii) When this inductor is used in series with a capacitor of unknown value and a resistor of 10 Ω *at* 300 s^{-1}*, maximum power dissipation occurs in the circuit. Calculate the capacitance of the capacitor.*
OR
(a) A conductor of length l is rotated about one of its ends at a constant angular speed

ω in a plane perpendicular to a uniform magnetic field B. Plot graphs to show variations of the emf induced across the ends of the conductor with (i) angular speed ω and (ii) length of the conductor l.

(b) Two concentric circular loops of radius 1 cm and 20 cm are placed coaxially.

(i) Find mutual inductance of the arrangement.

(ii) If the current passed through the outer loop is changed at a rate of 5 A/ms, find the emf induced in the inner loop. Assume the magnetic field on the inner loop to be uniform.

[2020 • Set 55-5-1]

SOLUTION

(a) Power dissipated by an ideal inductor: For an ideal inductor (zero resistance) carrying an alternating current

$$I = \mathrm{I}_0 \omega\, t$$

the voltage across it is

$$V = \mathrm{L}\frac{\mathrm{dI}}{\mathrm{dt}} = L\,\mathrm{I}_0 \omega\omega\, t = \mathrm{V}_0 \omega\, t = \mathrm{V}_0 \sin(\omega\, t + \frac{\pi}{2})$$

where $\mathbf{V_0 = \omega\, L\, I_0}$. Thus the voltage leads the current by $\frac{\pi}{2}$, so the phase difference is $\boldsymbol{\phi = \frac{\pi}{2}}$.

The average power over one cycle is

$$P = \mathrm{V}_{\mathrm{rms}} \mathrm{I}_{\mathrm{rms}} \phi$$

$$P = \mathrm{V}_{\mathrm{rms}} \mathrm{I}_{\mathrm{rms}} \cos\frac{\pi}{2} = \mathrm{V}_{\mathrm{rms}} \mathrm{I}_{\mathrm{rms}} \times 0 = 0$$

Hence an ideal inductor does not dissipate any power; it only stores energy in its magnetic field in one quarter cycle and returns it to the source in the next.

(b)(i) Self-inductance from the graph: The inductive reactance is $\mathbf{X_L = \omega\, L = 2\pi f\, L}$, so the graph of X_L against f is a straight line through the origin with slope $2\pi L$. Reading a point from the line, at $\boldsymbol{f = 300\ s^{-1}}$ the reactance is $\mathbf{X_L = 60\ \Omega}$. Hence

$$L = \frac{\mathrm{X}_L}{2\pi f} = \frac{60}{2\pi \times 300} = \frac{60}{600\pi} = \frac{1}{10\pi}\ H$$

$$L \approx 0.032\ H$$

(b)(ii) Capacitance for maximum power: Maximum power dissipation occurs at resonance, when $\mathbf{X_L = X_C}$, here at $\boldsymbol{f = 300\ s^{-1}}$. At resonance

$$\omega_0 L = \frac{1}{\omega_0 C} \Rightarrow C = \frac{1}{\omega_0{}^2 L}$$

with $\boldsymbol{\omega_0 = 2\pi f = 2\pi \times 300 = 600\pi\ rad/s}$. Substituting,

$$C = \frac{1}{(600\pi)^2 \times \frac{1}{10\pi}} = \frac{10\pi}{(600\pi)^2} = \frac{10\pi}{360000\pi^2} = \frac{10}{360000\pi}$$

$$C = \frac{1}{36000\pi} = 8.8 \times 10^{-6}\ F \approx 8.8\ \mu F$$

Hence $\boldsymbol{L \approx 0.032\ H}$ and $\boldsymbol{C \approx 8.8\ \mu F}$.

OR

(a) Emf of a rod rotating in a magnetic field: A conductor of length l rotating about one end with angular speed ω perpendicular to a uniform field B develops an emf

$$\varepsilon = \frac{1}{2} B \omega\ l^2$$

Graph (i) ε versus ω: With B and l fixed, $\boldsymbol{\varepsilon \propto \omega}$, so the graph is a straight line through the origin.

Graph (ii) ε versus l: With B and ω fixed, $\boldsymbol{\varepsilon \propto l^2}$, so the graph is an upward-opening parabola through the origin.

(b)(i) Mutual inductance of the coaxial loops: For a small inner loop of radius $\mathbf{r_1 = 1\ cm = 0.01\ m}$ placed at the centre of a large outer loop of radius $\mathbf{r_2 = 20\ cm = 0.20\ m}$, the field at the centre due to the outer loop is taken as uniform over the inner loop. The mutual inductance is

$$M = \frac{\mu_0 \pi\ r_1{}^2}{2\ r_2}$$

$$M = \frac{(4\pi \times 10^{-7})\pi \times (0.01)^2}{2 \times 0.20}$$

$$M = \frac{4\pi^2 \times 10^{-7} \times 10^{-4}}{0.40} = \frac{4\pi^2 \times 10^{-11}}{0.40}$$

$$M \approx 9.87\times 10^{-10}\ H$$

(b)(ii) Induced emf in the inner loop: The rate of change of current is $\frac{dI}{dt} = 5\ A/ms = \frac{5}{10^{-3}} = 5\times 10^{3}\ A/s$. Hence

$$\varepsilon = M\frac{dI}{dt} = 9.87\times 10^{-10}\times 5\times 10^{3}$$

$$\varepsilon \approx 4.93\times 10^{-6}\ V \approx 5\ \mu V$$

Hence the mutual inductance is about $9.87\times 10^{-10}\ H$ and the induced emf is about 5 μV.

TIP

When a graph supplies the data, always read a clear grid point and state it (here X_L = 60 Ω at f = 300 Hz); examiners award the reading mark only if the chosen point is written down.

Final answer. (a) P_avg = 0 ($\phi = \pi/2$); (b) L ≈ 0.032 H, C ≈ 8.8 μF; OR (a) $\varepsilon \propto \omega$ (line) and $\varepsilon \propto I^2$ (parabola); (b) M ≈ 9.87×10^{-10} H, ε ≈ 5 μV.

← *Back to Q147*

Q148

In a series LCR circuit connected across an ac source of variable frequency, obtain the expression for its impedance and draw a plot showing its variation with frequency of the ac source.

[2019 • Set 55-1-1]

SOLUTION

Impedance of a series LCR circuit: Consider a resistor R, an inductor L and a capacitor C connected in series across an ac source of variable frequency. The same current I flows through all three elements, so the current is taken as the reference phasor.

The voltage across the resistor, $V_R = IR$, is in phase with the current.

The voltage across the inductor, $\mathbf{V_L = IX_L}$, leads the current by $\frac{\pi}{2}$ (drawn upward).

The voltage across the capacitor, $\mathbf{V_C = IX_C}$, lags the current by $\frac{\pi}{2}$ (drawn downward).

Since V_L and V_C are in opposite directions, their resultant is $(V_L - V_C)$. The applied voltage is the phasor sum of V_R and $(V_L - V_C)$, which are at right angles:

$$V = \sqrt{V_R^2 + (V_L - V_C)^2}$$

$$V = \sqrt{(IR)^2 + (IX_L - IX_C)^2}$$

$$V = I\sqrt{R^2 + (X_L - X_C)^2}$$

The impedance is therefore

$$Z = \frac{V}{I} = \sqrt{R^2 + (X_L - X_C)^2}$$

Substituting $\mathbf{X_L = \omega L}$ and $\mathbf{X_C = \frac{1}{\omega C}}$,

$$Z = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

Variation of impedance with frequency: Plot Z on the vertical axis against the frequency f (or ω) on the horizontal axis. The curve is U-shaped. At very low frequency X_C is large, so Z is large; as f increases Z falls and reaches its minimum value $\boldsymbol{Z = R}$ at the resonant frequency $\mathbf{f_0 = \frac{1}{2\pi\sqrt{LC}}}$, where $\mathbf{X_L = X_C}$; beyond f_0, X_L dominates and Z rises again. Mark the bottom of the dip at $\boldsymbol{f = f_0}$ with $\boldsymbol{Z = R}$.

Final answer. $Z = \sqrt{(R^2 + (\omega L - 1/\omega C)^2)}$; the Z-versus-frequency graph is U-shaped with minimum $Z = R$ at $f_0 = 1/(2\pi\sqrt{(LC)})$.

← *Back to Q148*

Q149

What is the phase difference between the voltages across inductor and the capacitor at resonance in the LCR circuit?

[2019 • Set 55-1-1]

SOLUTION

In a series LCR circuit the same current flows through both the inductor and the capacitor. The voltage across the inductor, V_L, always leads the current by $\frac{\pi}{2}$, while the voltage across the capacitor, V_C, always lags the current by $\frac{\pi}{2}$.

Therefore the phase difference between V_L and V_C is

$$\Delta\phi = \frac{\pi}{2} - (-\frac{\pi}{2}) = \pi$$

This phase difference of π (that is 180°) between V_L and V_C holds at all frequencies, and so also at resonance. The two voltages are always exactly opposite in phase, which is why they tend to cancel each other.

Hence the phase difference between the voltages across the inductor and the capacitor at resonance is π (or 180°).

Final answer. The phase difference between V_L and V_C is π (180 °).

← *Back to Q149*

Q150

When an inductor is connected to a 200 *V dc voltage, a current of* 1 *A flows through it. When the same inductor is connected to a* 200 *V,* 50 Hz *ac source, only* 0.5 *A current flows. Explain, why? Also, calculate the self inductance of the inductor.*

[2019 • Set 55-1-1]

SOLUTION

Why the ac current is smaller: In a dc circuit the current is steady, so the rate of change of current is zero and the inductor offers no inductive reactance (X_L =

0). The inductor then behaves like an ordinary resistor of resistance R (the resistance of its coil), and the steady current is limited only by R.

In an ac circuit the current is continuously changing, so the inductor develops an inductive reactance $\mathbf{X_L = \omega L}$, which opposes the current in addition to R. The effective opposition is now the impedance $\mathbf{Z = \sqrt{R^2 + X_L^2}}$, which is greater than R. Hence for the same applied voltage the ac current is smaller than the dc current.

Calculation:

From the dc data, with no reactance,

$$R = \frac{V_{dc}}{I_{dc}} = \frac{200}{1} = 200\ \Omega$$

From the ac data, the impedance is

$$Z = \frac{V_{ac}}{I_{ac}} = \frac{200}{0.5} = 400\ \Omega$$

Now using $\mathbf{Z^2 = R^2 + (\omega L)^2}$,

$$(\omega L)^2 = Z^2 - R^2 = (400)^2 - (200)^2 = 160000 - 40000 = 120000$$

$$\omega L = \sqrt{120000} = 2003 = 346.4\ \Omega$$

The angular frequency is

$$\omega = 2\pi f = 2\pi \times 50 = 100\pi\ \text{rad/s}$$

Therefore

$$L = \frac{\omega L}{\omega} = \frac{2003}{100\pi} = \frac{23}{\pi}$$

$$L \approx \frac{3.464}{3.14} \approx 1.1\ H$$

Hence the self-inductance of the inductor is about **1.1 H**.

Final answer. The ac current is smaller because of the inductive reactance X_L = ωL; the self-inductance is L ≈ 1.1 H.

← Back to Q150

Q151

Draw the diagram of a device which is used to decrease high ac voltage into a low ac voltage and state its working principle. Write four sources of energy loss in this device.

[2019 • Set 55-1-1]

SOLUTION

The device used to decrease a high AC voltage into a low AC voltage is a **step-down transformer**.

Labelled diagram (described in words): Draw a rectangular closed loop representing the laminated soft-iron core. On the left limb wind the primary coil P with a large number of turns N_p, and connect its two ends to the AC input source. On the right limb wind the secondary coil S with a smaller number of turns N_s, where $\mathbf{N_s < N_p}$, and connect its two ends to the load (output). Label the input voltage V_p across the primary and output voltage V_s across the secondary. Shade the core to show it is a closed laminated soft-iron frame so that the magnetic flux Φ stays confined to the core and links both coils.

Working principle: A transformer works on the principle of **mutual induction**. When an alternating voltage is applied across the primary, the changing current produces a continuously changing magnetic flux in the soft-iron core. This same flux links the secondary coil, and by Faraday's law of electromagnetic induction an alternating emf is induced in the secondary.

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

Since $\mathbf{N_s < N_p}$, we get $\mathbf{V_s < V_p}$, so the output voltage is smaller than the input voltage, which is exactly the action of a step-down transformer.

Four sources of energy loss in a transformer:

1. Copper loss: heat produced as I^2R in the resistance of the primary and secondary windings.

2. Eddy current loss: induced circulating currents in the core dissipate energy as heat; minimised by using a laminated core.

3. Hysteresis loss: energy lost in repeatedly magnetising and demagnetising the core; minimised by using a soft magnetic material (soft iron) with a narrow hysteresis loop.

4. Flux leakage: some of the flux produced by the primary does not link the secondary, reducing the energy transferred.

Final answer. The device is a step-down transformer working on mutual induction, with copper, eddy current, hysteresis and flux-leakage losses.

← *Back to Q151*

Q152

A small town with a demand of 1200 kW *of electric power at* 220 *V is situated* 20 km *away from an electric plant generating power at* 440 *V. The resistance of the two wire line carrying power is* 0.5 Ω *per km. The town gets the power from the line through a* 4000-220 *V step-down transformer at a sub-station in the town. Estimate the line power loss in the form of heat.*

[2019 • Set 55-1-1]

SOLUTION

The power is transmitted at the high-voltage side of the step-down transformer in the town, that is at the line voltage of 4000 V.

The line current is found from $\boldsymbol{P = VI}$, where P is the power to be delivered and V is the line (transmission) voltage.

$$I = \frac{P}{V}$$

$$I = \frac{1200 \times 10^3\ W}{4000\ V}$$

$$I = 300\ A$$

The line consists of two wires, each 20 km long, with resistance 0.5 Ω per km. The total resistance of the line (both wires) is:

$$R = 2 \times 20 \text{ km} \times 0.5 \ \Omega \text{ km}^{-1}$$

$$R = 20 \ \Omega$$

The power lost as heat in the line is given by $\mathbf{P_{loss} = I^2R}$.

$$P_{loss} = (300)^2 \times 20$$

$$P_{loss} = 90000 \times 20$$

$$P_{loss} = 1.8 \times 10^6 \ W = 1800 \text{ kW}$$

Hence the line power loss in the form of heat is **1800 kW**. (This is why power is transmitted at high voltage: a higher voltage means a smaller current and so much smaller I^2R losses.)

COMMON MISTAKE

Always use the voltage on the line side of the transformer (4000 V), not the town's 220 V, when finding the transmission current. Using 220 V is the most common mistake here.

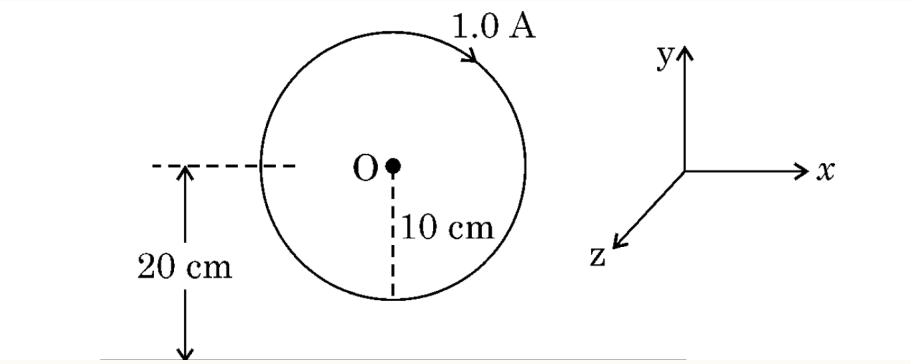
Final answer. The line power loss is $\mathbf{1.8 \times 10^6 \ W = 1800 \text{ kW}}$.

← Back to Q152

Q153

(a) What do you understand by 'sharpness of resonance' for a series LCR resonant circuit? How is it related with the quality factor 'Q' of the circuit? Using the graphs given in the diagram, explain the factors which affect it. For which graph is the resistance (R) minimum?
(b) A 2 μF capacitor, 100 Ω resistor and 8 H inductor are connected in series with an ac source. Find the frequency of the ac source for which the current drawn in the circuit is maximum.

If the peak value of emf of the source is 200 *V, calculate the (i) maximum current, and (ii) inductive and capacitive reactance of the circuit at resonance.*



OR

(a) Draw a schematic diagram of an ac generator. Explain its working and obtain the expression for the instantaneous value of the emf in terms of the magnetic field B, number of turns N of the coil of area A rotating with angular frequency ω. *Show how an alternating emf is generated by a loop of wire rotating in a magnetic field.*

(b) A circular coil of radius 10 cm *and* 20 *turns is rotated about its vertical diameter with angular speed of* 50 rad s^{-1} *in a uniform horizontal magnetic field of* 3.0×10^{-2} *T.*

(i) Calculate the maximum and average emf induced in the coil.

(ii) If the coil forms a closed loop of resistance 10 Ω, *calculate the maximum current in the coil and the average power loss due to Joule heating.*

[2019 • Set 55-4-1]

SOLUTION

(a) Sharpness of resonance: In a series LCR circuit the current is maximum at the resonant frequency ω_0 and falls on either side. The sharpness of resonance describes how quickly the current falls off as the frequency moves away from ω_0. It is measured by the ratio $\frac{\omega_0}{2\Delta\omega}$, where $2\Delta\omega$ is the bandwidth, the width of the resonance curve between the two frequencies at which the power drops to half its peak value.

$$\frac{\omega_0}{2\Delta\omega} = \frac{\omega_0 L}{R} = Q$$

Thus the sharpness of resonance is equal to the quality factor Q of the circuit. A larger Q means a narrower curve and a sharper resonance.

Factors affecting it: Since $Q = \frac{\omega_0 L}{R} = \frac{1}{R}\sqrt{\frac{L}{C}}$, the sharpness increases when R is decreased (and when L is large or C is small). In the given graphs, the tallest and narrowest curve corresponds to the smallest resistance, because small R gives a high peak current $\frac{V}{R}$ and a large Q. The curve with the minimum resistance R is therefore the sharpest (narrowest, tallest) curve.

(b) Given $\mathbf{C = 2\ \mu F = 2 \times 10^{-6}\ F}$, $\mathbf{R = 100\ \Omega}$, $\mathbf{L = 8\ H}$, peak emf $\mathbf{V_0 = 200\ V}$.

The current is maximum at resonance, where the resonant angular frequency is:

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$\omega_0 = \frac{1}{\sqrt{8 \times 2 \times 10^{-6}}}$$

$$\omega_0 = \frac{1}{\sqrt{16 \times 10^{-6}}} = \frac{1}{4 \times 10^{-3}}$$

$$\omega_0 = 250 \text{ rad s}^{-1}$$

$$f_0 = \frac{\omega_0}{2\pi} = \frac{250}{2\pi} \approx 39.8 \text{ Hz}$$

(i) At resonance $\mathbf{X_L = X_C}$, so the impedance is minimum and equals R. The maximum current is:

$$i_0 = \frac{V_0}{R} = \frac{200}{100} = 2\ A$$

(ii) The inductive and capacitive reactances at resonance are equal:

$$X_L = \omega_0 L = 250 \times 8 = 2000\ \Omega$$

$$X_C = \frac{1}{\omega_0 C} = \frac{1}{250 \times 2 \times 10^{-6}} = 2000\ \Omega$$

So $\mathbf{X_L = X_C = 2000\ \Omega}$ at resonance.

OR

(a) Schematic diagram of an AC generator (described in words): Draw a rectangular coil ABCD of N turns and area A placed between the curved pole-pieces N and S of a strong permanent magnet, so the field B is uniform and

horizontal. The coil is mounted on an axle and rotates about an axis perpendicular to $\vec{B}$ with angular speed ω. The two ends of the coil are joined to two separate slip rings R_1 and R_2, which press against two carbon brushes B_1 and B_2; the brushes connect to the external load R.

Working: As the coil rotates in the magnetic field, the magnetic flux linked with it changes continuously. By Faraday's law a changing flux induces an emf, and because the slip rings rotate with the coil, the induced emf reverses direction every half rotation, giving an alternating emf in the external circuit.

Derivation: When the normal to the coil makes an angle $\theta = \omega t$ with $\vec{B}$, the flux through one turn is:

$$\Phi = B\,A\,\omega\,t$$

For N turns the total flux linkage is $N\Phi = NBA\omega t$. By Faraday's law the induced emf is:

$$\varepsilon = -\frac{d(N\Phi)}{dt} = -NBA\frac{d}{dt}(\omega t)$$

$$\varepsilon = NBA\,\omega\omega\,t$$

$$\varepsilon = \varepsilon_0\,\omega\,t,\ \varepsilon_0 = NBA\omega$$

This shows that a loop of wire rotating in a magnetic field generates a sinusoidal alternating emf of peak value $\varepsilon_0 = NBA\omega$.

(b) Given radius $r = 10\ \text{cm} = 0.1\ m$, $N = 20$ turns, $\omega = 50\ \text{rad s}^{-1}$, $B = 3.0 \times 10^{-2}\ T$.

Area of the coil:

$$A = \pi r^2 = \pi\,(0.1)^2 = 3.14 \times 10^{-2}\ \text{m}^2$$

(i) Maximum (peak) emf:

$$\varepsilon_0 = NBA\omega$$

$$\varepsilon_0 = 20 \times (3.0 \times 10^{-2}) \times (3.14 \times 10^{-2}) \times 50$$

$$\varepsilon_0 \approx 0.94\ V$$

The average emf over one full cycle is zero, because the sinusoidal emf is positive for half the cycle and equally negative for the other half.

(ii) Maximum current in the closed loop of resistance $R = 10\ \Omega$:

$$I_0 = \frac{\varepsilon_0}{R} = \frac{0.94}{10} = 0.094\ A \approx 9.4\ \text{mA}$$

Average power dissipated as Joule heating:

$$P_{avg} = \frac{\varepsilon_0\, I_0}{2} = \frac{0.94 \times 0.094}{2}$$

$$P_{avg} \approx 0.044\ W = 4.4 \times 10^{-2}\ W$$

Final answer. Sharpness equals the quality factor Q; for the LCR circuit $f_0 \approx 39.8$ Hz, $i_0 = 2\ A$, $X_L = X_C = 2000\ \Omega$; for the rotating coil $\varepsilon_0 \approx 0.94\ V$, $I_0 \approx 9.4$ mA, $P_{avg} \approx 4.4 \times 10^{-2}\ W$.

← *Back to Q153*

Q154

A device X is connected across an ac source of voltage $V = V_0 \sin \omega t$. *The current through X is given as* $I = I_0 \sin(\omega t + \frac{\pi}{2})$.

(a) Identify the device X and write the expression for its reactance.
(b) Draw graphs showing variation of voltage and current with time over one cycle of ac, for X.
(c) How does the reactance of the device X vary with frequency of the ac ? Show this variation graphically.
(d) Draw the phasor diagram for the device X.

[2018]

SOLUTION

(a) The current is $I = I_0 \sin(\omega t + \frac{\pi}{2})$ while the voltage is $V = V_0\, \omega t$, so the current leads the voltage by $\frac{\pi}{2}$. In an AC circuit, the current leads the applied voltage by a quarter cycle only for a pure capacitor. Hence the device X is a **capacitor**.

Its reactance is the capacitive reactance:

$$X_C = \frac{1}{\omega C}$$

(b) Voltage and current versus time (described in words): Plot both as sine curves of the same period on one set of axes with time t along the horizontal axis. The voltage curve $V = V_0 \omega t$ starts from zero at $t = 0$ and rises. The current curve $I = I_0 \sin(\omega t + \frac{\pi}{2})$ is shifted a quarter cycle to the left, so it is already at its positive peak at $t = 0$. The current peak occurs a quarter period before the voltage peak, clearly showing the current leading the voltage by $\frac{\pi}{2}$.

(c) Variation of reactance with frequency: Since $X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$, the reactance is inversely proportional to the frequency.

$$X_C \propto \frac{1}{f}$$

Graph (described in words): Plot X_C on the vertical axis against frequency f on the horizontal axis. The curve is a rectangular hyperbola: X_C is very large at low frequencies and decreases, approaching zero as f becomes large.

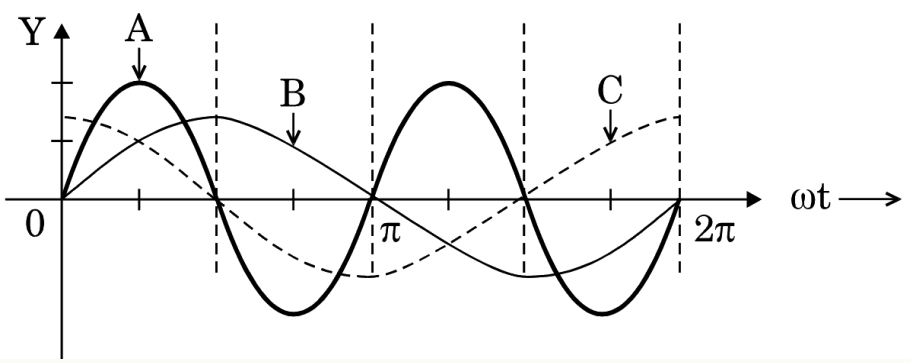
(d) Phasor diagram (described in words): Represent the voltage by a phasor $\vec{V}$ drawn along the horizontal reference direction. The current phasor $\vec{I}$ is drawn at 90° ahead of $\vec{V}$ (rotated anticlockwise), since the current leads the voltage by $\frac{\pi}{2}$ in a pure capacitor. Both phasors rotate anticlockwise with angular speed ω, with $\vec{I}$ always a quarter turn ahead of $\vec{V}$.

Final answer. The device X is a capacitor with reactance $X_C = \frac{1}{\omega C}$, which decreases hyperbolically with frequency, and its current leads the voltage by $\frac{\pi}{2}$.

← *Back to Q154*

Q155

A device X is connected to an ac source $V = V_0 \sin \omega t$. The variation of voltage, current and power in one cycle is shown in the following graph:



(a) Identify the device 'X'.
(b) Which of the curves A, B and C represent the voltage, current and the power consumed in the circuit? Justify your answer.
(c) How does its impedance vary with frequency of the ac source? Show graphically.
(d) Obtain an expression for the current in the circuit and its phase relation with ac voltage.

[2017]

SOLUTION

(a) The power curve in the graph is symmetric about the time axis, oscillating equally above and below zero so that its average over a cycle is zero. A device that consumes zero average power is a purely reactive element. Since the current leads the voltage by $\frac{\pi}{2}$ (the current curve reaching its peak a quarter cycle before the voltage curve), the device *X* is a **capacitor**.

(b) Identifying the curves A, B and C:

The voltage curve is $V = V_0 \omega t$, a pure sine starting from zero. The current curve $I = I_0 \sin(\omega t + \frac{\pi}{2})$ is also sinusoidal at the same frequency but shifted a quarter cycle ahead of the voltage. The power curve is the product $P = VI$; it oscillates at twice the source frequency (2ω) and is symmetric about the time axis with zero average value.

So, the curve that is a plain sine through the origin is the voltage; the sine curve leading it by a quarter cycle is the current; and the curve oscillating at double frequency with zero mean is the power. The justification is that for a capacitor the current leads the voltage by $\frac{\pi}{2}$, and the average power over a complete cycle is zero.

(c) Impedance versus frequency: For a pure capacitor the impedance is the capacitive reactance:

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

$$X_C \propto \frac{1}{f}$$

Graph (described in words): Plot X_C (vertical axis) against frequency f (horizontal axis). The curve is a rectangular hyperbola, large at low frequency and falling towards zero as f increases.

(d) Expression for current and phase relation: Let the applied voltage be $V = V_0 \omega t$. For a capacitor the charge is $q = CV = CV_0 \omega t$.

$$I = \frac{dq}{dt} = CV_0 \,\omega\omega\, t$$

$$I = \omega C V_0 \sin(\omega t + \frac{\pi}{2})$$

$$I = \frac{V_0}{X_C} \sin(\omega t + \frac{\pi}{2}),\ X_C = \frac{1}{\omega C}$$

This shows the current leads the applied voltage by a phase angle of $\frac{\pi}{2}$.

Final answer. The device X is a capacitor whose impedance $\mathbf{X_C = \frac{1}{\omega C}}$ falls hyperbolically with frequency, and $\mathbf{I = \frac{V_0}{X_C} \sin(\omega t + \frac{\pi}{2})}$ leads the voltage by $\frac{\pi}{2}$.

← Back to Q155

Q156

(i) Draw a labelled diagram of a step-down transformer. State the principle of its working.
(ii) Express the turn ratio in terms of voltages.
(iii) Find the ratio of primary and secondary currents in terms of turn ratio in an ideal transformer.
(iv) How much current is drawn by the primary of a transformer connected to 220 *V*

supply when it delivers power to a 110 *V* − 550 *W refrigerator?*

OR

(a) Explain the meaning of the term mutual inductance. Consider two concentric circular coils, one of radius r_1 *and the other of radius* r_2 $(\mathbf{r_1 < r_2})$ *placed coaxially with centres coinciding with each other. Obtain the expression for the mutual inductance of the arrangement.*

(b) A rectangular coil of area A, having number of turns N is rotated at f revolutions per second in a uniform magnetic field B, the field being perpendicular to the coil. Prove that the maximum emf induced in the coil is $2\pi f N B A$.

[2016]

SOLUTION

(i) Labelled diagram of a step-down transformer (described in words): Draw a closed rectangular laminated soft-iron core. On the left limb wind the primary coil *P* of N_p turns connected to the AC input V_p; on the right limb wind the secondary coil *S* of N_s turns connected to the output V_s. For a step-down transformer $\mathbf{N_p > N_s}$. Show the magnetic flux Φ circulating around the closed core, linking both coils.

Principle of working: A transformer works on the principle of **mutual induction**. The alternating current in the primary sets up a changing magnetic flux in the soft-iron core. This same changing flux links the secondary coil and, by Faraday's law, induces an alternating emf in it.

(ii) Turn ratio in terms of voltages: Because the same flux per turn links both coils, the induced emf in each coil is proportional to its number of turns. Therefore:

$$\frac{V_p}{V_s} = \frac{N_p}{N_s}$$

(iii) Ratio of currents: For an ideal (loss-free) transformer the input power equals the output power.

$$V_p I_p = V_s I_s$$

$$\frac{I_p}{I_s} = \frac{V_s}{V_p} = \frac{N_s}{N_p}$$

So the current ratio is the inverse of the turn ratio: when voltage is stepped down, the current is correspondingly stepped up.

(iv) The refrigerator is rated 110 V, 550 W, so this is the secondary side; the primary is connected to 220 V.

Secondary current:

$$I_s = \frac{P}{V_s} = \frac{550}{110} = 5\ A$$

For an ideal transformer, $\mathbf{V_p\, I_p = V_s\, I_s}$, so the primary current is:

$$I_p = \frac{V_s\, I_s}{V_p} = \frac{110 \times 5}{220}$$

$$I_p = 2.5\ A$$

Hence the primary draws a current of 2.5 A.

OR

(a) Mutual inductance: The mutual inductance of a pair of coils is the property by which a changing current in one coil induces an emf in the neighbouring coil. If a current I_1 in coil 1 produces a flux linkage $\mathbf{N_2\Phi_2 = M\, I_1}$ in coil 2, then M is the mutual inductance; the induced emf in coil 2 is $\boldsymbol{\varepsilon_2 = -M\frac{dI_1}{dt}}$.

Two concentric coplanar circular coils of radii r_1 and r_2 with $r_1 \ll r_2$ (described in words: the small coil of radius r_1 lying at the centre of the larger coil of radius r_2, both in the same plane). Pass a current I_2 through the larger (outer) coil. The magnetic field at its centre is:

$$B_2 = \frac{\mu_0\, I_2}{2\, r_2}$$

Because $r_1 \ll r_2$, this field is nearly uniform over the area of the small inner coil. The flux through the small coil is:

$$\Phi_1 = B_2 \times (\pi\, r_1^{\,2}) = \frac{\mu_0\, I_2}{2\, r_2} \times \pi\, r_1^{\,2}$$

By definition $\mathbf{\Phi_1 = M\, I_2}$, so:

$$M = \frac{\Phi_1}{I_2} = \frac{\mu_0\, \pi\, r_1^{\,2}}{2\, r_2}$$

(b) A rectangular coil of area A and N turns rotating at f revolutions per second in a uniform field B. When the coil has turned so that its normal makes angle θ

$= \omega t$ with $\vec{B}$, the flux linkage is:

$$N\Phi = NBA\omega\, t$$

$$\varepsilon = -\frac{d(N\Phi)}{dt} = NBA\ \omega\omega\, t$$

The emf is maximum when $\omega t = 1$:

$$\varepsilon_{max} = NBA\omega$$

Since the coil makes f revolutions per second, $\omega = 2\pi f$. Substituting:

$$\varepsilon_{max} = 2\pi f N B A$$

Hence the maximum induced emf is $\varepsilon_{max} = 2\pi f N B A$, as required.

Final answer. Turn ratio $\frac{V_p}{V_s} = \frac{N_p}{N_s}$, current ratio $\frac{I_p}{I_s} = \frac{N_s}{N_p}$, and the primary current is $2.5\ A$ (alternative: $M = \frac{\mu_0 \pi r_1^{\,2}}{2r_2}$ and $\varepsilon_{max} = 2\pi f N B A$).

← *Back to Q156*

Q157

In a series LCR *circuit connected to an a.c. source of voltage* $v = v_m \omega t$*, use phasor diagram to derive an expression for the current in the circuit. Hence obtain the expression for the power dissipated in the circuit. Show that power dissipated at resonance is maximum.*

[2016]

SOLUTION

Consider a series LCR circuit with resistance R, inductance L and capacitance C connected to $v = v_m \omega t$. Since the same current i flows through all three elements, take the current phasor $\vec{I}$ as the reference along the horizontal.

Phasor diagram (described in words): Draw $\vec{I}$ along the $+x$ direction. The voltage across the resistor, $\vec{V}_R$, is in phase with the current, so draw it along $+x$ with magnitude IR. The voltage across the inductor, $\vec{V}_L$, leads the current by $\frac{\pi}{2}$, so

draw it along $+y$ with magnitude IX_L. The voltage across the capacitor, $\vec{V}_C$, lags the current by $\frac{\pi}{2}$, so draw it along $-y$ with magnitude IX_C.

The net reactive voltage is $(V_L - V_C)$ along the y-axis. The resultant source voltage $\vec{V}$ is the vector sum of $\vec{V}_R$ and $(\vec{V}_L + \vec{V}_C)$:

$$V = \sqrt{V_R^2 + (V_L - V_C)^2}$$

$$V = \sqrt{(IR)^2 + (IX_L - IX_C)^2}$$

$$V = I\sqrt{R^2 + (X_L - X_C)^2}$$

Therefore the peak current is:

$$i_m = \frac{v_m}{Z},\ Z = \sqrt{R^2 + (X_L - X_C)^2}$$

The source voltage makes a phase angle ϕ with the current, where:

$$\phi = \frac{X_L - X_C}{R}$$

So the instantaneous current is:

$$i = i_m \sin(\omega t - \phi)$$

Power dissipated: The instantaneous power is $\boldsymbol{p = vi = v_m \omega t \cdot i_m \sin(\omega t - \phi)}$. Averaging over one full cycle gives the average power:

$$P_{avg} = V_{rms}\, I_{rms} \phi$$

$$P_{avg} = \frac{v_m}{\sqrt{2}} \cdot \frac{i_m}{\sqrt{2}} \cdot \phi = \frac{v_m\, i_m}{2} \phi$$

Here $\boldsymbol{\phi = \frac{R}{Z}}$ is the power factor. Substituting $\boldsymbol{i_m = \frac{v_m}{Z}}$:

$$P_{avg} = \frac{v_m}{2} \cdot \frac{v_m}{Z} \cdot \frac{R}{Z} = \frac{v_m^2\, R}{2\, Z^2}$$

Power at resonance: At resonance $\boldsymbol{X_L = X_C}$, so the impedance is minimum and equals R.

$$Z = \sqrt{R^2 + (X_L - X_C)^2} = R$$

Putting $Z = R$ in the average power:

$$P_{max} = \frac{v_m^2 R}{2R^2} = \frac{v_m^2}{2R}$$

Since Z has its least possible value R at resonance, $P_{avg} = \frac{v_m^2 R}{2Z^2}$ attains its greatest value here. Hence the power dissipated is **maximum at resonance** and equals $\frac{v_m^2}{2R}$.

Final answer. The current is $i = \frac{v_m}{Z}\sin(\omega t - \phi)$, the average power is $P_{avg} = \frac{v_m^2 R}{2Z^2}$, and it is maximum at resonance equalling $\frac{v_m^2}{2R}$.

← *Back to Q157*

Q158

(a) Show that in an a.c. circuit containing a pure inductor, the voltage is ahead of current by π/2 in phase.
(b) A horizontal straight wire of length L extending from east to west is falling with speed v at right angles to the horizontal component of Earth's magnetic field B.
(i) Write the expression for the instantaneous value of the e.m.f. induced in the wire.
(ii) What is the direction of the e.m.f.?
(iii) Which end of the wire is at the higher potential?

[2011]

SOLUTION

(a) Pure inductor: Let the alternating current through a pure inductor of inductance L be:

$$I = I_0 \omega t$$

As this current changes, a back-emf is induced across the inductor. By Kirchhoff's voltage law the applied voltage must balance this back-emf, so the voltage across the inductor is:

$$V = L\frac{dI}{dt}$$

$$V = \mathrm{L}\frac{d}{\mathrm{dt}}(\mathrm{I}_0\omega\, t)$$

$$V = L\,\mathrm{I}_0\,\omega\omega\, t$$

$$V = \omega\, L\,\mathrm{I}_0 \sin(\omega\, t + \frac{\pi}{2})$$

$$V = \mathrm{V}_0 \sin(\omega\, t + \frac{\pi}{2}),\ \mathrm{V}_0 = \omega\, L\,\mathrm{I}_0$$

Comparing $V = \mathbf{V_0 \sin(\omega\, t + \frac{\pi}{2})}$ with $I = \mathbf{I_0 \omega\, t}$, the voltage is ahead of the current by a phase of $\frac{\pi}{2}$. Hence in a pure inductor the voltage leads the current by $\frac{\pi}{2}$.

(b) A horizontal wire of length L lying east to west falls vertically with speed v, cutting across the horizontal component B of the Earth's magnetic field (which points towards geographic north).

(i) The motional emf induced is the product of the field, the length and the velocity, all mutually perpendicular:

$$\varepsilon = B\, v\, L$$

(ii) Direction of the emf: The force on a positive charge in the rod is $\vec{F} = q(\vec{v} \times \vec{B})$. With $\vec{v}$ pointing vertically downward and $\vec{B}$ pointing north, the cross product $\vec{v} \times \vec{B}$ points towards the east. So positive charges are pushed towards the east end, and the emf drives conventional current from the west end to the east end inside the rod.

(iii) Because positive charge accumulates at the east end, the east end of the wire is at the higher potential.

Final answer. In a pure inductor the voltage leads the current by $\frac{\pi}{2}$; the induced emf is $\varepsilon = BvL$, directed so that the east end of the wire is at the higher potential.

← *Back to Q158*

Q159

(i) With the help of a labelled diagram, describe briefly the underlying principle and working of a step up transformer.
(ii) Write any two sources of energy loss in a transformer.
(iii) A step up transformer converts a low input voltage into a high output voltage. Does it violate law of conservation of energy? Explain.

[2011 • Set 55-1-1]

SOLUTION

(i) Labelled diagram of a step-up transformer (described in words): Draw a closed rectangular laminated soft-iron core. On one limb wind the primary coil P of N_p turns connected to the AC input V_p; on the other limb wind the secondary coil S of N_s turns connected to the output V_s. For a step-up transformer $\mathbf{N_s > N_p}$. Show the flux Φ circulating in the closed core and linking both coils.

Underlying principle: A transformer works on the principle of **mutual induction**, where a changing current in one coil induces an emf in the neighbouring coil through a shared magnetic flux.

Working: When AC flows in the primary, it produces a continuously changing magnetic flux in the soft-iron core. This flux is guided by the core to link the secondary coil. By Faraday's law, the changing flux induces an alternating emf in the secondary.

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

Since $\mathbf{N_s > N_p}$, we get $\mathbf{V_s > V_p}$, so the output voltage is greater than the input voltage, which is the action of a step-up transformer.

(ii) Two sources of energy loss:

1. Copper loss: heat dissipated as I^2R in the resistance of the primary and secondary windings.

2. Iron loss in the core, comprising eddy current loss (reduced by laminating the core) and hysteresis loss (reduced by using soft iron).

(iii) Does a step-up transformer violate the law of conservation of energy? No, it does not. A transformer only transfers energy from the primary to the secondary; it does not generate energy. For an ideal transformer the input power equals the output power:

$$V_p I_p = V_s I_s$$

When the voltage is stepped up, that is $\mathbf{V_s > V_p}$, the secondary current is correspondingly reduced so that the product $V_s I_s$ still equals $V_p I_p$. Thus the increase in voltage is exactly compensated by a decrease in current, the power remains the same, and the law of conservation of energy is fully obeyed.

Final answer. A step-up transformer works by mutual induction with $\frac{\mathbf{V_s}}{\mathbf{V_p}} = \frac{\mathbf{N_s}}{\mathbf{N_p}}$; copper and iron losses occur, and it does not violate energy conservation since $\mathbf{V_p I_p = V_s I_s}$.

← *Back to Q159*

Q160

Derive an expression for the impedance of a series LCR circuit connected to an AC supply of variable frequency.
Plot a graph showing variation of current with the frequency of the applied voltage.
Explain briefly how the phenomenon of resonance in the circuit can be used in the tuning mechanism of a radio or a TV set.

[2011 • Set 55-1-1]

SOLUTION

Consider a series LCR circuit with resistance R, inductance L and capacitance C connected to an AC supply $\boldsymbol{v = v_m \omega t}$ of variable frequency. The same current i flows through all elements, so take the current phasor $\vec{I}$ as the reference.

Phasor diagram (described in words): Draw the current phasor $\vec{I}$ along the $+x$ axis. The resistor voltage $\mathbf{\vec{V}_R = IR}$ is in phase with $\vec{I}$ (along $+x$). The inductor

voltage $\vec{\mathbf{V}}_L = \mathbf{IX}_L$ leads the current by $\frac{\pi}{2}$ (along $+y$). The capacitor voltage $\vec{\mathbf{V}}_C = \mathbf{IX}_C$ lags the current by $\frac{\pi}{2}$ (along $-y$).

The net reactive voltage along the y-axis is $(V_L - V_C)$. The applied voltage $\vec{V}$ is the resultant of $\vec{V}_R$ and $(\vec{V}_L - \vec{V}_C)$:

$$V = \sqrt{V_R^2 + (V_L - V_C)^2}$$

$$V = \sqrt{(IR)^2 + (IX_L - IX_C)^2}$$

$$V = I\sqrt{R^2 + (X_L - X_C)^2}$$

By Ohm's law for AC, $V = IZ$, so the impedance is:

$$Z = \frac{V}{I} = \sqrt{R^2 + (X_L - X_C)^2}$$

Substituting $X_L = \omega L$ and $X_C = \frac{1}{\omega C}$:

$$Z = \sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}$$

Variation of current with frequency (graph described in words): Plot the current $I = \frac{v_m}{Z}$ on the vertical axis against frequency f on the horizontal axis. The current is small at low and high frequencies and rises to a sharp peak at the resonant frequency f_0, where $X_L = X_C$ and Z is minimum and equal to R. At this peak the current is maximum, $I_{max} = \frac{v_m}{R}$. The resonant frequency is:

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

Application to tuning a radio or TV: The receiving aerial picks up signals of many frequencies from different broadcasting stations, and these act as AC sources in a series LCR circuit whose capacitance can be varied by the tuning knob. By adjusting C, the natural resonant frequency $f_0 = \frac{1}{2\pi\sqrt{LC}}$ of the circuit is made equal to the broadcast frequency of the desired station. At that setting the impedance for the chosen station's signal is minimum and its current is maximum, so that station is received strongly. Signals from other stations, having frequencies far from f_0, meet a high impedance and produce only weak

currents. In this way the resonance of the LCR circuit lets the set respond selectively to just the one tuned station.

Final answer. The impedance is $Z = \sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}$, the current peaks at $f_0 = \frac{1}{2\pi\sqrt{LC}}$, and matching f_0 to a station's frequency tunes a radio or TV.

← Back to Q160

Q161

(a) An alternating voltage $v = v_m \sin \omega t$ *applied to a series LCR circuit drives a current given by* $i = i_m \sin(\omega t + \phi)$. *Deduce an expression for the average power dissipated over a cycle.*
(b) For circuits used for transporting electric power, a low power factor implies large power loss in transmission. Explain.
(c) Determine the current and quality factor at resonance for a series LCR circuit with $L = 1.00$ mH, $C = 1.00$ nF *and* $R = 100\ \Omega$ *connected to an a.c. source having peak voltage of 100 V.*

[2011 • Set 55-2-1]

SOLUTION

(a) In a series LCR circuit the applied voltage is $v = v_m \omega t$ and the resulting current is $i = i_m \sin(\omega t + \phi)$.

The instantaneous power supplied is $p = vi = v_m i_m \omega t \sin(\omega t + \phi)$.

Expanding $\sin(\omega t + \phi)$, $p = v_m i_m \omega t (\omega t\phi + \omega t\phi)$.

$p = v_m i_m (\sin^2\omega t\, \phi + \omega t \omega t\, \phi)$.

The average power over one full cycle is $P_{avg} = v_m i_m ({}^2\omega t\phi + \omega t \omega t\phi)$.

Over a complete cycle ${}^2\omega t = \frac{1}{2}$ and $\omega t \omega t = 0$.

Therefore $P_{avg} = \frac{1}{2} v_m i_m \phi$.

Writing $\frac{v_m}{\sqrt{2}} = V_{rms}$ and $\frac{i_m}{\sqrt{2}} = I_{rms}$, we get $P_{avg} = V_{rms}\, I_{rms}\, \phi$, where ϕ is the power factor.

(b) For transmission of power P at line voltage V with power factor ϕ, $P = VI\phi$, so the current required is $I = \frac{P}{V\phi}$.

The heat lost in the transmission line of resistance R is $P_{loss} = I^2R = \frac{P^2R}{V^2\cos^2\phi}$.

Hence for fixed P and V, $P_{loss} \propto \frac{1}{\cos^2\phi}$. A low power factor makes ϕ small, which makes the line current large and the power loss large, so a low power factor causes heavy transmission losses.

(c) Given $L = 1.00\ \text{mH} = 10^{-3}\ H$, $C = 1.00\ \text{nF} = 10^{-9}\ F$, $R = 100\ \Omega$ and peak voltage $v_m = 100\ V$.

Resonant angular frequency $\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{10^{-3} \times 10^{-9}}} = \frac{1}{\sqrt{10^{-12}}} = 10^6\ \text{rad/s}$.

At resonance $X_L = X_C$, so the impedance is minimum and equals R. The peak current is $i_m = \frac{v_m}{R} = \frac{100}{100} = 1.0\ A$.

The rms current is $I_{rms} = \frac{i_m}{\sqrt{2}} = \frac{1.0}{\sqrt{2}} \approx 0.71\ A$.

The quality factor is $Q = \frac{\omega_0 L}{R} = \frac{(10^6)(10^{-3})}{100} = \frac{10^3}{100} = 10$.

Hence at resonance the rms current is about $0.71\ A$ and the quality factor is 10.

> **TIP**
>
> At resonance always reduce Z to R first, then I = v_m/R; do not keep X_L and X_C in the impedance since they cancel.

Final answer. P_avg = V_rms I_rms cos phi; low PF raises loss as 1/cos^2 phi; at resonance I_rms ≈ 0.71 A and Q = 10.

← *Back to Q161*

Q162

A series LCR circuit is connected to an ac source having voltage $v = v_m \sin \omega t$. *Derive the expression for the instantaneous current I and its phase relationship to the applied voltage.*
Obtain the condition for resonance to occur. Define 'power factor'. State the conditions under which it is (i) maximum and (ii) minimum.

[2010]

SOLUTION

Consider a series LCR circuit with resistance R, inductance L and capacitance C connected to a source $v = v_m \omega t$. Let the current be $i = i_m \sin(\omega t - \phi)$.

Using a phasor diagram, the voltage across R is $V_R = i_m R$ (in phase with i), across L is $V_L = i_m X_L$ (leads i by 90°) and across C is $V_C = i_m X_C$ (lags i by 90°), where $X_L = \omega L$ and $X_C = \frac{1}{\omega C}$.

Since V_L and V_C are opposite in direction, the resultant of the reactive voltages is $(V_L - V_C)$, drawn perpendicular to V_R.

By the parallelogram (phasor) law, $v_m = \sqrt{V_R^2 + (V_L - V_C)^2} = i_m\sqrt{R^2 + (X_L - X_C)^2}$.

Therefore the peak current is $i_m = \frac{v_m}{\sqrt{R^2 + (X_L - X_C)^2}} = \frac{v_m}{Z}$, where $Z = \sqrt{R^2 + (X_L - X_C)^2}$ is the impedance.

The phase angle between voltage and current is given by $\phi = \frac{V_L - V_C}{V_R} = \frac{X_L - X_C}{R}$.

Hence the instantaneous current is $i = \frac{v_m}{\sqrt{R^2 + (X_L - X_C)^2}} \sin(\omega t - \phi)$.

If $X_L > X_C$, ϕ is positive and the current lags the voltage; if $X_L < X_C$, the current leads the voltage.

Condition for resonance: resonance occurs when $X_L = X_C$, i.e. $\omega L = \frac{1}{\omega C}$, which gives $\omega_0^2 = \frac{1}{LC}$, so $\omega_0 = \frac{1}{\sqrt{LC}}$ and resonant frequency $f_0 = \frac{1}{2\pi\sqrt{LC}}$. At resonance $Z = R$ (minimum) and the current is maximum.

Power factor: the power factor is $\phi = \frac{R}{Z}$, the cosine of the phase angle between the applied voltage and the current.

(i) It is maximum, $\phi = 1$, when the circuit is purely resistive or at resonance ($X_L = X_C$, so $Z = R$).

(ii) It is minimum, $\phi = 0$, when the circuit is purely reactive (a pure inductor or pure capacitor, $R = 0$), so that the phase angle is 90°.

Final answer. i = (v_m/Z) sin(ωt − ϕ) with tan ϕ = (X_L − X_C)/R; resonance at $\omega_0 = 1/\sqrt{(LC)}$; power factor cos ϕ = R/Z, max at resonance, min for a purely reactive circuit.

← *Back to Q162*

Q163

(a) Derive an expression for the average power consumed in a series LCR circuit connected to a.c. source in which the phase difference between the voltage and the current in the circuit is ϕ.
(b) Define the quality factor in an a.c. circuit. Why should the quality factor have high value in receiving circuits? Name the factors on which it depends.

[2009]

SOLUTION

(a) In a series LCR circuit let the applied voltage be $v = v_m \omega t$ and the current be $i = i_m \sin(\omega t + \phi)$, where ϕ is the phase difference between them.

The instantaneous power is $p = vi = v_m i_m \omega t \sin(\omega t + \phi)$.

Using $\sin(\omega t + \phi) = \omega t\phi + \omega t\phi$, $p = v_m i_m (\sin^2\omega t \phi + \omega t\omega t \phi)$.

Averaging over one full cycle and using $^2\omega t = \frac{1}{2}$ and $\omega t\omega t = 0$, $P_{avg} = \frac{1}{2} v_m i_m \phi$.

Substituting $V_{rms} = \frac{v_m}{\sqrt{2}}$ and $I_{rms} = \frac{i_m}{\sqrt{2}}$, we obtain $P_{avg} = V_{rms} I_{rms} \phi$, where ϕ is the power factor of the circuit.

(b) Quality factor: the quality factor Q of a series resonant circuit is the ratio of the resonant frequency to the bandwidth, and it measures the sharpness of resonance.

It is defined as $Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 C R} = \frac{1}{R}\sqrt{\frac{L}{C}}$, where $\omega_0 = \frac{1}{\sqrt{LC}}$.

Equivalently $Q = \frac{\omega_0}{\Delta\omega}$, where $\Delta\omega$ is the bandwidth between the half-power points.

A high Q value is desirable in receiving circuits (such as radio and TV tuners) because a high Q gives a sharp, narrow resonance curve. This makes the circuit highly selective, so it responds strongly to the desired frequency while rejecting nearby unwanted frequencies, giving better tuning.

The quality factor depends on the values of inductance L, capacitance C and resistance R of the circuit; a larger L or smaller R and C give a higher Q.

Final answer. P_avg = V_rms I_rms cos ϕ; Q = ω_0L/R = (1/R)√(L/C); a high Q gives sharp, selective tuning in receiving circuits and depends on L, C and R.

← *Back to Q163*

Q164

(a) Derive the relationship between the peak and the rms value of current in an a.c. circuit.
(b) Describe briefly, with the help of a labelled diagram, working of a step-up transformer.
(c) A step-up transformer converts a low voltage into high voltage. Does it not violate the principle of conservation of energy? Explain.

[2009]

SOLUTION

(a) The rms (root mean square) value of an alternating current is the square root of the average of the square of the current over one full cycle.

Let the current be $I = I_0 \omega t$, where I_0 is the peak value.

The mean of the square over one cycle is $I^2 = I_0^2 \sin^2\omega t = I_0^{22}\omega t$.

Over a full cycle $^2\omega\, t = \frac{1}{2}$, so $I^2 = \frac{I_0^2}{2}$.

Therefore $I_{rms} = \sqrt{I^2} = \sqrt{\frac{I_0^2}{2}} = \frac{I_0}{\sqrt{2}} = 0.707\, I_0$.

Hence the rms value of an alternating current is $\frac{1}{\sqrt{2}}$ times its peak value.

(b) Diagram (described): two coils wound on a common soft-iron laminated core. The primary coil P with N_p turns is connected to the a.c. input source, and the secondary coil S with N_s turns (where $N_s > N_p$ for a step-up transformer) is connected to the load/output. The core is shown as a closed rectangular laminated loop linking both coils.

Principle: a transformer works on the principle of mutual induction. An alternating current in the primary produces a continuously changing magnetic flux in the core, which links the secondary and induces an alternating emf in it.

Working: the changing flux Φ through each turn is the same for both coils. The induced emfs are $\varepsilon_p = -N_p \frac{d\Phi}{dt}$ and $\varepsilon_s = -N_s \frac{d\Phi}{dt}$.

Dividing, $\frac{V_s}{V_p} = \frac{N_s}{N_p} = k$ (transformation ratio). Since $N_s > N_p$, $V_s > V_p$, so the output voltage is stepped up.

(c) A step-up transformer does not violate conservation of energy. For an ideal transformer the input power equals the output power: $V_p I_p = V_s I_s$.

So $\frac{I_s}{I_p} = \frac{V_p}{V_s} = \frac{N_p}{N_s}$. When the voltage is increased, the current is decreased in the same ratio, so the product (power) stays the same. Thus the gain in voltage is exactly offset by the loss in current, and energy is conserved.

COMMON MISTAKE

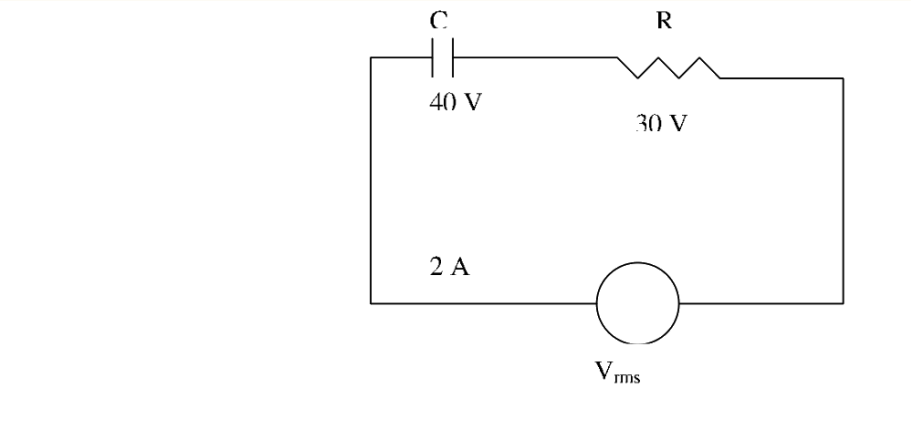
In part (c) examiners want the explicit relation V_p I_p = V_s I_s; just saying 'energy is conserved' without it loses the mark.

Final answer. I_rms = $I_0/\sqrt{2}$; a step-up transformer raises voltage by mutual induction with N_s > N_p, and since V_p I_p = V_s I_s the rise in voltage lowers the current, so energy is conserved.

← *Back to Q164*

Q165

State the condition for resonance to occur in a series LCR a.c circuit and derive an expression for the resonant frequency.
Draw a plot showing the variation of the peak current (i_m) with frequency of the a.c source used. Define the quality factor, Q of the circuit.
Calculate the (i) impedance, (ii) wattless current of the given a.c circuit.



[2008]

SOLUTION

Condition for resonance: in a series LCR circuit resonance occurs when the inductive reactance equals the capacitive reactance, $\mathbf{X_L = X_C}$.

Then $\omega L = \frac{1}{\omega C}$, which gives $\omega^2 = \frac{1}{LC}$.

So the resonant angular frequency is $\omega_0 = \frac{1}{\sqrt{LC}}$ and the resonant frequency is $f_0 = \frac{1}{2\pi\sqrt{LC}}$.

At resonance the impedance $Z = \sqrt{R^2 + (X_L - X_C)^2}$ becomes minimum and equals R, so the current in the circuit is maximum.

Plot of i_m versus frequency: the graph is a symmetric bell-shaped (resonance) curve. The horizontal axis is the source frequency f and the vertical axis is the peak current i_m. The current rises with frequency, reaches a sharp maximum $i_m^{max} = \frac{v_m}{R}$ at $f = f_0$, and falls off on either side. A smaller R gives a taller, sharper peak.

Quality factor: $Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 C R} = \frac{1}{R}\sqrt{\frac{L}{C}}$. It measures the sharpness of resonance, i.e. the ratio of the resonant frequency to the bandwidth $\Delta\omega$ between the half-power points, $Q = \frac{\omega_0}{\Delta\omega}$.

Numerical part (read the values from the given circuit figure). The general method is:

(i) Impedance: the impedance is the ratio of the rms voltage to the rms current, $Z = \frac{V_{rms}}{I_{rms}}$. (For a valid circuit Z can never be smaller than R, since $Z = \frac{}{\sqrt{R^2+(X_L-X_C)^2}} \geq R$; if a figure appears to give $Z < R$ the figure values have been misread, so take R, V and I directly from the printed diagram.)

(ii) Wattless current: it is the component of the current that is 90° out of phase with the applied voltage and therefore dissipates no power. It is $I_{wattless} = I_{rms}\phi$, where the phase angle satisfies $\phi = \frac{R}{Z}$, so $\phi = \sqrt{1-\frac{R^2}{Z^2}}$.

Substituting the impedance $Z = \frac{V_{rms}}{I_{rms}}$ and the resistance R read from the figure gives the numerical value of $I_{wattless} = I_{rms}\sqrt{1-\frac{R^2}{Z^2}}$.

Final answer. Resonance at X_L = X_C gives omega_0 = 1/sqrt(LC), f_0 = 1/(2 pi sqrt(LC)); the i_m vs f curve peaks at f_0 with i_m = v_m/R; Q = (1/R) sqrt(L/C); impedance Z = V_rms/I_rms (with Z >= R always), and the wattless current is I_wattless = I_rms sin phi = I_rms sqrt(1 - R^2/Z^2) using values read from the figure.

← *Back to Q165*

Q166

Draw a labeled circuit arrangement showing the windings of primary and secondary coil in a transformer. Explain the underlying principle and working of a step-up transformer. Write any two major sources of energy loss in this device.
How much current is drawn by the primary coil of a transformer which steps down 220 *V to* 22 *V to operate a device with an impedance of* 220 ohm*?*

[2008]

SOLUTION

Circuit arrangement (described): two coils are wound on a common closed rectangular soft-iron laminated core. The primary coil P with N_p turns is on one limb and connected to the a.c. input; the secondary coil S with N_s turns is on the other limb and connected to the output device. For a step-up transformer $\mathbf{N_s > N_p}$.

Principle: a transformer works on the principle of mutual induction. An alternating current in the primary sets up a continuously varying magnetic flux in the iron core, which links the secondary coil and induces an alternating emf in it.

Working: the same flux Φ passes through each turn of both coils. The induced emfs are $\boldsymbol{\varepsilon_p = -N_p \frac{d\Phi}{dt}}$ and $\boldsymbol{\varepsilon_s = -N_s \frac{d\Phi}{dt}}$.

Dividing gives $\boldsymbol{\frac{V_s}{V_p} = \frac{N_s}{N_p}}$. For a step-up transformer $\mathbf{N_s > N_p}$, so $\mathbf{V_s > V_p}$ and the output voltage is raised.

Two major sources of energy loss: (1) copper loss, the I^2R heating in the resistance of the windings; (2) iron (core) loss, made up of eddy-current loss (reduced by using laminated cores) and hysteresis loss (reduced by using soft iron). Flux leakage is another acceptable loss.

Numerical: given $\mathbf{V_p = 220\ V}$, $\mathbf{V_s = 22\ V}$, load impedance $\mathbf{Z = 220\ \Omega}$.

The secondary current is $\boldsymbol{I_s = \frac{V_s}{Z} = \frac{22}{220} = 0.1\ A}$.

For an ideal transformer $\mathbf{V_p I_p = V_s I_s}$, so $\boldsymbol{I_p = \frac{V_s I_s}{V_p} = \frac{22 \times 0.1}{220} = 0.01\ A}$.

Hence the primary draws $\mathbf{I_p = 0.01\ A = 10\ mA}$.

Final answer. A step-up transformer (N_s > N_p) works by mutual induction with V_s/V_p = N_s/N_p; the main losses are copper (I^2R) loss and iron (eddy-current + hysteresis) loss; the primary current is I_p = 0.01 A.

← *Back to Q166*

Q167

Explain the term 'inductive reactance'. Show graphically the variation of inductive reactance with frequency of the applied alternating voltage.
An a.c. voltage $E = E_0 \sin \omega t$ *is applied across a pure inductor of inductance L. Show mathematically that the current flowing through it lags behind the applied voltage by a phase angle of* $\pi/2$.

[2007]

SOLUTION

Inductive reactance: it is the opposition offered by a pure inductor to the flow of alternating current. It arises because the changing current induces a back emf that opposes the change. It is given by $X_L = \omega L = 2\pi f L$, and its SI unit is the ohm (Ω).

Graph (described): a plot of X_L against frequency f is a straight line passing through the origin, with positive slope $2\pi L$. As the frequency increases, the inductive reactance increases linearly.

Phase relation for a pure inductor: let $E = E_0 \omega t$ be applied across an inductor L.

The induced back emf is $L\frac{dI}{dt}$, and applying Kirchhoff's voltage law, $E_0 \omega t = L\frac{dI}{dt}$.

So $dI = \frac{E_0}{L}\omega t\, dt$.

Integrating, $I = \frac{E_0}{L}\left(-\frac{\omega t}{\omega}\right) = -\frac{E_0}{\omega L}\omega t$ (taking the constant of integration as zero).

Writing $-\omega t = \sin(\omega t - \frac{\pi}{2})$, we get $I = \frac{E_0}{\omega L}\sin(\omega t - \frac{\pi}{2}) = \frac{E_0}{X_L}\sin(\omega t - \frac{\pi}{2})$.

Comparing with $E = E_0 \omega t$, the current has the same frequency but its phase is $\frac{\pi}{2}$ less than that of the voltage.

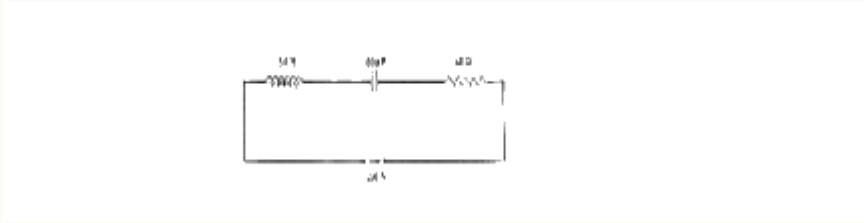
Hence the current through a pure inductor lags behind the applied voltage by a phase angle of $\frac{\pi}{2}$.

Final answer. X_L = ωL (in ohm) rises linearly with f; for E = E_0 sin ωt across a pure inductor, I = (E_0/X_L) sin(ωt − π/2), so the current lags the voltage by π/2.

← Back to Q167

Q168

The given circuit diagram shows a series LCR circuit connected to a variable frequency 230 V source:



(a) Determine the source frequency which drives the circuit in resonance.
(b) Obtain the impedance of the circuit and the amplitude of current at the resonating frequency.
(c) Determine the rms potential drops across the three elements of the circuit.
(d) How do you explain the observation that the algebraic sum of the voltages across the three elements obtained in (c) is greater than the supplied voltage?

[2006]

SOLUTION

For the standard NCERT data of this problem, $R = 40\ \Omega$, $L = 5.0\ H$, $C = 80\ \mu F = 8010^{-6}\ F$ and source voltage $V_{rms} = 230\ V$.

(a) At resonance $X_L = X_C$, so $\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{5.0\times 8010^{-6}}} = \frac{1}{\sqrt{410^{-4}}} = \frac{1}{0.02} = 50$ **rad/s**.

The source frequency is $f_0 = \frac{\omega_0}{2\pi} = \frac{50}{2\pi} \approx 7.96$ **Hz**.

(b) At resonance the inductive and capacitive reactances cancel, so the impedance is minimum and equals the resistance: $Z = R = 40\ \Omega$.

The rms current is $\mathbf{I_{rms}} = \frac{\mathbf{V_{rms}}}{Z} = \frac{230}{40} = 5.75\,A$, and the amplitude (peak) of the current is $\mathbf{i_m} = \sqrt{2}\,\mathbf{I_{rms}} = \sqrt{2} \times 5.75 \approx 8.13\,A$.

(c) The rms potential drops across the three elements:

Across R: $\mathbf{V_R} = \mathbf{I_{rms}R} = 5.75 \times 40 = 230\,V$.

Across L: $\mathbf{V_L} = \mathbf{I_{rms}}\,\omega_0\,L = 5.75 \times 50 \times 5.0 = 1437.5\,V$.

Across C: $\mathbf{V_C} = \frac{\mathbf{I_{rms}}}{\omega_0\,C} = \frac{5.75}{50 \times 8010^{-6}} = \frac{5.75}{410^{-3}} = 1437.5\,V$.

(d) The algebraic sum $V_R + V_L + V_C$ is much greater than 230 V, but this is not a contradiction. The three voltages are not in phase: V_L leads the current by 90° and V_C lags it by 90°, so V_L and V_C are exactly 180° out of phase and cancel each other ($\mathbf{V_L} = \mathbf{V_C}$ at resonance).

Hence the correct (phasor) sum is $V = \sqrt{\mathbf{V_R}^2 + (\mathbf{V_L} - \mathbf{V_C})^2} = \mathbf{V_R} = 230\,V$, which equals the supplied voltage. Voltages in a.c. circuits must be added as phasors, not algebraically.

CONCEPT

Always add a.c. voltages as phasors: V_L and V_C oppose each other, so the source voltage equals $\sqrt{(V_R^2 + (V_L - V_C)^2)}$, not the plain sum.

Final answer. $f_0 \approx 7.96$ Hz; at resonance Z = R = 40 Ω, I_rms = 5.75 A (i_m ≈ 8.13 A); V_R = 230 V, V_L = V_C = 1437.5 V; the large algebraic sum is explained because V_L and V_C are 180 ° out of phase and cancel, leaving the phasor sum equal to 230 V.

← *Back to Q168*

Q169

The primary coil of an ideal step-up transformer has 100 *turns and the transformation ratio is also* 100. *The input voltage and power are* 220 *V and* 1100 *W respectively. Calculate:*
(i) number of turns in the secondary
(ii) the current in the primary

(iii) voltage across the secondary
(iv) the current in the secondary
(v) power in the secondary

[2006]

SOLUTION

Given for the ideal step-up transformer: primary turns $N_p = 100$, transformation ratio $\frac{N_s}{N_p} = 100$, input voltage $V_p = 220\ V$ and input power $P_p = 1100\ W$.

(i) Number of turns in the secondary: from $\frac{N_s}{N_p} = 100$, $N_s = 100 \times N_p = 100 \times 100 = 10000$ turns.

(ii) Current in the primary: since power $P_p = V_p I_p$, $I_p = \frac{P_p}{V_p} = \frac{1100}{220} = 5\ A$.

(iii) Voltage across the secondary: using $\frac{V_s}{V_p} = \frac{N_s}{N_p}$, $V_s = 100 \times V_p = 100 \times 220 = 22000\ V = 22\ kV$.

(iv) Current in the secondary: for an ideal transformer $V_p I_p = V_s I_s$, so $I_s = \frac{V_p I_p}{V_s} = \frac{220 \times 5}{22000} = \frac{1100}{22000} = 0.05\ A$.

(v) Power in the secondary: $P_s = V_s I_s = 22000 \times 0.05 = 1100\ W$.

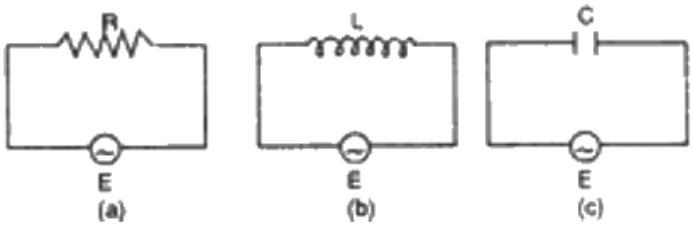
The secondary power equals the input power, confirming that for an ideal transformer no power is lost and energy is conserved.

Final answer. N_s = 10000 turns, I_p = 5 A, V_s = 22000 V, I_s = 0.05 A, and P_s = 1100 W (equal to the input power).

← *Back to Q169*

Q170

What is a choke coil? Why is it preferred to resistance in a.c. circuits?
In figures (a), (b) and (c) are shown three a.c. circuits with equal currents. If the frequency of e.m.f. be increased, then what will be the effect on the currents flowing in them? Explain with reason.



[2003]

SOLUTION

Choke coil: a choke coil is a coil of insulated copper wire wound on a soft-iron laminated core, having a large inductance L and a very small resistance. It is used to limit the current in an a.c. circuit without dissipating energy.

Why it is preferred to a resistance: a resistance also limits the current but it wastes energy as heat at the rate I^2R. A choke coil limits the current by its inductive reactance $\mathbf{X_L = \omega L}$ instead of resistance.

In a pure inductor the current and voltage differ in phase by $\frac{\pi}{2}$, so the power factor $\boldsymbol{\phi = \cos 90° = 0}$, and the average power $\boldsymbol{P = V_{rms} I_{rms} \phi = 0}$.

Thus a choke coil reduces the current with almost no power loss, whereas a resistance wastes power as heat. This is why a choke is preferred.

Effect of increasing the frequency in the three circuits (all carry equal currents initially):

Circuit (a), pure resistance R: the current is $\boldsymbol{I = \frac{V}{R}}$, which does not contain frequency. So on increasing f the current remains unchanged.

Circuit (b), pure inductor L: the reactance is $\mathbf{X_L = \omega L = 2\pi f L}$, which increases with frequency, so $\boldsymbol{I = \frac{V}{X_L}}$ decreases. On increasing f the current decreases.

Circuit (c), pure capacitor C: the reactance is $\mathbf{X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}}$, which decreases with frequency, so $\boldsymbol{I = \frac{V}{X_C} = 2\pi f C V}$ increases. On increasing f the current increases.

Hence with rising frequency the resistive current stays the same, the inductive current falls and the capacitive current rises.

Final answer. A choke is a high-inductance, low-resistance coil that limits a.c. current with zero power loss ($\cos\phi = 0$), unlike a resistor; on raising the frequency the resistive current is unchanged, the inductive current decreases and the capacitive current increases.

← Back to Q170